

Envision Dutch Bay 150 MW + 23.8 Bess Renewable Energy Wind Power Project - Kalpitiya



Environmental Impact Assessment (EIA)

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Project Proponent:

Envision Energy Co., Ltd

EIA Consultant:

Lanka Hydraulic Institute Ltd



Abbreviations

AC	Alternating Current
AEP	Annual Energy Production
AoI	Area of Influence
BCR	Benefit-Cost Ratio
BESS	Battery Energy Storage System
BOD	Biochemical Oxygen Demand
BRMS	Bar Reef Marine Sanctuary
BS	British Standards
CapEx	Capital expenditure
CBA	Cost-Benefit Analysis
CBD	Convention on Biological Diversity
CC&CRMD	Coast Conservation and Coastal Resource Management Department
CCA	Coast Conservation Act
CCTV	Closed-Circuit Television
CEA	Central Environmental Authority
CEB	Ceylon Electricity Board
CFD	Computational fluid dynamics
CH ₄	Methane
CO	Carbon Monoxide
CO ₂	Carbon Dioxide
COD	Chemical Oxygen Demand
CR	Critically Endangered
CRI	Coconut Research Institute
CRMP	Coastal Resources Management Project
CZ&CRMP 2024	Coastal Zone and Coastal Resource Management plan 2024-2029
dB(A)	A-weighted decibels
DC	Direct Current
DFAR	Department of Fisheries & Aquatic Resources
DO	Dissolved Oxygen
DSD	Divisional Secretariat Division



DWC	Department of Wildlife Conservation
EC	Electrical Conductivity
ECBA	Extended Cost Benefit Analysis
EDA	Energy Development Areas
EDRR	Early detection and rapid response
EIA	Environmental Impact Assessment
EIRR	Economic Internal Rate of Return
EMoP	Environmental Monitoring Plan
EMP	Environmental Management Plan
EN	Endangered
ENPV	Expected Net Present Value
EOI	Expression Of Interest
EPCC	Engineering, Procurement, and Construction Cost
EPL	Environmental Protection License
ERP	Emergency Response Plan
ESSD	Environmental Studies and Services Division
FGD	Focus Group Discussions
FID	Fisheries Inspector Divisions
FS	Feasibility Study
GHG	Greenhouse gas
GIIP	Good International Industry Practice
GIS	Geographic Information System
GND	Grama Niladhari Division
GPS	Global Positioning System
GRC	Grievance Redress Committee
GRM	Grievance Redress Mechanism
GSMB	Geological Survey and Mines Bureau
GSS	Grid Sub Station
GWh	Gigawatt-hours
HC	Hydrocarbon
IBA	Important Bird Area



IDAY	Inboard Single-day Boats
IEC	International Electro technical Commission
IEE	Initial Environmental Examination
ILO	International Labour Organization
IMO	International Maritime Organization
IMUL	Inboard Multi-day Boats
IRR	Internal Rate of Return
ISO	International Organization For Standardization
IUCN	International Union for Conservation of Nature
KII	Key Informant Interviews
KITRP	Kalpitiya Integrated Tourism Resort Program
kV	Kilovolts
L90	Background Noise Levels
LA	Land Acquisition
LC	Least Concerned
Leq	Residual Noise Levels
LKR	Sri Lankan Rupee
MARPOL	International Convention for The Prevention Of Pollution From Ships
MEA	Millennium Ecosystem Assessment
MEMS	Micro-Electro-Mechanical Systems
MEPA	Marine Environment Protection Authority
MOE	Ministry of Environment.
MSL	Mean Sea Level
MT	metric tons
MtCO ₂ e	million tonnes of CO ₂ equivalent
MTRB	Motorized Traditional Boats
MVA	Megavolt-amperes
MW	Megawatts
MWh	Megawatt Hour
NBRO	National Building Research Organization
NBSB	Non-Motorized Beach Seine Crafts



NCS	National Conservation Status)
NDC	Nationally Determined Contributions
NE	Northeast
NEA	National Environmental Act
NO2	Nitrogen Dioxide
NPV	Net Present Value
NREL	National Renewable Energy Laboratory
NT	Near threatened
NTRB	Non-motorized Traditional Boats
O&M	Operation and maintenance
OFRP	Out-board engine Fiberglass Reinforced Plastic Boats
OSRP	Oil Spill Response Plan
PAA	Project Approving Agencies
PAC	Project Approving Committee
PAP	Project-Affected Persons
PCF	Plant Capacity Factors
PGRC	Project-Level Grievance Redress Committee
PM10 & PM2.5	Particular Matter
PMU	Project Management Unit
PP	Project Proponent
PS	Pradeshiya Sabha
PUCSL	Public Utilities commission of Sri Lanka
RO	Reverse Osmosis
SCADA	Supervisory Control and Data Acquisition
SCR	Short Circuit Ratio
SDG	Sustainable Development Goals
SDS	Safety Data Sheets
SEP	Stakeholder Engagement Plan
SIA	Social Impact Assessment
SLPA	Sri Lanka Ports Authority
SLSEA	Sri Lanka Sustainable Energy Authority



SLTDA	Sri Lanka Tourism Development Authority
SMS	Short Message Service
SO ₂	Sulphur Dioxide
SPL	Sound Pressure Levels
STATCOM	Static Synchronous Compensator
SW	Southwest
TA/TB	Ambient temperature/ blade temperature or tip speed ratio
TEEB	The Economics of Ecosystems and Biodiversity
TEV	Total Economic Value
TOR	Terms of References
TS	Taxonomic status
TVOC	Total Volatile Organic Compounds
UDA	Urban Development Authority
URL	Uniform Resource Locator
USA	United States of America
USD	United States Dollar
VES	Visual Encounter Survey
VOC	Vehicle Operating Costs
VSL	Value of Statistical Life
VU	Vulnerable
WACC	Weighted Average Cost Of Capital
WHO	World Health Organization
WTGS	Wind turbine generator system



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EXECUTIVE SUMMARY

The Dutch Bay Onshore Wind Farm Project marks a significant step towards a sustainable energy future for Sri Lanka. This ambitious project aims to harness the region's abundant wind energy resources to generate 150 MW of clean electricity daily for the national grid. The proposed wind farm will feature 15 state-of-the-art 10 MW wind turbines, primarily situated within Dutch Bay area (GN Division Area).

Upon completion, the project is projected to deliver an Annual Energy Production (AEP) of 464,510 MWh¹, meeting approximately 2.7% of the nation's current electricity demand. This directly supports the government's policy goal of achieving 70% electricity generation from renewable energy sources by 2030. Beyond its energy contributions, the project is expected to inject significant economic vitality into the Kalpitiya, Dutch Bay area, a region currently identified as underdeveloped within Sri Lanka.

Environmental Assessment and Project Scope

This report details the environmental assessment of the proposed wind power project, encompassing its design, construction, and operational phases. The primary objective of this assessment is to ensure all project activities are environmentally sound and sustainable. This Environmental Impact Assessment (EIA) report was prepared to inform decisions based on a thorough understanding of the project's potential environmental consequences, aiming to protect, restore, and enhance the environment. The report's structure aligns with the Terms of Reference (TOR) provided by the Coast Conservation and Coastal Resource Management Department (CC&CRMD).

The proposed development activities have been examined in detail to identify the construction methods, technology, and other factors that could influence their environmental impact. These activities span from pre-construction to decommissioning, with each phase potentially generating distinct environmental, social, economic, and technical impacts. The following development activities are taken into consideration while preparing the potential impacts and corresponding risk ratings;

- Construction of wind towers (foundation, turbine transport and assembly)

¹ Ref: Dutch Bay Onshore Wind Farm Project Feasibility study report (p. 215)



- Ground formation (land clearing and grading) and compaction
- Construction of access roads.
- Grid infrastructure and other installations / infrastructure facilities
- Operation phase of the wind farm

During the construction phase, significant areas around each wind turbine location will be utilized. This will involve vegetation clearing, excavation, filling, compacting, levelling, and other related activities.

Baseline Data and Impact Assessment

This impact assessment was conducted based on internationally accepted principles, where impact significance is determined by the intensity of the impacts and the likelihood of their occurrence. The anticipated environmental impacts related to the proposed development activities have been thoroughly studied, considering the physical environment, marine biological and ecological resources, and socio-economic environment.

Relevant baseline data was collected from both primary and secondary sources to understand the existing environmental conditions of the site. All project construction activities will occur within the Dutch Bay Grama Niladari Division, covering approximately 600 hectares of land. The project area's landscape currently features a diverse mosaic of natural and semi-natural plant communities, including mangroves, salt-tolerant shrubs, grasses, and scattered terrestrial flora, particularly remnants of thorny shrub-mixed dry evergreen forest. These contribute significantly to local biodiversity and ecological balance.

However, the project footprint, including the wind turbines, proposed roads, and other facilities, will require a total area of approximately 13 hectares of land acquired through leasing or purchasing specific plots by the project proponent. This includes a hardstand area of about 0.7 hectares where turbine wind tower construction will take place. The proposed roads, mostly 20 feet wide, will span approximately 15 km.

Great care has been taken to strategically locate almost all wind turbines by providing the required coastal setbacks (60m from permanent vegetation, with a 25m reservation and 35m restricted area) and lagoon-side reservations. Only Wind Turbine 1 is situated very close to the coastal setback limits. Therefore, all turbine installations will adhere to a minimum coastal setback of 60 meters and lagoon reservation, fully complying with setback standards and planning regulations.



The planning and design stage of the project prioritized avoiding all potential environmental disturbances by implementing proper coastal setbacks and steering clear of lagoon reservation areas. This reflects the project developer's goal of promoting sustainable power generation in harmony with the environment.

The presence of surface water bodies (Odai/s) and groundwater sources within the area is seasonal. Potential impacts on natural drainage patterns and surface water areas could arise mainly from haphazard soil stockpiling and wastewater disposal. However, a severe impact due to construction activities is not anticipated.

A detailed marine biological and ecological survey was conducted within and around the project area to identify species composition and diversity. Furthermore, an avifauna study is underway to observe the behaviour of migratory birds in the surrounding area.

During the initial phase of the project, consultation meetings were held with public bodies and the local community. Their concerns and requests have been integrated into the development proposal for the site wherever practicable.

Mitigation Measures

The findings of this EIA Report provide comprehensive information on the nature and extent of potential environmental impacts arising from the proposed project activities. It also identifies suitable mitigation measures to minimize or avoid these impacts, ensuring compliance with environmental legislation and standards. An environmental monitoring plan has been developed to track identified parameters across categories, ensuring the effectiveness of proposed mitigation measures. By adopting a robust management plan, it is highly feasible to mitigate the project's possible significant impacts and prevent any residual environmental impacts.

Project Contribution and Commitment

Overall, the project is expected to supply 2.7% of the national current electricity demand via wind power, thereby significantly contributing to the government's policy goal of achieving electricity generation from renewable energy sources. From a regional development perspective, the project promises to inject economic vitality into the Kalpitiya, Dutch Bay area, a region currently recognized as underdeveloped within Sri Lanka.

The project promoter is fully committed to developing the project in a sustainable manner while thoroughly considering the EIA recommendations. However, the proponent requires EIA approval to proceed with the proposed development.



1 INTRODUCTION

1.1 Background of the proposed project

Sri Lanka's energy landscape is undergoing a transformation driven by the twin goals of achieving energy security and addressing climate change. With electricity demand rising at an average rate of 6% annually, there is an urgent need to diversify the country's energy mix. Renewable energy, particularly wind, offers a sustainable solution that aligns with the government's commitment to achieving 70% renewable energy by 2030.

The Dutch Bay Wind Farm project represents a pivotal advancement in Sri Lanka's renewable energy sector. Positioned on the Dutch Bay Peninsula, North West to Kalpitiya and across the Puttalam Lagoon, the project capitalizes on the region's exceptional wind resources to generate clean energy. With a proposed installed capacity of 150 megawatts (MW), this wind farm is set to reduce the country's dependence on fossil fuels, lower greenhouse gas emissions, and contribute to sustainable socio-economic development in the area. The Dutch Bay Peninsula is a prime location for wind energy development, with average wind speeds of 7–9 metres per second (m/s) and proximity to existing energy infrastructure. The project is strategically positioned to deliver environmental and socio-economic benefits, including the creation of local jobs and the promotion of eco-tourism in Kalpitiya.

The proposed Dutch Bay Wind Farm project with a capacity of 150 MW is estimated to cost US\$ 200.89 million. The proposed wind farm will comprise 15 state-of-the-art wind turbines of 10 MW capacity placed mainly within Dutch Bay area (GN Division Area). Upon implementation of the project, the project will deliver 464,510 MWh² of Annual Energy Production (AEP).

² Ref: Dutch Bay Onshore Wind Farm Project Feasibility study report (p. 215).



1.2 Objectives and justification of the proposed project

The Dutch Bay Onshore Wind Farm Project represents a significant step towards a sustainable energy future for Sri Lanka, aiming to harness the abundant local wind energy resources to generate 150 MW of clean electricity daily for the national grid. Upon completion, this crucial infrastructure project will not only bolster the nation's power supply but also catalyze local economic development while delivering substantial social and environmental advantages. The Dutch Bay Wind Farm, featuring 15 wind turbine generator systems (WTGSs) each with a capacity of 10,000 kW (10 MW), is strategically designed to achieve this goal.

This wind farm is a critical element in Sri Lanka's ambitious strategy to aggressively reduce carbon emissions and transition towards a resilient, low-carbon economy. It directly supports the government's policy framework goal of achieving 70% electricity generation from renewable energy sources by 2030. In the context of Sri Lanka's estimated total greenhouse gas (GHG) emissions of approximately 38.4 million tonnes of CO₂ equivalent (MtCO₂e) in 2023 (a modest increase from 35.6 MtCO₂e in 2021) and specifically around 20.5 million tonnes of CO₂ emissions in 2023 (a 2.69% increase from 2022³), the implementation of this project offers a tangible solution. It is projected to yield an Annual Energy Production (AEP) of 464.51 GWh, meeting approximately 2.7% of the nation's current electricity demand. This clean energy generation is anticipated to result in annual fuel savings of around 8 billion rupees and prevent the emission of approximately 0.3⁴ million metric tons of CO₂ / Greenhouse Gases (GHG) annually.

Unlike conventional power plants, wind power generation produces negligible carbon emissions during operation (with significantly lower emissions associated with manufacturing). This inherent characteristic directly contributes to mitigating climate change. Consequently, the proposed project aligns strongly with global climate action goals, including the United Nations Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action).

The North-western province has been identified as an optimal region for wind energy harvesting, a finding supported by the National Renewable Energy Laboratory (NREL) of the USA, which highlights the consistent winds over the flat coastal landscapes during both the Southwest (SW) and Northeast (NE) monsoons. The geographical location of the proposed Dutch Bay 150 MW +

³ Ref: countryeconomy.com.

⁴ Considering grid emission factor of 0.7044 kgCO₂/kWh



23.8 Bess Renewable Energy Wind Power Project strategically positions it to enhance national energy sovereignty through reliable and uninterrupted power generation. Furthermore, the selection of the Envision doubly fed Wind Turbine Generator (WTG), known for its higher power generation efficiency in low-speed wind regimes, offers an inherent advantage in bolstering grid stability.

Developing indigenous renewable energy resources like this wind farm directly reduces Sri Lanka's dependence on imported fossil fuels, thereby strengthening energy security and independence. Beyond environmental and climate benefits, such projects spur economic growth and create valuable employment opportunities. Onshore wind farms also promote sustainable land use by allowing for the coexistence of wind turbines with existing agricultural practices and livestock farming.

From a regional development perspective, the project promises to inject economic vitality into the Kalpitiya, Dutch Bay area – a region currently recognized as underdeveloped within Sri Lanka. This will be achieved through the creation of direct and indirect employment opportunities and the improvement of local infrastructure. The potential integration of eco-tourism initiatives further aligns with the government's objectives of fostering sustainable tourism in the region. Moreover, the project will contribute to rural economic revitalization by providing landowners with a stable source of income through land leases.

1.3 Objectives of the EIA report

According to the Gazette No 772/22 dated 24th June 1993, of the National Environmental Act, all renewable energy-based electricity generation stations exceeding 50 MW are categorized as Prescribed Projects which requires an EIA/ IEE prior to implementation. As the project approving agency, Coast Conservation and Coastal Resource Management Department (CC&CRMD) has decided to conduct an Environmental Impact Assessment (EIA) for the proposed Envision Dutch Bay 150 MW + 23.8 Bess Renewable Energy Wind Power Project –Kalpitiya and has issued a Terms of References (**Refer Annexure I**).

The objective of this report includes the environmental assessment of the proposed wind power project with respect to the design, construction and operation phases of it. The main purpose of this assessment is to ensure the project activities under consideration are environmentally sound and sustainable. This EIA report is prepared to facilitate the decisions that are based on understanding the environmental consequences of this Project and the decisions taken will



facilitate to protect, restore and enhance the environment. The report structure is developed based on the given Final TOR.

The EIA study for the Dutch Bay Wind Farm is designed/structured to achieve the following:

- Assimilate baseline data and information relating to physical, biological and social environment in and around the project site;
- Have a series of dialogues with the line agencies, local communities / households living in and around the project site as well as other stakeholders of the project to obtain their views;
- Assess the positive impacts and potential negative environmental and social impacts that might emanate during construction and operation phases of the project;
- Formulate necessary countermeasures against the potential adverse impacts to avoid, minimize or remedy the possible negative impacts due to project implementation and to enhance potential positive impacts;
- An integrated Impact–Mitigation–Monitoring matrix is included as a summary tool to track all identified risks, actions taken, and verification protocols.
- Identify practical approaches in implementation of mitigatory measures and monitoring throughout the implementation of the project and contribute to the overall process of project monitoring and auditing;
- Prepare an environmental monitoring plan (EMoP), so that the project proponent can take timely action to prevent negative environmental impacts before they become irreversible and also to achieve overall environmental sustainability of the project. Monitoring responsibilities are divided among the project developer, CC&CRMD, SLSEA, and third-party auditors. (Refer EMP Monitoring Details in **Table 6-1**)
- The EMP will be strengthened by a structured monitoring framework with monthly performance indicators on habitat regrowth, fauna sightings, and invasive species recurrence. Community Eco guards will support quarterly reviews.

1.4 Methodologies and technologies adopted in EIA preparation

To the maximum possible extent, environmental considerations have governed the planning and layout of the proposed development and incorporated it into the design process from its inception. This has allowed the early identification of any environmental issues, which may impose constraints on the design. This approach of “Mitigation by Design” has reduced the need



for a range of specific mitigation measures. This concept is adopted for all sectors including the physical environment, biological environment and socio-cultural environment of the area.

Several methods have been adopted in assessing the environmental impacts of the proposed project. The impacts arising from the construction of different structures and changing land use along the site (landside/ seaside) are evaluated with respect to the existing environment (baseline). Therefore, establishing the baseline condition at the project area is extremely important and baseline data gathering was done using following methodologies.

1.4.1 Background Information

Baseline data related to the physical and marine environment was primarily obtained through comprehensive literature surveys, site inspections and discussions with stakeholder agencies. Baseline data includes secondary (existing) data, which was obtained from published and unpublished sources (e.g. books, papers, consultancy reports, etc.), and provided by the local or national government and other stakeholders. The data gaps were identified and relevant investigations were carried out to obtain the site-specific data.

1.4.1.1 Reconnaissance Survey at the Site

Reconnaissance surveys were carried out by the consultants, to become familiarized with the project site, by understanding the project environmental settings, identification of stakeholders and developing a detailed plan for the field assessments. The results of the reconnaissance surveys are used for planning further investigations, laboratory testing and sampling strategy to acquire the required additional data.

1.4.1.2 Desk Studies

The EIA study was constructed upon the desk studies of available reports or any other supporting documents such as; Feasibility Study Reports, Preliminary Design Study Reports, EIA reports completed for other development projects in the area, project layout maps, etc. In addition, the land use and sensitive area maps were also referred in identification of any overlaps of the project area on environmentally sensitive areas. These studies were allowed, early identification of engineering, environmental and sociological issues in the area due to the proposed project.

1.4.1.3 Stakeholder and Community Meetings

Detailed discussions were held with the project proponent and community for the better understanding of their expectations. The information gathered through third party discussions



are communicated with the project proponent, and where applicable, with other relevant parties for finding the possibility of incorporating suitable measures during the initial stage of the project to minimize any impact.

1.4.2 Assessment of the Existing Environment

Field surveys, measurements and laboratory testing were carried out to fill the gaps of the available data/information and to establish the baseline condition of the project area.

1.4.2.1 Topographic Surveys

The site has been topographically surveyed along the waterfront strip between the inner edge of the coast road and the water line within the public beach area. All topographic surveys have been conducted using calibrated GPS equipment and the existing vertical datum pegs (Bench Marks) established by the Survey Department of Sri Lanka have been used to determine all values of elevation. Based on such data, the relevant Survey Plan has been prepared.

1.4.2.2 Noise and Air Quality Measurements

Noise Measurements

The baseline noise levels in the project area were measured to understand the existing levels of noise. Noise monitoring program was conducted on Dutch Bay Island for the ambient conditions on 31st March and 1st of April 2025 by Environmental Studies and Services Division (ESSD) of National Building Research Organisation (NBRO). Noise levels were measured at five (5) selected locations (Figure 1.1) where sensitive receptors are available in the project area during day and night (from 12:00 hr to 17:00 hr during the day time on 31st March 2025 and 21:00 hr on 31st March 2025 to 02:00 hr on 01st April 2025). The measurements were performed in accordance with the methods laid down in the BS:4142:2014+A1:2019 as National Environmental Noise Control Regulations stipulated under the Extraordinary Gazette No. 924/12 dated on Thursday, May 23 1996 of Sri Lanka. The existing noise levels are provided as Measured Residual Noise Levels L_{eq} and Background Noise Levels L_{90} . (**Refer Annexure III A**).

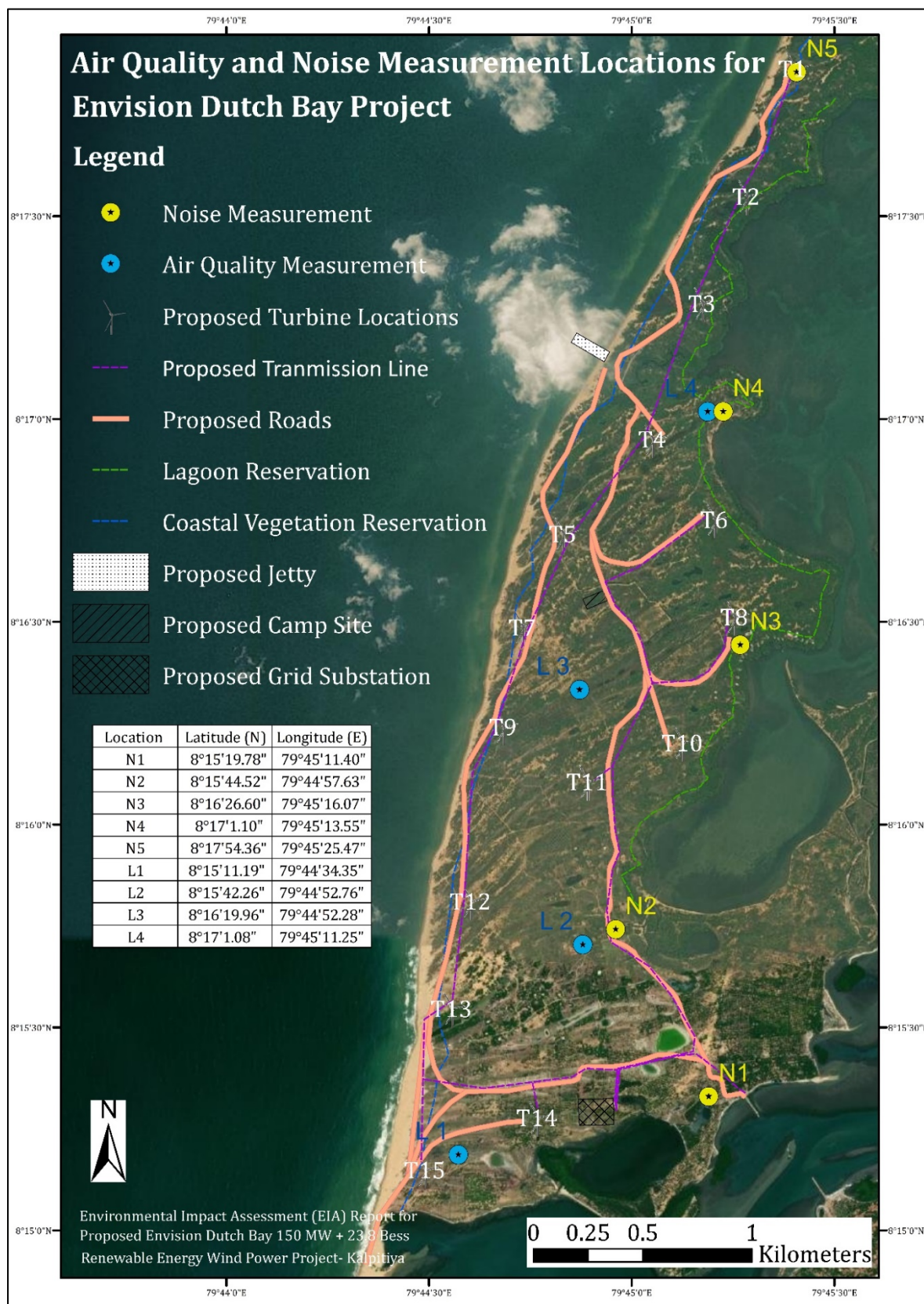


Figure 1.1: Noise and Air Quality Sampling Locations



Air Quality Measurements

Ambient air quality levels in the Dutch Bay Island were measured by the Environmental Studies and Services Division (ESSD) of National Building Research Organisation (NBRO) on 31st March and 1st of April 2025. Measurements were carried out at the four (4) selected locations (**Figure 1.1**) covering the entire Dutch Bay Island. Air samples were collected from selected locations on One (1) hour basis for the analysis of Carbon Monoxide (CO), Total Volatile Organic Compounds (TVOC) and Methane (CH₄), while eight (08) hour basis for the analysis of Carbon Dioxide (CO₂), Sulphur Dioxide (SO₂) and Nitrogen Dioxide (NO₂). Sampling duration for Particular Matter (PM₁₀ & PM_{2.5}) analysis were 24 hrs. **Refer Annexure III B.**

1.4.2.3 Water Quality Measurements

Water samples were collected at thirteen (10 inland and 3 sea) sensitive locations of the project site for water quality testing and establishing baseline parameters. Water quality testing was conducted on 04.04.2025 and 10.04.2025 for collected samples at 13 Locations (**Figure 1.2**) on Dutch bay Island by the team from Environmental Laboratory of University of Moratuwa. The Standard Methods for Examination of Water and Wastewater (2005) has been followed during the analysis.

The main objective of the study was to perform an analysis of water quality parameters of Dissolved Oxygen, Salinity, Turbidity, Chlorophyll content, Nitrates, Phosphates, Ammonia, Oil and Grease contents of the given samples. **Refer Annexure III C.**

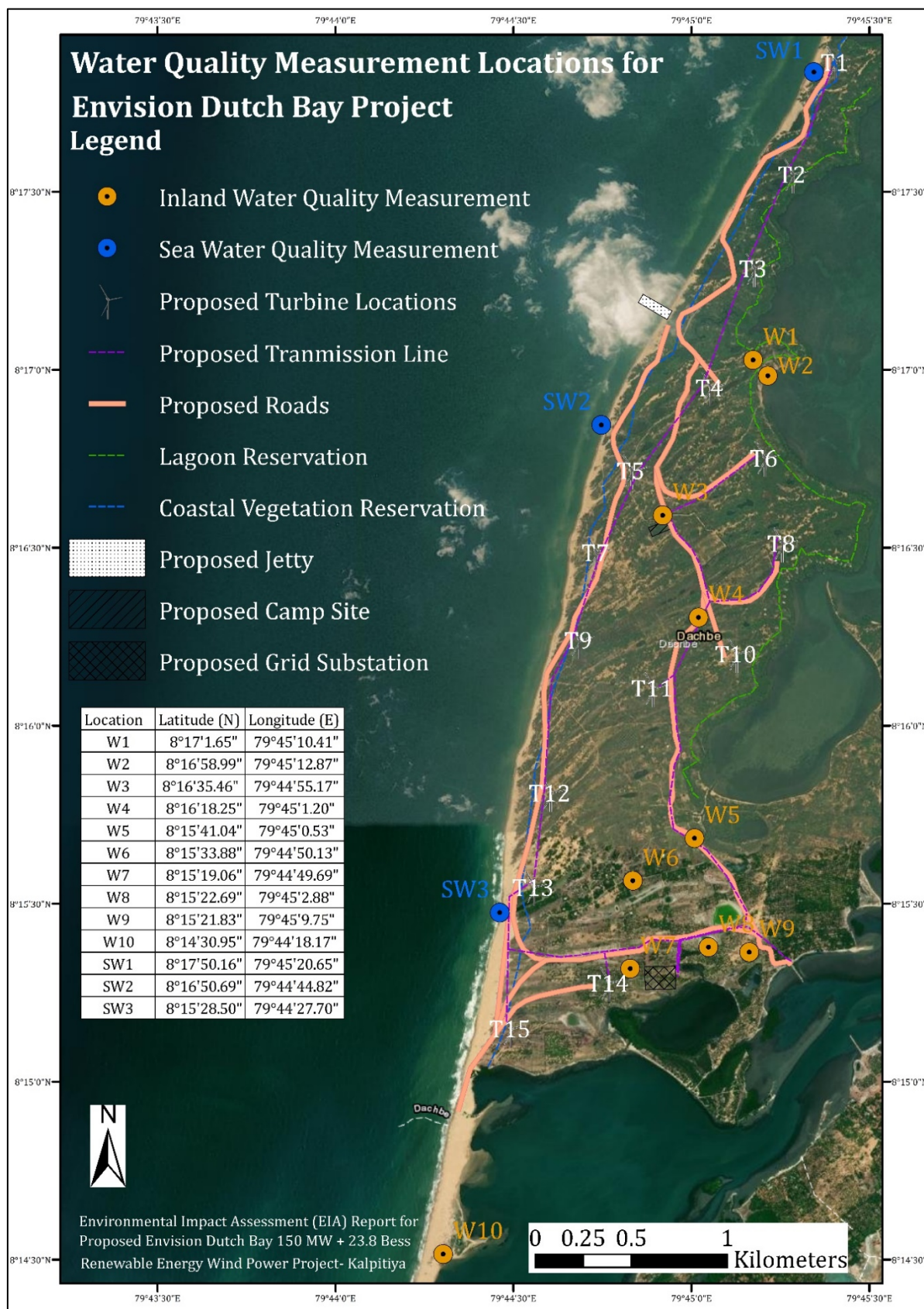


Figure 1.2: Water Sampling Locations



1.4.2.4 Ecological Surveys

An ecological study was performed within the project footprint encompassing all areas potentially impacted (both directly and indirectly) by the construction and operation of the proposed onshore wind farm project, as outlined in the preliminary invitation survey.

The followed methodology includes rapid assessment methodologies to gather ecological information from both:

- Primary sources, and
- Secondary sources.

The study area is ecologically complex, comprising a mosaic of marine and near-coastal habitats around the peninsula, lagoon, and shallow continental shelf. These habitats include sea, nearshore, coastal, lagoon, and terrestrial ecosystems.

An initial reconnaissance survey was conducted in September 2024 to gain a preliminary understanding of the ecological context. This three-day visit facilitated familiarization with the terrain and environmental conditions and provided a broad overview of the area's ecological diversity.

Subsequently, two more intensive field surveys (lasting three and four days, respectively) were conducted in February and March 2025. These surveys aimed to systematically document the area's biodiversity and assess its conservation significance. Traditional knowledge and experiences gathered through stakeholder consultations further enriched the field assessments.

In addition to fieldwork, secondary data sources were extensively consulted. These included scientific literature and reports, land use maps, aerial photographs, and geological and geographical maps. Historical data on flora, fauna, and climate were also reviewed to develop a comprehensive understanding of both natural and anthropogenic habitat diversity and prevailing environmental conditions.

1.4.2.5 Avifauna Study

The main objective of this assignment is to study avifaunal species, including nocturnal species, as required for the environmental study in the designated area within the given time frame. The study aimed to collect essential data and information on both terrestrial and aquatic avifauna, including nocturnal species, their habitats, migration corridors, feeding grounds, and more, within the study area. Further, it focused on the following objectives to meet these requirements:

- Compile a species inventory, including their conservation status.



- Record significant or critical habitats, feeding areas, migratory routes, etc., and map them using GPS locations.
- Identify potential impacts on these areas resulting from the construction and operation phases of the proposed project.
- Propose necessary mitigation measures to entirely avoid, alleviate, or compensate for potential adverse effects on these areas during the project's construction and operation phases.
- Suggest a monitoring mechanism as needed

Time Frame for Avifauna Study

Surveys were conducted from April 22nd to 25th, 2025, covering the end of the 2024/2025 migratory season. Four time slots, identified as periods of higher bird activity based on a preliminary survey, were selected for sampling.

- Slot 1 5:30 -7:30 h
- Slot 2 7:30 – 9:30h
- Slot 3 15:30 – 16:30 h
- Slot 4 16:30 – 18:30 h

Sampling Methods -Point counts

To assess avian use of vertical space and movement patterns, point counts were conducted at the locations of the proposed wind turbines. The dimensions for the point count survey were determined based on the specifications of the proposed turbines. Horizontally, the survey focused on the area swept by the rotor blades, within a 25-meter radius. Vertically, observations were categorized into three strata: Stratum 1 (ground level to 25 m) represented the area below the rotor; Stratum 2 (25–135 m), the rotor-swept zone, was considered the most critical; and Stratum 3 (above 135 m) encompassed the space above the rotor. During each point count, the species observed, number of individuals in each stratum, flight direction, and flight pattern were recorded.

Opportunistic observations were conducted throughout the study period to enhance the identification of avian diversity in the area (**Figure 1.3**). General surveys were carried out across all identified habitat types, including coastal grasslands, beaches, coconut plantations, shrublands, Odai (seasonal streams), mangroves, lagoons, and other water bodies. Night surveys were also undertaken to detect nocturnal species.



All available sources of existing background information on bird populations in the areas proposed for wind park development in Kalpitiya were compiled to provide an ornithological assessment of the potential impacts and associated risks to the avifauna.

An inventory of bird species inhabiting the proposed wind park sites on Kalpitiya Island was prepared, with particular attention given to the presence of endangered or restricted-range species.

Baseline environmental conditions on Kalpitiya Island were documented to serve as a reference for future monitoring and to assess actual impacts resulting from the proposed development. Diurnal and seasonal patterns in bird behaviour were also examined, along with environmental factors such as wind and rainfall that may influence these patterns.

Potential impacts of the proposed development were identified, and recommendations were formulated to minimize any harmful effects on the bird population.



Figure 1.3: Data collection on avifauna during the survey period

The avifaunal species were identified and classified using the latest published field guides. Threaten status of species were obtained from the National Red List Data for Avifauna (MOE, 2021).

1.4.2.6 Socio-Economic Assessments

The socio-economic assessment of the surrounding area of development sites was carried out using both published and unpublished data. Statistical data by the Commission of Fisheries was collected on fish production, fishing population and types of fishing crafts operated in the area. A structured questionnaire survey was conducted to identify the living status of fishing communities, fishery-related activities, income and expenditure patterns and seasonal



migrations. Structured interviews with fishers and semi-structured interviews with key informants of the fishing community were held near the project area. Baseline data on the social and economic status of the project area was established based on the gathered information.

1.4.3 Prediction of Environmental Impacts

Potential impacts of the project on the existing physical biological and social environment have been identified concerning the proposed development during the a.) construction and b.) operational phase. This will include identification and evaluation of the potential impacts from the project components and project related activities i.e., wind turbines, transmission lines, access roads and ancillary structures.

1.4.4 Formulation of the Mitigation Measures and Monitoring Activities

Possible mitigation measures were formulated to avoid or mitigate the identified significant impacts of the project. A comprehensive Environmental Monitoring Plan (EMoP) was formulated to ensure the long-term sustainability of the project while providing the less environmental impacts to the surrounding area.

1.5 Government policy, legal and administrative framework with reference to the Project

The successful implementation of the Dutch Bay Wind Farm depends on a robust understanding of the policy, legal, and administrative framework governing renewable energy projects in Sri Lanka. This Section outlines the national regulations, international agreements, and institutional structures that guide the project, ensuring compliance and sustainability.

1.5.1 National Policies and Regulations

Sri Lanka's legal framework emphasizes sustainable development and environmental conservation. The following national policies and regulations are relevant to the Dutch Bay Wind Farm project:

➤ **National Environmental Act (NEA) No 47 of 1980 and its Amendment Acts no 56 of 1988 and Act No 53 of 2000**

National Environmental (Amendment) Act No. 56 of 1988 of Sri Lanka stipulates EIA as a mandatory requirement for establishment of various large-scale developmental projects in Sri Lanka and assigns regulatory functions to the Central Environmental Authority (CEA). According



to Part IV C of the above-mentioned amendment act, and the regulations published in Gazette (Extra Ordinary) No 722/22 dated June 24, 1993 requires all "prescribed" development projects, which are wholly or partly located outside the Coastal Zone to be subjected to Environmental Impact Assessment.

The National Environmental Act further stipulates that approval for all prescribed projects shall have to be obtained from the appropriate project approving agencies (PAA) concerned or connected with such prescribed project. Since located within the coastal zone, Coast Conservation and Coastal Recourse Management Department (CC&CRMD) is the PAA in present case.

However, the regulatory instruments set out under NEA to control Water Pollution (Surface, Ground and Coastal) and air/noise/vibration emissions that needs to be considered during all construction activities are listed below:

- National Environmental [Protection & Quality] Regulations No. 01 of 2008: Issue of Environmental Protection Licence for Emission or Disposal of Waste and Management of Waste (Gazette Notification No. 1534/18 dated 01/02/2008) and activities for which a license is required (Gazette Notification No. 1533/16 dated 25.01.2008)
- National Environment [Ambient Air Quality] Regulation 1994. (Gazette Notification Number 1562/22 dated 15th August 2008.)
- National Environmental [Noise Control] Regulations No. 01 of 1996 (Gazette Notification No. 924/12 dated 23.05.1996)
- National Environmental [Air Emission, Fuel and Vehicle Importation Standards] Regulations – 2000 (Gazette No. 1137/35 dated 23rd June 2000)
- National Environmental [Air Emissions, Fuel & Vehicle Importation standards] Amended Regulation No. 01, 2003 (Gazette No. 1295/11 dated 30th June 2003)
- Amendment to National Environmental [Air Emissions, Fuel and Vehicle Importation Standards] Regulations No 01, 2003 (Gazette Notification Number 1557/14 dated 9th July 2008).
- Motor Traffic [Emission Control] Regulation of 1994 (Gazette Notification Number 817/6 dated 3rd May 1994)
- Regulation published under Section 23W [Prohibition of Use of Ozone Depleting Substances] (Gazette Notification Number 1309/20 dated 10th October 2003)
- Interim standards on vibration control being imposed as per CEA notice dated 4/12/2008, on the following vibration causing activities until such time the final vibration control standards are gazetted.



- Interim standards for vibration of the Operation of Machinery, Construction Activities and Vehicle Movements Traffic
- Interim standards on Air Blast over Pressure and Ground Vibration for Blasting Activities
- Standards for Inconvenience of the occupants of buildings

➤ **Coastal Conservation and Coastal Resource Management Act**

The Coast Conservation Act No. 57 of 1981 together with the Coast Conservation (Amendment) Act, No. 64 of 1988 and Coast Conservation (Amendment) Act, No. 49 of 2011 governs the Coastal Zone. This Zone comprises mainly “the area lying within a limit of three hundred meters (300m) landwards of the Mean High-Water line and a limit of two kilometres (2km) seawards of the Mean Low Water line”. Any person desiring to engage in a development activity within the Coastal Zone will be required to obtain a permit issued by the Department prior to commencing the activity.

The EIA process is part of the permit procedure mandated in Part II of the Coast Conservation Act (CCA). Section 16 of the CCA confers on the Director General of Coast Conservation and Coastal Recourse Management Department (CC&CRMD), the discretion to request a developer applying for a permit (to engage in a development activity within the Coastal Zone) to furnish an Initial Environmental Examination or Environmental Impact Assessment relating to the proposed development activity. The CCA does not however specify how and when this discretion should be exercised. The Coast Conservation and Coastal Recourse Management Department (CC&CRM) interprets this provision as requiring an EIA when the impacts of the project are likely to be significant.

However, the list of “prescribed projects” published in the Gazette Notification No. 772/22 dated 24.06.1993 under the National Environmental Act states that the CCA applies to those prescribed projects only if they are located wholly within the Coastal Zone.

Nevertheless, as the Coast Conservation Act states that it is established to regulate and control development activities within the Coastal Zone, the Coast Conservation and Coastal Recourse Management Department is the final authority to determine whether to permit a development activity in terms of the CCA, even though such activity may be required to go through the approval process laid down in the NEA.

Since the Dutch Bay Wind Farm, located at the Dutch bay Island adjacent to the Puttalam Lagoon, mostly within the coastal zone, must adhere to this act’s guidelines for coastal development.



Upon the EIA, process being concluded satisfactorily, as the Project Approving Agency, the CC&CRMD will under Part III Section 14 of the Coast Conservation & Resource Management Act No 57 of 1981, issue a Development Permit for the proposed development activities.

➤ **Sri Lanka Sustainable Energy Authority Act No 35 of 2007**

This act establishes the Sri Lanka Sustainable Energy Authority (SLSEA), which oversees the development and promotion of renewable energy projects. The act mandates that all renewable energy projects align with national energy policies and achieve compliance with technical and environmental standards.

➤ **Fauna and Flora Protection Ordinance No 2 of 1937 and its Amendment Acts Nos. 49 of 1993 and no 22 of 2009**

EIA provisions are also included in the Fauna and Flora (Amended) Act No. 49 of 1993. According to this Act, any development activity of any description what so ever proposed to be established within one mile from the boundary of any National Reserve, is required to be subject to EIA, and written approval should be obtained from the Director General, Department of Wild Life Conservation prior to implementation of such projects.

➤ **Forest Ordinance no 16 of 1907 and its Amendment Act no 65 of 2009**

The Forest Reserves gazetted under the provisions of the ordinance and all proposed reserves that are not gazetted under these provisions but selected for conservation based on biological and hydrological importance should be considered in implementation of this project.

➤ **Urban Development Authority Act 41 of 1978 as amended by Act No. 04 of 1982**

Kalpitiya was declared an Urban Development Area, leading to the Urban Development Authority (UDA) preparing a comprehensive development plan; hence, a Preliminary Planning Clearance from UDA is required in execution of this project.

➤ **Pradeshiya Sabha Act No. 15 of 1987**

Activities falling under this Project such as construction of Grid Sub Station (GSS), waste disposal etc. come under the purview of the Pradeshiya Sabha (PS) as local authority responsibility for health and sanitation and public thoroughfares and as such need its approval.

➤ **Fisheries and Aquatic Resources Act amended (2016)**

Provisions for Management, regulation, conservation and development of fisheries and aquatic resources which restricts detrimental or risk activities for aquatic fauna and flora.



➤ **Land Acquisition Act No 9 of 1950 and LA Regulations of 2008**

Acquisition of lands for public need is guided by the provisions and procedures outlined under the Land Acquisition Act.

➤ **Sri Lanka Electricity Act No. 20 of 2009 and 2013 (as amended)**

Regulates of the generation, transmission, distribution, supply and use of electricity in Sri Lanka with a view to enabling Sri Lanka to meet the increasing demands for electricity in the future.

➤ **The Marine Pollution Prevention Act, No. 35 of 2008.**

An Act to provide for the prevention, control and reduction of pollution in the Territorial Waters of Sri Lanka or any other Maritime Zone, its fore-shore and the Coastal Zone of Sri Lanka and for matters connected therewith or incidental thereto.

➤ **Felling of Trees Act No. 9 of 1951**

Jak, Breadfruit and female Palmyra trees can be felled only with a valid permit issued by an authorized officer according to the order of 1962 Felling of Trees (Control) Act which is applicable for this project on land clearance for project footprint.

➤ **Public Utilities Commission of Sri Lanka Act, No.35 of 2002**

The electricity (Application for Licenses and exemptions) regulations requiring License for power generation covered by the act.

➤ **National Energy Policy**

The primary objective of the energy policy is to ensure energy security through supplies that are cleaner, secure, economical and reliable, to provide convenient, affordable energy services to support socially equitable development of Sri Lanka.

1.5.2 Institutional and Administrative Framework

The Dutch Bay Wind Farm requires coordination among multiple governmental and non-governmental entities to ensure compliance and stakeholder alignment. Key institutions include:

➤ **Sri Lanka Sustainable Energy Authority (SLSEA)**

SLSEA is responsible for granting approvals for renewable energy projects and ensuring alignment with national energy policies. It serves as the primary regulatory body for the Dutch Bay Wind Farm.



➤ **Central Environmental Authority (CEA)**

The CEA oversees the EIA process, ensuring that all environmental assessments meet national standards. The authority is also responsible for monitoring the project's compliance with environmental regulations during its construction and operational phases.

➤ **Provincial Environmental Authority of the North Western Province (PEA-NWP)**

Commencement of any development project which affects the environment within the North Western province should obtain the approval by the Provincial Environmental Authority in accordance with the regulations set forth under the Provincial Environment Statute (PES) as the service provider and law enforcement institution for Natural Resources Management and Environmental Protection.

➤ **Coast Conservation & Coastal Resources Management Department (CC&CRMD)**

Screening of Preliminary Information of the proposed 150 MW + 23.8 Bess Renewable Energy wind power generation project led to the selection of Coast Conservation & Coastal Resource Management Department (CC & CRMD) as the Project Approving Agency (PAA) since most of the proposed wind turbines located within the coastal zone.

➤ **Department of Wildlife Conservation (DWC)**

Given the ecological sensitivity of the Dutch Bay area, the DWC's involvement is crucial for protecting wildlife and ensuring compliance with conservation regulations.

➤ **Local Government Authorities**

Local authorities play a key role in stakeholder engagement, land use approvals, and addressing community concerns. Collaboration with local government entities (such as Kalpitiya pradeshiya Sabha, North Western Province Environmental Authority) ensures smooth project implementation and community support.

➤ **Marine Environment Protection Authority (MEPA)**

Clearance was sought from the Authority entrusted with the sole responsibility of preventing, controlling, and managing the pollution of Sri Lanka's marine environment especially during construction period.

➤ **Forest Department**

Conservation, protection and management of forest and forest resources for control of felling and transport of timber regulated by the department. Provision for the prohibition, regulation felling of trees relevant to type and the Compensation as Dutch bay area consist of Palmyra trees.



➤ **Department of Fisheries and Aquatic Resources**

Formulating and implementing national policy on fisheries and aquatic resources development comes under the department with information of fishing community

➤ **Public Utilities commission of Sri Lanka (PUCSL)**

Regulate all the utilities within the purview of the Public Utilities Commission of Sri Lanka, to ensure safe, reliable and reasonably priced infrastructure services for existing as well as future consumers in the most equitable and sustainable manner.

1.5.3 International Agreements and Commitments

Sri Lanka is a signatory to several international agreements that influence the planning and execution of renewable energy projects. The Dutch Bay Wind Farm aligns with the following:

➤ **Paris Agreement**

As part of its Nationally Determined Contributions (NDCs) under the Paris Agreement, Sri Lanka has committed to reducing greenhouse gas emissions by 20% by 2030. The Dutch Bay Wind Farm contributes directly to this target by offsetting around 0.3⁵ million metric tons of CO₂ / Green House Gas (GHG) reduction annually.

➤ **Convention on Biological Diversity (CBD)**

The CBD obligates Sri Lanka to conserve biodiversity and sustainably manage its natural resources. The project adheres to CBD principles by minimizing impacts on sensitive ecosystems and implementing mitigation measures like habitat restoration.

➤ **Ramsar Convention**

The Puttalam Lagoon and associated wetlands are recognized for their ecological importance under the Ramsar Convention. However, the project footprint is located outside the protected areas under Ramsar Convention. Moreover, the project ensures no adverse impacts on these wetlands by employing best practices in site selection, construction, and monitoring.

⁵ Considering grid emission factor of 0.7044 kgCO₂/kWh



1.5.4 Compliance Process for the Dutch Bay Wind Farm

In order to fulfil the requirement of CC&CRMD act and the given TOR, a comprehensive EIA study has been conducted for the current project concerning the potential environmental impacts that could be arising during the construction and operation stages. The main purpose of this assessment is to ensure the project activities under consideration are environmentally sound and sustainable. This EIA report is prepared to facilitate the decisions that are based on understanding the environmental consequences of this Project and the decisions taken will facilitate to protect, restore and enhance the environment. The report structure is developed based on the EIA guidelines provided by the project-approving agency in order to comply the requirements under the National Environmental Act.

The Dutch Bay Wind Farm's adherence to Sri Lanka's legal and administrative framework, as well as international commitments, ensures its sustainability and regulatory compliance. By integrating robust policies and institutional coordination, the project serves as a model for renewable energy development in the region.

1.6 Compatibility with other development projects/ programmes/ plans in the area

1.6.1 Kalpitiya Development Plan 2021 – 2030

Development plan for the Kalpitiya Pradeshiya Sabha area was prepared by the Puttalam District Office, UDA accordance with the provisions of the Section 08 A (1) and (2) of the Urban Development Authority (Amendment) Act No. 04 of 1982 brought as an amendment to the Urban Development Authority Act No. 41 of 1978 and duly approved by the Parliament of the Democratic Socialist Republic of Sri Lanka.

Proposed project area lies within the Dutch bay Islands where the development plan envisage to develop for tourism, construction of a bridge from Thoradiya to Dutch Bay Island to facilitate the transport of people and goods and implementation of fisheries housing scheme at Mohonthuwarama.

The Kalpitiya council area was identified under Nine Development zones and Dutch Bay comes under Development Zone 7- Beach Based Tourism Development Zone. Strategy 03 of Proposed Tourism Development Plan-Tourism Island corridor covers the Dutch Bay area. Proposed project will enhance the tourism attractions of the islands. Dutch bay area however has not been identified under Wind Power Generation plan of Kalpitiya Development Plan 2021 – 2030.



1.6.2 Kalpitiya Integrated Tourism Resort Program (KITRP)

Dutch Bay Island is one of the 14 islands in the Kalpitiya peninsula have been identified for the Kalpitiya Integrated Tourism Resort Program (KITRP) which is one of the most extensive projects undertaken in the country to generate environmentally friendly sustainable tourism. Accordingly, Dutch Bay Resorts have been constructed on the Eastern edge on the Island, which had commenced its operations, yet temporary closed at the moment. Footprint of the proposed wind power generation project does not fall within the resort property and it would be a scenic value addition to the tourism with improved road infrastructure.

1.6.3 Renewable Energy Resource Development Plan 2021-2026

Renewable Energy Resource Development Plan Prioritize the renewable energy development sites in the country. First priority was given to wind power development and potential land-use types considered for it are sparsely used croplands, homesteads, open forest, grassland, scrub land, barren land, palmyrah, coconut and sand areas. Areas where wind speed above 7 m/s of at 100m height for wind resource were considered having elevations from sea level only up to 270 m with low gradients of slope less than 30%.

Proposed project area positioned outside the earmarked sites for planned energy park project in Puttalam for wind power generation. Yet proposed wind power generation project location in Dutch Bay area is compatible with the criteria imposed on national Renewable Energy Resource Development Plan 2021-2026.

1.6.4 Sri Lanka Revised Coastal Zone and Coastal Resource Management Plan 2024 - 2029

Coastal Zone and Coastal Resource Management plan 2024-2029 (CZ&CRMP 2024) has been prepared by the Coast Conservation and Coastal Resource Management Department (CC&CRMD) in conformity with the legal provisions of the CC&CRM Amendment Act No.49 of 2011. Updated CZ&CRMP 2024 has broadened the scope of the previous management plan by incorporating climate change resilience aspects and streamlining with the international conventions, protocols, targets, and goals.

1.6.5 National Physical Planning Policy & the Plan – 2048

Dutch Bay Island in Kalpitiya has been identified as a potential land located within the zone under Coastal Belt Wind for energy and power development in Physical Infrastructure Development



Strategy, National Physical Planning Policy & the Plan – 2048. However, the plan is still not gazetted.

1.6.6 Preliminary Clearances, Approvals needed for the project from state agencies and any conditions laid down by the state agencies

The details of the obtained preliminary clearances (**Table 1-1**), pending clearances and the correspondences / request letters sent to each and every authority are provided in **Annexure III D**.



Table 1-1: Approvals Needed

No	Approving Agency	Approval status with specific conditions
11	Department of Wildlife Conservation	Request letter for preliminary clearances was sent. Pending approval/ responses The extents of the Bar Reef Sanctuary and permissible set backs are to be further discussed with the Department of Wildlife Conservation. Another request letter was sent in August 2025 after the TEC meeting.
2	Department of Forest	Request letter for preliminary clearances was sent. Pending approval /Responses
3	Central Environment Authority	
4	North Western Province Environmental Authority	
5	Department of Fisheries & Aquatic Resources	
6	National Aquaculture Development Authority	Received the preliminary clearance letter dated 27 -05-2025
7	Marine Environment Protection Authority (MEPA)	
8	Sri Lanka Sustainable Energy Authority	
9	Ceylon Electricity Board	
10	Sri Lanka Tourism Development Authority	Request letter for preliminary clearances was sent. Pending approval /Responses
11	Palmyra Development Board	
12	Urban Development Authority	Request letter for preliminary clearances was sent. Pending approval /responses Potential future development zones for the kite surfing area to be further discussed with UDA and obtain the clearances.
13	Road Development Authority	Request letter for preliminary clearances was sent. Pending approval /responses. Another letter sent in August 2025 after the TEC meeting, Annexure III D
14	Department of Archeology	Request letter for preliminary clearances was sent. Pending approval /responses. Another letter sent in August 2025 after the TEC meeting, Annexure III D
15	Land Commissioner General's Department	
16	Divisional Secretary – Kalpitiya and Pradeshiya Sabawa-Kalpitiya	Has granted clearance through the letter dated 01 st Sep 2025. Provided in Annexure III D
17	Ministry of Defense	Request letter for preliminary clearances was sent.



2 DESCRIPTION OF THE PROPOSED PROJECT AND REASONABLE ALTERNATIVES

The Dutch Bay Wind Farm is a landmark renewable energy initiative designed to harness the region's exceptional wind resources. This chapter provides a detailed description of the project (location, extents etc.), covering its technical specifications, infrastructure components, and design considerations. It also outlines alternative configurations evaluated during the planning phase to ensure the project's optimal alignment with environmental and socio-economic objectives.

2.1 Description of the Proposed Project

The proposed **Envision Dutch Bay 150 MW +23.8 Bess Renewable Energy Wind Power Project** (Figure 2.1) is situated on the Dutch Bay Peninsula, North West of Kalpitiya, across the Puttalam Lagoon. Spanning approximately 600 hectares, the project will generate 150 megawatts (MW) of clean energy, contributing significantly to Sri Lanka's renewable energy capacity.

The wind farm comprises 15 (Figure 2.2) state-of-the-art turbines, each with a capacity of approximately 10 MW. These turbines are strategically positioned to maximize energy output while minimizing environmental disruption. The project also includes essential infrastructure such as access roads, a booster station, and grid integration facilities/ Grid substation.

The anticipated project lifecycle includes a two-year / 24 months construction phase followed by an operational period of 20 years. Comprehensive decommissioning plans are included to restore the site to its natural state at the end of the project's life.

The Project falls within the Duechbay GN division of Kalpitiya Pradesheeya Sabha (Figure 2.3).

Province	: Northern Western Province
District	: Puttalam
DS Division	: Kalpitiya
Pradesheeya Sabha	: Kalpitiya
GN Divisions	: Duechbay (Dutch Bay –Original name)

None of the turbines/ footprints is located within any of the protected areas or reservations. Some of the turbine locations also fall at the borders of the mangrove environment and around 60m offset from the coastal vegetation line.



Figure 2.1: Study Area for Envision Dutch Bay Wind Farm Project

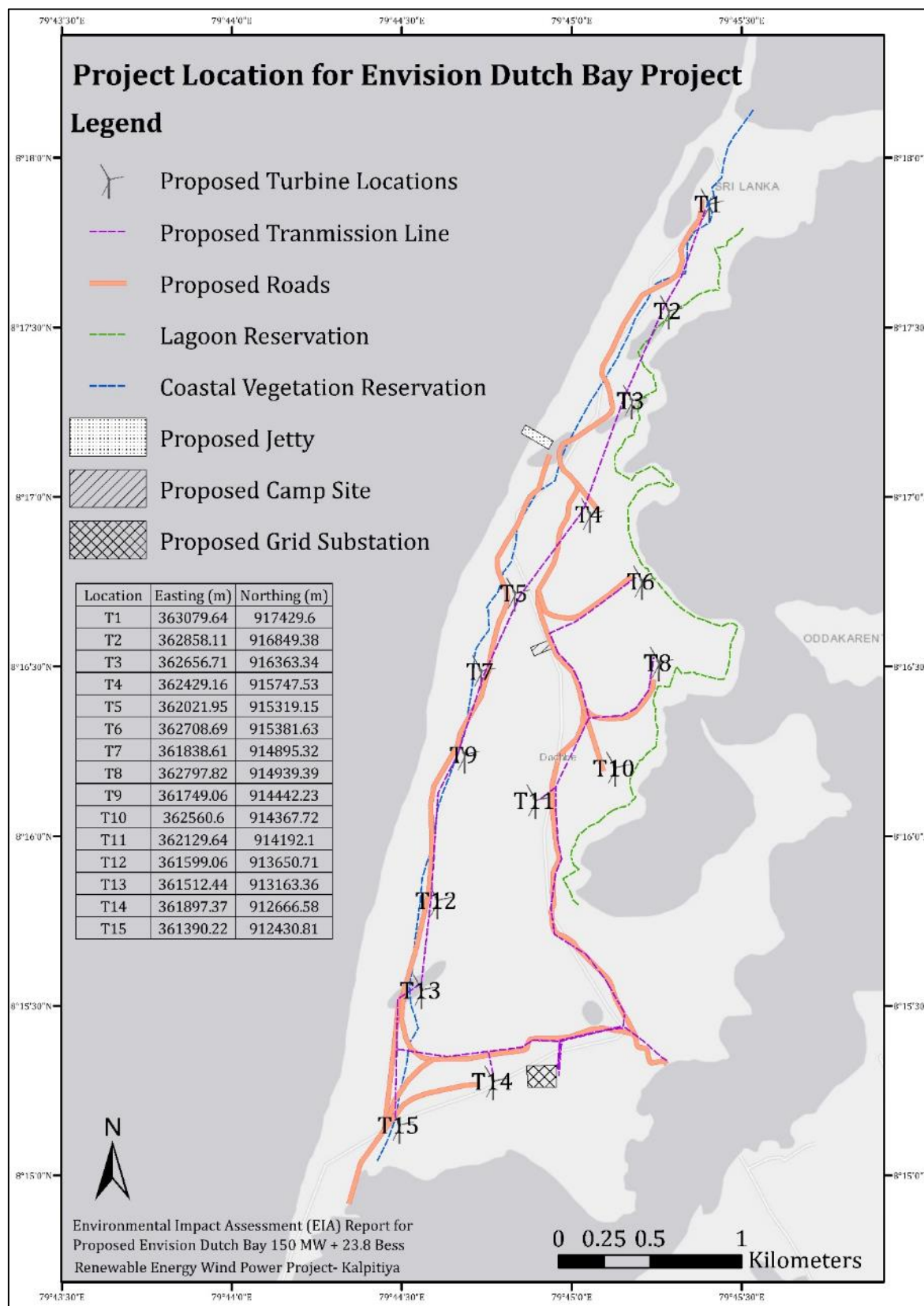


Figure 2.2: Details of Proposed Wind Turbine and Reservations

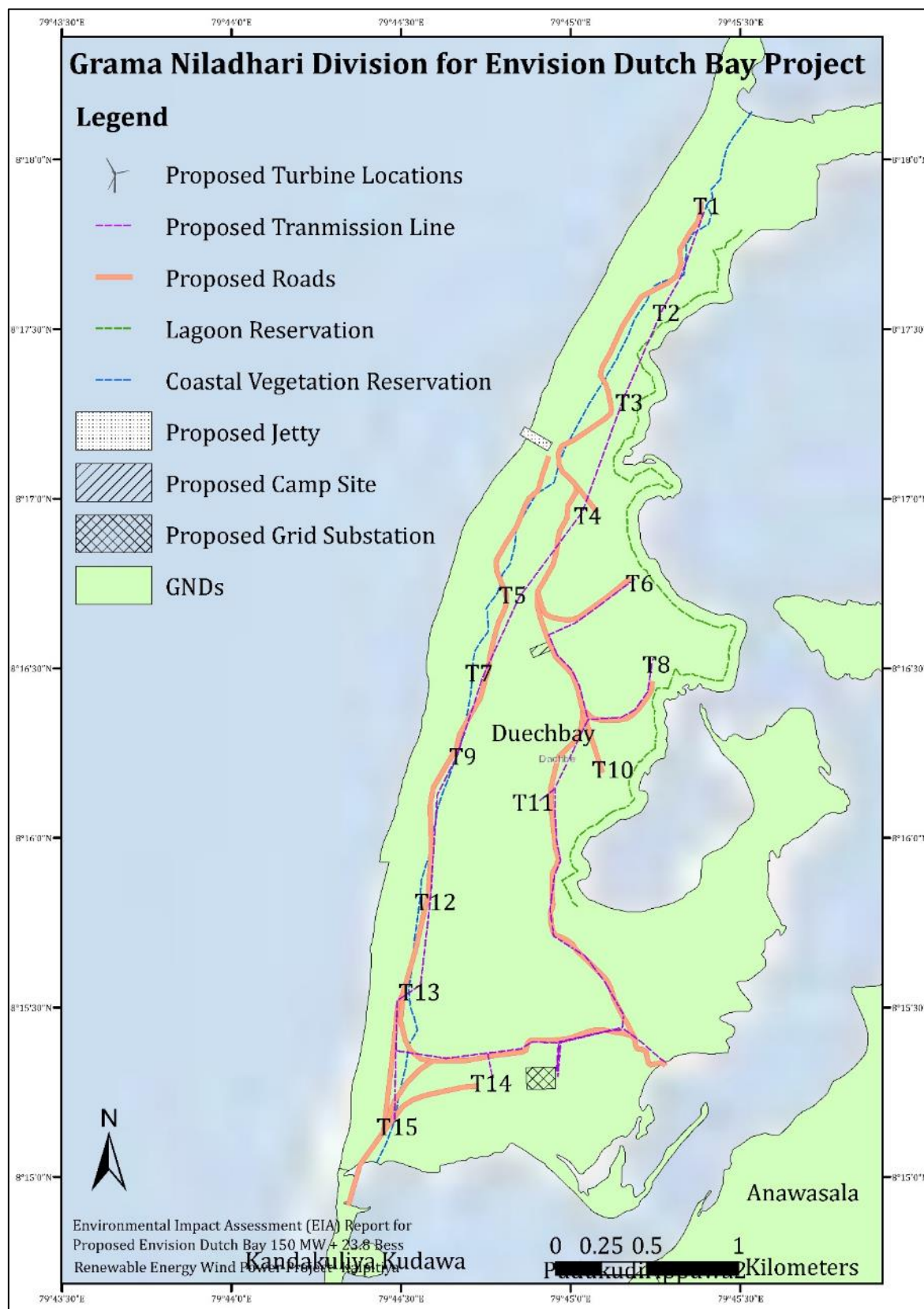


Figure 2.3: Administrative Boundaries of the Project Area



Details of Project Area and the vicinity

The Dutch Bay Peninsula is characterized by its coastal terrain, comprising sandy soils, mangrove forests, and wetlands. The site's average wind speeds range from 7 to 9 metres per second (m/s), making it ideal for wind energy generation.

The project's proximity to the Puttalam Lagoon, a critical habitat for migratory birds and marine life, necessitates careful planning to mitigate environmental impacts. Sensitive ecosystems, including mangroves and seagrass beds, are located near the project area and are integrated into the project's mitigation strategies.

A comprehensive overview of the project area and in the vicinity are provided in following section. It does include photographs taken from Kudawa entrance, sand bar and project area covering Wind Turbines 1-15.



Figure 2.4: Selected Stretches for the representation of area details



Figure 2.5: The details / photographs for the Stretch 01

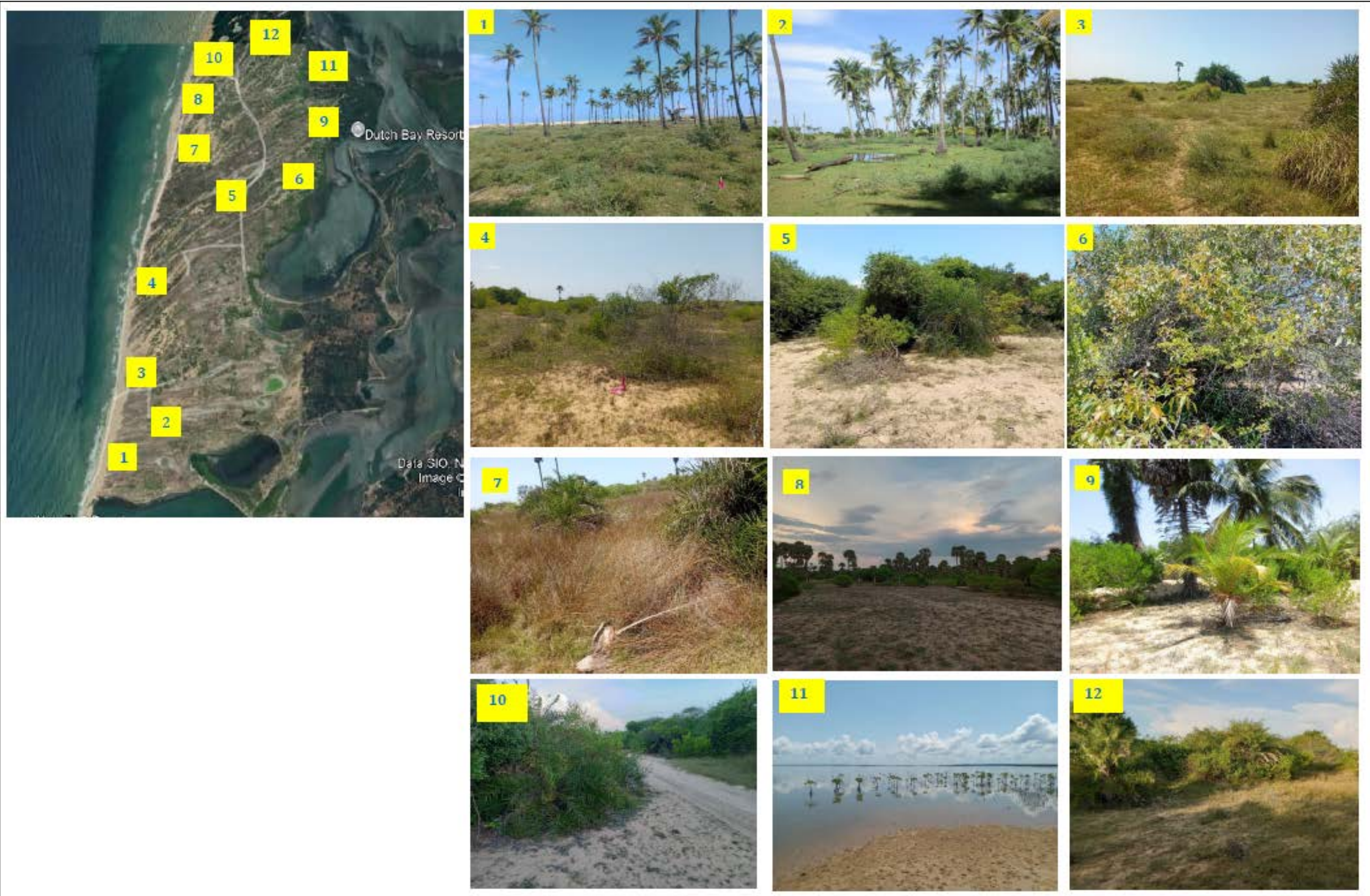


Figure 2.6: The details / photographs for the Stretch 02



Figure 2.7: The details / photographs for the Stretch 03



Details of land ownership of the project

The lands that are required for the project will be acquired by the project proponent (Envision Energy International Trading (Pvt.)) Limited through Leasing and or purchase of certain selected lands. All the required lands are **private owned** (*none of the lands is state owned*) lands and relevant plans are annexed to the report (**Refer Annexure III E**). None of the project locations has any residential places. The project footprint spreads over approximately 10 hectares of land that has different private land ownerships. Hence, private owned lands will be utilised for the development of proposed Wind Farm Project and relevant plot details are provided in **Figure 2.8**. The land plots for each wind turbine has been carefully selected to avoid socially, environmentally and economically sensitive areas. Almost all the selected lands are uninhabited and mostly exist as scrub forests.

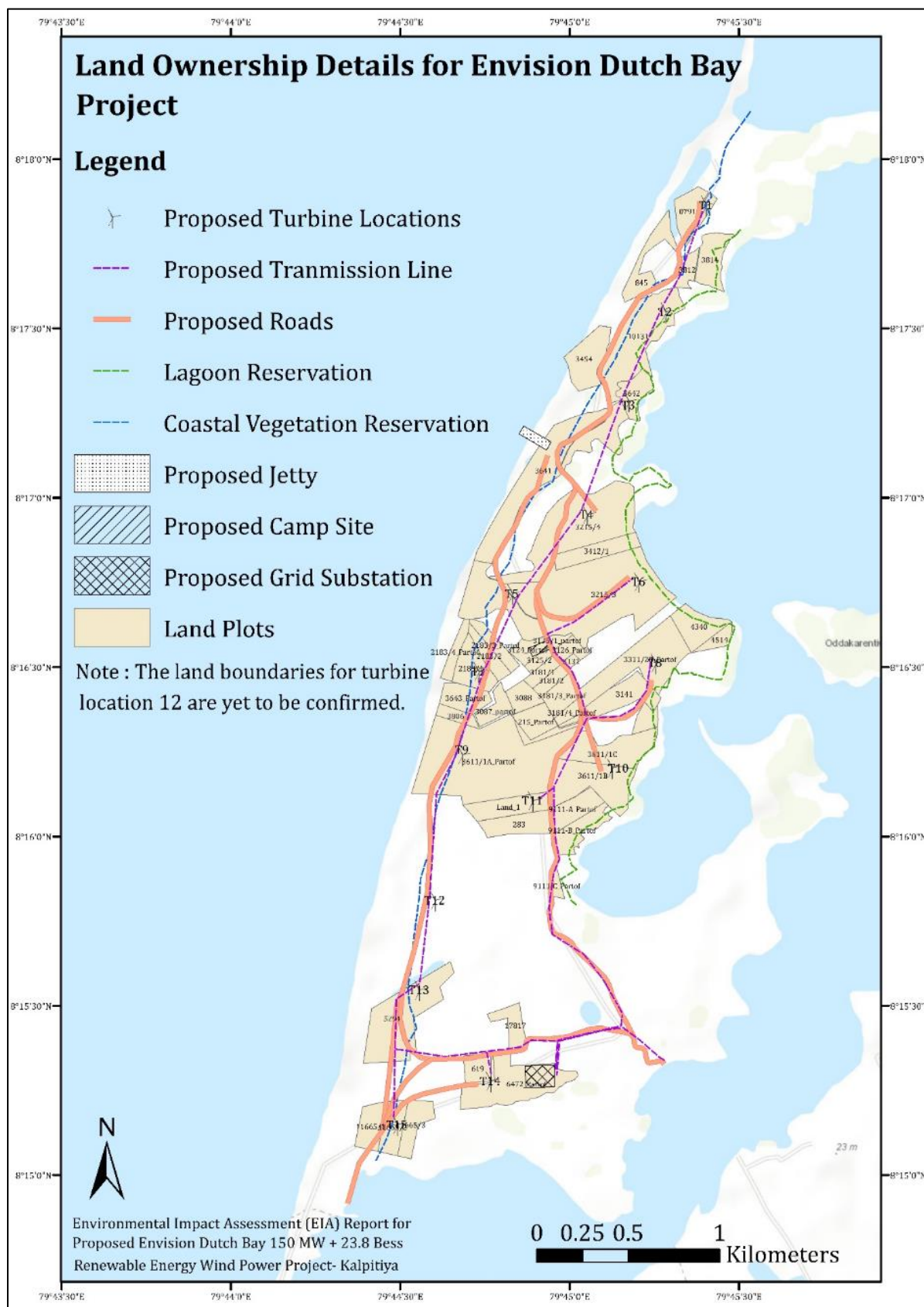


Figure 2.8: Land ownership details of the project



The land requirements for the wind turbines, proposed road network and other facilities

The required land extent to install 15 numbers of wind turbines has been earmarked by the project proponent. Each wind turbine requires land for the wind turbine foundation (permanent land use), hardstand (temporary land use during the construction period), and the reservation for the rotor (No interventions on the ground) in the sky. The area allocated for wind turbine foot prints will be around 0.7 ha.

The proposed road (20ft in width) lengths extend around 14.8 km and the land requirement will be around 9 ha. Refer **Table 2-1**.

Table 2-1: Required Land Area Details

Item Description	Area (m ²)	Area (ha)
Area of Wind Turbine Foot Prints (15 Nos.) (Diameter of foundation base 23.7m)	6,620	0.66
Clearance Area of Wind Turbines (15 Nos.) during Construction (Diameter of Clearance circle 50m)	29,450	2.95
Area of Proposed Roads (consider 20ft / ~6m width) -length of 14.8 km	90,220	9.02
Area for other facilities including camp sites, sub stations accommodating 250 workers (as per ILO standards 1927 m ²)	3,885	0.39

2.2 Details of Proposed Power Project

The installed capacity of the Dutch Bay Onshore Wind Farm Project is 150MW, and the WTGS wiring model is recommended to be one machine and one transformer unit. This project will build a new 220kV booster station, install an on-load voltage-regulating transformer with a capacity of 150MVA, and reserve a location for later construction. The booster station will be equipped with an 11MW/23.8 MWh battery energy storage system (BESS), designed to provide frequency regulation, voltage support, and short-duration peak shaving will be co-located at the Dutch Bay substation with seamless SCADA integration.

2.2.1 Details of Permanent Structures

2.2.1.1 Wind power plant

A wide range of wind turbine technology is available worldwide. Details for options and the most suitable one for the proposed project is detailed under **Section 2.2**. The proposed wind farm will comprise 10 MW scale 15 wind turbines and the installed capacity of the Dutch Bay Onshore Wind Farm Project is 150MW.



Selection criteria of wind turbines capacity

The selection of individual wind turbine, design and testing was based on the classification system outlined in *IEC (International Electro technical Commission) 61400 -1:2005*.

This standard provides a classification system for wind turbines. The IEC61400 standard divides WTGSs into three types of I, II, and III based on the maximum wind speed that occurs once in fifty years. This standard, titled "Wind turbines - Part 1: Design requirements," defines the various parameters and characteristics of wind turbines and categorizes them based on certain criteria.

Further details are provided in following sub sections;

Wind Energy Resource Analysis

Dutch Bay Onshore Wind Farm Project is located in Kalpitiya, Puttalam, North West Province, Sri Lanka. According to Global Wind Atlas database, the annual average wind speed at the height of 130m of the wind farms are estimated (**Figure 2.9**).



Source: Envision Dutchbay Onshore Wind Farm Project Feasibility Study Report

Figure 2.9: Variation of Wind Velocity in Dutch Bay onshore wind farm Area



Based on layering surface roughness and terrain map, Envision simulate a CFD model collaborating with the world's two largest supercomputing centre — Sunway Taihu Light (Wuxi, Jiangsu) and Tianhe-2 (Guangzhou, Guangdong) to calculate generating capacity. The outcomes are as follows (**Table 2-2**).

Table 2-2: WTGs siting of Dutch Bay onshore wind farm project _during Feasibility Stage*

WTG NO.	X (°)	Y (°)	Turbine type	Hub height** (m)	Wind speed (m/s)	Wake (%)	P50 equivalent full load hours (h)
T1	79.755	8.296	EN-233/12.5	130	8.41	14.05	3006
T2	79.752	8.291	EN-233/12.5	130	8.43	17.02	2915
T3	79.749	8.286	EN-233/12.5	130	8.4	16.3	2935
T4	79.746	8.282	EN-233/12.5	130	8.41	10.06	3164
T5	79.749	8.276	EN-233/12.5	130	8.43	11.94	3106
T6	79.758	8.277	EN-233/12.5	130	8.38	10.95	3101
T7	79.753	8.274	EN-233/12.5	130	8.39	14.96	2966
T8	79.753	8.268	EN-233/12.5	130	8.29	5.51	3233
T9	79.748	8.268	EN-233/12.5	130	8.4	10.88	3125
T10	79.744	8.274	EN-233/12.5	130	8.45	13.41	3063
T11	79.743	8.268	EN-233/12.5	130	8.45	10.24	3179
T12	79.742	8.252	EN-233/12.5	130	8.34	2.83	3368

*It should be noted that it had been considered **12 Nos** of wind turbines with hub height of **130 m** during the **feasibility stage**. However, **no of turbines has been modified and 15 Nos of Turbines has been considered during the EIA**.

**** modified/increased to 140m and this has been considered during EIA**

Wind Turbine Generation System (WTGS) Selection

Following facts has been considered during the WTGS selection.

- Current manufacturing level, technological maturity and price of WTGS
- Topographic and geological conditions of wind farm, wind characteristics, equipment transportation and installation conditions
- Safety level of WTGS
- Regional adaptation of WTGS
- Convenience of operation and maintenance after completion of wind farm
- The WTGSs must have primary frequency regulation and high/low voltage ride-through function.

Based on the conditions of this project, it was proposed to select the EN-233/12.5 wind turbine with the following parameters (Table 2-3) during the feasibility stage and refer **Annexure III F**.



Table 2-3: Energy Production Summary_Dutchbay Wind Farm

Proposal	Dutch Bay
Turbine model	EN-233/12.5
Wind Turbine Type Code	TA/TB
Individual Wind Turbine Capacity (MW)	10
Number of units (sets)	15
Total capacity (MW)	150
Hub height (m)	140
Average wind speed (m/s)	8.40 (Project-specific customization)
Wake loss (%)	11.53
P50 Net AEP (MWh/year)	464,700
P75 Net AEP (MWh/year)	427,500
P90 Net AEP (MWh/year)	385,800
P50 Equivalent full load hours (h)	3098
P75 Equivalent full load hours (h)	2850
P90 Equivalent full load hours (h)	2572

However, based on site-specific analysis and climatic validation, the proposed turbine model for the Dutch Bay project is the **EN220/10.0**, manufactured by Envision Energy. This model offers a rated capacity of 10 MW, a rotor diameter of 220 metres, and a hub height of approximately 140 metres. It is specifically designed for near-shore resilience with high humidity and salt ingress protection, including C5-H corrosion-rated coatings and pressurized, filtered ventilation systems. The turbine selection is aligned with coastal wind resource optimization and international best practices for Class IIA wind conditions.

Expected plant factor of wind power plant.

Wind plant capacity factor was obtained after estimating the net AEP.

Plant Capacity Factor P75 : 36 %

Gross, net AEP and capacity factors

Table 2-4: Turbine Details

Model	Capacity (MW)	Wake Losses (%)	Net Energy (GWh/year) P75	Capacity Factor P75
EN-233/12.5.5 10 MW/140 m Hub Height	150	11.53	427.500	36%



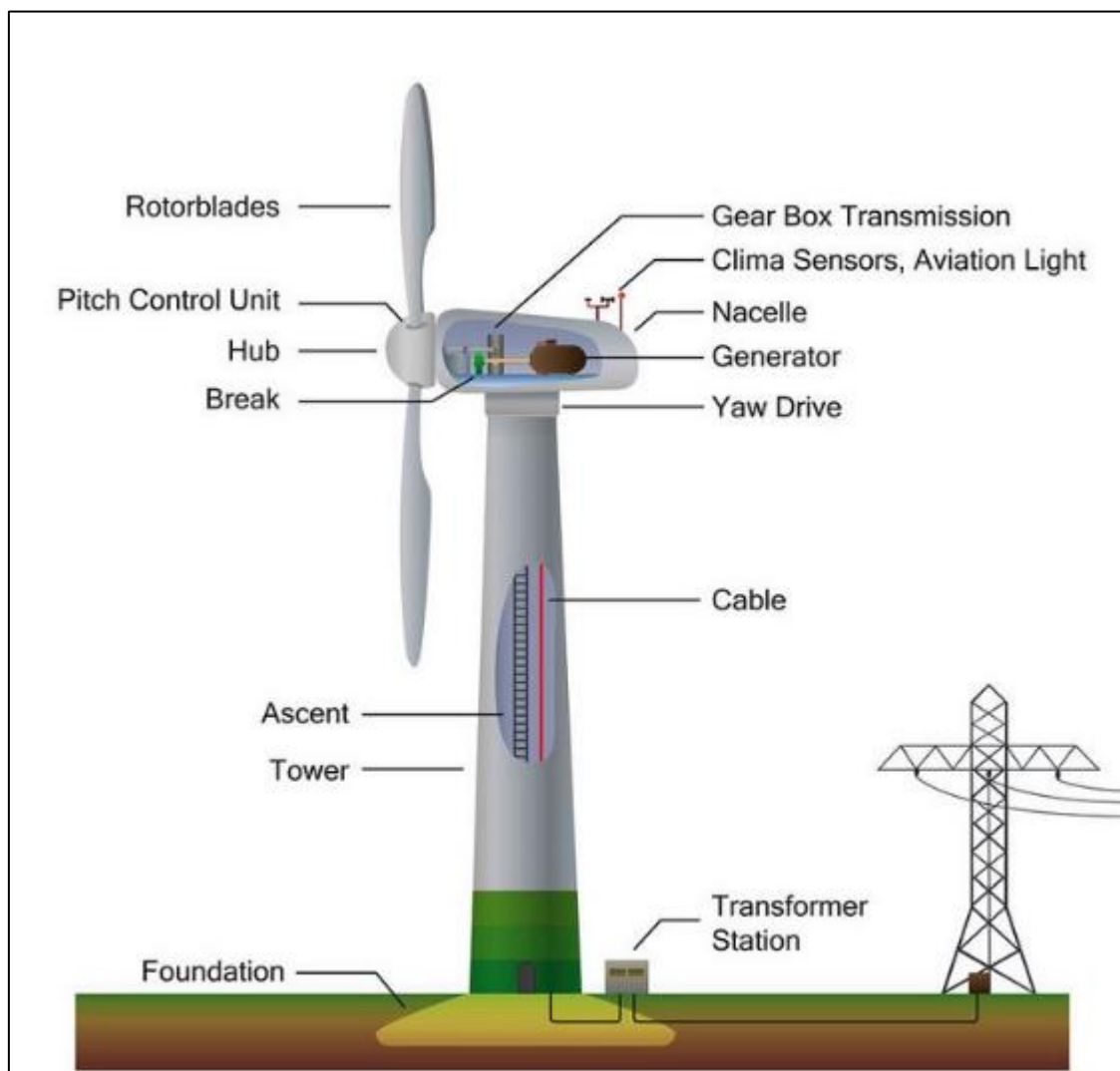
Major components of the wind power plant

The main project components of the wind power project consist of the following (**Figure 2.2**).

- The turbines 15 Nos. These include the reinforced concrete foundation, turbine tower, turbine blades and the nacelle.
- Access Roads to the turbines
- Cable Route/Transmission lines (overhead)
- Temporary yards for vehicles and to temporarily store turbine parts

Further, the major components of a Wind Tower are as presented below in **Figure 2.10**.

Additional details on heights and rotor diameters are depicted in **Figure 2.11** and **Figure 2.12**.



Source: <https://www.windustry.com/what-components-comprise-a-wind-power-system.htm>

Figure 2.10: Major component of Wind Power Plant

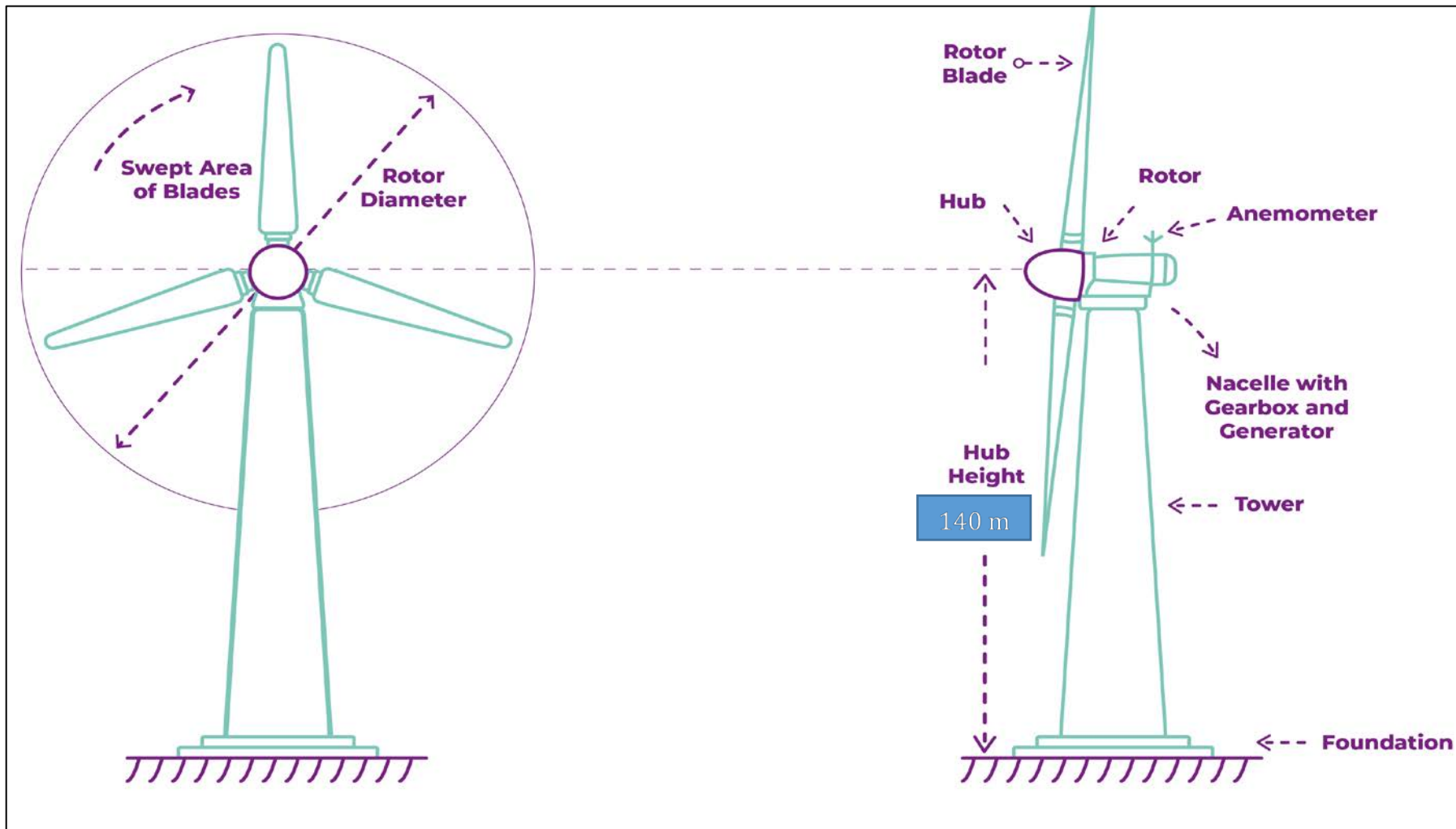


Figure 2.11: Components of a Wind Tower and different dimensions

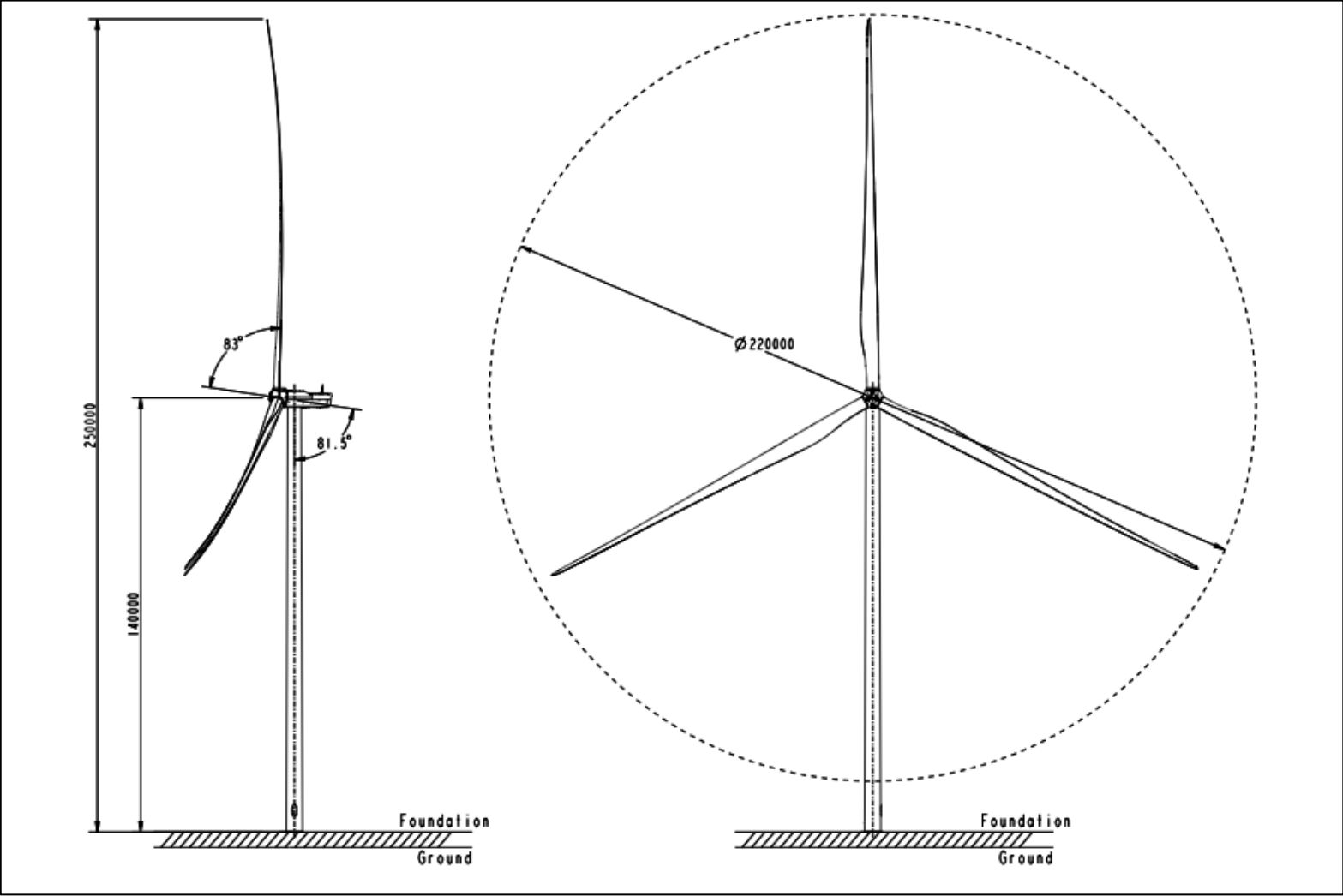


Figure 2.12: Hub Height and Rotor Diameter



Summary Design details of the proposed wind farm

A total of 15 wind turbines will be installed and each turbine has a maximum electric power output of 10 MW. The rotor consists of three rotor blades, the rotor hub, three geared bearings and three-phase drives for adjusting the pitch of the rotor blades. The rotor diameter of a turbine is 220m, and the hub height is 140m. In brief,

Total capacity of power plant: 150 MW

The design lifetime is 20 years.

Capacity of a turbine: 10 MW

Number of wind towers: 15

Type : TA/TB

Plant Factor : 36 %

Hub height: 140 m

Rotor diameter: 220 m

Spacing between turbines: 350m Minimum

Land area required for tower installation: 0.7 ha

Land area required for access roads: 9 ha

Land area requirement for the wind farm

Refer **Table 2-1**. The foundation of the wind turbine will be a circular area with 8.8 m radius (17.6 m Diameter). All the relevant drawings are provided in **Annexure III G**.

Dimension of the area that need to be excavated for a tower footing

About 23m diameter area will be excavated for tower footing of which the diameter is 18.6m considering the slopes to be maintained during the excavation process.

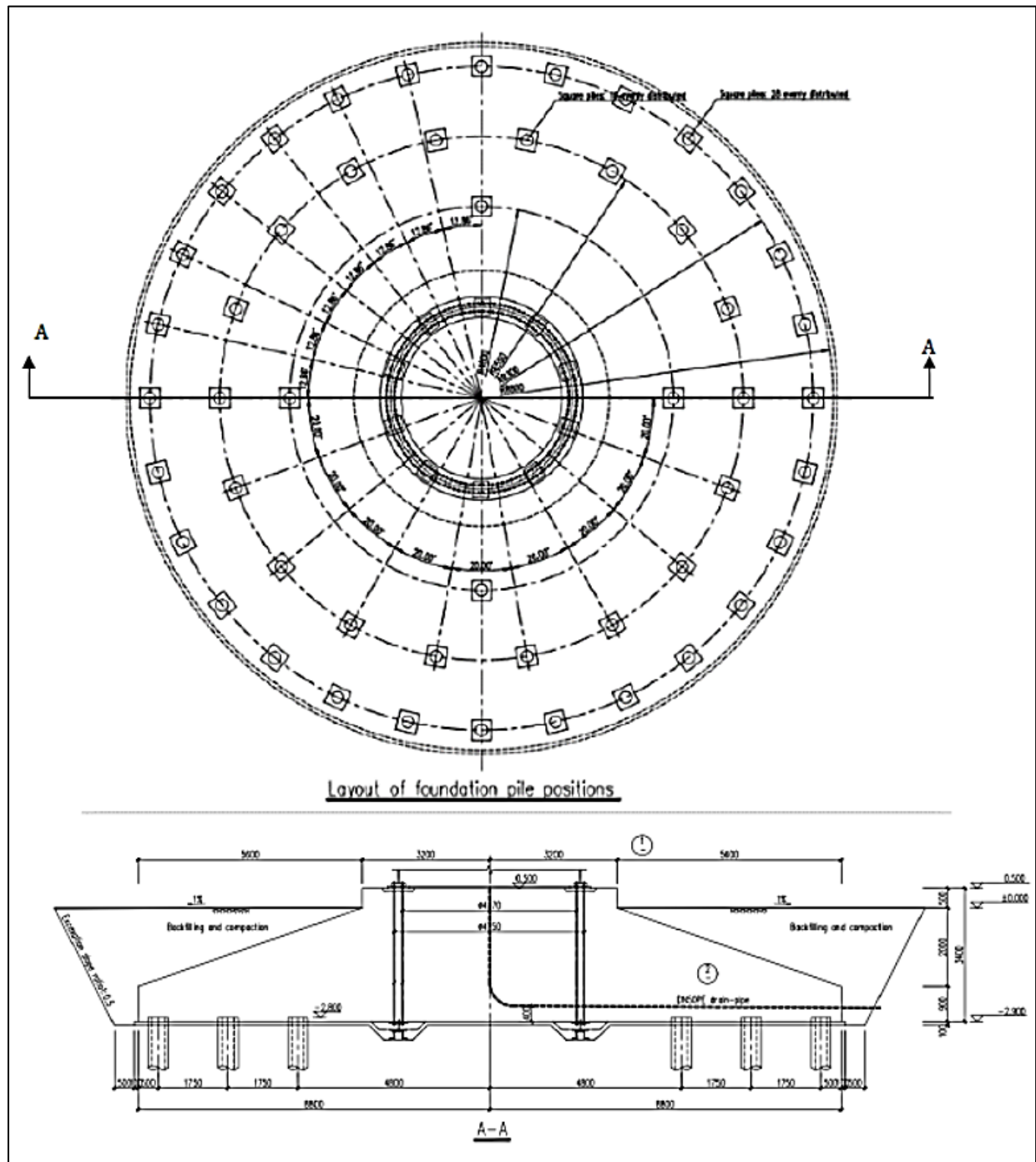


Figure 2.13: Wind Tower Foundation Details

Noise emission from the wind turbines

Modern wind turbines, including models in the 10 MW, as those manufactured by Envision Energy, typically generate broadband aerodynamic noise due to blade-air interaction and mechanical components. Published data and benchmarked turbine models suggest the following Sound Pressure Levels (SPL) as presented in **Figure 2.14**:



- ~113 dB(A) at the nacelle (source level)
- ~60 dB(A) at 100 metres distance from the turbine
- ~55 dB(A) at 300 metres
- ~45 dB(A) at 500–750 metres, depending on terrain and atmospheric conditions

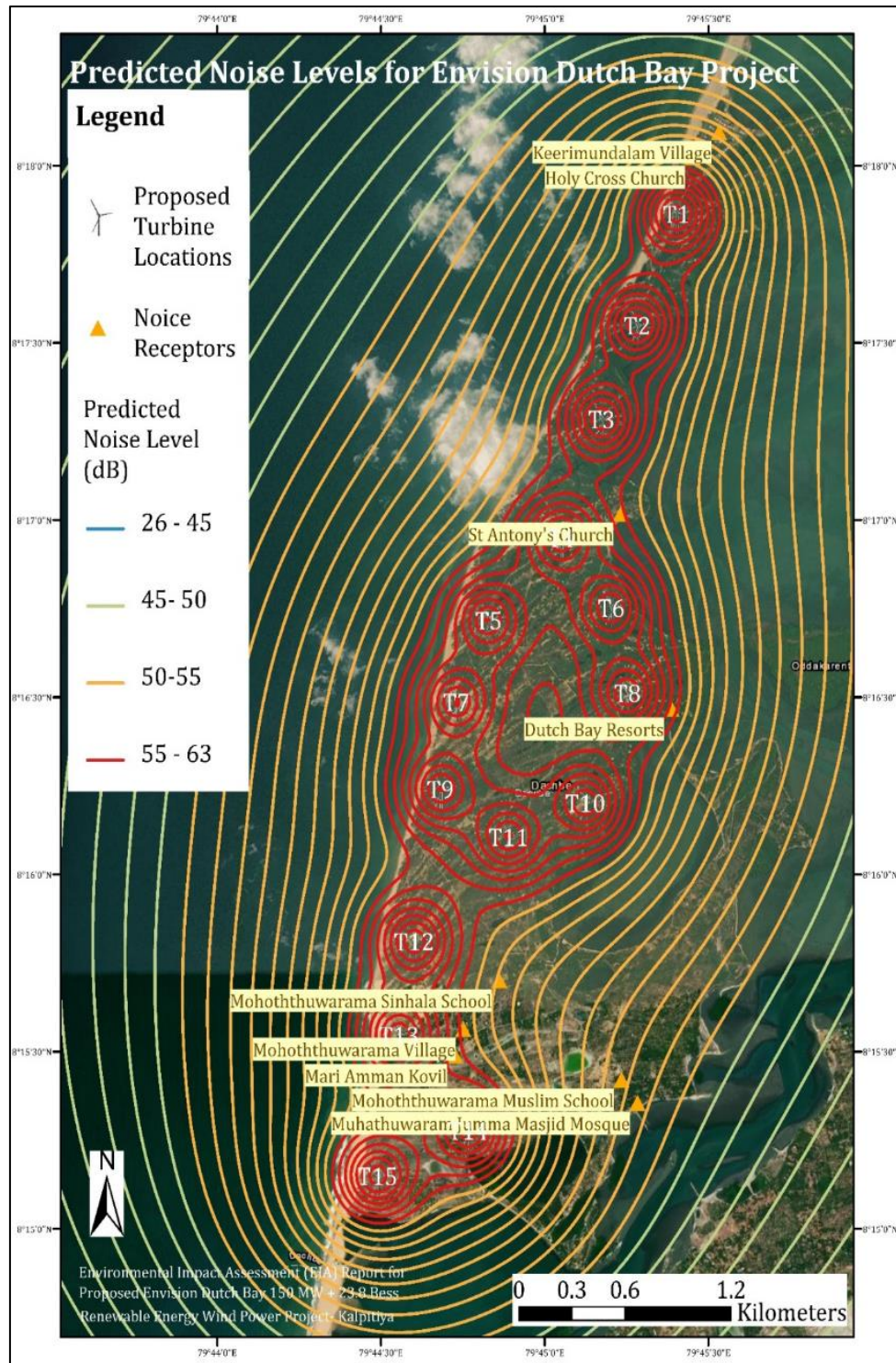


Figure 2.14: Predicted Noise Levels by Turbines



Internal power harnessing system / network incorporated.

The internal power harnessing system is the grid station. From the grid, station power will be transmitted to the national grid through the proposed transmission line.

Collector substation and it's control, metering, and monitoring systems

The collector substation enables efficient collection and distribution of electricity generated by multiple turbines. They are equipped with control, metering, and monitoring systems to ensure safe and reliable operation, optimize power generation, and allow for real-time monitoring and management. The system is an integral part of maximizing the efficiency and overall performance of wind power plants. The collector substation consists of transformers, switchgear, protective relays, control systems, and monitoring equipment. It is equipped with automation and control systems that manage the operation of the substation equipment. The system ensures safe and efficient operation and is remotely accessible. SCADA (Supervisory Control and Data Acquisition) will be used to monitor and control collector substations. They allow operators to supervise the substation's performance, remotely operate equipment, and respond to alarms or issues. It will be equipped with a monitoring system that tracks the performance of wind turbines and the substation equipment. They can detect issues or inefficiencies in real-time. The monitoring systems can generate alarms and alerts when equipment malfunctions, grid conditions change, or performance deviates from expected norms. This allows for timely responses to issues.

Transformers, switch yard

Transformers and switchyards will be established during the detailed design stage.

2.2.1.2 Security and Safety Measures

Security / safety measures such as fencing, CCTV, fire protection, prevention of bird strike and damage to the animals.

For the prevention of bird strikes, a Radar system, turbine blade colouring, and boom boxes are proposed. Turbine blade colouring and boom boxes are proposed on an experimental basis. Further, details are provided in sections related to Avifauna Detecting Mechanism.

2.2.1.3 Avifauna Detecting Mechanism

This section presents details about the proposed Bird and Bat Repellent System by Envision to be applied to the Dutch Bay Wind Farm. It integrates findings from the project's avifaunal impact assessment, international best practices, and scientific research to support ecological mitigation strategies under the Environmental Management Plan (EMP).



The avifauna detection mechanism (**Figure 2.15**) will be implemented at suitable locations within the site. The parameters related to the Avifauna (birds and bats) repellent system are provided in **Table 2-5**.

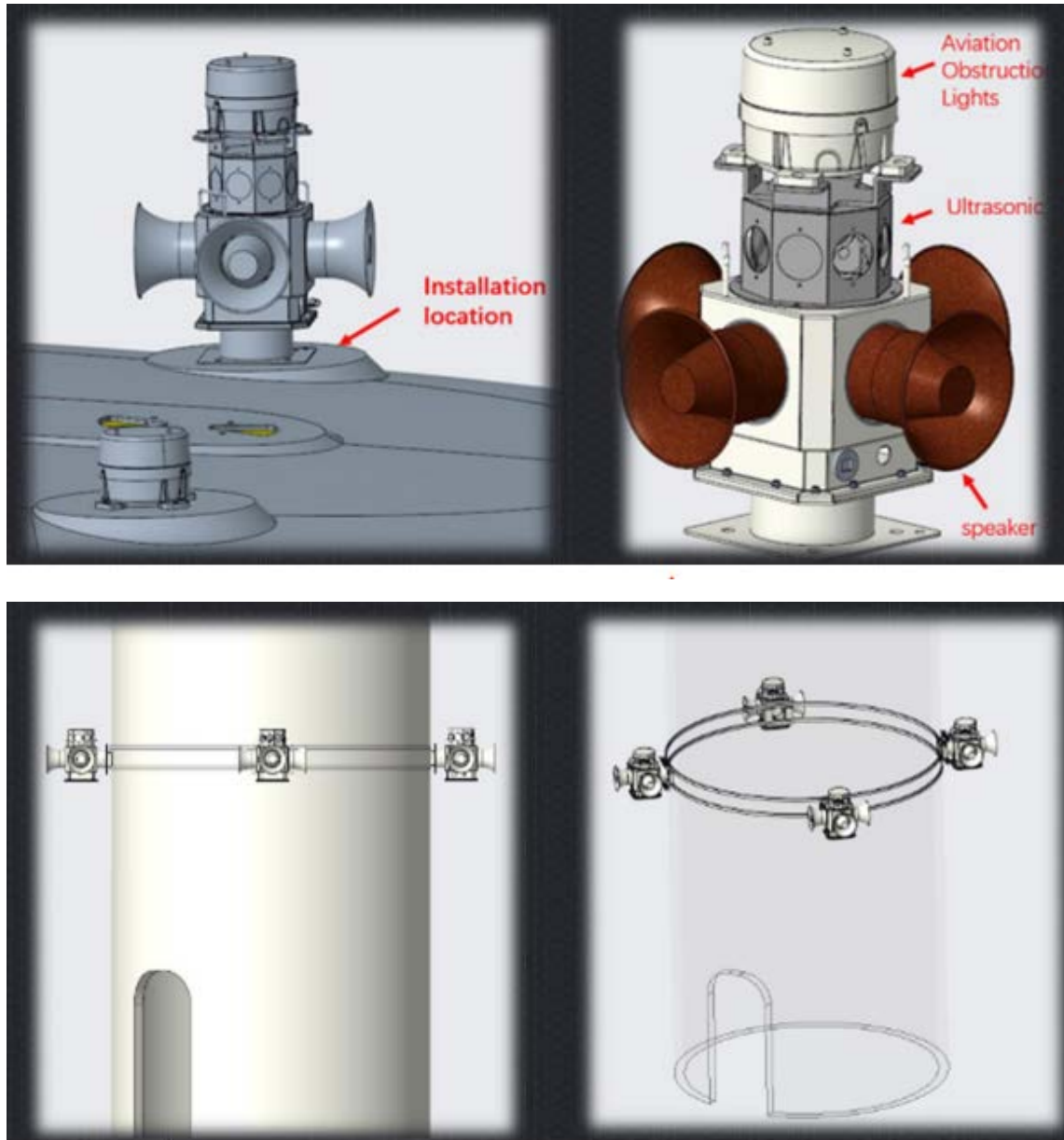


Figure 2.15: Details of Avifauna Repellent system

The general operational phenomena is as provided below,

- Daytime – Repelling birds through high decibel phonetics (Background or time controlled start/stop)
- Night – Repelling bats through ultrasonic with variable frequency (Background or time controlled start/stop)



Table 2-5: Parameters related to Avifauna Detection System

Parameter	Value
Voltage	24 V DC or 220 V AC
Power	<= 40 W
Weight	6 kg
IP Grade	IP66
Anti-Corrosion Grade	C5-M
Operating Temperature	-30°C to +55°C
Survival Temperature	-40°C to +70°C
Audio Radius	150m
High decibel phonetics	120 dB
Working life / warranty	20 a/5 a

Envision Repellent System Overview is provided below

Model: Envision Bird and Bat Repellent System

Specifications:

- Output: 120 dB sound pressure
- Repellent Range: 150 metres
- Modes: Day (phonetic bird calls), Night (ultrasonic bat deterrents)
- Power: 24 V DC / 220 V AC
- Durability: IP66, Anti-corrosion rating C5-M

Bat Mitigation Effectiveness

Mechanism: Ultrasound (>20 kHz) interferes with echolocation, deterring approach

Field Results: Peer-reviewed studies report up to 54% reduction in bat fatalities when ultrasonic deterrents are used near wind turbines (NRG Systems, 2024)

Deployment Guidance:

- Mount at nacelle or hub level for maximum range
- Program to operate during high bat activity periods (night-time, low wind speed)

Bird Deterrence Limitations and Augmentation

Most birds do not perceive ultrasonic sound; the system uses audible phonetic deterrents during daylight hours.

Visual Augmentation:

- Blade painting (e.g. one black blade) reduces collision risk by up to 72% (May et al., 2020)
- Turbines WT6 and WT9 sited with enhanced visual contrast due to proximity to avian habitats



Operational Safeguards and Compliance

Health & Safety:

- Deterrents must be deactivated during maintenance to prevent hearing damage
- Placement should comply with ISO acoustic exposure thresholds for field technicians

Environmental Safeguards:

- Device operation scheduled to avoid overlap with community activities
 - o Bird repellent system needs to be turned off in advance when site personnel are working
- Located away from human settlements and domestic animal paths
 - o High decibel phonetics and ultrasounds have negative influence on health. Therefore, the installation location should be far away from the crowd.

Monitoring and Adaptive Management

- Weekly visual inspections during migratory seasons
- Review of carcass data (if any) to assess deterrent effectiveness
- Adaptive shift in activation timings based on seasonal monitoring
- Coordination with local wildlife departments and CC&CRMD

2.2.1.4 Access Roads

On site Road Surveys

Based on the outcome of onsite road surveys, it was observed that the overall road condition of the wind farm area is not good, and it is necessary to carry out the construction of the transportation road and clear the obstacles, etc., in order to meet the needs of the wind turbine equipment transportation and backward transportation. The original road condition of the site is as follows (**Figure 2.16**).



Figure 2.16: Original Condition of the Existing Roads at Site

The existing road network at the site is depicted in **Figure 2.17** while the proposed roads are presented in **Figure 2.18**. Almost all the roads are generally 20ft in width.

Wind farm backhauls road construction

Most of the original/existing roads will be widened, reinforced and connected to main road for the transportation of equipment in the site. These newly built roads connect the wind turbine platform to the main road, with a length of about 3.4 kilometres and a width of 20 feet. All these proposed roads would accommodate a length of around 16km. These main roads will be used mainly for the transportation of equipment to the site.

The new and reinforced roadway needs to have a bearing capacity of not less than 12 tons per square meter, and shrubs and trees along the roadway need to be cleared.

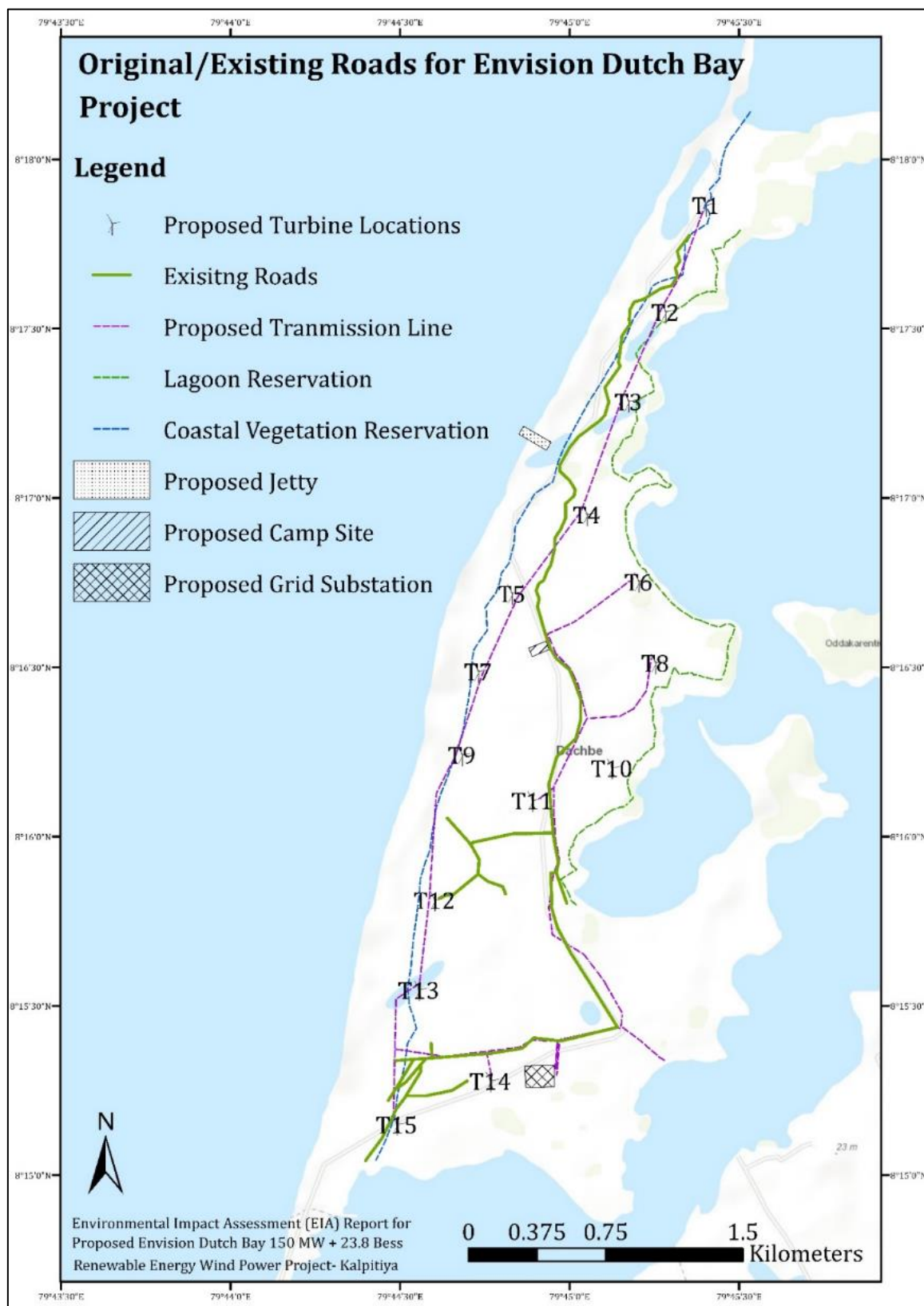


Figure 2.17: Original/Existing Roads at Site

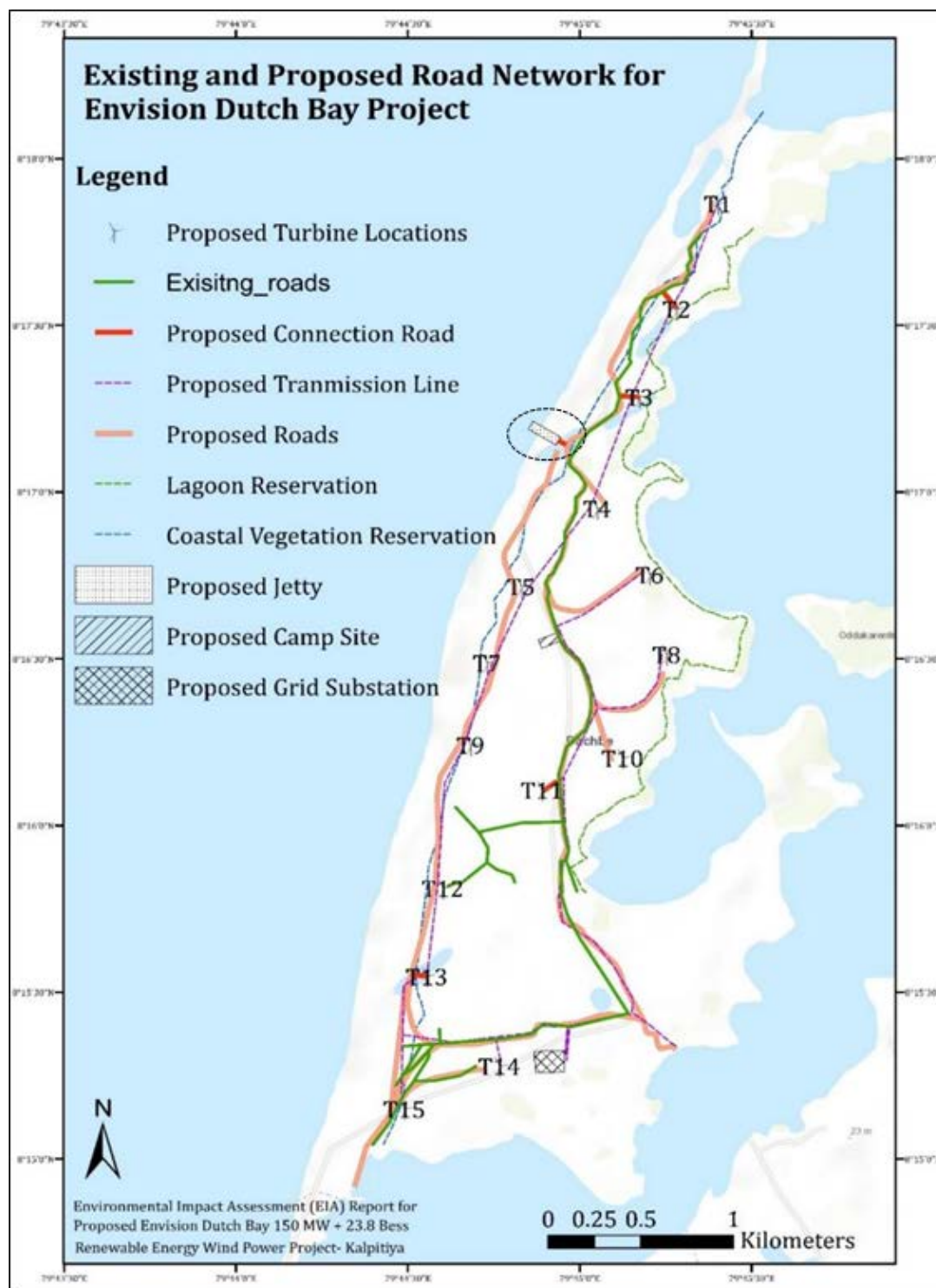


Figure 2.18: Existing and Proposed Road Network to wind power plant site

Proposed sections of the main roads are depicted in following **Figure 2.19** and **Figure 2.20**. The gradient of the road is $\leq 8^\circ$. The difference in height between the highest and lowest part of the road surface is ≤ 100 mm over a length of 35 m.

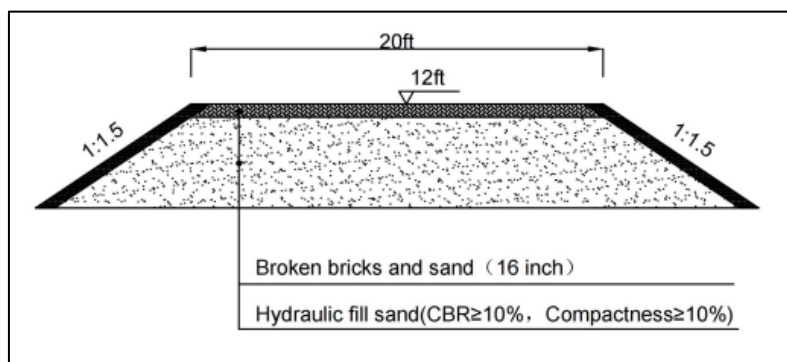


Figure 2.19: Proposed New Roadway Cross Section

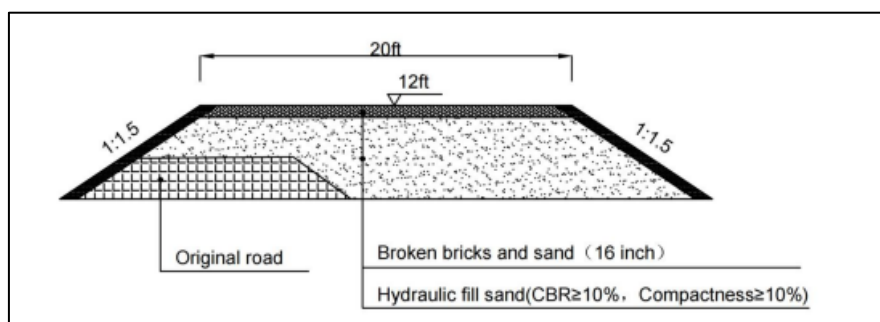


Figure 2.20: Proposed Expansion of (New Reinforced) Existing Roadway Cross Section

Specification size of road turning widening

- (1) No obstacles on the outside of the curve, Specification size of road turning widening is shown **Figure 2.21**.

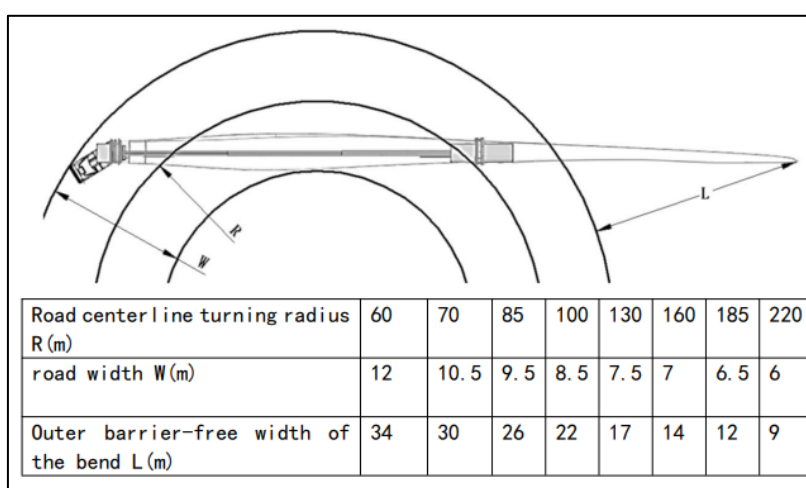


Figure 2.21: Proposed Road turning widening when No Obstacles on the outside of the curve



- (2) Obstacles on the outside of the curve, Specification size of road turning widening is shown
Figure 2.22.

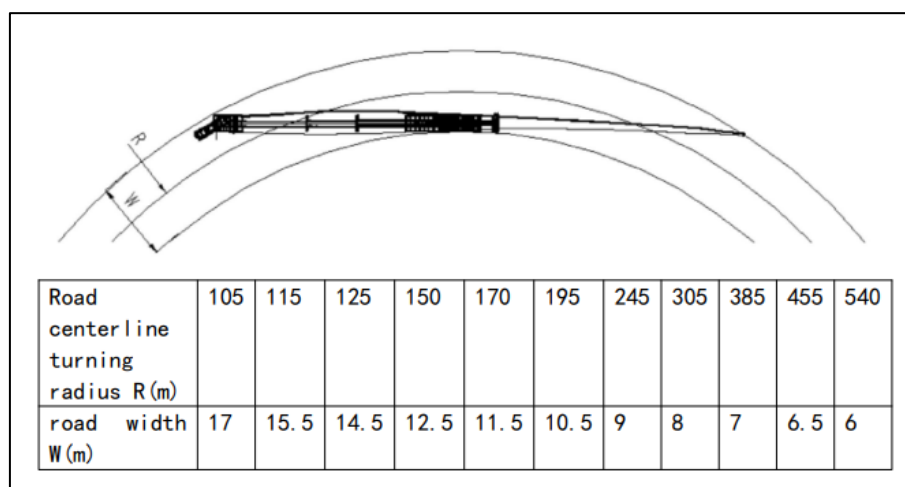


Figure 2.22: Proposed Road turning widening when No Obstacles on the outside of the curve

Further details on the turbine locations and the proposed access roads are provided in **Annexure III Q**.

2.2.1.5 Piers / Unloading facilities/ Options of wind turbines/blades

Heavy-lift operations for nacelle and blade assembly will utilise 600-tonne crawler cranes stationed on either pre-compacted platforms or floating platform. Crane access paths are reinforced with geotextile and gravel subbase to prevent soil subsidence.

Unloading facility: Under this project, it is proposed to build a new temporary jetty (**Figure 2.23**), located near the Wind Turbine No. 4 (the longitude and latitude is 8°17'4.82"N 79°44'54.09"E) as depicted in **Figure 2.18** and **Figure 2.24** beside seaside. This temporary jetty will be constructed with standardized floating box modules as needed, and can be motorized by mounting outboard engines on the side or assisted by other vessels. The length and width of the jetty are 110 m, and 45 m respectively and the jetty will be designed for bearing requirements to meet the 600-tonne crane unloading.

This jetty will be used for unloading large turbine parts, such as long blades for wind turbines. Moreover, this will be equipped to handle the oversized components ensuring safe, efficient, and environmentally responsible operations is essential to the success of wind energy projects.



Figure 2.23: Project Case Images for a Temporary Jetty

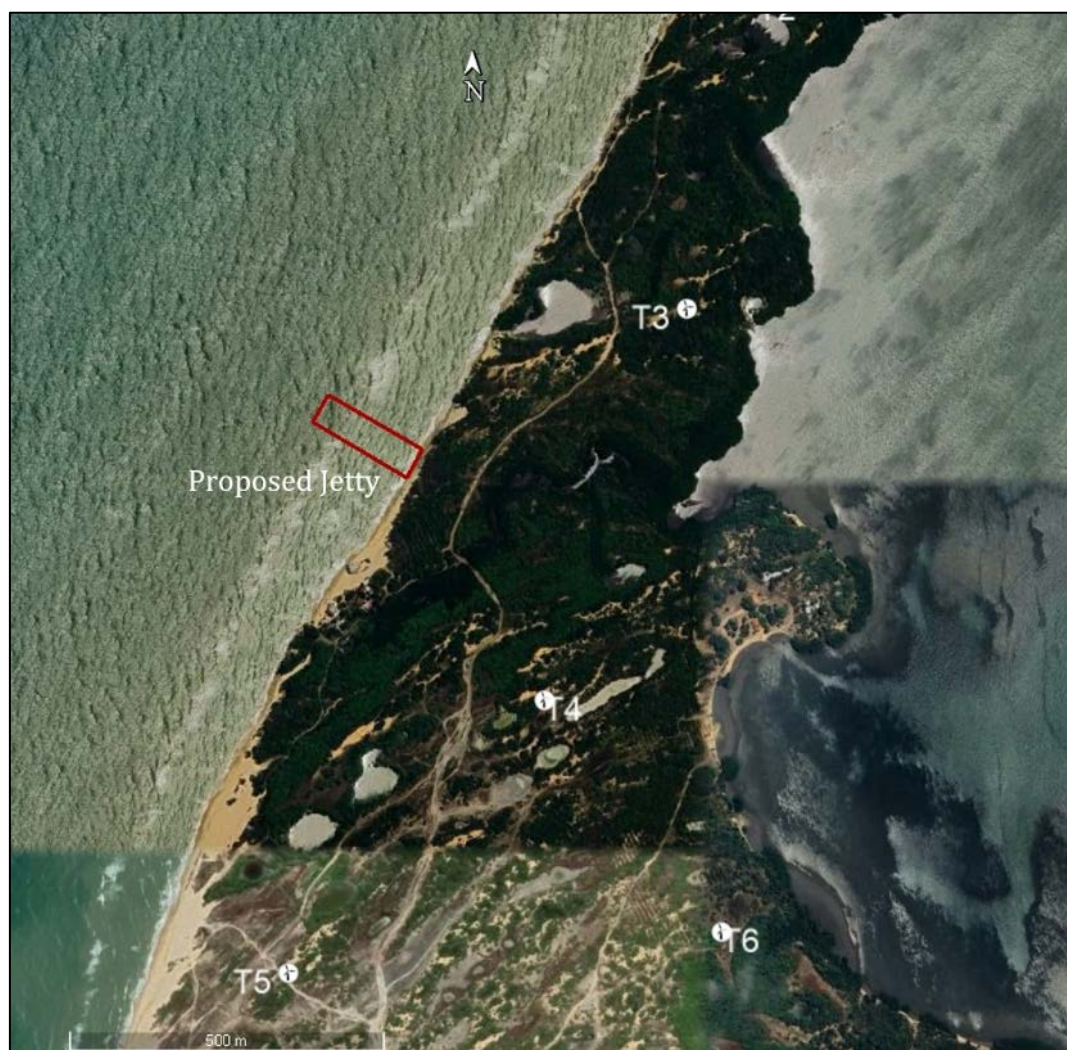


Figure 2.24: Location Map of Proposed new Jetty



Construction conditions of the temporary jetty

(1) **Site topography and geomorphology:** the project site is located in Kalpitiya, Puttalam, Northwest Province of Sri Lanka; the location of the proposed jetty is a coastal beach geomorphology.

(2) **Stratigraphic structure:** the sandy soil in the site has not been found to pass through the new active faults, and from the analysis of the geological structure and seismic activity, the site is a relatively stable area, which is suitable for the construction of the temporary jetty.

(3) **Hydrogeological conditions:** under long-term immersion in water and alternating wet and dry environments, seawater and seasonal rainwater are corrosive to steel piles.

Refer **Figure 2.25** for the temporary Jetty arrangement.

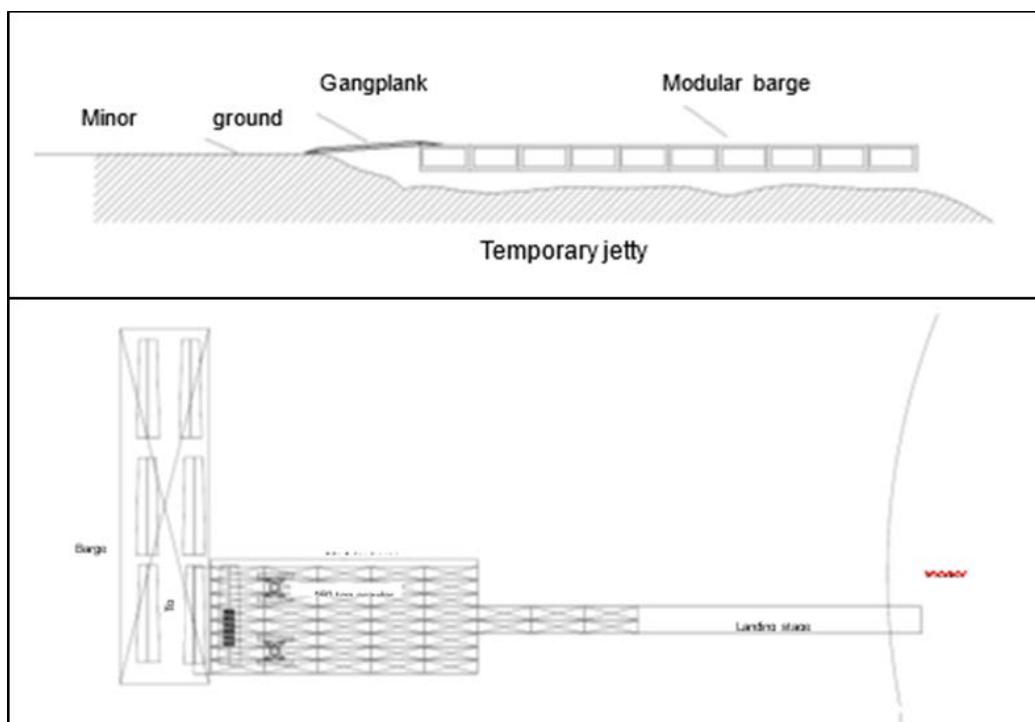


Figure 2.25: Proposed Floating Jetty for the Loading /unloading Operations

Temporary jetty unloading program

In view of the size and weight of the project wind power equipment, it is planned to take double cranes temporary jetty unloading program, to complete the large pieces of rotor, towers and nacelles unloading barge loading transporter work. The specific operations of unloading the rotor, towers and nacelles are provided in **Annexure III H**.

The staffing requirement the temporary jetty operations for unloading and receiving are as follows;



Table 2-6: Temporary Jetty Staffing for unloading and receiving

Regional	Staff	Responsibilities
Barge	2-4 persons	Responsible for berthing and de-berthing of barges at the dock and monitoring cargo safety on board.
Jetty unloading	8-10 persons	Responsible for the preparation of dock unloading materials and staff
Crane related personnel	6-8 persons	crane operator, and responsible for the use of equipment security and maintenance and testing personnel
Transportation vehicles	6-8 persons	Driver, vehicle maintenance and security, and tie-down personnel
Safety officer	4-6 persons	Responsible for the safety of the vehicle transportation process

2.2.1.6 Storm water Drainage plan

A specific storm water drainage plan is not proposed / presented herewith as the present project does not disturb natural storm water flow paths. Moreover, the design of proposed road layouts will be done ensuring that the existing storm water drainage paths will not be blocked additionally. Consequently, safe conveyance of the storm water is ensured, when culverts or water paths are crossed with road layout.

2.2.1.7 Any other infrastructure required/provided by the project

Temporary yards will be used to store turbine parts, vehicles etc. It is expected to select the sites during the construction stage as per requirement.

2.2.2 Details of the Transmission Line

External Transmission Line

The proposed 150 MW wind power generation farm will be connected to the national grid from the wind power collector grid substation at Kalpitiya to the CEB Wariyapola switching station through a 220 kV double circuit overhead transmission line, with an approximate route length of 104.77 km.

A simplified schematic showing the transmission line path (along with the topographical variations) from location of the Kalpitiya peninsula to the grid injection point, and the proposed 220 kV twin Zebra overhead line connecting to the Wariyapola switching station is presented in **Figure 2.26** below.

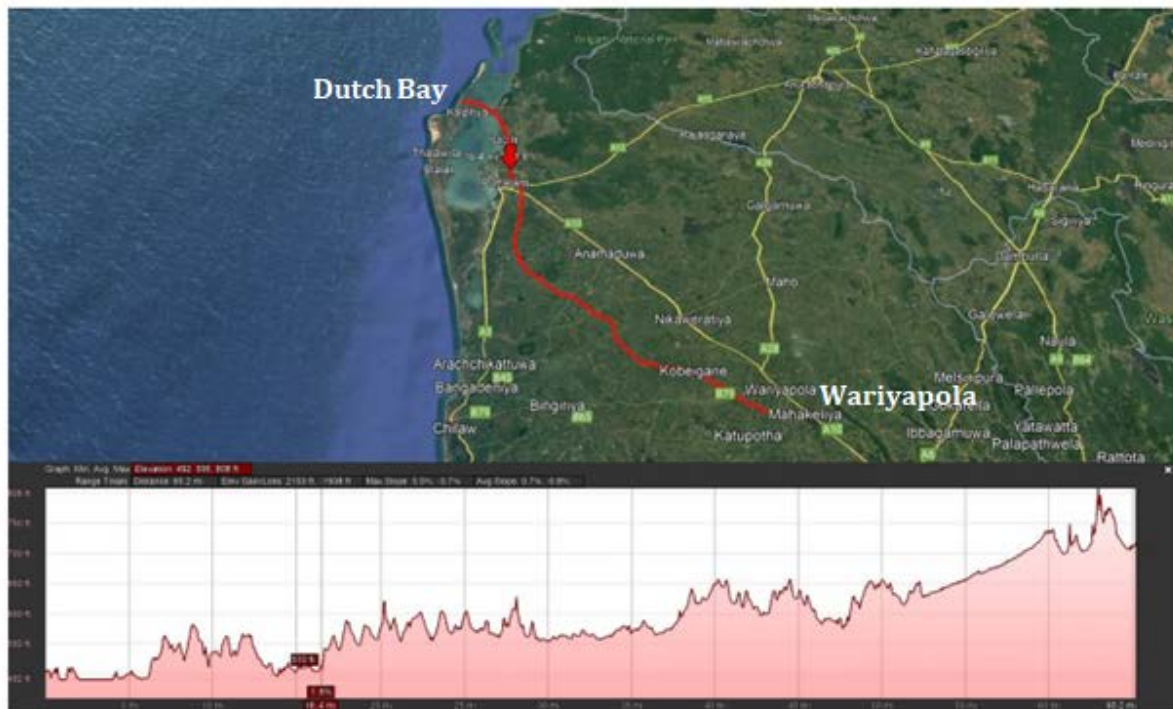


Figure 2.26: Proposed external transmission line path from Kalpitiya to Wariyapola

A feasibility study has been carried out and separate EIA will be carried out for this proposed external transmission line component. Moreover, the Report on **“Transmission Line Engineering Report for Transmission Infrastructure Development of 150 MW Wind Collector Sub at Dutch Bay “**is provided in Annexure III I For reference.

Internal Transmission Lines

The cable route/transmission line layout is presented in **Figure 2.18 / Figure 2.27** . Typical overhead cable details (33kv armoured cables) are given with insulated/coating. The proposed transmission line paths do follow the proposed road paths mostly.

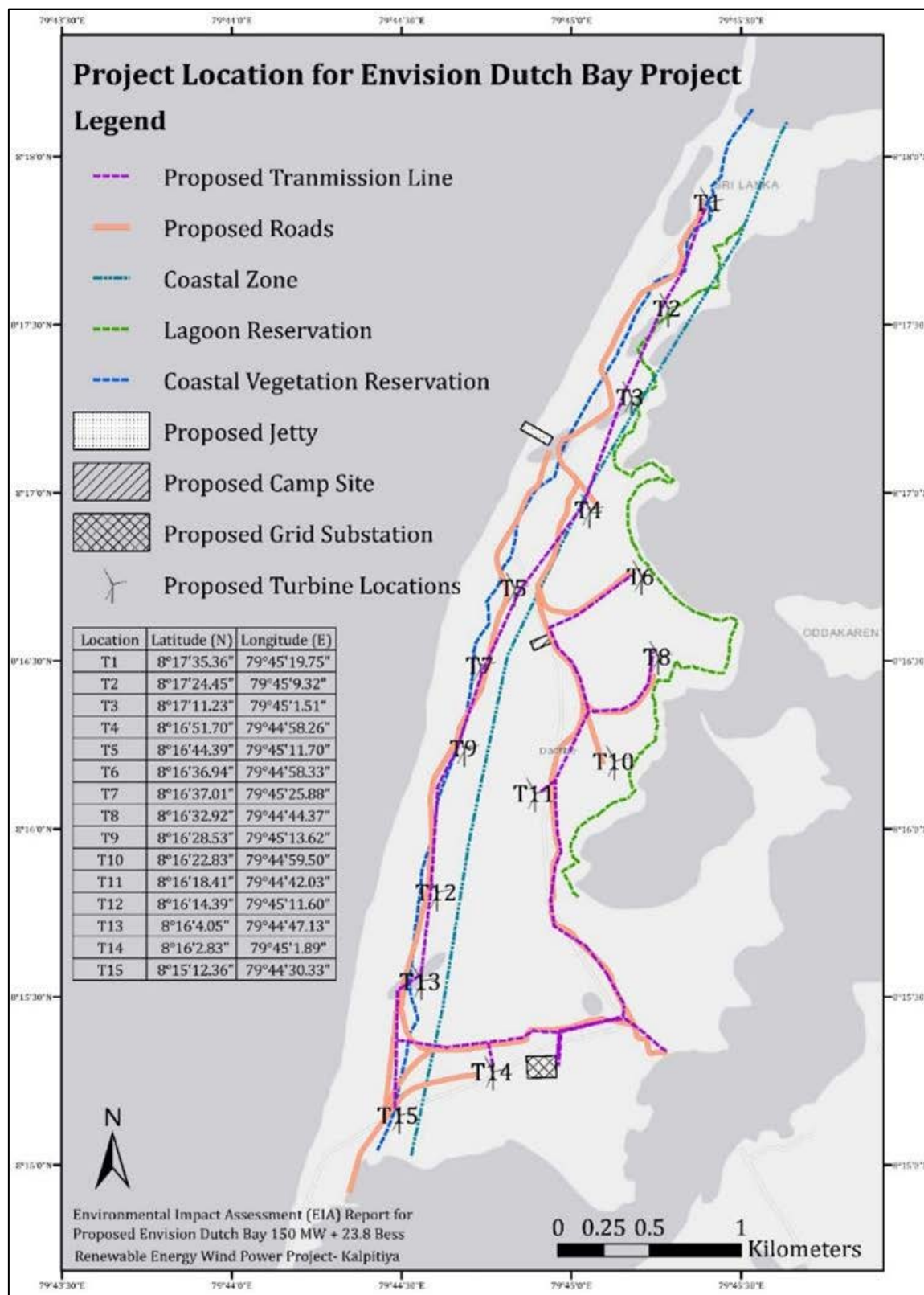


Figure 2.27: Arrangement of Internal Transmission Lines



2.2.2.1 Land Preparation

Removal of trees/ vegetation along the transmission line trace, wind tower hardstand area, access road, collector substation, grid substation

The clearance of vegetation is required within the hard standing area with a 25m radius area around each wind turbine site and as well as for the access roads and transmission lines. The total area estimated to be affected by clearing of vegetation for erection of wind turbines is indicated in **Table 2-1**. In addition, detailed descriptions are included in Ecological Impact section.

Land clearing and filling etc.

Land clearing involves the removal of trees, vegetation, rocks, and other obstacles from the project area. This is done to create a level and clear surface for the installation of wind turbines and associated infrastructure. Grading and filling will be done after land clearing. This process involves reshaping the terrain to create a level and stable foundation for the turbines and access roads. Fill material, often comprising soil, gravel, or other suitable materials, will be brought in to achieve the desired grades, and elevations. The clearance of vegetation is required within the hard-standing area (approximately 25 m radius circle) around each wind turbine site and as well as for the access roads and transmission lines. The total area estimated to be affected by clearing of vegetation for erection of wind turbines is approximately 3 ha while the total land area to be cleared will be approximately around 13 ha including all the proposed roads, site camps etc.

Earth / soil material / debris disposal during land preparation

The land clearing process will generate a substantial quantity of plant material and sand. These will be collected and disposed of on-site or off-site as appropriate. The process will not generate any artificial material or hazardous material. The ultimate disposal strategy will be decided at the project planning and lay-out setting using the following strategies.

- Some plant material has a timber value and those will be dealt with prevailing regulatory process.
- Other plant parts with firewood values will be allowed to be taken by the interested parties (residents).
- The balance material is all biodegradable and will be returned to the environment as on-site or off-site disposal.
- If on-site disposal is considered specific designated areas will be chosen for disposal, which are sufficiently away from sand dunes, water bodies and flow paths.
- Off-site disposal will be done only at the local authority designated sites through registered waste depositors.



- Excavated sand will also be deposited on-site at designated areas and will not be taken out-site the project area.

2.2.3 Details of Temporary Structures

2.2.3.1 Piers and any other temporary structures for transportation of material/equipment to the site

Some temporary structures will require to be constructed specially for the transportation of large turbine parts if the transportation by sea option is adopted. A temporary jetty will be constructed and all the details related to it are provided in **Section 2.2.1.5**.

2.2.3.2 Contractors and Engineers' site offices

These will be decided by the Contractor in collaboration with developer, ENVISION. Suitable private lands will be leased for this purpose.

2.2.3.3 Contractors' store building and store yards

These will be decided by the Contractor in collaboration with developer, ENVISION. Suitable private lands will be leased for this purpose.

2.2.3.4 Dumping sites

The construction of the turbine will be on a rigid foundation resting on piles. The Dutch bay area is with Sandy Soil. The foundation excavated material such as sand will be separated and reused for concrete works. The separated limestone will be re used. Dumping sites for the construction waste will not be necessary, as all remaining construction material will be recycled and reused.

2.2.3.5 Temporary quarry sites

Quarry sites will be needed to obtain fine and coarse aggregate for concrete and soil for backfilling. Sand sources also will be needed to obtain sand for concrete. Aggregates will be obtained from the quarry sites approved by the GSMB, and with a valid Environmental Protection Licence (EPL) from the CEA.



2.2.3.6 Concrete batching plant

No concrete batching plant will be installed within the site. Ready mixed concrete will be transported to the site from a batching plant outside. ENVISION ensure that the outside concrete batching plant is an approved one with a valid Environmental Protection Licence (EPL).

2.2.3.7 Labour camps and access to camps

An area has been allocated to be used as the camping site for about 250 workers. All approvals for the establishment of the labour camp will be obtained at the time of its establishment.

2.3 Methodology of Construction of Wind Towers

Construction workforce logistics will include temporary accommodation, sanitation, and meal areas located beyond 100 m from sensitive coastal habitats. Work will be phased to avoid nesting and fishery peaks.

2.3.1 The details on pre-construction and construction activities, phased implementation schedule, staffing/workforce, future development /expansion etc.

The expected construction period of the project is 2 years (24 months) and there is no phased implementations are planned.

The key construction sequence for the main project includes “road hydraulic filling construction (including drainage consolidation) → hydraulic filling of lifting platform and fan foundation working face → pile foundation construction → foundation excavation → anchor bolt assembly installation and foundation concrete → tower installation → wind turbine lifting → box-type substation construction → commissioning and trial operation”. The construction implementation focuses mainly on road and wind turbine construction.

The overall work schedule as per the feasibility study report is provided in **Annexure III J**. Considering the production and operational needs, combined with the modern operational characteristics of wind farms, and adhering to the principles of being streamlined, unified, and efficient, the wind farm’s staffing is considered under the principle of minimal manning, based on the specific circumstances of the wind farm. The detailed staffing table for the wind farm is shown in **Table 2-7**:



Table 2-7: Wind Farm Staffing Table

Personnel category (management organization)		Number of personnel (person)
Management	Wind farm manager	1
	Technical supervisor	1
	Shift leader	2
Production	Operating personnel	4
	Maintenance personnel	1
	Safety specialist	(also serving as the duty manager)
	Logistics management personnel	1
Total		10

2.3.2 Construction of wind tower foundations, switchgear building, and transformer rooms, erection of wind towers and other buildings etc.

The foundation details/drawings are provided in **Figure 2.13** and **Figure 2.28**. Detailed foundation drawings are given in **Annexure III G**. Standard construction techniques will be used for the turbine construction. Large cranes will be needed for this type of construction.



Figure 2.28: Typical WT Foundation Construction _From Mannar Windpower Plant



Each wind turbine should have 4 settlement observation points, and observations and records should be made for these 4 points; settlement observations and analyses should also be made for the midpoint between every two adjacent observation points.

Switch Gear Building: Switchgear is used to control and protect transformers and other electrical equipment. It allows operators to switch the power flow and isolate equipment for maintenance or in case of electrical faults. It serves to connect or disconnect the turbines from the grid, protect against electrical faults, and ensure efficient power distribution. Switchgear buildings house the equipment and controls necessary for these functions, ensuring the safe and reliable operation of the wind farm's electrical system. The location of this unit will be decided at the final layout setting.

Transformer Rooms: Transformers are used efficiently to convert the electricity generated by wind turbines into a form that can be transmitted and integrated into the electrical grid. They are housed in transformer rooms with switchgear, and other equipment necessary for the collection, transformation, and distribution of electrical power. Transformer rooms are equipped with cooling systems, fire suppressors and adhere to strict safety regulations and standards to protect workers and the equipment. This includes appropriate signage, safety barriers, and protective gear. The location of the transformer room within the wind power plant will be carefully planned to minimize transmission losses and ensure efficient power distribution, accessibility and safety at the final planning and lay-out setting.

The erection procedure of wind towers involves several key steps to ensure the safe and efficient installation of the tower structure. First, the construction site is prepared, which includes clearing the area, creating a level foundation, and installing any necessary infrastructure like access roads and crane pads. Once the site is ready, a large crawler crane or a mobile crane with sufficient lifting capacity is positioned near the tower's base. Tower sections are transported to the site and assembled on the ground. These sections are then lifted and carefully aligned, often using precision laser guidance systems, to ensure proper vertical alignment. The tower sections are bolted or welded together as they are lifted into position. Next, the nacelle, which houses the wind turbine's generator and other components, is hoisted to the top of the tower and attached. Finally, the rotor with its blades is mounted on the nacelle, completing the assembly. Throughout the entire process, safety measures are of utmost importance, and experienced construction crews follow strict guidelines and procedures to minimize risks and ensure a successful and reliable wind tower installation.



2.3.3 Method of wind tower and Blade transportation at the site

Transportation to Site

There are currently two main transportation methods: traditional transportation and special transportation:

1) Traditional transportation of tower barrels

Refer **Figure 2.29** for the traditional transportation scheme. Advantages and disadvantages are as follows:

- Number of tower sections transported per vehicle: 1
- No need for secondary transshipment

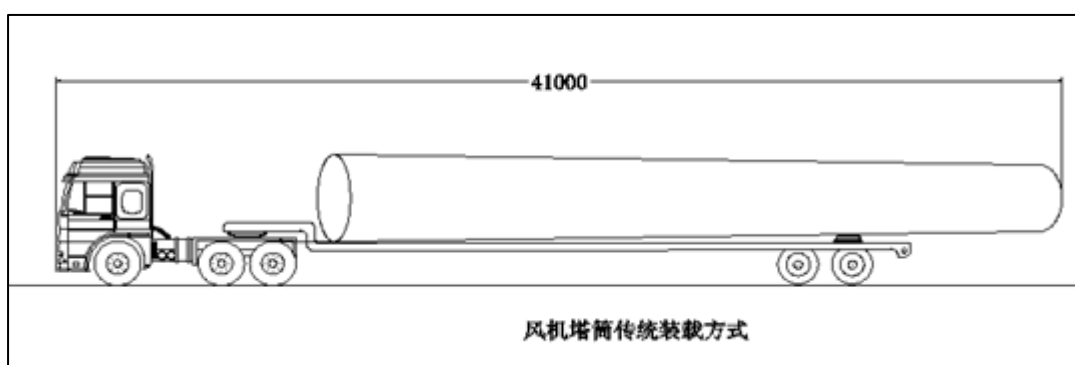


Figure 2.29: Traditional Transportation Scheme for Tower Barrels

2) Special transportation for tower barrels

See **Figure 2.30** for the special transportation scheme. Advantages and disadvantages are as follows:

- Number of tower sections transported per vehicle: 1
- Secondary transshipment is required, with a transshipment site requirement of 55m×55m

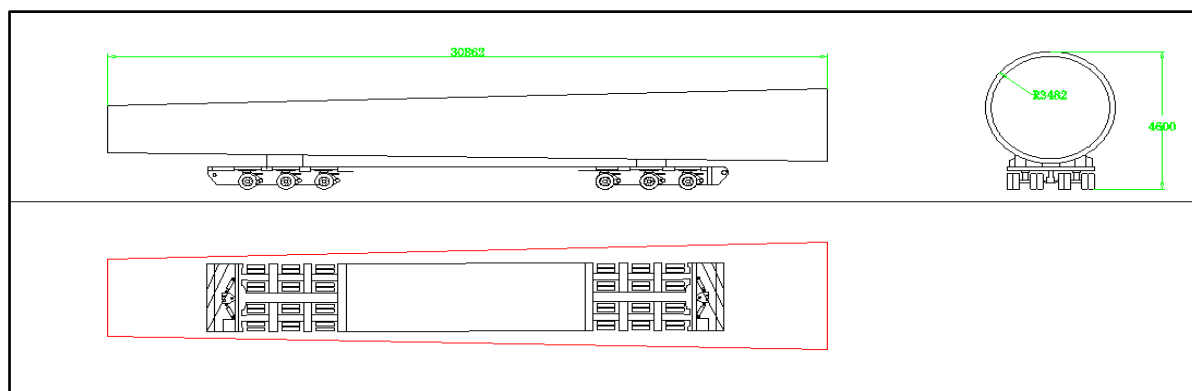


Figure 2.30: Special Transportation Scheme for Tower Barrels



Turning indicates, as detailed in **Figure 2.31**, that: both front and rear have driving force ==> enhances climbing capability.

Both front and rear have steering capability ==> reduces the turning radius.

No vehicle head ==> reduces the turning radius and minimizes the sweep range during operation.

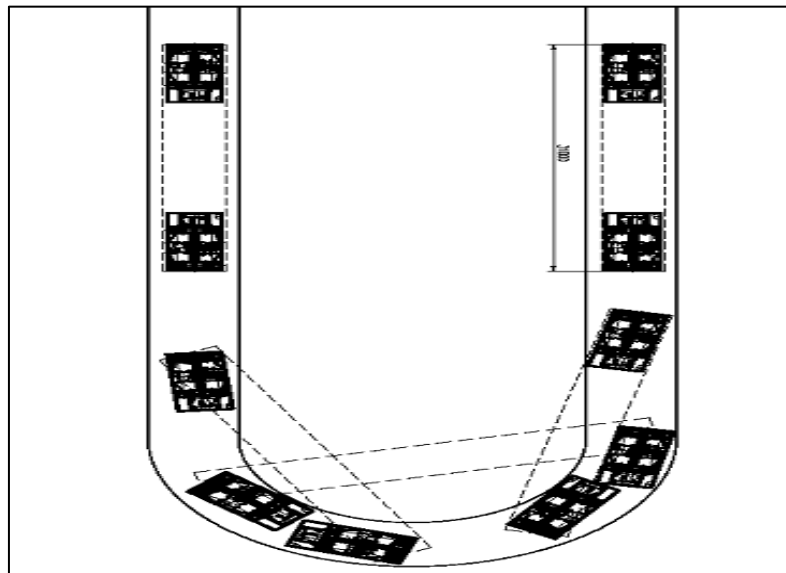


Figure 2.31: Special Transportation Turning Indication

3) Traditional transportation of blades

See **Figure 2.32** for the conventional transportation scheme of blades. Advantages and disadvantages are as follows:

- (1) Number of blades transported per vehicle: 2;
- (2) No need for secondary transshipment

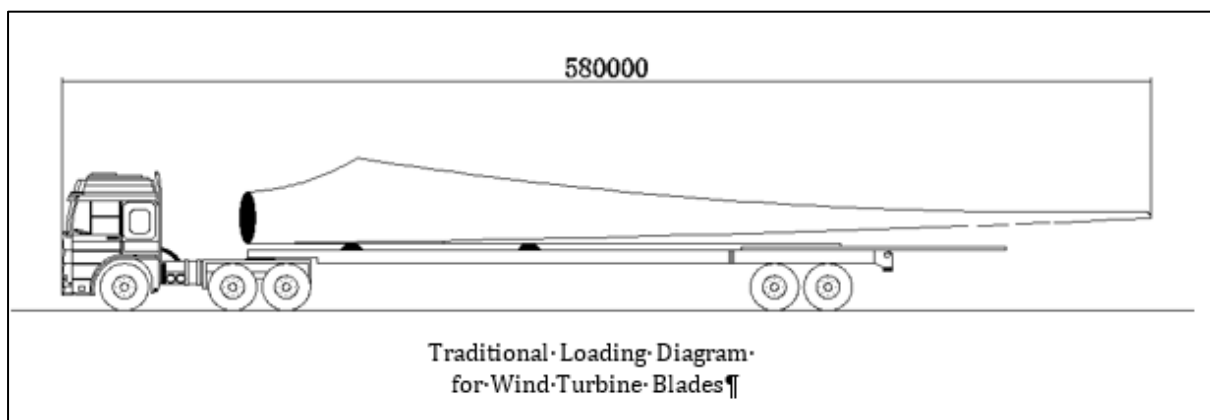


Figure 2.32: Traditional Transportation Example for Wind Turbine Blades



4) Special transportation of blades

Refer Figure 2.33 and Figure 2.34 for the special transportation scheme of blades. Advantages and disadvantages are as follows:

- Number of blades transported per vehicle: 2
- Secondary transshipment is required, with a transshipment site requirement of 55m×55m;

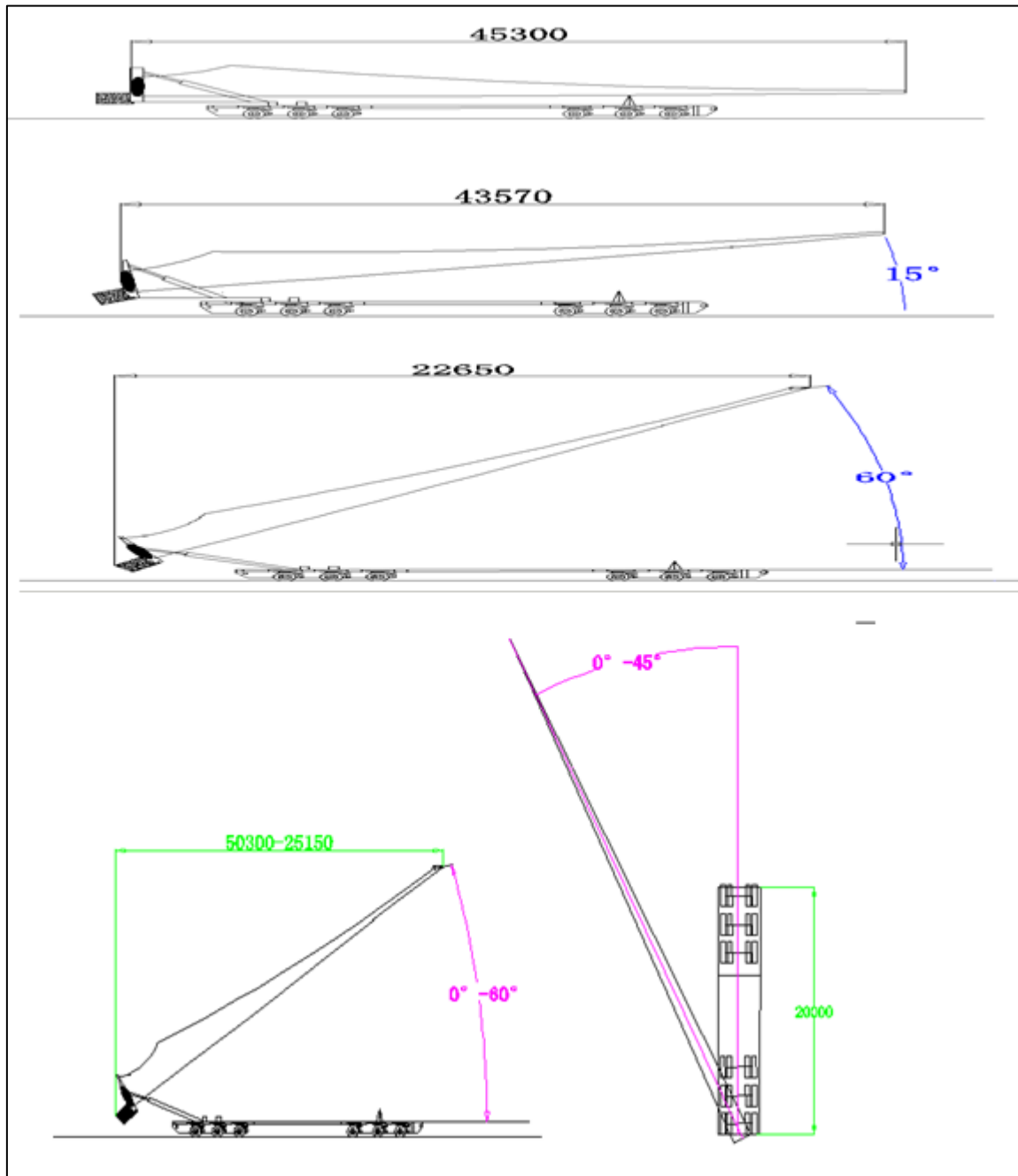


Figure 2.33: Special Transportation Loading Diagram for Blades



Figure 2.34: Special Transportation Example for Blades

Both methods have their advantages and disadvantages, but from a technical perspective, special transportation effectively reduces the turning radius, making road layout more flexible and convenient. It significantly reduces the height of roadbed excavation and fill, fundamentally benefiting local soil and water conservation. Although the cost of special transportation is higher than traditional transportation, a comprehensive comparison shows that in terms of road construction costs, earthwork excavation volume, and slope height, special transportation is superior to traditional transportation. Therefore, this project recommends the use of special transportation.

A preliminary Logistic study will be conducted to find the most suitable blade transport routes once the EPC for the project be awarded. Findings from this logistic study will be made available to the necessary authorities through the EMP.



2.3.4 Installation of wind turbine and rotor blades

Rotor blades will be brought to the site (after unloading at temporary jetty) using a special vehicle and the blades will be installed using cranes, observing standard safety measures. A turbine tower will be assembled at site, and it will be lifted to position using cranes (**Figure 2.35**). 600-700 tones capacity cranes will be used to erect wind turbines



Figure 2.35: 600-700 tones capacity cranes which will be used to erect wind turbines

2.3.5 Laying of power and instrumental cables

The instrument and power main cable between the turbines will be overhead cables.

2.3.6 Method of construction of substations /switch yard, transmission towers and method of Stringing

The switchyard building construction will adopt standard building construction methods, using small machinery and mainly manual labour. The electrical circuits will be installed after completing the main building works. Building plans along with electrical layout plans will be prepared before construction.

Most of the cables will be overhead cables, where stringing is required. Stringing will be needed for the overhead transmission line. Overhead transmission line stringing will be carried out using machinery.



2.3.7 Infrastructure Developments (if any)

The project will be connected to the national grid via a newly constructed 220 kV double-circuit line extending from the Dutch Bay Wind Collector Substation to the Wariyapola South Grid Substation. This configuration has been validated by EMT studies and is supported by a 100 Mvar STATCOM installation for voltage and reactive power stabilization. This replaces earlier assumptions of a 132 kV connection to Puttalam Grid Sub Station (GSS).

The construction and operation of a wind tower plant involves distinct sets of infrastructure requirements to support the installation, maintenance, and efficient functioning of wind turbines. During the construction phase, the critical infrastructure components are the transportation infrastructure for the delivery of turbine components, including tower sections, nacelles, and rotor blades. Adequate road access and temporary construction of roads or paths will be established for safe and efficient transportation. Moreover, the construction site will have electrical and grid connections to power construction equipment, communication facilities, temporary yards, sanitary facilities, security units, health and safety and emergency management facilities for the workforce and on-site waste and scrap management system.

In the operation phase, infrastructure needs to be shifted towards long-term operational efficiency. The central infrastructure element is the electrical substation, which connects the turbines to the grid. Grid connectivity is vital for exporting electricity generated by wind turbines. Access roads, both for maintenance personnel and emergency services, will be well-maintained to ensure timely response and regular upkeep of the turbines. Additionally, maintenance facilities and storage areas for spare parts and equipment will be set up. Facilities will be established to store segregated waste including hazardous waste. A meteorological mast or remote sensing devices will be set up to monitor wind conditions continuously for optimizing turbine performance. Communication infrastructure will also be in place for effective communication. sanitary facilities, security units, health and safety and emergency management facilities for the workforce will also be provided.

2.3.8 Details of any permanent / temporary access roads to be built / improved (existing condition and anticipated improvements)

Refer **Section 2.2.1.4 Access Roads**.

2.3.9 Techniques and equipment to be used

The construction of wind turbines for the Envision Dutch Bay Onshore Wind Power Project begins with detailed site investigations and geotechnical assessments, which have confirmed the



suitability of reinforced concrete gravity foundations for each turbine. As described in the EIA, these foundations will typically consist of a circular base pad approximately 21–23 metres in diameter and 3–3.5 metres in depth, depending on soil bearing capacity. Reinforced with high-tensile steel and designed to resist both vertical loads and overturning moments from rotor operation, the foundations are cast in-situ using imported formwork and local aggregates. All excavation and earthworks will follow best-practice erosion control protocols, with excess spoil used in site rehabilitation or stabilisation of nearby embankments.

Once foundations are completed and cured, the transportation phase commences. Envision deploys a marine-first logistics strategy for Dutch Bay, with blades (110 m), nacelles (138 tons), and tower sections (five tower sections with weights of 63.7 tons, 72.8 tons, 84 tons, 99 tons and 128 tons) delivered via barge from Colombo Harbour to a modular temporary jetty constructed at the shoreline near Turbine No. 4. From there, specialised multi-axle hydraulic trailers transport the components along an upgraded 3.4 km internal road network reinforced to a bearing capacity of 12 t/m², with careful curve widening and sand-brick sublayers to accommodate long cargo and cranes. The entire logistics flow is designed to minimize environmental disturbance and avoid congested public highways.

At each turbine location, Envision’s construction crews mobilize crawler cranes and telescopic boom cranes capable of lifting up to 250 tonnes at hub heights exceeding 140 metres. The tower sections are installed sequentially using precision alignment systems and digital plumb monitoring. The nacelle, pre-assembled with drive train and yaw system, is then hoisted and mounted atop the tower. Finally, rotor blades are lifted either individually or pre-assembled in a “bunny ear” or “star” configuration depending on wind conditions and terrain. Real-time SCADA calibration ensures safe commissioning and functional integration with the turbine’s AI-based EnOS™ platform.

Throughout the construction phase, Envision implements strict environmental and occupational safeguards. Oil spill risks are mitigated through sealed hydraulic systems and pre-fuelled crane operations, with no on-site refuelling at the jetty. Spill kits, booms, and sediment traps are maintained at all active sites. Dust suppression, noise barriers, and daily compliance monitoring are conducted in accordance with the Environmental Management Plan. All construction activity is conducted under the oversight of certified engineers, with reporting aligned to CCD and SLSEA protocols to ensure minimal impact on the local ecology and community.

2.3.10 Method of material / equipment transport and installations

This has been discussed in other sub sections of Section 2.3.3, Section 2.3.7, and Section 2.4.4.2. .



2.3.11 Time Schedule, staffing and support facilities, technologies

For the time Schedule, refer to Section 2.3.1. All support facilities and the required technology will be provided by ENVISION. The latest technology will be used for turbine manufacturing, transportation, and erection.

The estimated time schedule is provided below in **Table 2-8**. All support facilities and the required technology will be provided by ENVISION. The latest technology will be used for turbine manufacturing, transportation, and erection.

Table 2-8: Estimated Project Time Schedule

Phase	Description	Duration	Schedule Window
1. Mobilization & Enabling Works	Project office setup, material yards, workforce mobilization	1 month	Month 1
2. Jetty Construction & Coastal Works	Floating jetty assembly, environmental safeguards	3 months	Months 1–3
3. Internal Road & Platform Construction	3.4 km road upgrades, turbine hardstand preparation	6 months	Months 2–7
4. Foundation Construction (All Turbines)	Rebar, in-situ casting, inspections (staggered)	9 months	Months 3–11
5. Substation & Grid Infrastructure	220/33 kV substation, STATCOM, 220 kV line	12 months	Months 3–14
6. BESS Installation & Commissioning	Site grading, containerized units, interconnection	5 months	Months 10–14
7. Component Delivery & Storage	Marine delivery, storage, haulage from jetty	8 months	Months 6–13
8. Wind Turbine Erection (15 Units)	Tower assembly, nacelle, rotor using cranes	7 months	Months 9–15
9. Electrical Cabling & SCADA	MV cabling, fiber, SCADA system setup	6 months	Months 11–16
10. Testing & Grid Synchronization	Functional tests, CEB compliance checks	3 months	Months 15–18
11. Demobilization & Restoration	Jetty removal, landscaping, final grading	3 months	Months 20–22
12. Commercial Operations Readiness	Final inspection, CCD & CEA sign-off	2 months	Months 23–24



2.3.12 Disposal of construction debris and solid waste

Construction material remnants (steel pieces, paint cans) will be handed over for recycling. Shuttering material, sheet piles used for shoring could be reused. Sand mixed with other impurities will be sifted at site and reused. Hence there will not be a significant amount of construction material which will have to be disposed of as waste. Removed large trees will be handed over to the Timber Corporation. Removed Palmyra and Coconut trees will be disposed of according to the instructions of the Palmyra Development Board and Coconut Cultivation Board. Other bushes, shrubs, twigs will be given to the locals for firewood, fencing etc. The balance leaf load will be disposed onsite for bio degradation.

2.3.13 Other construction activities

Apart from the turbine construction, access road rehabilitation and dedicated access route construction will be done.

2.4 Methodology of Operation, Repair, and Maintenance

2.4.1 Operational Activities related to the wind plant and transmission line

Operational activities are related to energy harnessing through wind power. Most of the operations will be automatic with minimum human interventions.

2.4.2 Operational activities of avifauna detection and collision preventing system

Refer section 2.2.1.3 Avifauna Detecting Mechanism.

2.4.3 Repair and maintenance activities with frequencies

Complex systems such as wind turbines require regular inspections to guarantee trouble-free operation. A differentiation needs to be made between inspection, maintenance, and repair. The inspection of a wind turbine requires a visual evaluation of the entire structure without any remedial work being performed. Such an inspection of a wind turbine can take approximately one day.

Repair and maintenance activities of the turbines will be done based on the standard procedure. The maintenance of wind turbines takes place at predetermined intervals, usually once or twice a year whereby all important mechanical and electrical assemblies are checked. In addition,



minor repairs can be performed, and consumables such as grease, oils and filters can be exchanged.

More serious damage, above all to the rotor blades, is rectified during repairs. Depending on the extent and complexity of the damage, this can take up to a few days. While mechanical assemblies tend to be less prone to damage, areas such as the electrics, electronics, sensors, and hydraulics require more frequent repair. However, such repairs normally result in little down-time and are simple.

Repairs for gearboxes, rotor blades or the generator are extremely complex. Spare parts are expensive, and the repair work requires significantly longer down-times. Damage also occurs to the tower's foundation, flanges, and the tower, in the form of corrosion. Damage to the rotor blades is usually the result of corrosion and lightning strikes.

The manufacturer will provide an operation and maintenance (O&M) manual which will be followed for the turbine maintenance. There will be similar maintenance manuals for the grid sub stations and underground/overhead cables.

2.4.3.1 Lightning protection

Envision Energy places paramount importance on lightning protection in its wind turbine designs, integrating advanced systems to safeguard all components from potential lightning-induced damages. This comprehensive approach ensures the structural integrity and operational reliability of their turbines, particularly in lightning-prone areas.

At the forefront of Envision's lightning protection strategy is the incorporation of lightning receptors on turbine blades. These receptors are meticulously designed to capture lightning strikes and channel the electrical energy safely through conductive paths within the blade structure. This design minimizes the risk of structural damage and ensures the longevity of the blades.

Beyond the blades, Envision's turbines feature a continuous conductive path that extends from the nacelle down through the tower to the grounding system. This path is engineered to direct lightning currents away from critical components, such as the generator and control systems, thereby preventing potential malfunctions or failures.

The grounding system itself is a critical component of the lightning protection scheme. Envision employs robust grounding techniques, including the use of copper conductors and grounding electrodes, to dissipate lightning energy safely into the earth. This system is designed in accordance with international standards to ensure optimal performance and safety.



To maintain the effectiveness of the lightning protection system, Envision implements regular inspection and maintenance protocols. These protocols involve checking the integrity of lightning receptors, conductive paths, and grounding systems, ensuring that any potential issues are identified and addressed promptly. This proactive approach underscores Envision's commitment to safety and reliability in their wind turbine operations.

2.4.3.2 Integration of AI-Driven Predictive Maintenance

Envision Energy has incorporated advanced AI algorithms into their wind turbines, enabling real-time analysis of sensor data to detect anomalies in vibration, temperature, and power output. This proactive approach allows for early identification of potential faults, facilitating timely interventions and minimizing unplanned outages. By continuously learning from historical and real-time data, these AI systems enhance diagnostic accuracy over time, ensuring optimal turbine performance.

2.4.3.3 Deployment of MEMS Sensor Technology

To bolster the structural integrity monitoring of wind turbines, Envision Energy employs Micro-Electro-Mechanical Systems (MEMS) sensors developed by Analog Devices. These sensors provide precise measurements of vibration and tilt, crucial for assessing the health of critical components like the tower barrel. The integration of MEMS technology facilitates real-time monitoring, enabling predictive maintenance and enhancing the safety and reliability of wind turbine operations.

2.4.3.4 Utilization of Deep Learning for Failure Prediction

Recent studies have demonstrated the efficacy of deep learning models in predicting failures in wind turbine components. By analyzing SCADA data, these models can forecast anomalies in critical subsystems, such as gearboxes and generators, weeks in advance. Implementing such predictive analytics allows for strategic maintenance scheduling, reducing downtime and extending the lifespan of turbine components.

2.4.3.5 Adoption of Centralized Data Management Platforms

Envision Energy leverages centralized platforms like the 3DEXPERIENCE by Dassault Systèmes to manage the lifecycle of their wind turbines. This integration ensures seamless data flow across design, engineering, and maintenance processes, facilitating informed decision-making and



efficient resource allocation. By centralizing data management, Envision enhances the traceability and responsiveness of their maintenance operations.

Incorporating these advanced maintenance strategies into the EIA will align the Dutch Bay Wind Farm project with cutting-edge industry practices, promoting operational excellence and sustainability.

2.4.3.6 Safety System

The safety system is designed with particular care and fulfils the strict requirements of the guidelines in ISO 13849-1 (Safety of machinery – safety-related parts of control systems). The purpose of the safety system is to ensure that the turbine remains in safe operational condition under any circumstances, particularly if the supervisory control, individual components, or systems would fail. (See Annexure III N).

2.4.4 Material and utility requirements

2.4.4.1 Raw materials for construction activities

According to the conceptual designs of wind turbine foundations, raw material requirement for wind turbine foundation construction is as follows.

Concrete	=	3085 m ³
Cement	=	1234 tons
Water	=	497 m ³
Sand	=	2089 m ³
Coarse aggregates	=	4185 m ³

Table 2-9: Construction material Statistics per 10 MW wind turbine

No	Type	Concrete (m ³)	Cement (ton)	Water (m ³)	Sand (ton)	Coarse aggregates (ton)
1	WTG foundation	1430 (40 MPa)	572 (M42.5)	229	973	1945
2	Foundation concrete piles	1580 (40 MPa)	632 (M 42.5)	253	1075	2145
3	Foundation layer	75 (20 MPa)	30 (M32.5)	15	41	95
4	Total	3085	1234	497	2089	4185



2.4.4.2 Water requirement during construction period

As ready mixed concrete will be used for foundation, which is the most water consuming activity, the amount of water needed for construction work is limited as the ready-mix will be transported to location by mixture trucks from an existing mixing plant outside of the project area. During the construction period, all domestic water requirements for the labour camp will be transported in water bowsers from an existing external source and stored locally at the camp location.

As the turbines are spread and far between water will be purchased from an external source, which is approved whenever required.

2.4.4.3 Amount of water to be used for the project during operational period

The water requirement during operational phase will be limited only to domestic water and the quantity will be very much low. Domestic water will only use by the skeletal maintenance and security personnel who will be housed at the sub-station location.

Based on available data and international guidelines, the estimated daily water requirement during the operational phase of the Dutch Bay Wind + BESS project is approximately 1.2 cubic meters per day (m^3/day).

Estimated water consumption

- **Personnel Count:** Assuming a skeletal team comprising 6 personnel (4 maintenance staff and 2 security personnel) stationed at the substation.
- **Per Capita Water Usage:** The World Health Organization (WHO) recommends a minimum of 50 to 100 litres per person per day to meet basic needs, including drinking, sanitation, and hygiene. Additionally, a study conducted in Sri Lanka reported an average daily water consumption of approximately 119.44 litres per person.

Calculation:

- Using the conservative estimate of 50 litres per person per day:
- $6 \text{ persons} \times 50 \text{ litres/person/day} = 300 \text{ litres/day} = 0.3 \text{ m}^3/\text{day}$
- Using the higher estimate of 119.44 litres per person per day:
- $6 \text{ persons} \times 119.44 \text{ litres/person/day} \approx 716.64 \text{ litres/day} \approx 0.72 \text{ m}^3/\text{day}$

Considering potential variations and to ensure adequacy, it is prudent to plan for a slightly higher provision. Therefore, estimating a daily water requirement of $1.2 \text{ m}^3/\text{day}$ provides a reasonable buffer to accommodate any unforeseen needs.



2.4.4.4 Possible water sources in the area

Groundwater Availability and Quality in Dutch Bay

The Dutch Bay area, located in the Puttalam District of Sri Lanka's northwestern dry zone, is characterized by sandy soils and a shallow coastal aquifer system. Groundwater serves as the primary source of water for domestic and agricultural needs in this region. However, the quality of groundwater is significantly influenced by factors such as seawater intrusion and agricultural activities.

A study focusing on the Kalpitiya Peninsula, which shares similar hydrogeological characteristics with Dutch Bay, revealed that groundwater salinity levels vary considerably across the area. Electrical Conductivity (EC) measurements ranged from 103 $\mu\text{S}/\text{cm}$ to 33,016 $\mu\text{S}/\text{cm}$, corresponding to salinity levels between 0.1 ppt and 31.4 ppt. Notably, the Talavila area recorded an average EC of 33,016 $\mu\text{S}/\text{cm}$ and a salinity of 31.4 ppt, indicating significant seawater intrusion.

These findings suggest that groundwater in the Dutch Bay area may not be suitable for direct domestic consumption without treatment, especially during the dry season when salinity levels tend to increase.

Feasibility of Borehole Water Extraction

Considering the estimated daily water requirement of 1.2 m^3/day for the project's operational phase, it is technically feasible to extract this volume from a borehole. However, the high salinity levels observed in the region's groundwater necessitate the implementation of a water treatment solution to ensure the water meets potable standards.

Necessity for a Small-Scale Reverse Osmosis (RO) Plant

Given the elevated salinity levels, a small-scale Reverse Osmosis (RO) plant would be essential to treat the extracted groundwater and render it suitable for domestic use. RO systems are effective in removing dissolved salts and other impurities, making them a viable solution for desalinating brackish water.

For the project's scale, an RO plant with a capacity of approximately 1.5 m^3/day would suffice, providing a buffer above the estimated daily requirement. Such systems are compact, energy-efficient, and can be tailored to the specific water quality parameters of the source.

Recommendations

- Hydrogeological Survey: Conduct a detailed hydrogeological assessment to determine the optimal location for borehole drilling, ensuring the best possible water quality and yield.



- **Water Quality Testing:** Perform comprehensive water quality analyses, focusing on parameters such as salinity, hardness, and the presence of contaminants like fluoride and nitrates.
- **RO System Design:** Based on the water quality results, design an RO system tailored to the specific treatment requirements, ensuring compliance with Sri Lanka's potable water standards.
- **Monitoring and Maintenance:** Implement a regular monitoring and maintenance schedule for the RO system to ensure consistent performance and water quality.
- If tube wells are to be constructed to extract water such an activity will be done under the approval of the Water Resources Board.

By adopting these measures, the project can ensure a reliable and safe water supply for its operational needs, leveraging local groundwater resources while mitigating the challenges posed by salinity.

2.4.5 Construction of a fence, boundary wall around the land or wind towers

A fence will not be constructed. Although the land is acquired or leased to construct turbine footings and access roads after construction, these lands will be released for routine activities of the villagers such as cattle grazing, walking etc. and other uses permitted by the owners of the surrounding land areas. Only the ~18m diameter turbine foundation area will be permanently used and fenced for security purposes.

2.4.6 E-waste generation storage and disposal methods

Electronic waste, or e-waste, includes various electronic components and equipment used in wind power projects, such as sensors, control systems, communication devices, and more will require proper safe management to minimize environmental impact and comply with regulations. The e-waste generated from the power plant will be managed as follows:

Assessment and Inventory: Develop an inventory of all electronic equipment used in the wind power project. This includes identifying and documenting the type, quantity, and condition of each item.

Segregation: When the e-waste is generated, it will be sorted into categories based on its nature and potential hazards. Common categories include batteries, circuit boards, monitors, cables, and numerous electronic parts of turbines, control panels and other accessories. This step helps in organising the disposal process.



Labeling: Clearly label all e-waste items with information such as the date of decommissioning, the reason for disposal, and any hazardous materials contained within. This labeling is crucial for tracking and managing the waste.

Storage: Store e-waste in a secure and controlled environment to prevent damage and contamination. Ensure that the storage area is dry, well ventilated, and protected from the elements.

Data security: Before disposing of electronic devices, ensure that all sensitive data is securely erased or destroyed to prevent data breaches. Will use data wiping or physical destruction methods as appropriate.

Hazardous Materials Handling: Identify and handle hazardous materials found in e-waste (e.g., lead-acid batteries, mercury-containing components) according to local environmental regulations. Store these items separately and dispose of them responsibly.

Reuse and Recycling: Wherever possible, will consider refurbishing or recycling e-waste components. Some components may have reusable parts or valuable materials that can be extracted for reuse, reducing the overall environmental impact.

Disposal: Dispose of e-waste through authorized and certified e-waste recycling and disposal facilities that follow proper disposal procedures and comply with environmental regulations. CEA registered licenced E- waste collectors will be used. For the parts where local collectors are not available will be shifted to host country or else places outside Sri Lanka where facilities are available.

2.5 Decommissioning Phase

Envision Energy advocates for a sustainable and circular approach to wind turbine decommissioning, aligning with its commitment to environmental responsibility. After a typical operational lifespan of 20–25 years, Envision recommends a structured decommissioning process that emphasizes component reuse, material recycling, and site restoration. This approach not only minimizes environmental impact but also maximizes the value extracted from each turbine component.

A significant challenge in turbine decommissioning is the disposal of fiberglass-reinforced blades, which are not easily recyclable. Envision supports innovative repurposing strategies for these blades. For instance, retired blades have been transformed into architectural elements such as



pedestrian bridges and urban furniture. These applications extend the utility of the blades and contribute to sustainable urban development.

Beyond blade repurposing, Envision emphasizes the importance of recycling other turbine components. Steel towers and copper wiring can be efficiently recycled, while concrete foundations can be crushed and reused in construction. Envision also explores advanced recycling techniques for composite materials, aiming to recover valuable fibers and resins for use in new products.

Envision's decommissioning strategy includes thorough site restoration to its original state or better, ensuring that land can be repurposed for agriculture or other uses. This holistic approach to decommissioning reflects Envision's dedication to sustainability and its role in promoting a circular economy within the renewable energy sector.

2.6 Details of Resettlement, Land acquisition (if any)

The proposed wind power project has been carefully designed to avoid physical resettlement, which is a positive aspect of the project. Currently, there are no houses, buildings, or other permanent or temporary structures present on the lands, which would be utilised for wind turbines footprint, grid station and temporary labour camping areas and there will be no requirement for either voluntary or involuntary resettlement. As a result, no physical displacement of people is anticipated. Consequently, no resettlements are engaged to this project and no households are to be relocated due to the project.

2.7 Details of Investment and Funding Sources and Mechanisms (state the total cost of the project financial commitment)

The project will be financed through a 70:30 split of debt and capital - with senior debt being provided by a consortium of international banks who are long-term financing partners of Envision Energy Global. Financing commitments had been obtained and provided to the sustainable energy authority to obtain provisional approvals. Equity will be provided by Envision Energy Trading, a 100 subsidiary of Envision Energy Global.

2.8 Evaluation of Alternatives

This section describes the reasonable alternatives considered during the proposed project in a way the project is less harmful to the environment. The following options can be considered for analysis of alternatives.



2.8.1 Site alternatives / Sites with good wind resource

NREL's validated wind mapping for Sri Lanka shows that several areas have good or excellent potential wind resources. These areas are distributed in two main regions. The first one is the northwest coastal region, from the Kalpitiya Peninsula in the north to the Mannar Island and the Jaffna Peninsula (**Figure 2.36**). The second one is Sri Lanka's central highlands, located in the Central Province, but also involving parts of the provinces of Sabaragamuwa.

This would consist of five different area in the country, naming Southeast Coast, Kalpitiya Peninsula, Mannar Island, Jaffna, and Ambewela in a more detailed way. In addition, studies that are more refined suggest that the Kalpitiya Peninsula on the west coast, Mannar Island etc. as the promising sites for new wind power generation.

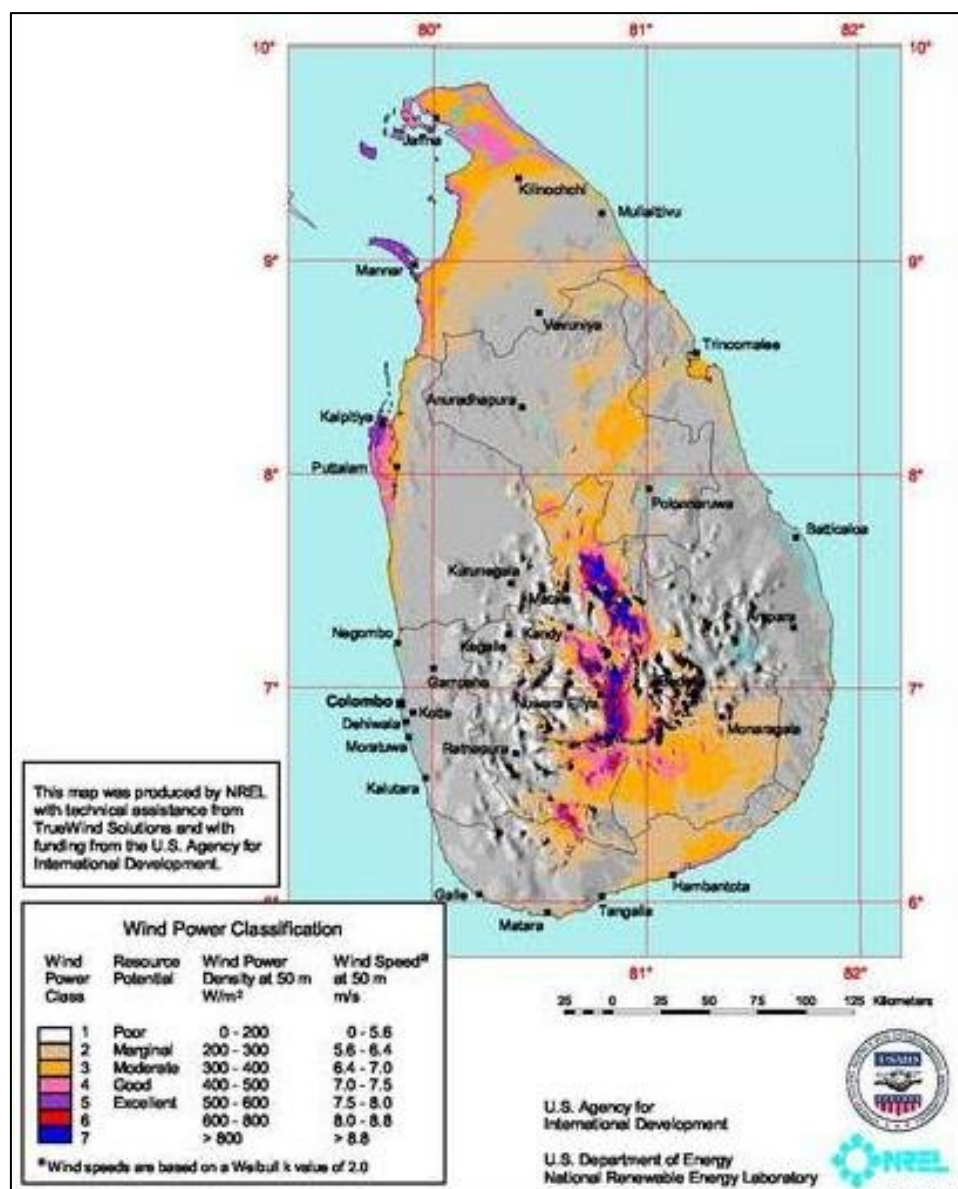


Figure 2.36: NREL's Wind Mapping for Sri Lanka



The proposed site in Kalpitiya satisfies the technical feasibility with respect to energy resource availability, terrain condition, vegetation character, soils, and also most of the socio-economic requirements such as land availability with good control of security in the area by the government, feasible site access (roads), resource availability in proximity (water, aggregates, soils) and transportation routes for power plant units (long blades and towers), material and construction machinery (feasible route is there from Colombo harbour) etc.

In selecting the Dutch Bay site for the development of the 150 MW Wind + 23.8 MWh BESS project, the project proponent (Envision Energy) has operated strictly within the formal policy and legal framework established by the Government of Sri Lanka. In accordance with Sections 5(c)(iii), 10, and 12–15 of the Sri Lanka Sustainable Energy Authority Act No. 35 of 2007, the Sri Lanka Sustainable Energy Authority (SLSEA), acting as the designated agency under the Ministry of Power and Energy, is empowered to identify, evaluate, and allocate sites suitable for on-grid renewable energy development. This includes the authority to declare areas as Energy Development Areas (EDAs), and to grant exclusive development rights to selected investors following expression of interest (EOI) procedures endorsed by Cabinet and reviewed by SLSEA's Project Approving Committee (PAC), which includes representatives from the Ceylon Electricity Board (CEB), Central Environmental Authority (CEA), and the Department of Wildlife Conservation (DWC).

Pursuant to these legal provisions, the Dutch Bay area in the Kalpitiya Peninsula was preliminarily assessed and reserved by SLSEA as a suitable wind development zone, based on technical feasibility, coastal wind yield, minimal environmental constraints, and proximity to available grid infrastructure. Envision responded to the EOI call, submitted detailed technical and financial proposals, and was granted Provisional Approval No. PA-243748 by SLSEA on 26 July 2024. This approval granted Envision the exclusive right to conduct environmental studies, secure grid capacity, and prepare project implementation documentation for the Dutch Bay site, subject to clearance from relevant authorities. This process followed all legal steps under Sections 16–19 of the SLSEA Act for on-grid renewable project permitting, with coordination between SLSEA, the Coast Conservation Department (CC&CRMD), and CEA to formalize the Environmental Impact Assessment pathway.

In terms of transmission feasibility, the Dutch Bay site is uniquely positioned with confirmed access to the national grid via a newly approved 220 kV interconnection to the Wariyapola South Grid Substation. This route, validated by the CEB in their official letter dated 25 September 2024, replaces earlier, less feasible options to Puttalam or Norochcholai substations. The proposed route ensures strong grid stability with a Short Circuit Ratio (SCR) of 10.73 at the Point of



Connection and includes a planned 100 Mvar STATCOM for reactive power support. The EMT and PSSE/PSCAD modelling requirements have been defined, and all preliminary concurrence milestones with CEB have been achieved. These conditions make Dutch Bay not only technically feasible but also one of the few viable grid-connected wind corridors remaining in Sri Lanka with open capacity for utility-scale integration.

As such, there are no practicable alternatives to the Dutch Bay site for Envision within the current legal and energy planning framework. The project's development rights were awarded through a legitimate, transparent, and structured process, and the technical prerequisites for successful integration—wind yield, grid access, logistical connectivity, and environmental compatibility—are uniquely satisfied by this location. The site is also not subject to overlapping energy developer claims or litigation, nor does it lie within protected areas, cultural heritage zones, or water catchments. Accordingly, the Evaluation of Alternatives must acknowledge that while Sri Lanka possesses other high-wind regions, none currently offer a legally reserved, environmentally compliant, and grid-feasible option for the proponent other than Dutch Bay. **(Refer Annexure III O).**

2.8.2 Design and Technology Selection

2.8.2.1 Wind turbine location alternatives (based on other studies, avifauna, reservations, wake factor etc.)

Wind turbine locations were determined through a site optimization analysis that accounted for energy losses, accessibility via existing and planned road infrastructure, land ownership constraints, environmental considerations, and potential disturbances to human activities. Considering uncertainties in site-specific conditions, wind turbines T5 to T14 were repositioned **Figure 2.37**. This decision was based on a comprehensive assessment that included energy losses (wake factor), road accessibility, land ownership, environmental sensitivities (such as waterlogged areas and low-vegetation zones), and potential disturbances to human activities. Taking all these factors into account, the turbine positions were adjusted to optimize both operational performance and environmental compatibility.

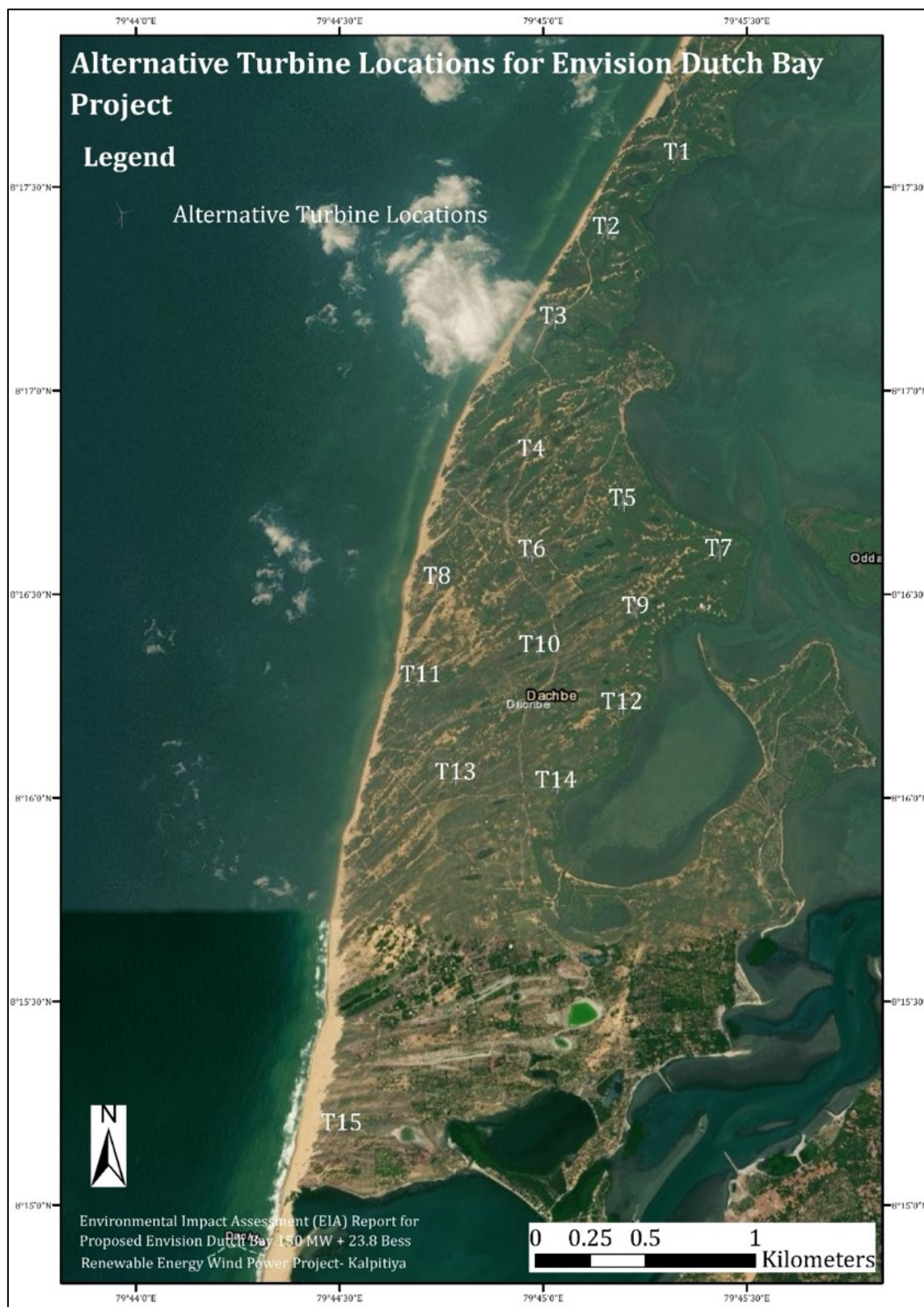


Figure 2.37: Wind Turbine Location Alternative



2.8.2.2 Construction technique and phasing

Wherever possible, existing roads and tracks were utilised for construction and maintenance access to the site.

2.8.2.3 Collector network alternatives

To minimize ground disturbance, the connection network was designed to follow the existing distribution line along the roads.

2.9 External Transportation Strategy

The external transportation strategy for the Dutch Bay Wind Power Project is centred on marine logistics, with the majority of heavy components—including turbine blades, nacelles, towers, and BESS containers—being transported by sea from the Colombo Harbour to a purpose-built temporary jetty near Turbine No. 4 (located at 8°17'4.82"N, 79°44'54.09"E). This jetty is designed as a modular floating platform of 110 metres in length and 45 metres in width, capable of supporting up to two 250-ton mobile cranes, facilitating efficient offloading of wind turbine components directly at the project site. Since the structure is floating, it will not disturb the existing hydrodynamics of the area and thereby not influence the sediment transport pattern of the area. The temporary nature of this infrastructure ensures minimal long-term impact, with complete removal and restoration programmed post-construction.

A total of approximately 25 barge trips will be used for transporting components such as blades (110 m), nacelles (138 tons), and tower sections (five tower sections with weights of 63.7 tons, 72.8 tons, 84 tons, 99 tons and 128 tons) with each barge optimized for safety and stability during nearshore berthing. The use of sea transportation minimises the burden on Sri Lanka's public road network, reduces carbon emissions from long-haul trucking, and improves schedule reliability during the monsoon transition period. All marine transport operations will be executed under strict tidal, weather, and safety protocols, including ballast control, site illumination, and emergency mooring procedures.

Once landed at the temporary jetty, components will be moved inland via an upgraded internal road network, purpose-built to withstand a bearing capacity of at least 12 t/m², with 20 ft road widths and curve-widening designed to accommodate blades and long cargo. The routes have been aligned to avoid ecologically sensitive areas, and pavement substructures use a sand-brick fill system with 50% compaction layering. This dual-mode logistics system—integrating optimised marine transport with reinforced inland access—ensures secure and efficient delivery



to each turbine foundation, while reducing environmental disruption compared to purely land-based transport.

The selection of this marine-first transportation model was a key criterion in the site's overall feasibility. Dutch Bay's natural shoreline contour, proximity to navigable channels, and open staging areas made it uniquely suitable for hosting such infrastructure. Combined with logistical access from Colombo and grid connectivity to Wariyapola South GSS via a new 220 kV double-circuit transmission line, the site offers unmatched integration potential. The transportation and delivery plan aligns with CC&CRMD, SLPA, and CEA regulatory safeguards and contributes to the Dutch Bay project's low-impact development profile.



3 DESCRIPTION OF THE EXISTING ENVIRONMENT

3.1 Study Area

3.1.1 Project Site

Project area is located in the North-western coastal region of the country, near Kalpitiya in the Puttalam District of the North Western Province. It is a part of the Kalpitiya Peninsula, which separates the Puttalam Lagoon from the Indian Ocean. Proposed wind turbine locations are shown in the google satellite image (**Figure 3.1**).

Figure 3.2 shows a context area map representing the special features in the vicinity of the project site covering an area around 2 km from project site. The area mainly covers the Dutch bay and the Bar reef marine sanctuary which is environmentally highly sensitive.

The Bar Reef Marine Sanctuary (BRMS) is a complex of offshore continental shelf patch reefs constituting one of the largest coral reef systems in Sri Lanka. Bar Reef Marine Sanctuary is located northwest of the Kalpitiya peninsular (**Figure 3.3**), and covers an area of 306.7 km², encompasses coral reefs, sandstone reefs, sea grass communities and soft bottom habitats⁶. The area was declared as a Marine Sanctuary in 1992 due to its rich species diversity and extensive coral reefs. However, the defined South-eastern BRMS boundary falls very closer to the project site.

All project components, including turbines, jetty, and access roads, are located outside the designated boundaries of the Bar Reef Marine Sanctuary. This has been confirmed through spatial overlays using CZMP 2024 GIS data and coastal conservation clearance boundaries.

3.1.2 Influenced Area (Area beyond the project site where there is potential for environmental impacts due to the project)

The potential influenced area has been defined in relevant sections considering the type of the impact assessment category (ecological, social, coastal, etc.). Most of the time it would be limited within the project area except instances when considering the impact in coastal environment etc. Therefore, a common map is not included under this section.

⁶ Bar Reef Marine Sanctuary, National Wetland Directory of Sri Lanka

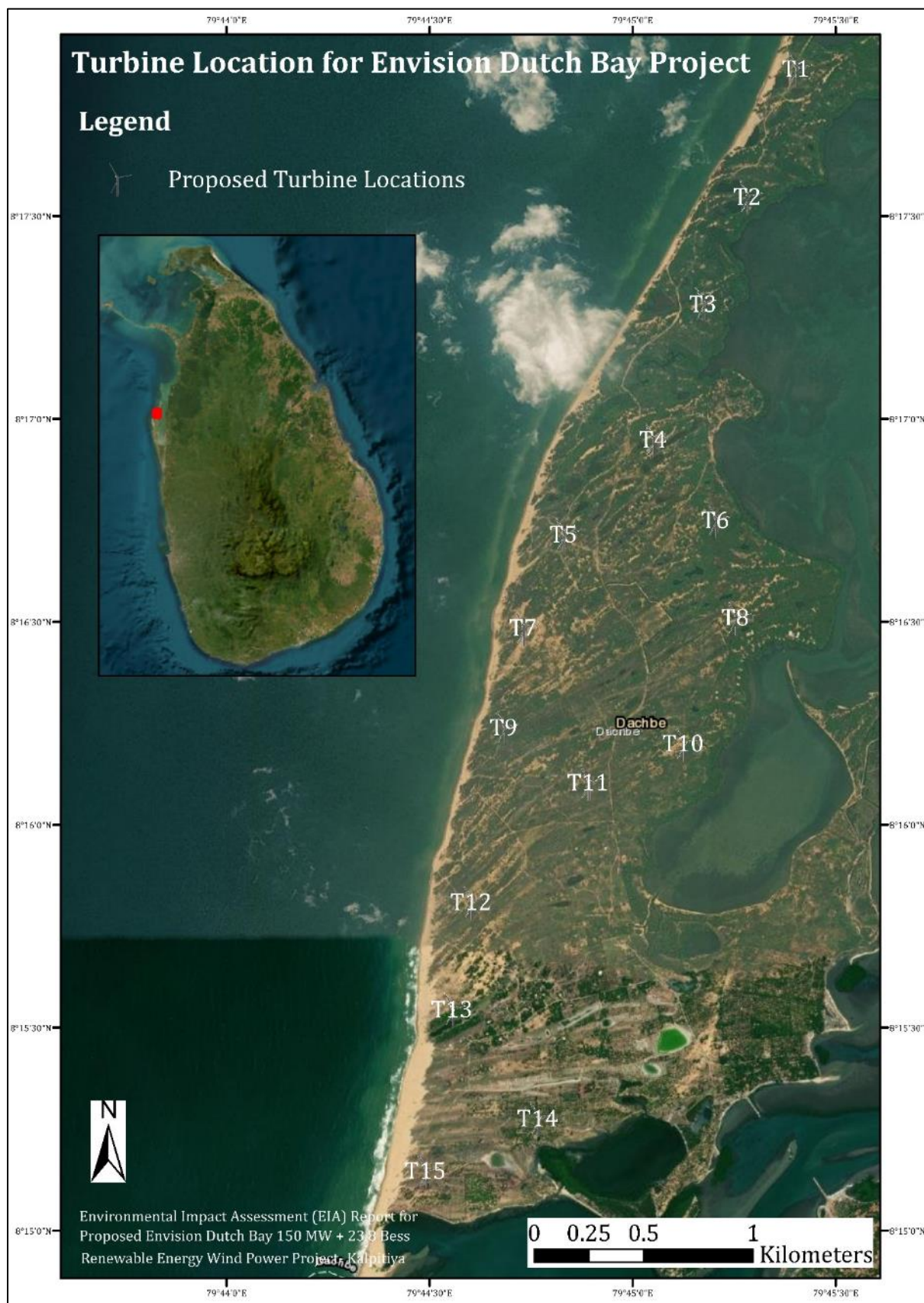


Figure 3.1: Proposed Turbine Locations

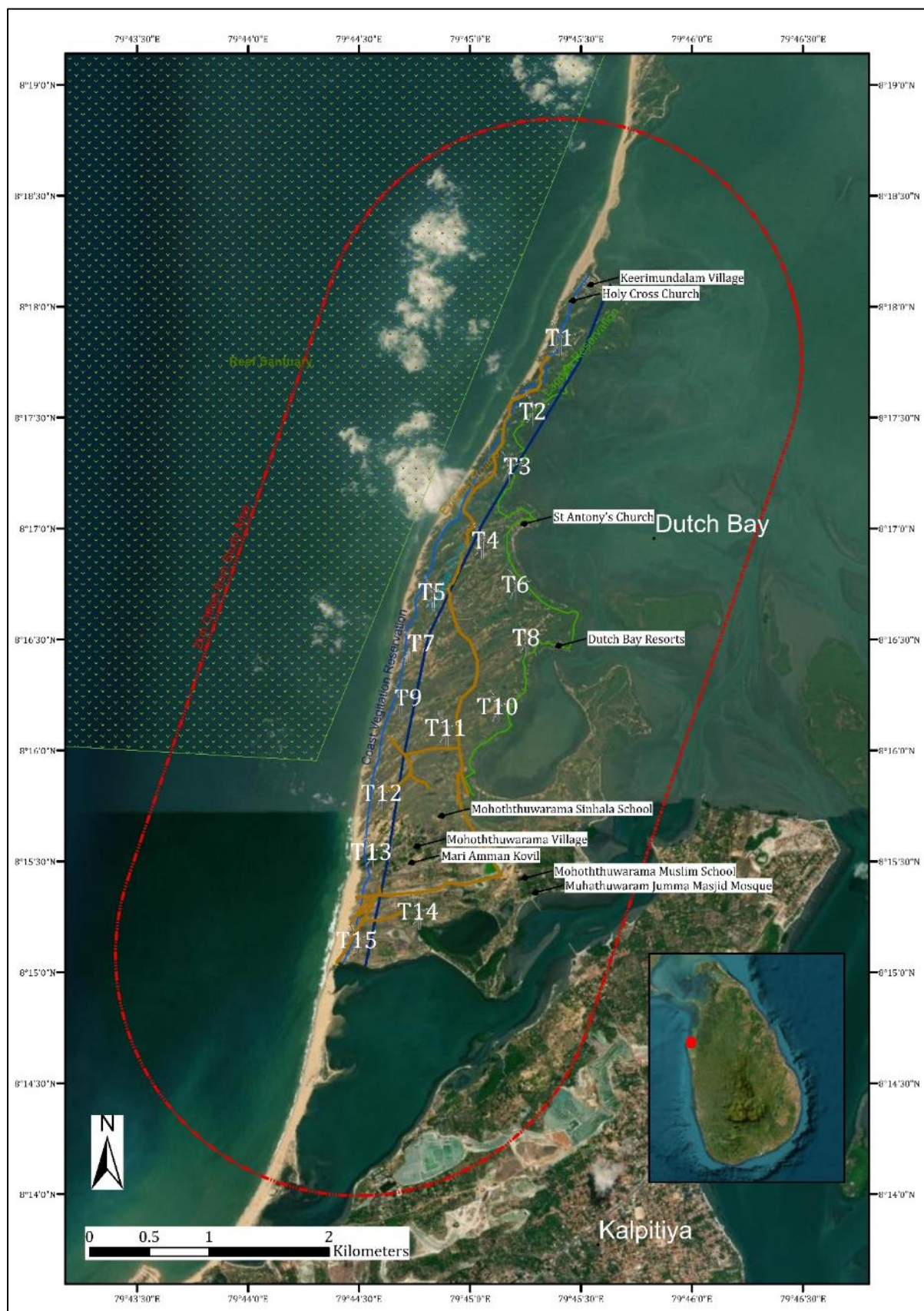


Figure 3.2: Special Features in the vicinity of project site

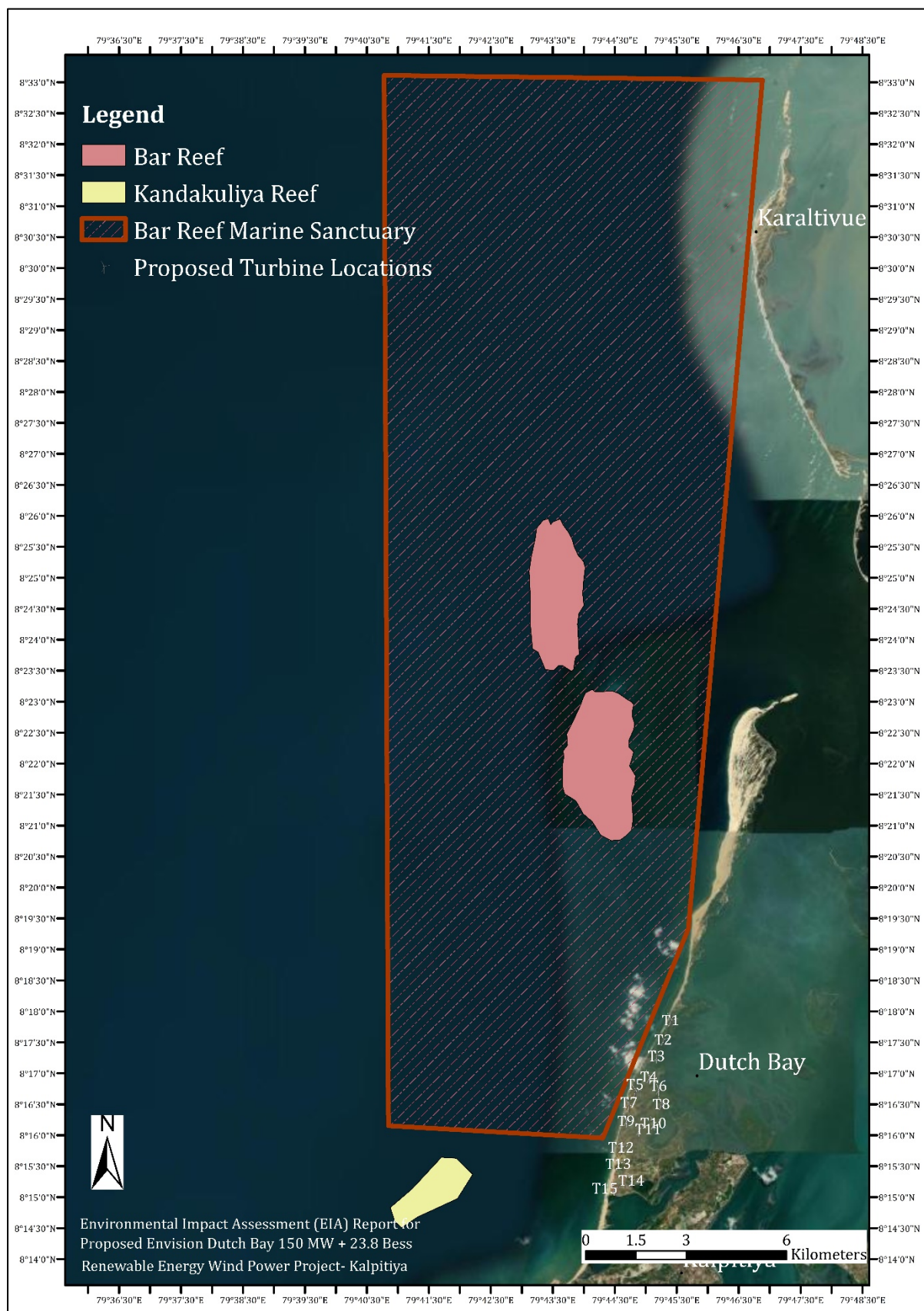


Figure 3.3: Bar Reef Marine Sanctuary Boundaries and Project Site



3.2 Physical Environment

3.2.1 Topography

The topography of the beach area with sand dunes typically shows a gentle rise in elevation from the shoreline inland. Starting at sea level, the flat beach area ranges from 0 to 5 metres in elevation, with contour lines on a topographic map either widely spaced or absent due to the very gradual slope. As you move further inland, small sand dunes begin to appear, reaching elevations of up to 10 metres. These dunes are represented by closed contour lines on the map, often in irregular or oval shapes, with the lines becoming closer together as the elevation increases. The tighter spacing of the lines around the dunes indicates a steeper, though still mild; incline compared to the flat beach. Overall, the topographic map of such an area reveals a low-lying coastal landscape with subtle variations in elevation caused by the natural formation of sand dunes as shown in **Figure 3.4**.

The study area is geographically bounded by the ocean on the western side and by a lagoon on the eastern side. The lagoon margin is predominantly characterized by the presence of mangrove forests and marshes, while the oceanfront consists primarily of sandy beach zones. The much of the area is covered by scrub vegetation, with localized patches of coconut cultivation. Additionally, several water holes are observed throughout the site. Scattered palm trees are also present, contributing to the region's mixed vegetation profile (**Figure 3.5**).

3.2.2 Geology and Soil

Sri Lanka is an island nation located in the Indian Ocean, with a diverse geological structure composed of five major geological zones: the Miocene to Quaternary formations, the Wannai Complex, the Highland Complex, the Vijayan Complex, and the Kadugannawa Complex. Each of these zones is characterized by distinct lithological features and geological origins (Indika et al., 2022). However, the study area does not fall under any of these complex categorizations. Instead, it is situated within a depositional environment. According to the geological map (**Figure 3.6**), the area is classified as consisting of beach and dune sands, which are unconsolidated sediments primarily formed by coastal processes such as wave and wind action. These sands typically consist of fine to medium quartz grains and shell fragments. The aquifer system present in the area is a shallow aquifer developed within the coastal sand, indicating the presence of groundwater stored and transmitted through the porous sandy layers near the surface (**Figure 3.7**).

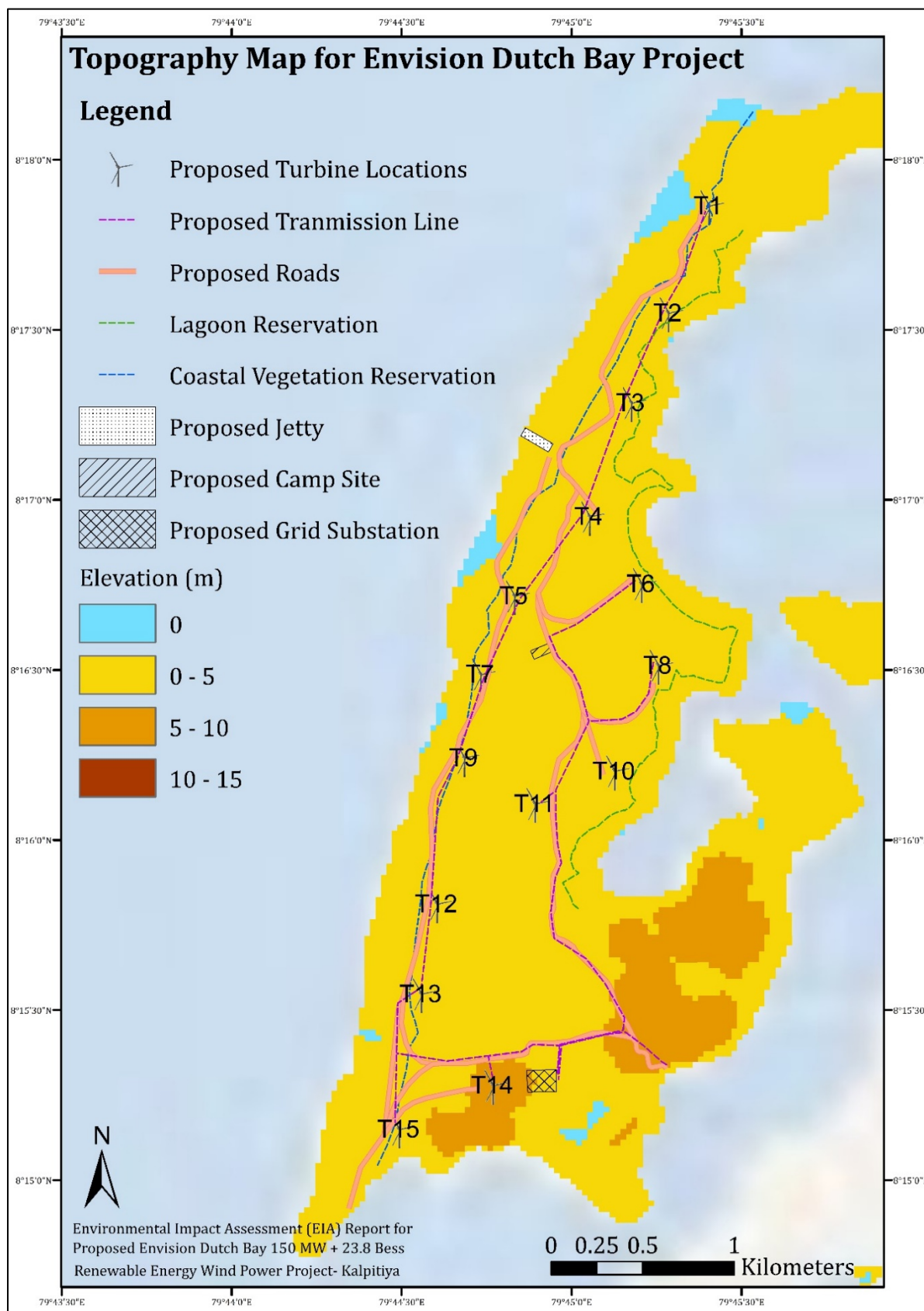


Figure 3.4: Topography Map

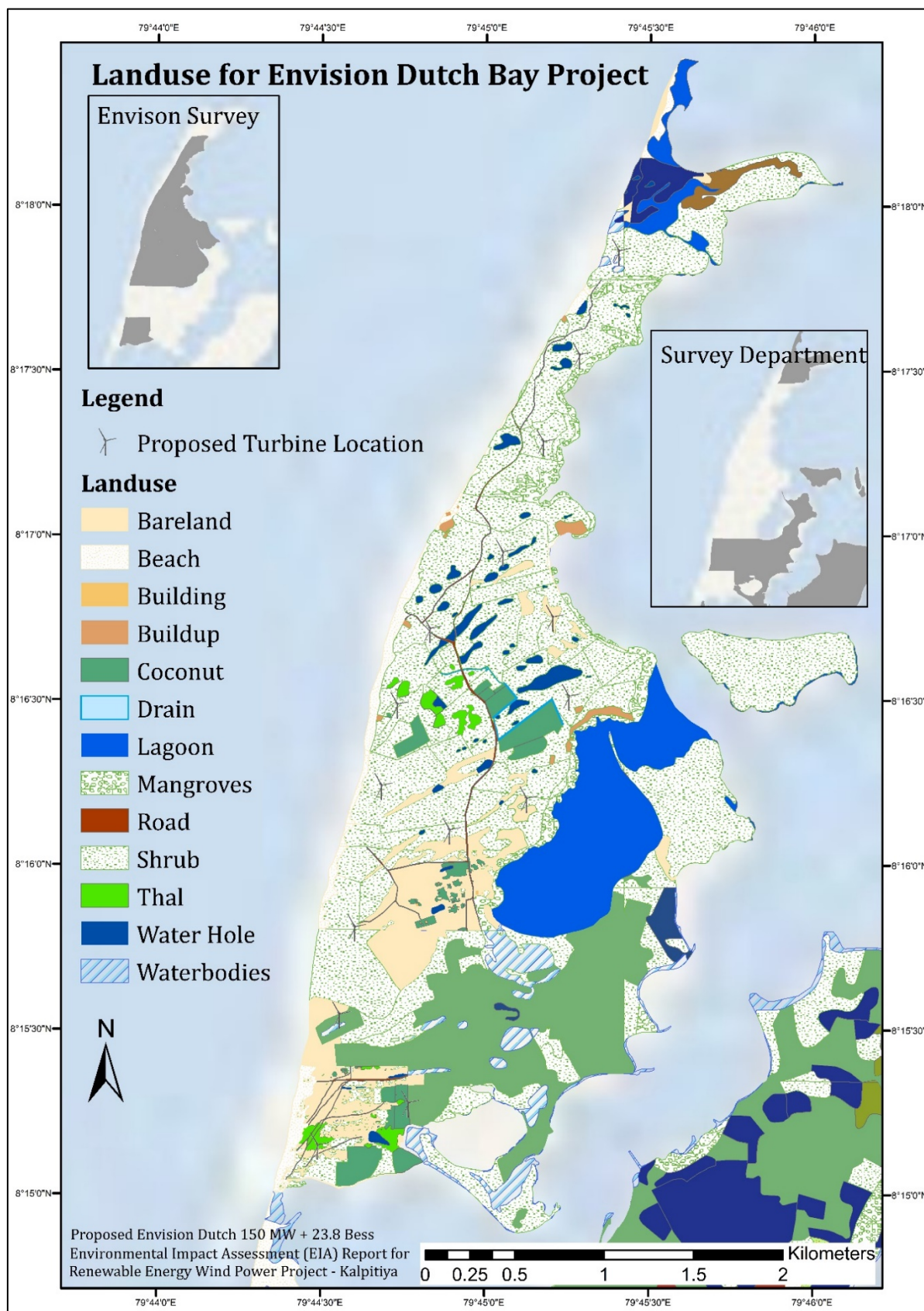
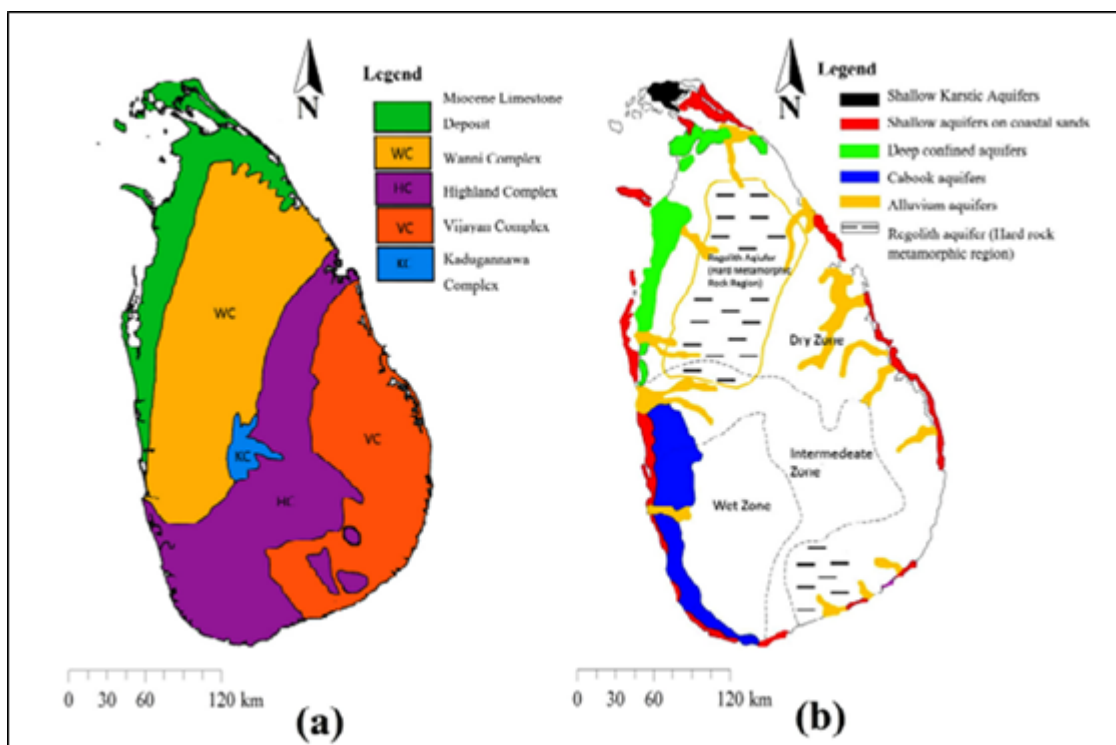


Figure 3.5: Land use details within the project site



Source: Indika et al., 2022

Figure 3.6: (a) Major geological zones in Sri Lanka (b) Different types of aquifer systems

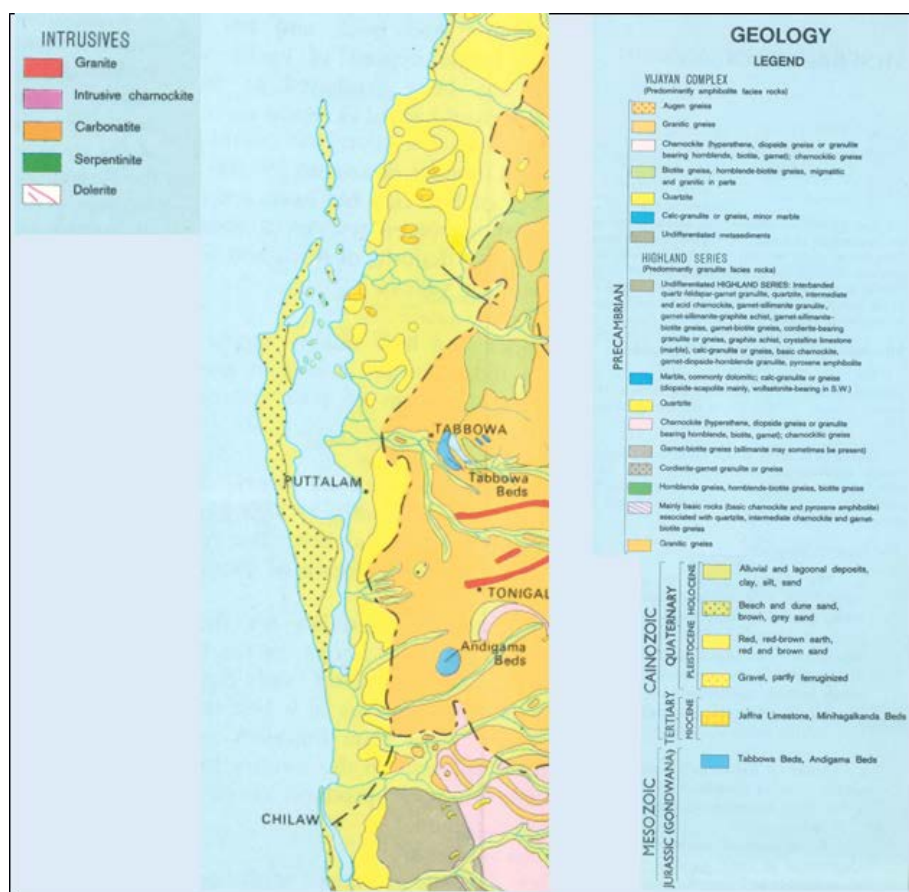


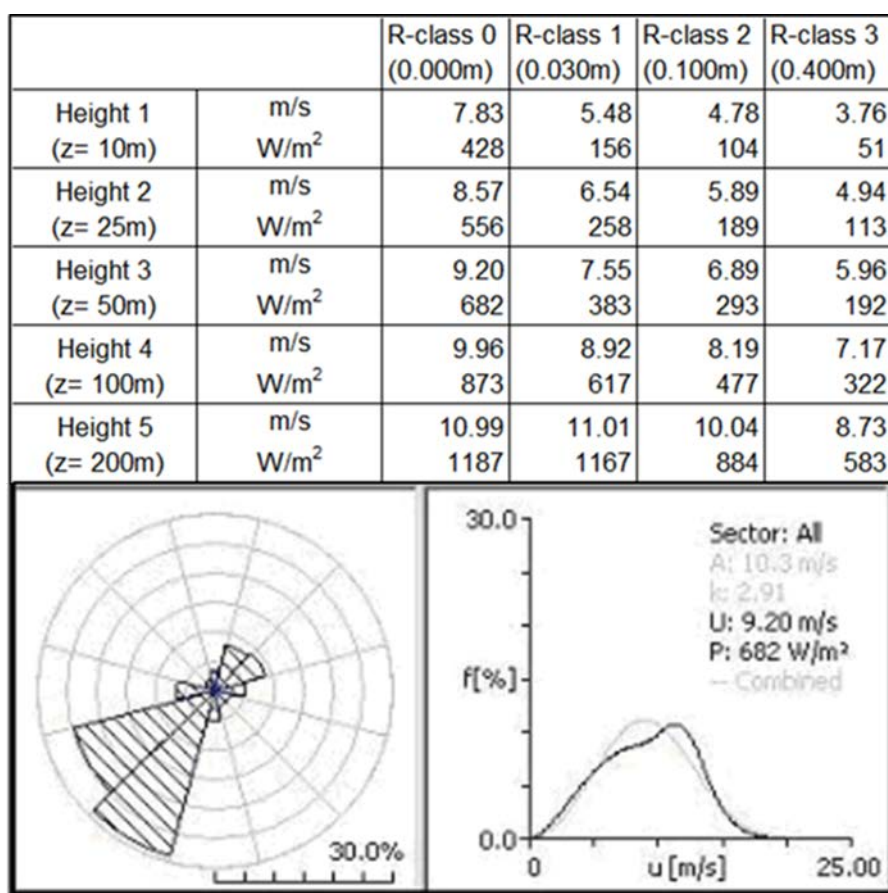
Figure 3.7: Geological Map



3.2.3 Climate and Meteorology

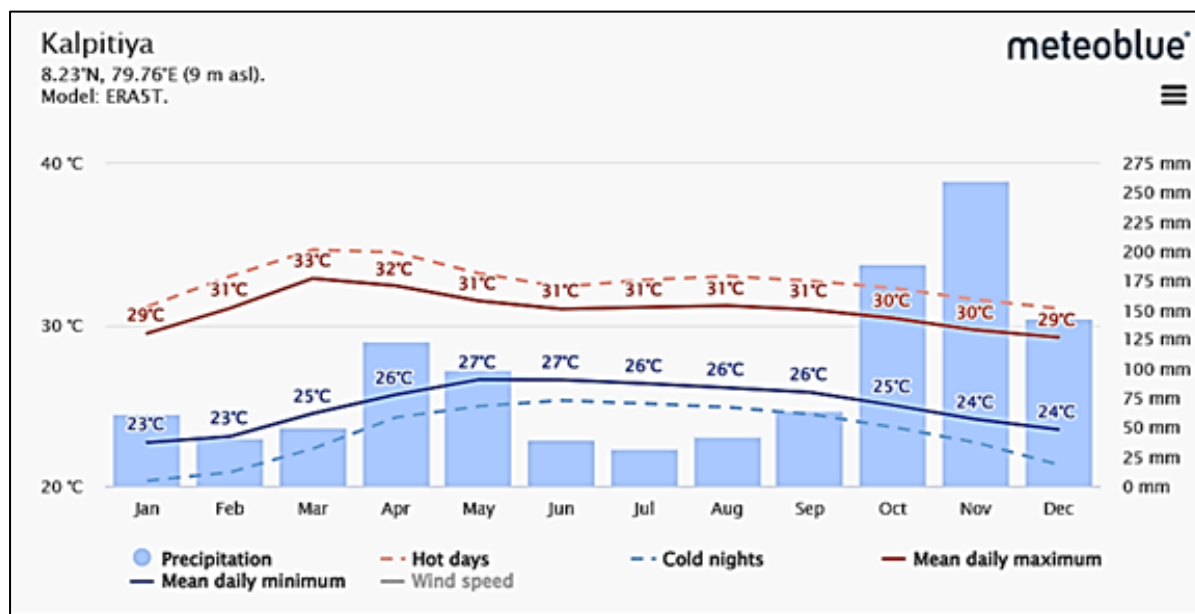
The wind climate of Sri Lanka is predominantly governed by the Asian monsoon system, which induces seasonal prevailing wind patterns lasting for several months annually. This system comprises two primary monsoon phases: the Northeast Monsoon (December to February) and the Southwest Monsoon (May to September). Of these, the Southwest Monsoon (**Figure 3.8**) is influence on the study area. In addition to the main monsoons, there are two inter-monsoon periods: the First Inter-Monsoon (March–April) and the Second Inter-Monsoon (October–November). Among these transitional periods, the Second Inter-Monsoon most significantly affects the study area.

According to meteorological data, the highest rainfall intensities are observed during the Inter-Monsoon periods, as illustrated in **Figure 3.9**. In addition, the area receives significant precipitation during the Southwest Monsoon season. Additionally, the months of March and April are characterized by elevated temperatures. Moreover, maximum RH stays relatively stable across the years, the minimum RH shows more distinct seasonal changes.



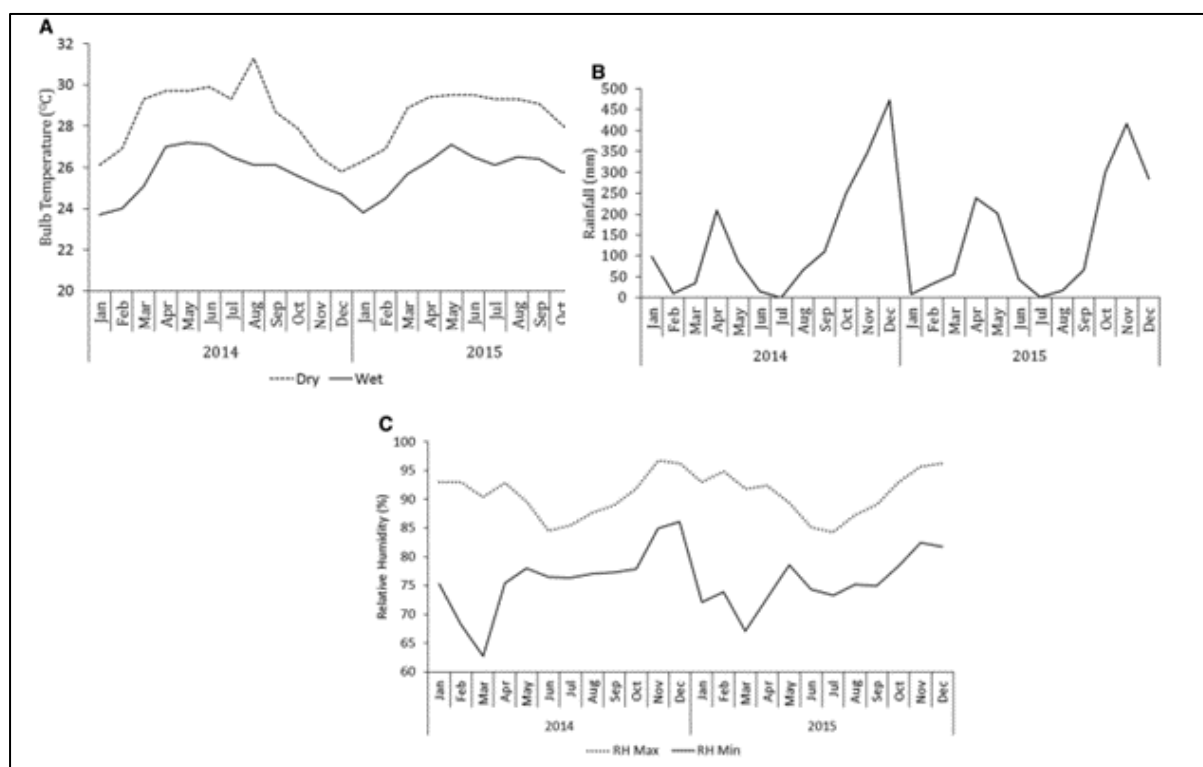
Source: Feasibility Study for a Wind Power Project in Sri Lanka, n.d.

Figure 3.8: Wind atlas of Kalpitiya region



Source: Meteoblue, n.d.

Figure 3.9: Annual climate (Meteoblue, n.d.)



Source: Attanayake et al., 2019

Figure 3.10: Annual climatology of Kalpitiya in 2014 and 2015 (A) Temperature (B) Rainfall (C) Relative humidity



3.2.4 Coastal Environment

3.2.4.1 Coastal Features

The definition of the Coastal Zone by Sri Lanka Revised Coastal Zone and Coastal Resource Management Plan 2024-2029 (CZ&CRMP 2024) means “an area lying within a limit of 300m landwards of the Mean High Water line (0.6 m MSL) and a limit of 2km seaward of the Mean Low Water Line and in the case of rivers, streams, lagoons, or any other body of water connected to the sea either permanently or periodically, the landward boundary shall extend to a limit of 2km measured perpendicular to the straight baseline drawn between the natural entrance points thereof and shall include the waters of such rivers, streams, and lagoons or any other body of water so connected to the sea, and shall also include the area lying within a further extended limit of 100m inland from the zero Mean Sea Level (MSL) along the periphery”. Accordingly proposed wind turbines T1, T2, T3, T5, T7, T9, T12, T13 and T15 lies within the coastal zone (**Figure 3.12**). Further, the temporary Jetty will also be located within the coastal zone.

Project site is located on the Dutch Bay Island bounded by the Indian Ocean on the west and rest of the area by the Dutch Bay inside the Puttalam lagoon (**Figure 2.2**). Main Coastal features observed on the Dutch bay island are primarily sandy beaches, sand dunes, salt marshes, mudflats, mangroves and lagoons (**Figure 3.11**). No bays, rock outcrops or headlands present on the ocean span of the island.



Figure 3.11: Coastal Features in the Dutch Bay Island



Figure 3.12: Wind Turbines within Coastal Zone

3.2.4.2 Beach Profile

Sandy beach stretches from Mohonthuwarema southern point to Keeimundalam village for about 6.5 Km on the north western coast of Dutch Bay Island. This stretch indicates characteristics of spits and considered as medium (+) level of vulnerability to erosion as defined by the CZ&CRMP 2024 and included in the coastal setback Segment No 2 Uchchamunai to Mohonthuwarema (Mohonthuwarema Split Northern Boundary) where total setback is 60m with 25m Reservation and 35m restricted area. Only wind turbine T1 is located closer to the setback area, while others are located considerable distance away from the designated setback zone (**Figure 3.13**).

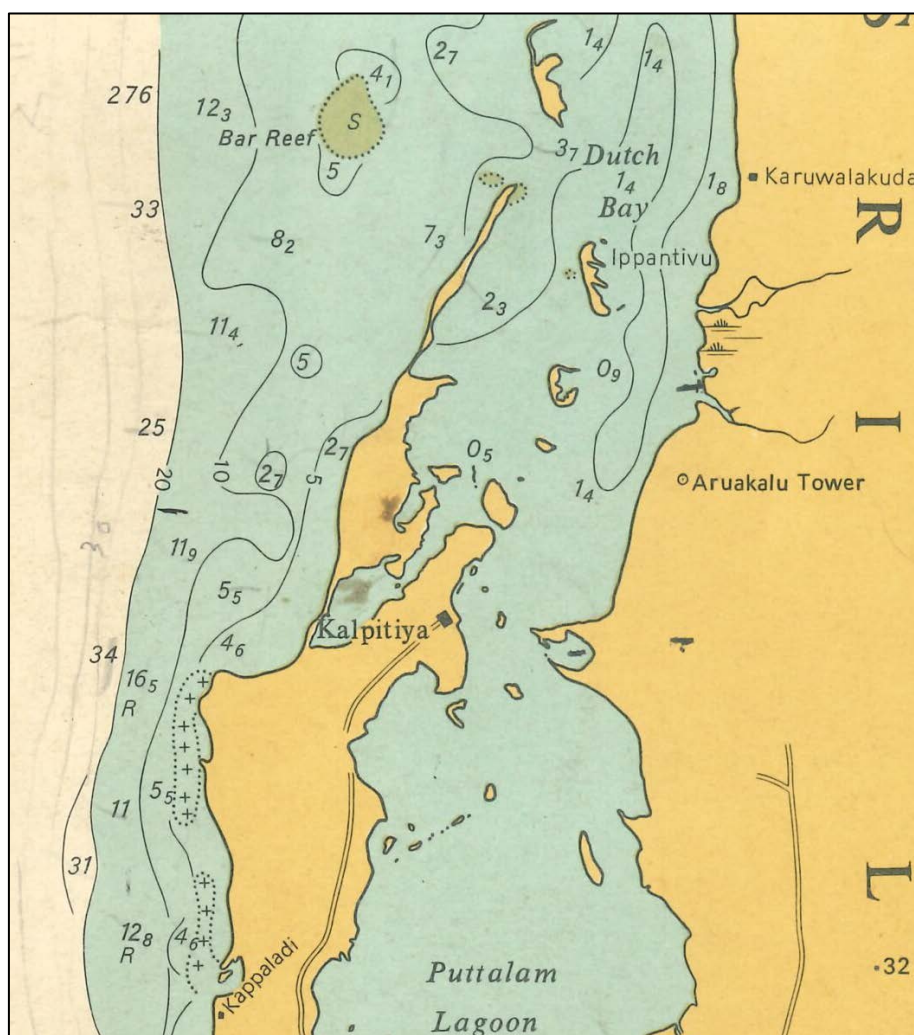


Figure 3.13: Wind Turbine locations and the defined coastal setback

Beach profile on the seaside consists of steep slope on the foreshore and mild backshore (Figure 3.14) due to strong wave action. Map no 1586 of admiralty chart shows (Figure 3.15) water depth on the seaside varies rapidly at near shore with the seabed slope of 1:200 up to about 5m depth while slope beyond 5m to 10m changes mildly with the slope of 1:400. 20m depth can be reach within 6km from the shoreline and beyond that, depth again increase rapidly due to slope of the continental shelf.



Figure 3.14: Beach Profile on the Seaside



Source: Admiralty Chart No. 1586, United Kingdom Hydrographic Office

Figure 3.15: Extract from Admiralty Chart



Bathymetry on the lagoon side is very shallow with gentle slope due to presence of shoals with the depth varying from 0 to 2m within the Dutch Bay area near Dutch Bay islands. There are intermittent canals in between the shoals which leads up to the shore which are used by the islanders for navigation such as Dutch canal for fishing and recreation purposes (Figure 3.16).



Figure 3.16: Presence of shoals & canals on Dutch bay

3.2.4.3 Sediment Transportation

Annual net sediment drift is approximately 0 to 10 in the order of $10^4 \text{ m}^3/\text{yr}$ towards north near Kalpitiya area as calculated by the several sediment transport studies conducted from 2001 to present. (Samarasekara RSM, 2022). Littoral drift along the coastline depends upon the supply of sediments from the rivers falls to the ocean southwards of Kalpitiya and linked with shoreline management activities.



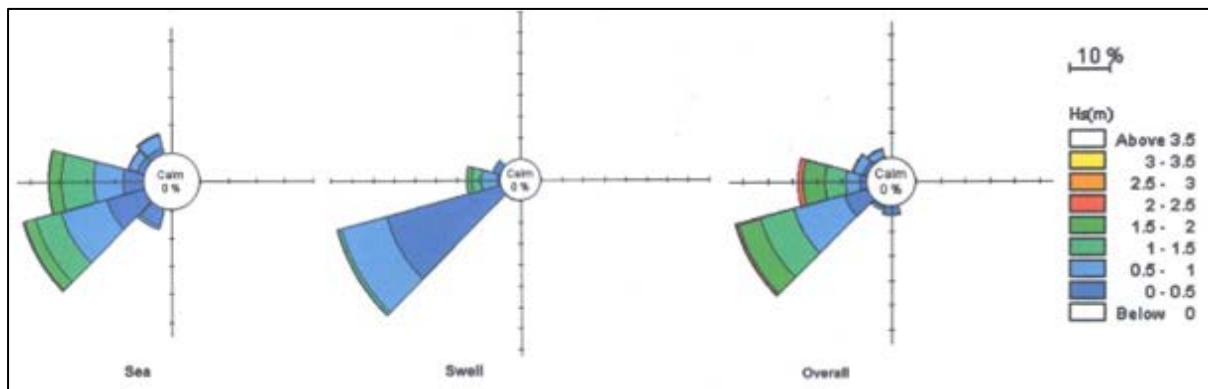
3.2.4.4 Coastal Morphology

Kalpitiya peninsula is a formation of a spit by the dominant northward sediment drift on the west coast of Sri Lanka and Dutch Bay Islands located at the northern tip of the peninsula. The coastal stretch from Kirimundalm to Uchchamunei consist of very fine beach materials and has identified as highly dynamic, unstable coastal islands with seasonal changes and frequently reshaping islands in CZ&CRMP 2024. Kalplitiya peninsula is slowly eroding with the intensified wave climate (Samarasekara RSM, 2022).

Lagoonal and estuarine deposits of the Holocene period are found in Dutch Bay area (Samarasekara RSM, 2022) where riverine estuary of Kala Oya open into the bay (CZ&CRMP 2024). Currents and waves are weak inside the lagoon area due to the presence of islands and shoal, hence lagoonal morphology is stable at the Dutch Bay Island.

3.2.4.5 Near shore wave Height & Direction

Annual Distribution of wave heights at 15m depth near Kalpitiya established by CRMP indicated dominant wave directions are from west to southwest and heights are ranging from 0 to 2.5m for overall waves. Swell waves are dominant from Southwest direction and sea waves are scattered from North North-West to South South-West directions with the magnitudes of 0 to 2m (Figure 3.17).



Source: CRMP- Component B

Figure 3.17: Wave Climate at 15m depth at Kalpitiya

Marine environmental baseline survey conducted at Puttalam Lagoon in May to mid-June 2018 showed waves are coming from Southwest direction with heights varying from 0.18m to 1.02m at the Location (MO-1) in between Dutch Bay and Baththalangunduwa Islands on the seaside at the depth of 4.7m, while measurements at the location inside the lagoon (MO-2)(Figure 3.19) are scattered around directionally with wave heights ranging from 0.03m to 0.51m at the depth of 4.2m (Figure 3.18) Waves approaching from a narrow direction band (SW) to the Location MO-



01 shows the swell wave characteristics, while waves at Location MO-02 are direction variable waves with short periods generated due to limited fetch inside the lagoon.

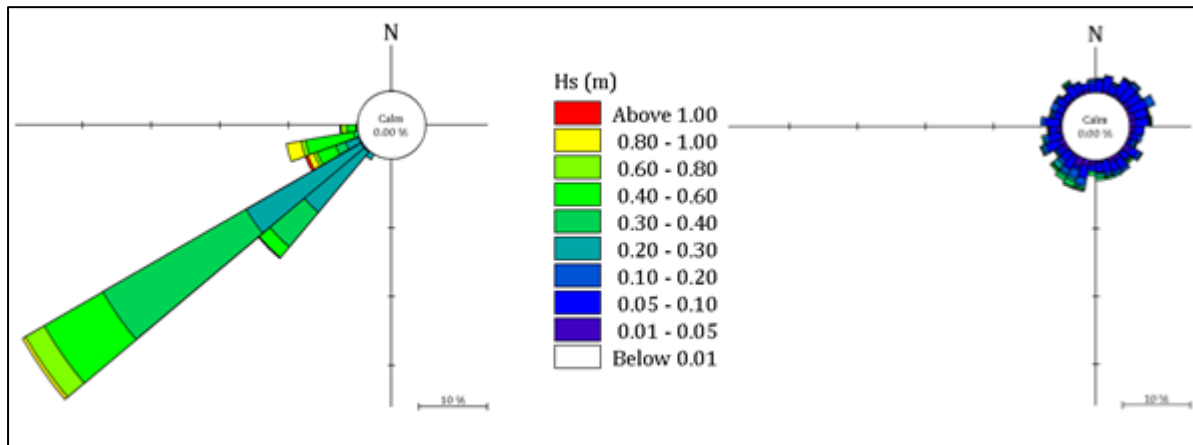


Figure 3.18: Wave Rose Plots for Location MO-1 (Left) and Location MO-02 (Right)

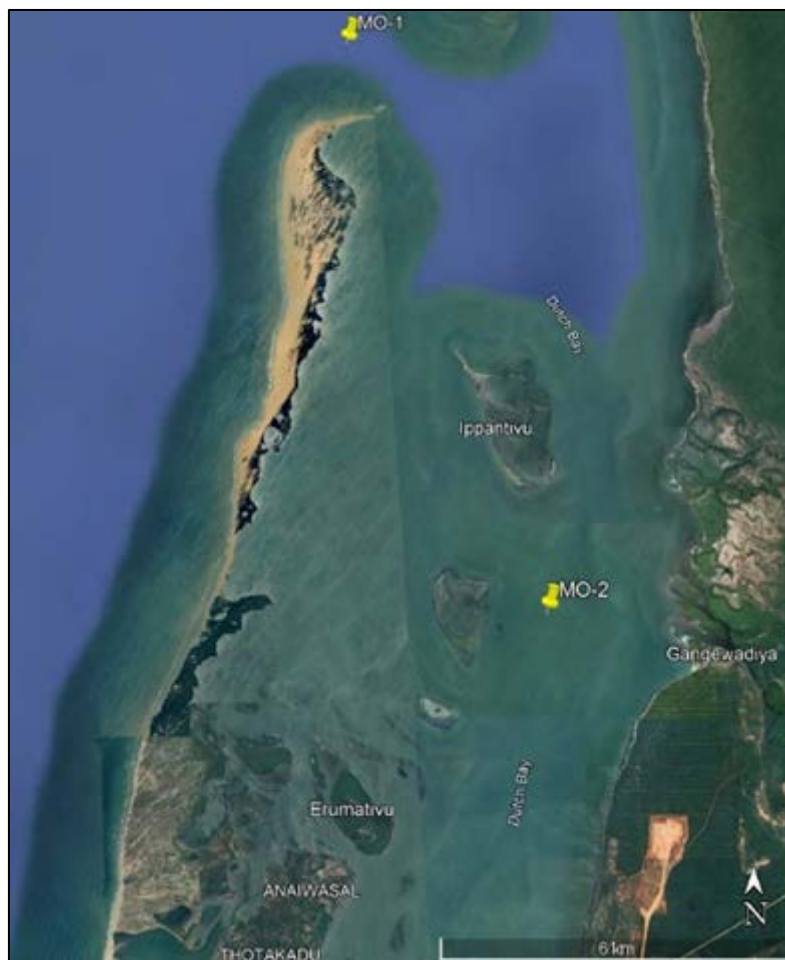


Figure 3.19: Measurement Locations inside Puttalam Lagoon



3.2.4.6 Near shore current velocity & Direction

Near shore current on the seaside is a combination of tidal currents, alongshore currents, and on-offshore (cross-shore) currents. Apart from alongshore currents mostly towards north, sediment movement due to cross-shore current is also significant in the coastline where formation of offshore underwater sandbar parallel to the beach is visible (**Figure 3.20**).



Figure 3.20: Offshore sandbar formation due to on-offshore currents

There are no measured current available on the coastal stretch at the Dutch bay island. However Currents measured at 4 locations near intake and outfall of Lakvijaya Power Plant-Norochcholai between depth of 10m to 1m during Southwest monsoon in 2008 indicated values ranging from 0 to 0.4 m/s while measurements during Northeast monsoon of the same year at 6 locations on the same vicinity indicated values ranging from 0 to 0.9 m/s. Current directions were variable during Northeast monsoon while currents were flowing towards north during Southwest monsoon. Measured current data during the period from 2020.10.11 to 2020.10.25 at (-10.4) m MSL in front of Lakvijaya Power Plant ranges from 0 to 0.55m/s towards North.

Variation of current speeds measured in May to mid-June 2018 for two locations (MO1 and MO2) inside the Lagoon are as per the (**Table 3-1**) below.



Table 3-1: Summarized Results for Measured Current Data at Lagoon (Source: LHI-Interim Report-05 (Met-Ocean Report 2018))

Location	Position	Parameter		
		Speed (m/s)		
		Minimum	Average	Maximum
MO-1	Top	0.026	0.427	0.798
	Middle	0.016	0.182	0.523
	Bottom	0.010	0.172	0.460
MO-2	Top	0.014	0.306	0.807
	Middle	0.003	0.170	0.449
	Bottom	0.001	0.145	0.390

3.2.4.7 Tidal & current characteristics

Tidal variation is semi-diurnal both in the open sea and within the lagoon area. The tidal range predicted at Kalpitiya Fishery harbor inside the Puttalam Lagoon from constituents published in Admiralty tide tables are within the range of +0.32m to -0.3m as given in **Figure 3.21**.

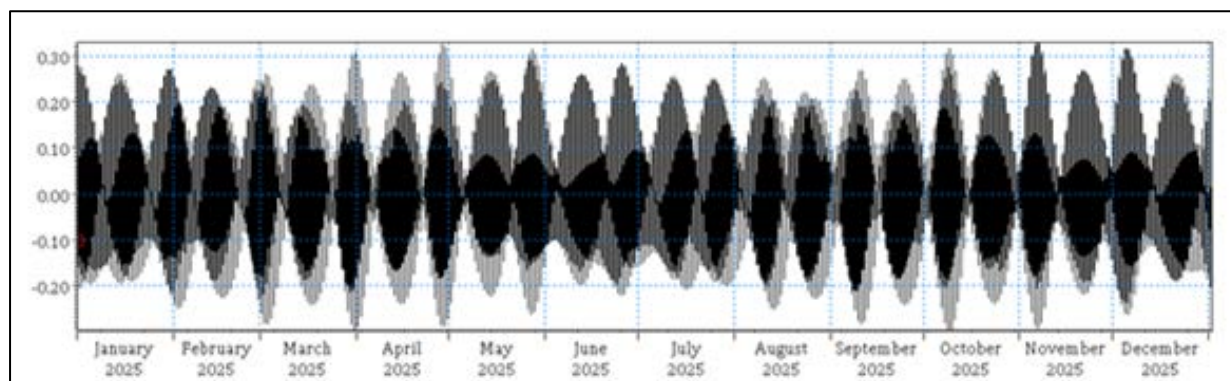


Figure 3.21: Predicted tide at Kalpitiya Fishery Harbour for 2025

Established tidal constituents for the open sea at Kalpitiya has been used to predict tide at open sea and ranges from +0.45m to -0.43m as per **Figure 3.22**.

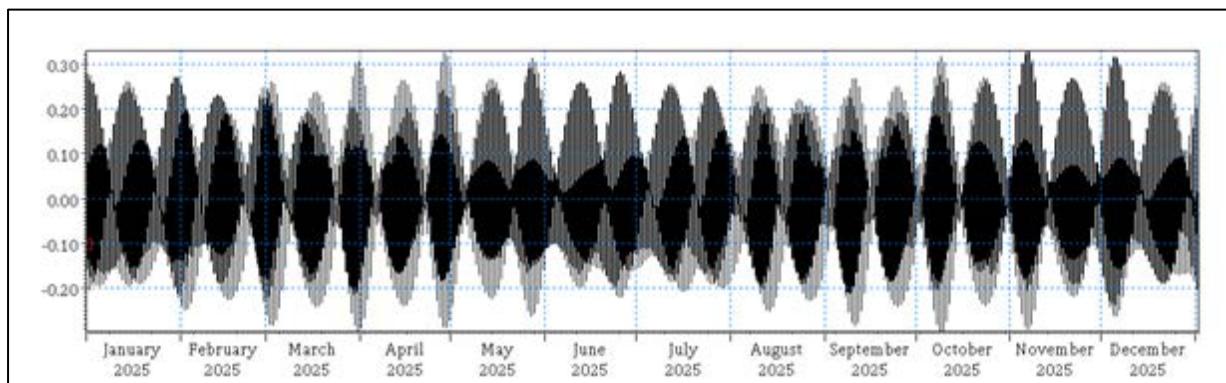


Figure 3.22: Predicted tide at Open sea-Kalpitiya for 2025

Tidal range derived from the measured water level at MO-2 location inside the Puttalam lagoon at Dutch bay area in 2018 varies from +0.3m to -0.27m as **Figure 3.23** below indicating lesser tidal range inside the lagoon.

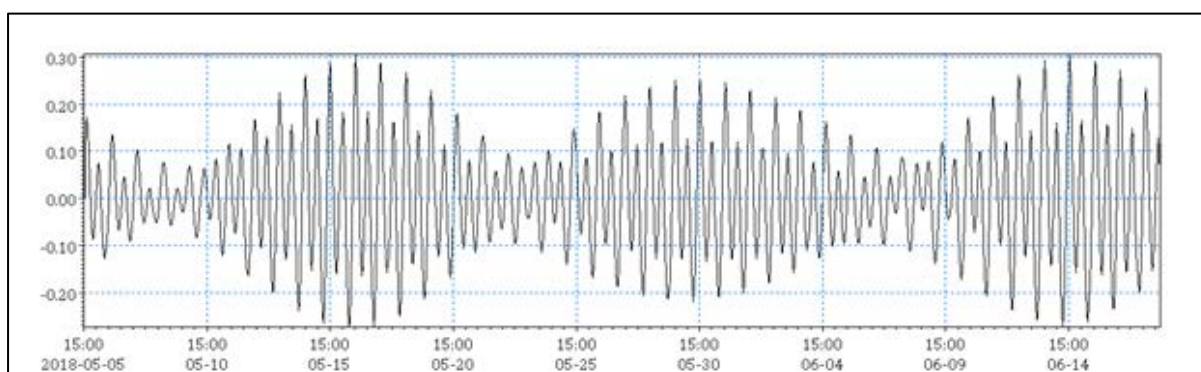


Figure 3.23: Derived tide at MO-2 from measured water level

3.2.4.8 Coastal erosion/accretion threats

Reduction of the net northward littoral drift due to the construction of detached breakwaters and groynes at Kandakuliya area before 2004 has caused erosion in the Dutch Bay on the northern side (Samarasekara RSM, 2022). Hence, there is a high possibility of further erosion on open coastline of Dutch Bay Island.

Dense mangrove cover on the edge of the Dutch bay side of the Island provides protection to the shoreline from erosion. Presence of shoals in the Lagoon also contributes to stable shoreline with the low tidal range, current speeds and calm wave conditions.

3.2.4.9 Coastal erosion records of last 10 years of both open sea & Dutch bay

Study conducted by Construction Technological Services on Google Earth Images indicated that shoreline has been receded by 200m at Mohonthuwarama & 75 m at Keerimundalam during past



20 years. 75m and 50m change in the shoreline is observed during recent 10 years from 2014 on the same locations respectively (**Figure 3.24**).



Figure 3.24: Shoreline change during past 20 Yrs

CZ&CRMP 2024 indicated that coastal stretch from Kirimundalm to Uchchamunei has reported emergency coastal erosion from 2014- 2022. However, no records indicated any erosion on the Dutch bay side of the Island.

3.2.4.10 Prior action taken to cope up with any severe erosion

There were no records on prior actions taken by the relevant authorities to cope up with the severe erosion happening at the Dutch bay Island. However, there were failed attempts by the locals to prevent the erosion by stacking the sand bags which are now in the sea.



3.2.4.11 Coastal structures, coastal protection systems (any plans)

There are no coastal structures or coastal protection systems exist on the coastal stretch of the Dutch bay Island at present. However Official website of the Coast Conservation and Coastal Resource Management Department indicated they had a discussion with the World Bank about Implementing Nature-based solutions in Kalpitiya Coastal stretch to mitigate long-term coastal erosion.

3.2.5 Water Quality

In the region, residents primarily rely on dug wells and rainwater harvesting systems (**Figure 3.25**) to meet their domestic water needs, particularly for non-potable uses such as washing and cleaning. Based on the local information, a centralized distribution system provides treated drinking water to the community, ensuring safer consumption and reducing the risk of waterborne diseases.



Figure 3.25: Rainwater harvesting

Water quality measurements were conducted on 04.04.2025 and 10.04.2025 at 13 Locations on Dutch bay Island. The water sampling locations (**Figure 3.26**) comprises a variety of water sources, including wells, lagoon water, and surface water collection points. Locations W1 and W6 are identified as abandoned wells, which are no longer in active use. In contrast, W9 and W10 are functional wells currently used by the local population for domestic purposes. Location W2 sources water from a nearby lagoon. The remaining sampling points are associated with surface water collection areas, including waterlogged zones and natural depressions where runoff



accumulates, potentially serving as supplementary sources for non-potable use. Location SW1, SW2 and SW3 are at the coastal stretch.

Water quality the lagoon side is of good condition except for relatively higher salinity of 39.45 mS/cm (25.133 ppt) with ammonia content is bit higher than level suitable for aquatic life. Oil & Grease too high for any of the ambient standards. (mg/l or µg/l) set by the National Environmental (Ambient Water Quality) Regulations, No. 01 of 2019.

Regarding the quality of the sea water samples taken at three locations on the sea side of the Island as indicated in the Table 3-3, DO, turbidity and Nitrates are in the prescribed range while phosphates indicated higher values for all three locations with Ammonia content is not suitable for aquatic life, yet suitable for water with minimum quality. Oil & Grease again too high for any of the ambient standards. (mg/l or µg/l). Detailed baseline water quality report is given in **Annexure III C.**

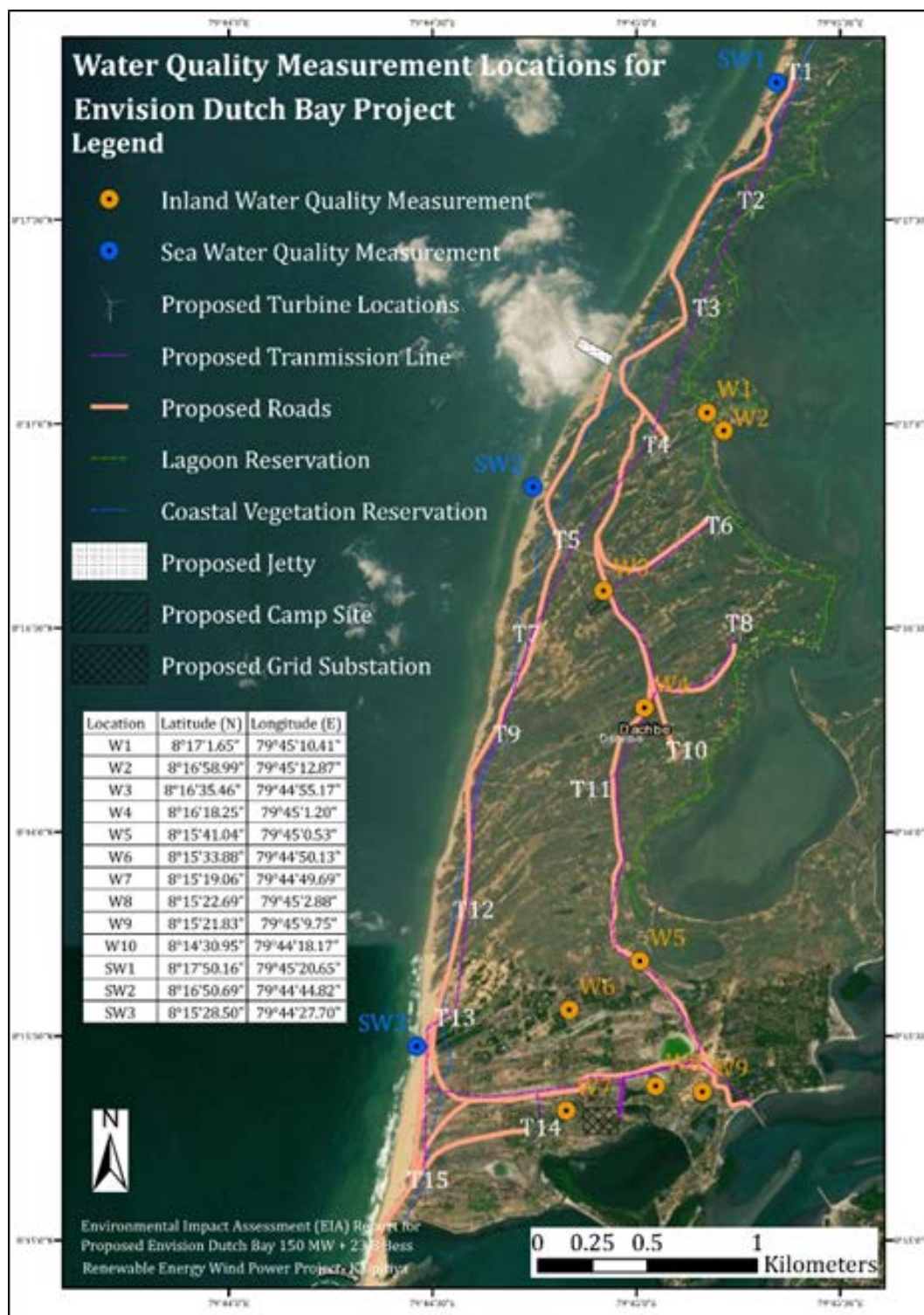


Figure 3.26: Surface and groundwater quality measurements locations



Table 3-2: Surface and groundwater quality measurements

Sample Number	Dissolved Oxygen (mg/l)	Salinity (mS/cm)	Turbidity (NTU)	Chlorophyll Content (µg/l)	Nitrate (mg/l)	Phosphates (mg/l)	Ammonia (mg/l)	Oil & Grease (mg/l)
W1	7.5	17.43	0.55	3.24	3.6	0.6	2	3
W2	7.42	39.45	0.5	4.25	2.8	0.34	1.5	3.6
W3	7.29	15.09	1.36	2.69	0.56	0.92	2.5	4.8
W4	7.26	25.5	0.37	4.44	0.84	1.25	2	2.4
W5	6.61	32.98	0.54	3.87	2.24	0.48	2	6.2
W6	9.1	7.84	40.6	0.45	1.28	0.7	1.5	2.2
W7	8.08	3.65	1.61	4.24	0.68	1.54	1.5	1.6
W8	7.03	17.62	4.04	3.58	1.28	0.86	2	340.4
W9	6.42	0.64	0.54	2.36	0.76	1.83	2.5	4.4
W10	6.27	3.11	0.3	3.14	0.6	1.16	1	3.6

Table 3-3: Coastal Water Quality Test Results

Sample Number	Dissolved Oxygen (mg/l)	Salinity (mS/cm)	Turbidity (NTU)	Chlorophyll Content (µg/l)	Nitrates (mg/l)	Phosphates (mg/l)	Ammonia (mg/l)	Oil & Grease (mg/l)
SW1	7.84	6.71	0.54	1.82	0.35	1.77	1.5	0.6
SW2	8.15	6.14	0.62	2.56	0.24	1.58	1.0	0.4
SW3	7.62	6.02	0.48	2.14	0.42	1.30	1.0	0.4

3.2.6 Hydrology and Drainage

Based on land use map (**Figure 3.5**) analysis and field observations, numerous waterlogged areas were identified within the study area, especially considerable number of Odais are present. Most of these areas drain towards the adjacent lagoon. During the rainy season, extensive flooding occurs in the low-lying regions of the study area, which gradually recede as the water drains or evaporates. In contrast, during the dry season, only the permanently waterlogged zones retain water.



Figure 3.27: Water log area near WT location 1



Figure 3.28: Water logged area and a drainage area near WT location 2

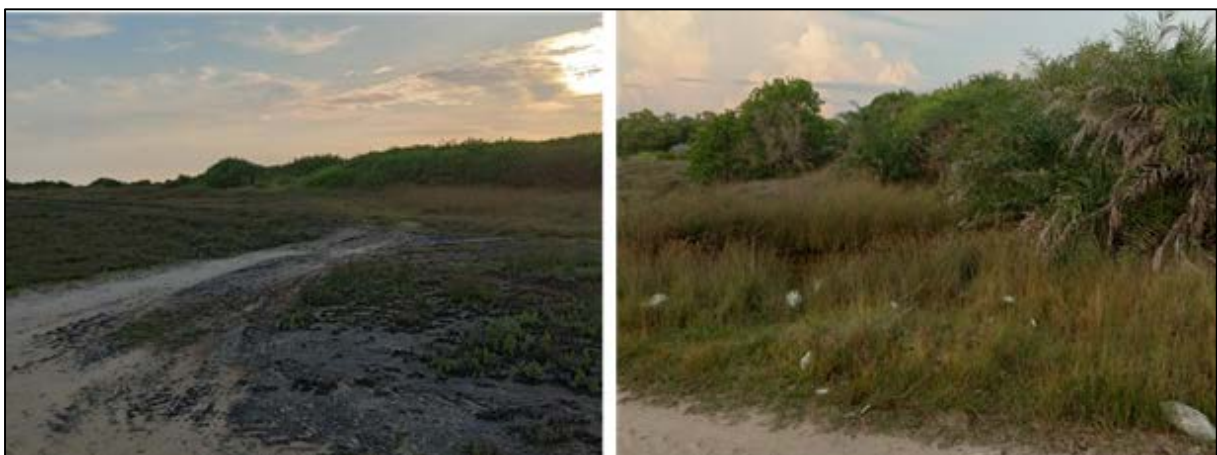


Figure 3.29: Drainage path and a water logged area near WT location 5



Figure 3.30: Water logged area near WT location 10 and 11



Figure 3.31: Drainage path and water logged area near WT location 15

3.2.7 Details of coastal hazard events in the past

Coastal erosion obviously is the most critical coastal hazard faced by the occupants on the Dutch bay Island. During past 20 years approximately 50m to 200m width of land has been lost to sea.

Cyclone occurred in 1964 caused surge height as high as 4 m at some places on the northwest coastline according to anecdotal evidence and model studies (Murthy et.al. 2004) (Coast Conservation and Coastal Resource Management Department, 2024). Apart from that tracks of past cyclones and storms indicated there are 2 storm tracks crossed the project site within 20 km in 1919 and 1922 on the Northern side and 2 cyclone tracks went closer to the Dutch bay islands in 1978 and 2000. Cyclone 1978 passed about 40km on the northern side and cyclone 2000 went



about 20km on the southern side respectively (**Figure 3.32**). These cyclones and storm events might have caused surge events on the Dutch Bay Islands, yet there were no records.

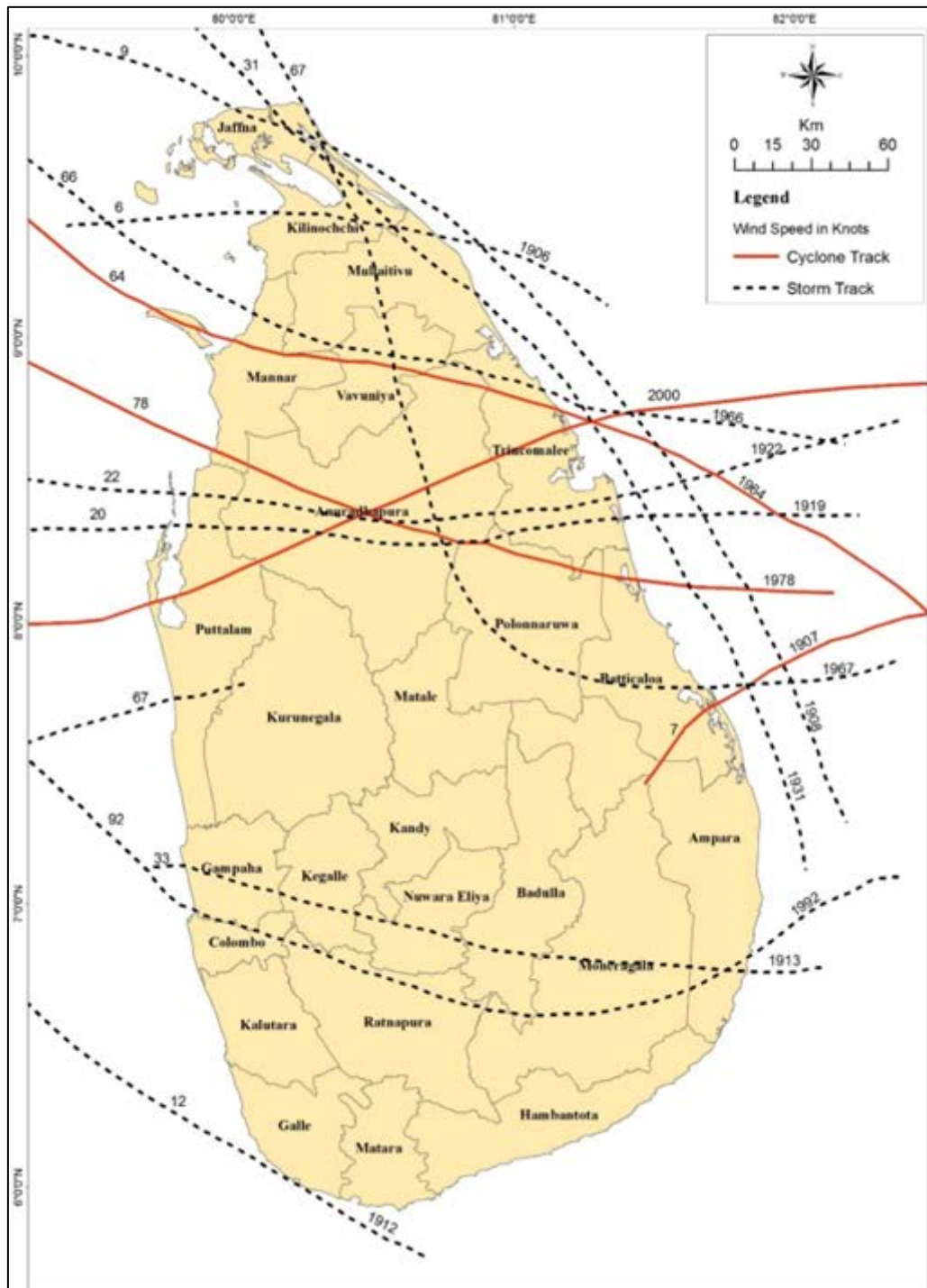


Figure 3.32: Tracks of Past Cyclones and Storms (source: www.dmc.gov.lk/)

It is known knowledge that the 2004 Tsunami did not have any impact on Kalpitiya area.

Both marine and lagoon fisheries serve as the primary sources of livelihood in the Puttalam Fisheries District. To facilitate effective fisheries management and administration, the district is subdivided into eight Fisheries Inspector Divisions (FIDs): Vanatavillu, Puttalam, Mangalaeliya, Baththuluoya, Palakuda, Kandakuliya, Kalpitiya Mainland, and Kalpitiya Island. The proposed project site, located in Dutch Bay Grama Niladhari Division (GND), falls within the southern part of the Kalpitiya Islands FI Division (one of the eight FI divisions), as illustrated in **Figure 3.33**

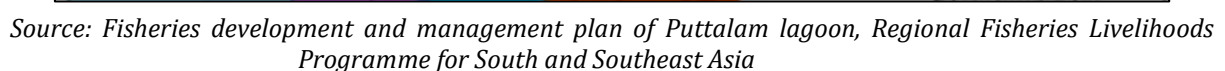


Figure 3.33: Map of Fisheries Inspector Divisions (FIDs) in Puttalam Fisheries District*



3.2.8.1 Number of Fishers (Fishing Population)

The Puttalam Fisheries District, comprising eight Fisheries Inspector (FI) Divisions, includes 9,530 households and a total fishing population of 43270. As of 2024, the district has an active fisher population of 13,350, with an average of 1.5 active fishers per household. According to the most recent data from the Puttalam Fisheries District Office (2024), 3,197 active fishers are currently engaged in fishing activities within the Kalpitiya Mainland and Kalpitiya Island FI Divisions. Of this number, 1,161 active fishers are based in the Kalpitiya Islands, where the proposed wind power project is expected to be implemented, underscoring the area's critical importance in sustaining local fisheries-based livelihoods. The details of the fisher population are presented in **Table 3-4**.

Table 3-4: Details of the Fisher Population

Kalpitiya mainland and Island FI Division	Number of Active Fishermen	Fisher Population
Kalpitiya mainland	2036	4454
Islands (including Dutch Bay)	1161	2568

Source: Assistant Director, DFAR, Puttalam

3.2.8.2 Number of types of fishing crafts operated and their sizes

The fishing fleet in the Puttalam fisheries district consisted of all types of fishing crafts, ranging from Multiday Boats (IMUL) to non-motorized beach seine crafts (NBSB). Refer Table 3-5.

Table 3-5: Number of types of fishing crafts operated

	IMUL	IDAY	OFRP	MTRB	NTRB	NBSB	TOTAL
Puttalam district	74	23	3892	288	1194	88	4459
Kalpitiya Islands			395	0	0	0	395
Kalpitiya Mainland			640	48	196	0	884

IMUL - Inboard Multi-day Boats, IDAY - Inboard Single-day Boats, OFRP - Out-board engine Fiberglass Reinforced Plastic Boats, MTRB - Motorized Traditional Boats, NTRB - Non-motorized Traditional Boats NBSB -Non Motorized Beach Seine Crafts



3.2.8.3 Number of types of fishing gears used

A diverse range of fishing gear is employed in the coastal and lagoon fisheries of the Kalpitiya area. These gears are predominantly used by fishers operating within the Social Area of Influence (AoI)⁷ too. While many of these gears are traditional and commonly accepted, some are classified as illegal and considered destructive due to their negative impacts on marine ecosystems and fish populations. The use of such harmful fishing methods poses a threat to the sustainability of local fisheries and biodiversity. The following **Table 3-6** provides a detailed overview of the types of fishing gears used in the region, including their target species, appropriate fishing seasons.

Table 3-6: Types of Fishing Gear Used in the Lagoon and Coastal Fishery

Gear	Target species	Seasonal	Lagoon or Coastal
Trammel nets	Shrimps and fish. This net is less selective than a gill net and catches non-target species.	All year round	
Gill nets	Mixed species. Targets small pelagic fish. Common for coral reef fish.		
Crab traps ('kakulu thatti')	Crabs.	Only during the south west monsoon, but during this time, about 80% of the fishers use this gear.	Used mainly in Lagoon during the monsoonal months, but occasionally used also in coastal areas.
Crab nets	Crabs.	Throughout the year.	Used both in lagoon and coastal fisheries.
Murali nets	Half-beak (called Muralla in Sinhala, hence the name of the net).	Only during peak monsoonal months.	Used in Lagoon during the monsoonal months, and also in coastal areas.
Soda dela	White Sardine (Sooda).	This is a seasonal fishery, ranging from Dec – April	Used both in lagoon and coastal fisheries
Monofilament trammel nets (Yaka del)	High mixed catch. A harmful fishing method as it catches fry and fingerlings. Illegal	All year	Used only in lagoon fisheries
Bottom long lines	For yellow fin tuna and scads and Liza spp.	Seasonal/small scale.	Used mainly in coastal fisheries, and very little, seasonally near the mouth of the lagoon
Hand lines	Used most commonly for cuttlefish.	Cuttlefish harvest is seasonal, from	Used only in coastal fisheries



Gear	Target species	Seasonal	Lagoon or Coastal
		January to March. Other small-scale fisheries are throughout the year.	
Push Nets	Used in shallow waters. A harmful fishing method; illegal. Push nets, like drag nets, are damaging to benthic and demersal flora and fauna.	Throughout the year/small scale. Used only in l	Used only in lagoon fisheries
Sangili del (chain nets)	A harmful fishing method; illegal.	Used in the Kalpitiya area for shrimps.	Used only in lagoon fisheries
Pull nets (Kadippu dela)	A harmful fishing method; illegal. These nets damage the Lagoon bed, disturb benthic fauna and catch all size of fish and shellfish. Used to	Throughout the year.	Used only in lagoon fisheries
Hand nets	A harmful fishing method; illegal. These nets damage the Lagoon bed, disturb benthic fauna and catch all size of fish and shellfish. Used to	November to April	Used only in coastal fisheries
Moxy nets	Ornamental fish. A harmful fishing method	November to April	Used only in coastal fisheries
Tungus nets	. Illegal	Throughout the year.	Used both in lagoon and coastal fisheries
Purse seine nets (Hambili del)	Illegal. Used to catch scad and tuna		Used only in coastal fisheries.
Lyla and Surukku Nets	Illegal.		

Source: Modified- IUCN (2012). An Environmental and Fisheries Profile of the Puttalam Lagoon System. Regional Fisheries Livelihoods Programme for South and Southeast Asia (GCP/RAS/237/SPA) Field Project Document 2011/LKA/CM/06. xvii+237 pp. Reviewed with AD Fisheries DFAR and Local Fishermen)

3.2.8.4 Details of Beach Seine in the area

Madel fishing (also known as beach seine fishing) differs significantly from other fishing methods due to its unique operational requirements. It necessitates a larger space for setting up and dragging the net from the sea to the shore. Typically, it is the only fishing practice that requires a specifically designated area for operation. According to the Madel Fishing Gazette, the seaward boundary for Madel fishing is fixed at 2,778 meters (2.78 km), exceeding Sri Lanka's standard coastal zone limit of 2 km. This fixed boundary does not vary by location, providing a consistent and unique demarcation across all Madel fishing areas. In contrast, the landward boundaries vary, as they extend along the beach, and the length of usable beach differs depending on the site.

Beach seine fishing is one of the most prominent and traditional practices in the Dutch Bay Grama Niladhari Division (GND). According to the District Fisheries Office, there are 13 Madal Padu



(beach seine sites) along the Mohonthuwaremma and Kirimundal coastlines. Of these, 10 sites are officially registered, but only 7 are currently operational, while three registered sites are inactive. Additionally, three other sites are active but not legally registered or designated in the gazette.

Unlike in the past, most beach seine operations today are conducted in the Fisheries Inspection Division (FID) of the Kalpitiya Islands using tractors and winches, with minimal hired labour mainly outside Kalpitiya to pull the large nets (**Figure 3.34**). This shift in practice has raised concerns about potential environmental impacts.



Figure 3.34: Beach seine operation at Mohonthuwaremma using Tractors

3.2.8.5 Seasons of Fishing Operations

The timing of fishing operations is primarily influenced by the weather conditions of the respective coastal regions and the specific fish species being targeted. According to the fishing community within the social Area of Interest (AoI), the main fishing season typically runs from October to May. This period deliberately avoids the southwest monsoon, which occurs from May to September and brings rough sea conditions that make coastal fishing hazardous.

During the monsoon months, when offshore fishing becomes difficult or unsafe, most coastal fishermen adapt by shifting to lagoon fisheries or migrating to the northern and eastern coastal districts, where conditions may be more favorable for fishing activities. This seasonal migration reflects both resilience and adaptability within the fishing community.

From September to April, fishermen in the AoI predominantly engage in shrimp fishing, employing gear such as trawl nets and trammel nets. In contrast, during the monsoon period, when sea fishing is limited, many fishermen turn to crab fishing within the lagoons using crab



traps. The calm and sheltered nature of lagoons during the monsoon makes them a viable alternative for sustaining livelihoods.

3.2.8.6 Species Diversity and Quantity of Fish Captured

The Puttalam Lagoon area, which encompasses the project AOI, is recognized for its rich diversity, particularly in finfish species. Among these, the most commercially important species include the sea bass (*Lates calcarifer*) and green chromide (*Etroplus suratensis*). In addition, the lagoon supports a variety of other fish species such as tarpon (*Megalops cyprinoides*), milkfish (*Chanos chanos*), ladyfish (*Albula vulpes*), halfbeaks (*Hemiramphus* spp.), garfish (family *Belonidae*), ponyfish (family *Leognathidae*), silversides (family *Atherinidae*), catfish (family *Siluridae*), and grey mullet (*Mugil cephalus*) (Mittapala S, 2013). Current scientific literature indicates that the Puttalam Lagoon hosts over 100 edible species of brackish water fish and shellfish. This diversity not only supports local subsistence and commercial fisheries but also reflects the ecological richness of the lagoon environment, which functions as a critical nursery and feeding ground for many aquatic species.

The coastal waters adjacent to the project area also exhibit significant fish biodiversity. These areas are dominated by small-meshed gill net fisheries, with mesh sizes typically ranging from 30 to 50 mm. This fishing method targets a wide range of commercially important pelagic and reef-associated species. Commonly harvested species include spotted sardinella (*Amblygaster sirm*), smoothbelly sardinella (*Amblygaster clupeioides*), other *Sardinella* species, barracuda (*Sphyraena* spp.), and mackerel (*Rastrelliger* spp.). In addition to finfish, the coastal fishery also targets valuable non-fish species such as shrimp and sea crabs (*Portunus* spp.), as well as flying fish (*Hirundichthys oxycephalus*). Furthermore, coral reef-associated species, including groupers (*Epinephelus* spp.) and snappers (*Lutjanus* spp.) contribute significantly to the fishery. The high diversity of species reflects the productivity of coastal ecosystems and underscores the importance of sustainable management practices to preserve this resource base.

According to the Assistant Director of the Department of Fisheries and Aquatic Resources, the annual fish production in 2023 was reported as 3,562 metric tons (MT) for the Kalpitiya Mainland Fisheries Inspector (FI) Division, while the Kalpitiya Islands FI Division produced 1,921 MT. In addition, approximately 299.86 MT of dried fish was also produced during the same period.

The following **Table 3-7** and **Table 3-8** present disaggregated data on the average daily and monthly fish catch, including quantity, value, and species type, as reported by the fishing communities of Mutwal, Uchchamunai, Kirimundal, and Sinnamunnakara in 2013. This



information highlights the critical role of fishing as a primary livelihood for these communities residing in the immediate coastal areas.

Table 3-7: Mutwal fishermen contribution

Product	Average quantity per day (kg)	Quantity Monthly(kg)	Value (Million)	Monthly
Fish	4000	120000	36	
Prawns	400	12000	6	
Sea Crabs	275	8250	4	
Mud Crabs	245	7350	3	
Cuttlefish	600	12000	3-4	
Sea cucumber	-			

Table 3-8: Uchchamunae, Kirimundal, and Sinnamunnakara Fishermen's Contribution

Product	Average quantity per day (kg)	Quantity Monthly(kg)	Value Monthly(Million)
Fish	5000	150000	45
Prawns			
Sea Crabs	500	15000-25000	6
Mud Crabs	40-50	1200	1
Cuttlefish	-		
Sea cucumber	300-400(Pieces)	8000-10000 Pieces	12=15
Sea Conch	300 (Number)	4000 Number	1-4

(Source: Investigative report on the looting of sustainable lands belonging to Kalpitiya inhabitants, 2013)

3.2.8.7 Income level of the fishers

According to a social survey conducted in the villages of Mohonthuwaramaa and Kirimundal, 94% of households are engaged in fishing or fishery-related activities. Although disaggregated income data specific to fishing households is not available, the monthly income figures provided in the Divisional Resource Profile of Kalpitiya for the Dutch Bay Grama Niladhari Division (GND) can serve as a proxy, as presented in **Table 3-9**. These figures indicate that 60.2% of families in the Dutch Bay GND earn less than LKR 16,000 per month, which is below the national poverty line of LKR 16,619 for 2024. This highlights the severe level of poverty faced by fishing communities in the Dutch Bay GND



Table 3-9: Divisional Resource Profile of Kalpitiya

Monthly Household Income	Number of Households	Percentage
0 - 3,999	65	18%
4,000 - 7,999	63	17%
8,000 - 11,999	43	12%
12,000 - 15,999	49	13%
16,000 - 19,999	44	12%
20,000 - 23,999	47	13%
24,000 - 27,999	23	6%
28,000 - 31,999	17	5%
32,000 above	14	4%
:	365	100%

3.2.8.8 No. of Dependents

According to the standard definition of the dependent population—comprising individuals younger than 15 years and those older than 64 years, approximately 56% of the population in the Dutch Bay Grama Niladhari Division (GND) falls within this category. This high dependency ratio indicates a significant burden on the working-age population (15–64 years), potentially affecting household income, economic productivity, and demand for social services such as education and elderly care. A detailed breakdown of the age-wise population distribution in the Dutch Bay GND is provided in **Table 3-10**.

Table 3-10: Age Group-wise population Distribution- Dutch Bay GND

Age Group (Years)		Population		Total
		Female	Male	
	0 - 5	123	91	214
	6 - 10	128	117	245
	11 - 16	168	121	289
	17 - 19	118	96	214
	20 - 25	96	71	167
	26 - 30	107	74	181
	31 - 60	102	77	179
	Above 60	139	83	222
		981	730	1711



3.2.8.9 Seasonal migration / Movements

During the monsoon months, when offshore fishing becomes challenging or unsafe due to rough sea conditions, most coastal fishermen adapt by shifting to lagoon fisheries or temporarily migrating to other regions with more favorable conditions. This seasonal adjustment specifically avoids the southwest monsoon, which spans from May to September and significantly hampers coastal fishing operations. Within the social Area of Influence (AOI), the majority of fishermen migrate to the northern coastal areas, particularly Mullaitivu and Jaffna, while others relocate to eastern coastal areas such as Trincomalee, Pulmudai, and Poduwakattu to continue their fishing activities.

3.2.8.10 Description and Assessment of the fishery and other communities / institutions affected due to the proposed project activities

The proposed project activities are expected to have both direct and indirect impacts—positive or negative—primarily affecting the residents of Mohonthuwaramaa and Kirimundal villages in the Dutch Bay Grama Niladhari Division (GND). These villages are home to multi-ethnic and multi-religious communities. According to the social survey, 94% of households in both villages are engaged in fisheries and fishery-related activities. The survey also revealed that the fishing community comprises approximately 64% Sinhalese, 18% Tamil, and 17% Moor households. Notably, over 95% of the residents have been living in these villages for more than twenty years.

The survey findings indicate that 45.7% of households receive government welfare assistance (Aswesuma), while 50% own fishing crafts, of which the majority are Outboard Fibre-Reinforced Plastic (OFRP) boats.

A notable finding from the survey was the discrepancy between the official population figures recorded for the Dutch Bay GND and the actual number of residents physically present. The current resident population is significantly lower than the registered population. This is primarily due to many individuals, though officially registered in Dutch Bay GND, residing in Kalpitiya mainland, where better access to social infrastructure is available. This trend is particularly pronounced in Kirimundal village.

Social receptors located within the project's Area of Influence (AoI) may experience minor disturbances during the construction period. These include St. Mary's Church in Palugahaturai, St. Anthony's Church in Mohonthuwaramaa, Holy Cross Church (Ave Maria Church) in Kirimundal, Mari Amman Hindu Kovil in Mohonthuwaramaa, Masjid Rahuman Mosque, and Masjith Mosque, both in Mohonthuwaramaa.



Educational facilities within the social AoI are also limited. The two local government schools—Penapidugama Vidyalaya and Mohonthuwaremaa Muslim School—operate with inadequate resources and staff, offering education only up to Grade 9. For Advanced Level studies, students must travel to schools in Kalpitiya Town. Minor impacts on schoolchildren may be anticipated during the construction phase, especially due to transportation of materials through local roads and the Thoradiya–Mohonthuwaremaa jetty.

Furthermore, the fishing community, individuals involved in tourism-related activities, and both local and foreign visitors may temporarily experience various impacts as a result of the project. Institutions such as schools, religious centers, and tourist establishments within the area may also be subject to temporary positive or negative effects, depending on the nature and scope of the project interventions

3.2.9 Noise

Dutch Bay Island is sparsely populated area with less anthropogenic activities except for fisheries and tourism. Hence natural noise sources prevail over the manmade noises on the island where wave breaking sound is dominant along the coastline with ad-hoc sounds from fishery activities. Day today activities in the Mohonthuwaremaa and Keerimundalam villages in the Island contributes to the anthropogenic noises.

3.2.9.1 Existing noise levels

Noise monitoring program was conducted on Dutch Bay Island for the ambient conditions on 31st March and 1st of April 2025 by Environmental Studies and Services Division (ESSD) of National Building Research Organization (NBRO). Noise levels were measured at 5 locations indicated on Figure 3.35 and the results are given in the **Table 3-11**. Measured noise levels are below the maximum permissible noise levels at all the locations during both day and night.

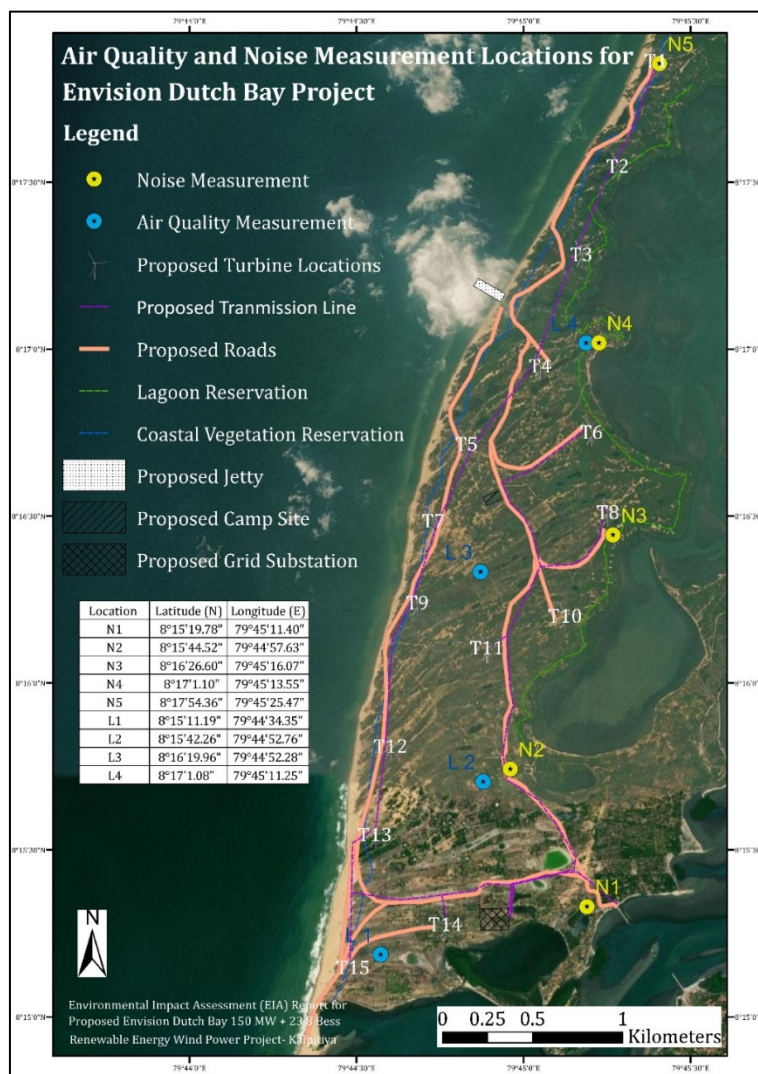


Figure 3.35: Noise sampling Locations_1

Table 3-11: Noise levels at Measurement Locations

Location	Time	Run Time	Measured Residual Noise Level L_{eq} (dB)	Background Noise Level L_{90} dB(A)
NLS1	Day	1 Hour	41	36
	Night	1 Hour	42	33
NLS2	Day	1 Hour	40	38
	Night	1 Hour	41	36
NLS3	Day	1 Hour	39	30
	Night	1 Hour	38	36
NLS4	Day	1 Hour	41	34
	Night	1 Hour	40	38
NLS5	Day	1 Hour	34	31
	Night	1 Hour	38	35



Humans and birds contribute to the noises occur during the day time on the Island while chirping from insects and barking of dogs contributes to the noises at night. Complete report on noise monitoring for the baseline condition is given in **Annexure III A**.

3.2.9.2 Distribution of existing noise receptors

Main noise receptors in the island are the two inhabited villages in the island (Mohonthuwarama and Keerimundalam) and religious places in the island such as St Antony's Church, Holy Cross Church, Mari Amman Hindu Temple and Mohonthuwaram Jumma Masjid mosque. Mohonthuwaramaa Sinhala School and Mohonthuwaramaa Muslim Maha Vidyala are the two schools in the Island where loud noises can disturb during daytime. Dutch Bay Resort would be also a sensitive receptor when it's in the functional condition.

3.2.9.3 Locations of noise sensitive receptors (if any) (distance of towers to such receptors to be indicated)

Locations of the sensitive noise receptors are illustrated in the **Figure 3.36** and distances from the nearest wind turbine are indicated in the **Table 3-12** below.

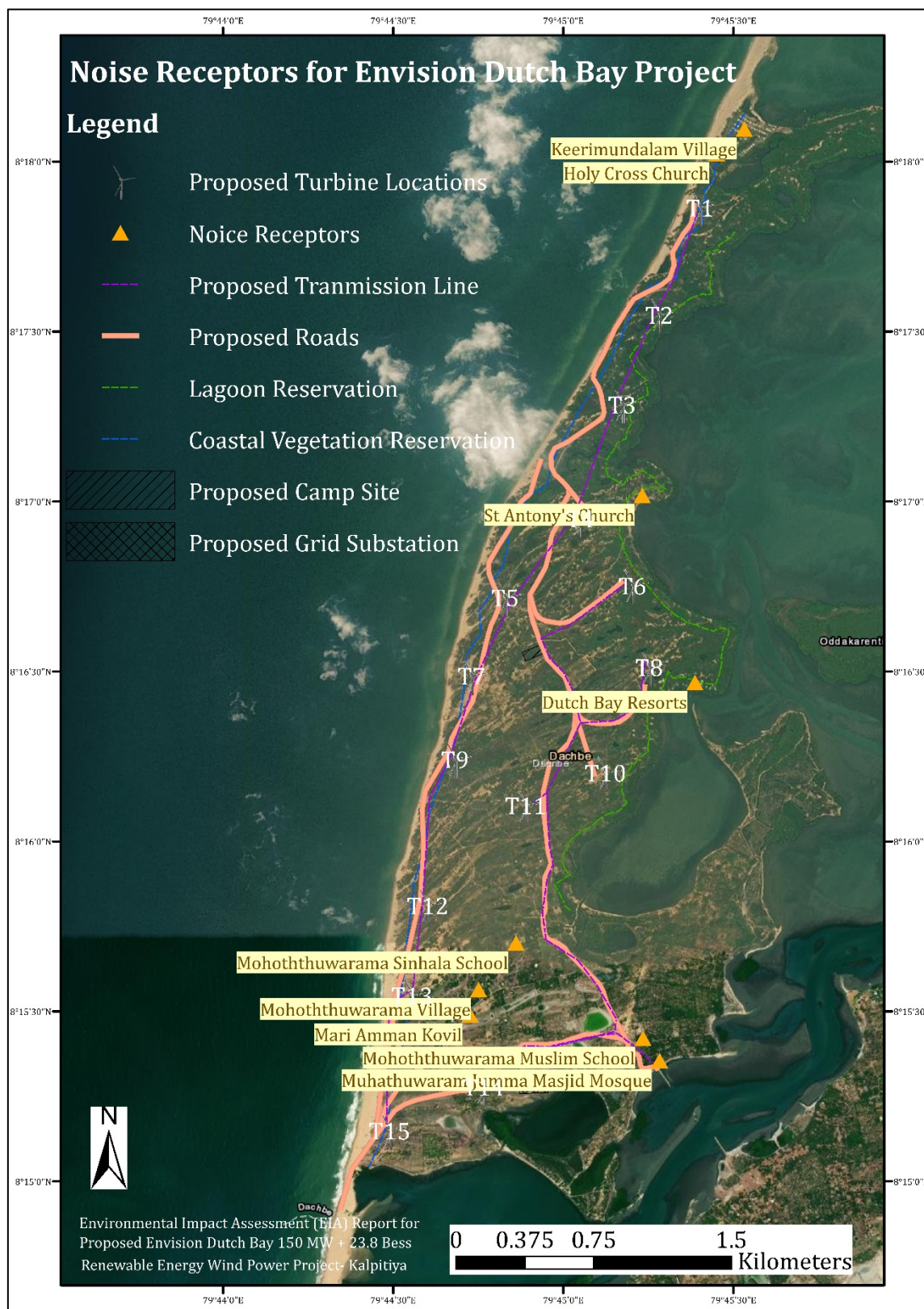


Figure 3.36: Locations of noise sensitive receptors



Table 3-12: Proximity from nearest wind turbines to sensitive noise receptors

Noise receptor	Closest Wind Turbine	Distance (m)
Keerimundalam village	T1	375
Holy Cross Church	T1	300
St Antony's Church	T4	350
Dutch Bay Resort	T8	100
Mohonthuwaramaa Sinhala School	T12	500
Muhuththuwarem Muslim Maha Vidyalaya	T14	900
Muhathuwaram Jumma Masjid mosque	T14	950
Muhathuwarama Village	T13	160
Mari Amman Hindu Temple	T13	300

3.2.10 Air Quality

Ambient air quality levels in the Dutch Bay Island were measured by the Environmental Studies and Services Division (ESSD) of National Building Research Organization (NBRO) on 31st March and 1st of April 2025. Measurements were carried out at the four locations covering the entire Island as illustrated in Figure 3.37 and results are given in **Table 3-13** below.

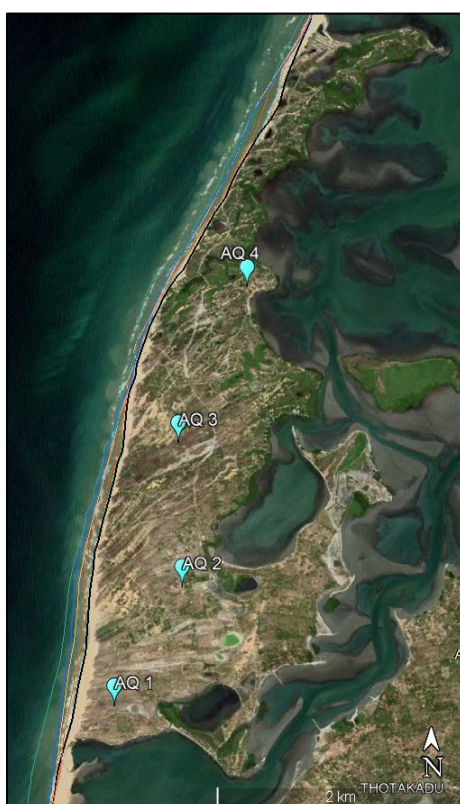


Figure 3.37: Air Quality Measurement Locations_1



Table 3-13: Concentration at sampling locations

Parameter	Date of Analysis	Time Average	Unit	Concentrations at Locations			
				L1	L2	L3	L4
CO	01/04/2025	1 hr	µg/m ³	<1000	<1000	<1000	<1000
TVOC	01/04/2025	1 hr	ppm	0.274	0.170	0.190	0.162
CH ₄	01/04/2025	1 hr	Vol%	1.25	1.25	1.25	1.48
CO ₂	01/04/2025	8 hrs	ppm	1165	1114	1134	1121
SO ₂	01/04/2025	8 hrs	µg/m ³	08	12	06	08
NO ₂	01/04/2025	8 hrs	µg/m ³	44	46	14	28
PM ₁₀	04/04/2025	24 hrs	µg/m ³	20	19	17	15
PM _{2.5}	04/04/2025	24 hrs	µg/m ³	11	10	09	08

Air quality of the Dutch Bay Island for the parameters NO₂, SO₂, CO, PM₁₀ and PM_{2.5} are below than the standard air quality levels stipulated for the Sri Lankan environment. No industrial emissions present in the Island except for the emissions from the motorised boats used for tourism and fishing and emissions from vehicles are insignificant as vehicular traffic are occasional in the Island.

Detailed Report of the Air quality measurements are given in Annexure III B.

3.3 Ecological Resources

3.3.1 Study Area

The ecological study encompassed all areas potentially impacted—both directly and indirectly—by the construction and operation of the proposed onshore wind farm project, as outlined in the preliminary invitation survey.

The designated project area, where all activities will be confined, is situated in Dutch Bay and extends approximately 6 kilometers along the coastal zone of the Kalpitiya Peninsula. It spans from Kalpitiya Beach (8.250912°; 79.741066°) to a point near Muhathuvaram Church (8.300573°; 79.757057°). All project activities will be conducted within the Coastal Zone, as defined by the Coast Conservation Act of 1981 and its subsequent amendments. This zone encompasses land extending 300 meters inland from the mean high-water line and seaward up to 2 kilometers from the mean low-water line.



The Environmental Impact Assessment (EIA) study area covers all locations that may be directly or indirectly affected by the project. This includes areas where project activities may be visible or audible, such as the beach, sea, lagoon, and surrounding land within a 1-kilometer radius of the project site boundaries in Dutch Bay.

3.3.2 Aims and Scope of the Study

As the island transitions to renewable energy to meet its climate goals, onshore wind power has emerged as a prominent solution. While wind farms offer a low carbon footprint, they also require substantial land areas, which can lead to ecological disruptions. Therefore, the expansion of onshore wind energy must carefully balance clean energy production with the preservation of sensitive ecosystems.

The primary objective of this ecological study is to understand the ecological characteristics of the project area, evaluate potential environmental impacts associated with development activities, and propose strategies to mitigate adverse effects while enhancing positive outcomes. This involves identifying existing habitats and species, assessing their ecological significance, and recommending appropriate conservation measures.

The assessment process was carried out in the following steps:

- Study the ecological setting of the entire project area, including both the core and buffer zones, through field surveys and literature reviews to establish baseline environmental conditions.
- Conduct a scoping exercise to identify potential environmental issues arising from the proposed project activities and land use changes, in consultation with local communities and relevant stakeholders.
- Assess potential impacts on environmental attributes resulting from the location, design, construction, and operation of the proposed wind energy project.
- Develop an Environmental Management Plan (EMP) outlining measures to protect the biological environment, including institutional arrangements and an environmental monitoring framework.

3.3.2.1 Comprehensive Ecological Survey

The comprehensive ecological survey was designed based on the land-use requirements of the proposed wind power project. Two major land-use categories associated with wind power development were identified:



- **Direct Impact Area:** This includes land that will be permanently altered due to the construction of physical infrastructure, such as turbine foundations, access roads, service areas, and operational facilities.
- **Total Project Area:** This encompasses the entire land area associated with the wind power project, including both the directly impacted areas and additional zones required for project operation, safety, and buffer purposes.

The project involves the use of land for several purposes:

- Permanent land use for turbine foundations, service areas, access roads, and other operational structures.
- Temporary land use for features such as hard stands and construction staging areas, which are required during the construction phase.
- Reserved airspace for the rotor sweep zone, where no physical ground interventions occur, but which must be factored into land-use planning (refer **Figure 2.11 and Figure 2.12**).

This classification informed the planning of the ecological assessment to ensure that all types of land impact—permanent, temporary, and spatial—were considered in the survey design and impact evaluation.

During both the rapid assessment and comprehensive ecological survey phases, the Visual Encounter Survey (VES) method was employed to document the general flora (flowering plants) and fauna across various ecosystems under different climatic conditions. Depending on site accessibility, two standard sampling approaches were utilized: opportunistic or randomized walks and transect-based surveys. All sampling activities were conducted during daytime hours.

In easily accessible and relatively uniform areas, both methods were applied to ensure comprehensive data collection. In more difficult-to-access locations, photographic documentation was used to record species that were unfamiliar or challenging to identify in the field. These photographic records were subsequently analyzed using standard taxonomic keys and relevant scientific literature, as referenced in this study, to ensure accurate species identification.

Field sampling methods were designed to document all major groups of vertebrates, invertebrates, and floral communities across habitats within the survey area. Faunal surveys covered both terrestrial and brackish-water habitats. Vertebrate groups—including mammals, birds, reptiles, and amphibians—as well as invertebrates such as butterflies, were sampled through random observations across various representative sites.



Assessment of Species Diversity and Abundance: The species diversity of fishery resources—including both finfish and non-fish species—exploited by lagoon fisheries and beach seine fishing in the adjacent shallow sea was assessed through systematic observations of commercial and artisanal fish catches. Data collection focused on identifying species composition, estimating relative abundance, and documenting gear types used.

Species diversity and abundance were estimated using visual estimation techniques, which included on-site catch monitoring at landing sites and direct observations during fishing operations. These methods provided qualitative and semi-quantitative insights into the richness and distribution of fishery resources across different seasons and habitats. The results contribute to understanding ecological dynamics and inform the sustainable management of fish stocks within the project area.

At each proposed wind turbine site, ecological assessments began with a detailed evaluation of habitat type and habitat condition. This involved surveying each site to characterize existing habitats, assess their suitability for supporting wildlife, and evaluate the potential impacts of turbine construction and operation.

A quantitative ecological assessment was conducted for land areas expected to be directly disturbed by turbine installation (**Figure 3.38**). This included:

- The turbine pad area (10 m × 10 m),
- A 50-meter radius surrounding each turbine (designated service area), and
- Where accessible, the rotor sweep zone, represented by a 200-meter radius from the turbine center. In some cases, assessments of this larger area were constrained by limited accessibility.

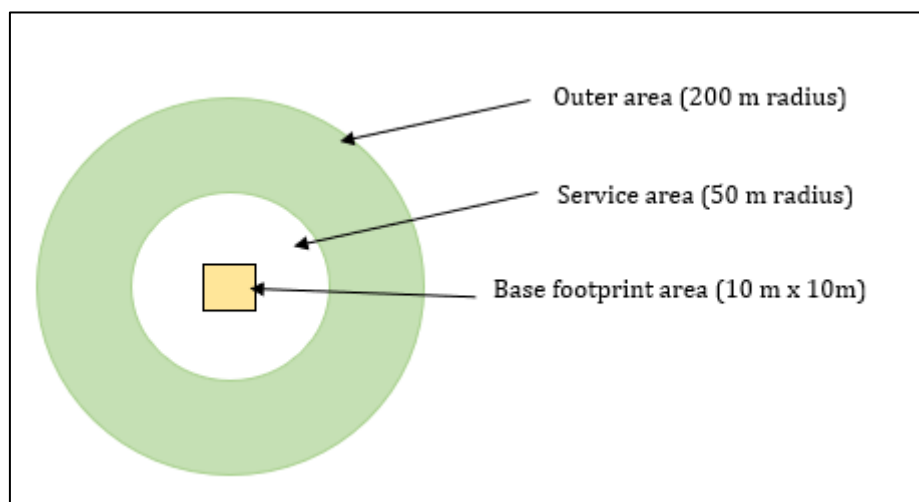


Figure 3.38: Area of Concern for Base Footprint, Service etc.



In addition to turbine-specific areas, other zones of permanent disturbance—such as approach roads (while the existing beach road is suitable for transporting wind turbines, approach roads to the site still need to be constructed), substations, service buildings, and other project infrastructure—were also assessed separately. These features were evaluated for their physical occupation of land and potential to create impermeable surfaces, thereby altering the local ecological conditions.

A quantitative ecological assessment of the construction footprint was carried out to document the flora and fauna present within the turbine installation areas. This included both the turbine pad area (10 m × 10 m) and the service area, defined as a circle with a 50-meter radius from the turbine center.

Within this zone, all plant species were identified and categorized into trees, shrubs, and ground covers (such as herbs and grasses). Trees with a trunk circumference (girth) greater than 25 centimeters were individually measured and documented.

In the rotor sweep zone—the area beyond the 50-meter service radius reserved for turbine blade rotation—plant species were also identified and recorded. Sampling within this zone was conducted using 5 to 6 line transects, each 150 meters in length, at all accessible locations, and results were extrapolated to estimate species abundance across the entire area.

For faunal assessment, the Point Count Method as described by Colin and Neil was employed. Counting points were systematically established at 200-meter intervals, with three points set up around each turbine installation site. Upon arrival at each point, a 5-minute settling period was observed to minimize the effects of disturbance and allow animals to resume natural behavior. This was followed by a 15-minute observation period, during which all visible or audible animal species were recorded.

3.3.3 Major Habitat Types in the Study Area

The study area in Dutch Bay, located on the Kalpitiya Peninsula, comprises a narrow strip of land facing the lagoon on its eastern side. This landscape is characterized by marshes, mangrove fringes, and mudflats. In contrast, the western shoreline features beaches and dune deposits. Permanent human settlements are limited and are primarily established on less waterlogged terrain, with access provided either by boat or via minor truck roads crossing the peninsula.

As a result, the area supports a diverse array of interconnected and interdependent natural habitats, forming a mosaic within a broader landscape shaped by variations in vegetation,



topography, hydrology, and soil conditions. Based on field surveys and ecological assessments, ten major habitat types have been identified. These are broadly classified into three categories:

- Natural wetland habitats
- Natural terrestrial habitats
- Managed habitats

3.3.3.1 Natural Wetland Habitats

Mangroves

Coastal intertidal forests dominated by salt-tolerant tree species. Mangroves provide critical nursery grounds for fish and invertebrates and act as natural buffers against coastal erosion and extreme weather events.

Mangroves within the study area can be broadly classified into two ecological groups: fringing mangroves and shrub mangroves.

- Fringing Mangroves

These are found along the intertidal coasts of the lagoon, where they are regularly inundated by tidal waters. The areas are characterized by well-developed mangrove vegetation, with *Rhizophora mucronata* and *Avicennia marina* being the dominant species.

- Shrub Mangroves

In contrast, shrub mangroves are located more inland, particularly in seasonally flooded zones and along temporary drainage paths. These mangroves are typically less developed, with scattered and stunted *Avicennia marina*, often growing alongside other species such as *Sonneratia alba*, *Excoecaria agallocha*, and *Lumnitzera racemosa*.

Additional mangrove species observed occasionally include:

- *Rhizophora apiculata*
- *Bruguiera cylindrica*
- *Bruguiera gymnorhiza*
- *Ceriops tagal*

Several mangrove associate species were also recorded in the area, including:

- *Thespesia populnea* (Suriya)
- *Derris trifoliolata* (Kalawel)
- *Caesalpinia bonduc* (Kalu Waulatiya)



These mangrove systems (**Figure 3.39**) are ecologically significant, serving as nursery habitats for marine species, stabilizing coastal shorelines, and supporting high biodiversity, especially in brackish-water-dependent fauna.



Figure 3.39: Mangrove Systems existing at Site

Lagoon, Seagrass Beds, and Tidal Mudflats

The lagoon serves as the main body of brackish to saline water, supporting both seagrass beds and mudflats. Seagrass Beds typically grow in shallow, clear areas of the lagoon where sunlight penetrates to the bottom, offering shelter, food, and nursery grounds for fish, crustaceans, and other marine organisms. Tidal Mudflats form in the shallower, intertidal zones, exposed during low tides, providing rich feeding grounds for migratory birds and habitats for invertebrates such as crabs, worms, and shellfish.

Dutch Bay is a key area within the Puttalam Lagoon, renowned for its rich biodiversity and ecological significance. Within this lagoon system, seagrass beds, tidal mudflats, and open lagoon waters create a mosaic of interconnected habitats that play vital ecological, economic, and social roles.

Due to the shallow nature of the lagoon in the study area, seagrass beds are primarily dominated by *Thalassia hemprichii*, *Halodule uninervis*, and *Cymodocea rotundata*. These seagrass meadows provide critical support for a wide range of marine life, contributing to sediment stabilization and water quality improvement. While some previous studies have reported dugongs feeding on seagrass in Dutch Bay, our study did not observe this behavior nor find any signs of it through community interviews.

Tidal flats, composed mainly of muddy or sandy substrates, are exposed during low tide and submerged during high tide, creating dynamic ecosystems. These areas provide essential habitats for various invertebrates (**Figure 3.40**) and fish and serve as crucial feeding grounds for both native and migratory bird species, such as plovers and sandpipers.



Figure 3.40: Presence of various invertebrates

Saltmarshes

Low-lying coastal habitats regularly inundated by saline or brackish water, dominated by halophytic (salt-tolerant) vegetation. These ecosystems serve as vital feeding and breeding grounds for a variety of fauna. The dominant vegetation includes chenopod succulent saltmarsh communities.

Salt marsh habitats in the study area are characterized by sandy and muddy flats that are periodically inundated with seawater. These coastal intertidal zones are regularly flooded by tides, supporting a variety of salt marsh plant species. In the peripheral margin of a salt marsh, a mix of succulent plants grass and sometimes mangrove species can be observed. The salt marsh vegetation includes species such as *Salicornia brachiata*, *Cynodon dactylon* is the most widespread grass species in the salt marsh, while *Cyperus rotundus* is primarily found in depressions within this habitat. These salt marsh habitats (Figure 3.41) are highly disturbed and degraded due to truck movement, which has contributed to significant habitat destruction.



Figure 3.41: Salt Marsh Habitats



Shallow Sea and Intertidal Zone

Shallow marine and tidal zones characterized by sandy substrates with sparse vegetation. These areas support marine biodiversity, including sand crabs, worms, snails, and clams, and serve as important feeding grounds for shorebirds.

The shallow sea adjacent to the coast is typically characterized by sandy substrates, gentle slopes, relatively high light penetration, and the absence of sensitive habitats. These areas often function as biological corridors for species migrating between reef systems and lagoon environments, forming a critical ecological interface that supports reef resilience through larval dispersal and trophic linkages.

However, the sea in Dutch Bay is biologically rich and ecologically sensitive. The project area lies close to, but slightly south of, the Bar Reef, Sri Lanka's largest and most ecologically significant coral reef complex. The Kandakuliya Reef, located just south of the Bar Reef, lies relatively close and slightly north of the project area (Figure 3.42).

The tidal zone supports a diverse range of fauna, including polychaetes, crustaceans, and mollusks that inhabit or burrow into the sediment. This shallow coastal zone plays a vital role in nearshore productivity and significantly contributes to local fisheries, particularly artisanal and small-scale commercial operations.



Figure 3.42: Kalpitiya, Kandakuliya Reef and Bar Reef



Shorebirds frequently forage in the tidal zone during low tide, and certain stretches of undisturbed beach—especially those free from fishing activity—serve as potential nesting sites for marine turtles, particularly Olive Ridley and Green turtles. However, no active turtle nesting was physically observed on the beaches within the study area during the survey period. Historical records of turtle nesting were clarified through community discussions and consultations with naval officials engaged in beach patrolling. These discussions confirmed that turtle nesting occasionally occurs along some stretches of coastline during the non-fishing season, typically from July to September.

Although the Kalpitiya region has traditionally been a key sediment accretion zone along Sri Lanka's northwestern coast, parts of the project area are now experiencing gradual coastal erosion. This erosion, driven partly by an increasingly intensified wave climate, poses challenges for local ecosystems, coastal infrastructure, and livelihoods.



Figure 3.43: Shallow Sea and inter tidal zone

Seasonally Inundated Grasslands

Sedge-dominated saltmarsh pastures that experience periodic flooding during the wet season. They support amphibians, aquatic invertebrates, and migratory birds and are also used for domestic livestock grazing.

The study area also contains patches of seasonally inundated maritime grasslands. The structure and floristic composition of these grasslands are strongly influenced by grazing pressure, salinity, and drought conditions. The dominant grass species include *Cynodon dactylon*, and *Panicum repens*, while sedges and other grasses such as *Schoenoplectus pungens*, *Eleocharis acuta*, and *Fimbristylis cymosa* occur intermittently across the landscape. Species dominance and overall plant composition are largely determined by site-specific moisture levels. Additionally, seasonally inundated waterholes are scattered throughout these grassland areas, providing important habitat diversity (Figure 3.44).



Figure 3.44: Existing Grasslands in the Area

3.3.3.2 Natural Terrestrial Habitat Types

Scrublands

Dry, open landscapes dominated by low-growing, thorny shrubs and stunted trees. These areas support a wide range of reptiles, birds, and small mammals.

Remnant patches of tropical dry mixed evergreen forests and mixed tropical thorn scrublands occur within the study area. These habitats are characterized by two distinct aerial strata. The uppermost tree layer includes species such as *Manilkara hexandra*, *Drypetes sepiaria*, and *Syzygium cumini*, which are often stunted due to environmental factors. These trees (**Figure 3.45**) typically reach heights of 5–7 meters and are noted for their deeply furrowed bark. The second layer is dominated by thorny treelets and shrubs, including *Dichrostachys cinerea*, *Salvadora persica*, *Phoenix pusilla*, *Catunaregam spinosa*, *Carissa spinarum*, *Ziziphus oenoplia*, *Canthium coromandelicum*, *Maytenus emarginata*, and *Azima tetracantha*.



Figure 3.45: Scrubs in the vicinity of project area

Sand Dunes

Coastal, low-elevated ridges of wind-deposited sand with sparse, hardy vegetation. Sand dunes provide a unique habitat for specialized flora and fauna adapted to these harsh conditions.

Sand dunes are present along the coastline of the study area, occurring above the zone of tidal influence (**Figure 3.46**). These features form a unique natural landscape that contrasts with the surrounding terrain. Sand dunes are sensitive and fragile ecosystems that provide habitat for a



range of specialized floral and faunal species adapted to sandy and exposed conditions. The dunes in the study area vary in shape and size, are generally low in elevation, and are formed through interactions with wind and water. They serve as vital natural barriers against coastal erosion, helping to protect life and property within the coastal zone by absorbing wave energy and acting as reservoirs of sand to replenish material lost during severe storms.

However, field investigations revealed that several dune segments have been irreversibly damaged, primarily due to human activities such as fishing operations and off-road vehicle movement.

The vegetation of the sand dunes is patchy, with open sandy areas interspersed among plant cover. Vegetation composition varies according to the terrain. On the beach terrace and in the valleys between dunes, ground-hugging, prostrate, or procumbent herbaceous and suffrutescent species dominate. The dune crests and areas above the beach terrace support scattered shrubs and small trees, typically reaching heights of 1–2 meters, forming small vegetative islands. Herbaceous vegetation is mostly found on the beach terrace and in dune valleys, while shrubs dominate higher flat areas and dune crests.

Common herbaceous species include *Spinifex littoreus*, *Mariscus pedunculatus*, *Ipomoea pes-caprae*, *Phyllanthus maderaspatensis*, and *Bulbostylis barbata*. Shrub species found above the beach terrace include *Calotropis gigantea*, *Scaevola taccada*, and *Pemphis acidula*.



Figure 3.46: Sand Dunes

Palmyrah Woodlands

Dry zone woodlands dominated by *Borassus flabellifer* (Palmyrah palm). The mixed understory contributes to both biodiversity and local livelihoods, offering habitat for various species.

Palmyrah woodlands refer to areas dominated by the Palmyrah palm (*Borassus flabellifer*), a robust, solitary, dioecious palm that grows between 25 to 40 meters tall. These woodlands are a common feature of the Kalpitiya Peninsula (Figure 3.47) and play a vital ecological and socioeconomic role. They provide critical habitats for various species, aid in water conservation



through their deep root systems, and offer local communities valuable resources such as food, materials, and income.

The ground vegetation in Palmyrah woodlands typically consists of low-lying shrubs such as *Dodonaea viscosa*, *Cassia auriculata*, fast-growing evergreen shrub, along with herbaceous plants, including grasses and sedges like *Imperata cylindrica*, which are well adapted to the arid and seasonal climate.



Figure 3.47: Palmyrah Woodlands

3.3.3.3 Managed Habitat Types

Coconut Plantations

Monoculture plantations of *Cocos nucifera*, with varying undergrowth. While these plantations are primarily cultivated for commercial purposes, they provide some habitat value to generalist species.

Due to arid conditions, only a few coconut plantations of varying ages are found in the southern part of the study area, where the water table remains high even during dry seasons. The ground vegetation primarily consists of grasses that support livestock farming (Figure 3.48).



Figure 3.48: Coconut Plantations



Home Gardens

Mixed-use areas around dwellings, typically featuring fruit trees, vegetables, and ornamental plants. These areas support diverse species that are adapted to human-altered environments.

Home gardens (Figure 3.49) are dynamic agroforestry systems that integrate crops and livestock to meet household needs and generate income. These systems are primarily composed of cultivated perennial species such as *Cocos nucifera* (coconut), *Mangifera indica* (mango), *Anacardium occidentale* and *Borassus flabellifer* (Palmyrah palm), with coconut being the dominant species in nearly all home gardens in the area. Beyond their economic role, home gardens contribute significantly to biodiversity conservation, climate regulation, and protection against natural hazards.



Figure 3.49: Presence of Home Gardens

3.3.4 Floral Diversity of the Project area

A total of 127 plant species (Table 3-14) belonging to 46 families were recorded across all habitats within the proposed project sites and their immediate surroundings. This includes one endemic species and 104 indigenous species. Additionally, 19 of the recorded plant species are introduced. None of the documented flora species are unique or restricted to the project area. All recorded plant species are listed in **Table 1 of Annexure III K**.

Table 3-14: The total number of species recorded, threatened (endangered, near-threatened and vulnerable), endemic, indigenous and introduced, by life forms.

Life form	Total species	Threatened	Endemic	Indigenous	Introduced
Tree	33	5	1	20	7
Shrub	35	3	0	31	1
Herb	33	1	0	21	11
Climber/creeper	19	0	0	18	1
Grass/Grass like	7	0	0	7	0
Total	127	9	1	97	20



3.3.4.1 Rare, threatened and endemic flora found in the project area

One plant species recorded within the project area is endemic to the country (Annex Table [X]), but it is not legally protected. Additionally, five near-threatened plant species, one endangered species, and one critically endangered species were recorded during the field survey (Table 3-15 and Table 3-16). None of the recorded endemic or indigenous flora species are unique to or restricted within the project area

Table 3-15: Endemic plant species recorded from the project area

Family	Species	Sinhala name	Habit	NCS
Celastraceae	Elaeodendron glaucum	Naralu	T	LC

Habit= T – Tree; S-Shrub, NCS=National Conservation Status- (LC – Least Concerned), VU-Vulnerable)

Table 3-16: Threatened plant species recorded from the project area

Family	Species	Sinhala name	Habit	NCS	TS
Celastraceae	<i>Pleurostyliya opposita</i>	Panakka	T	NT	N
Combretaceae	<i>Lumnitzera racemosa</i>	Bariya	T	NT	N
Fabaceae	<i>Indigofera oblongifolia</i>	Nari-Mun	S	NT	N
Lythraceae	<i>Pemphis acidula</i>		S	NT	N
Sapotaceae	<i>Manilkara hexandra</i>	Palu	T	NT	N
Fabaceae	<i>Vachellia planifrons</i>		T	EN	N
Solanaceae	<i>Solanum virginianum</i>	Katuwel-Batu	S	CR	N

Habit – T -Tree, H-Herbaceous, C-Creeper, NCS- National Conservation Status (NT- Near threatened, EN- Endangered, VU- Vulnerable, CR- Critically Endangered TS- Taxonomic status (N- Native, N- E- Endemic)

3.3.5 Fauna found in the Project Affected Area

The site selected for the proposed wind farm primarily consists of heavily disturbed coastal scrub-mixed forest, mangroves, and cultivated lands, including home gardens. The tidal zone and beach berm area are actively used for fishing activities. Consequently, faunal diversity in these areas is relatively low, with recorded species mainly comprising butterflies, reptiles, birds, and mammals, including domesticated animals.

Excluding avifauna, a total of 24 terrestrial faunal species were recorded during the field surveys conducted within the study area. All recorded species are classified as Least Concern, and none are endemic (Table 2 of Annexure III K).



3.3.6 Fishery Resources Exploited from the Lagoon and the Shallow Sea Bordering the Project Area

The lagoonal and coastal waters adjacent to the project site—including Dutch Bay, parts of Puttalam Lagoon, and the nearshore shallow sea areas bordering the Bar Reef Marine Sanctuary, support a diverse and economically significant fishery. These ecosystems provide essential habitats for a wide variety of finfish, shellfish, and invertebrate species, which are exploited by local artisanal and small-scale fishers for their livelihoods.

3.3.6.1 Lagoon Fisheries

The lagoon fisheries in Dutch Bay and surrounding areas are characterized by multispecies exploitation using traditional and low-impact methods. The key resources harvested include:

Finfish species: A total of 21 species from 15 families were identified (see Annex Table). Dominant species include *Etroplus suratensis* (green chromide), *Mugil spp.* (mullets), *Lates calcarifer* (barramundi), *Nematolosa nasus* (shad), and *Oreochromis mossambicus* (tilapia).

Crustaceans: Notably, *Metapenaeus dobsoni* and *Penaeus indicus* (white shrimp), as well as *Scylla serrata* (mud crab), which are commonly trapped or netted in shallow waters.

Bivalves: Edible clam species such as *Gafrarium tumidum*, *Marcia hiantina*, and *Marcia opima* are collected manually, primarily by women, from intertidal mudflats and shallow subtidal areas.

Fishing methods include cast nets, gill nets, crab traps, push nets, and hand collection. The availability and abundance of species are influenced by seasonal changes and salinity fluctuations due to tidal influence and freshwater inflows during the rainy season.

3.3.6.2 Coastal and Nearshore Fisheries

Coastal fishing activities in the shallow sea bordering the project area are primarily conducted using beach seines, which are deployed seasonally, from approximately August/September to May, when sea conditions are relatively calm.

A total of 38 species from 15 families were recorded from beach seine catches (**Refer Table 3 & Table 4 of Annexure III K**).

The most dominant species include:

Caranx sexfasciatus (scad)

Scomberoides spp. (queenfish)



Chirocentrus spp. (wolf herring)

Sardinella spp. (sardine)

Stolephorus spp. (anchovy)

Leiognathus spp. (ponyfish)

These fisheries contribute significantly to local food security and income generation and are deeply integrated into the socioeconomic fabric of surrounding coastal communities.

3.3.7 Proximity to Protected Areas to the Project Area

3.3.7.1 Wilpattu National Park

Wilpattu National Park is boarded south by the Kala Oya, and in the west by Portugal and Dutch Bay as well as the open sea. It is the largest protected area in Sri Lanka and was declared a National Park in 1938. Its proximity underscores the ecological connectivity between the estuary, lagoon and the park. However, the Wilpattu National Park is located around 7.5 km from the northern end of the project area.

3.3.7.2 Bar Reef Marine Sanctuary

The main protected marine area around the Kalpitiya Peninsula is the Bar Reef Marine Sanctuary, which is the largest marine protected area in Sri Lanka. It is located about 2 km off the coast of Kalpitiya and covers approximately 310 km². The Bar Reef is a system of coral reefs and was declared a marine sanctuary in 1992. Refer section 3.1.2.

3.3.7.3 Kala Oya Estuary

The Kala Oya riverine estuary opens to upper basin of the Dutch Bay. It features significant mangrove habitats, contributing to the ecological richness of the region. This Kala Oya estuary is located around 7 km away from the northern end of the project area.

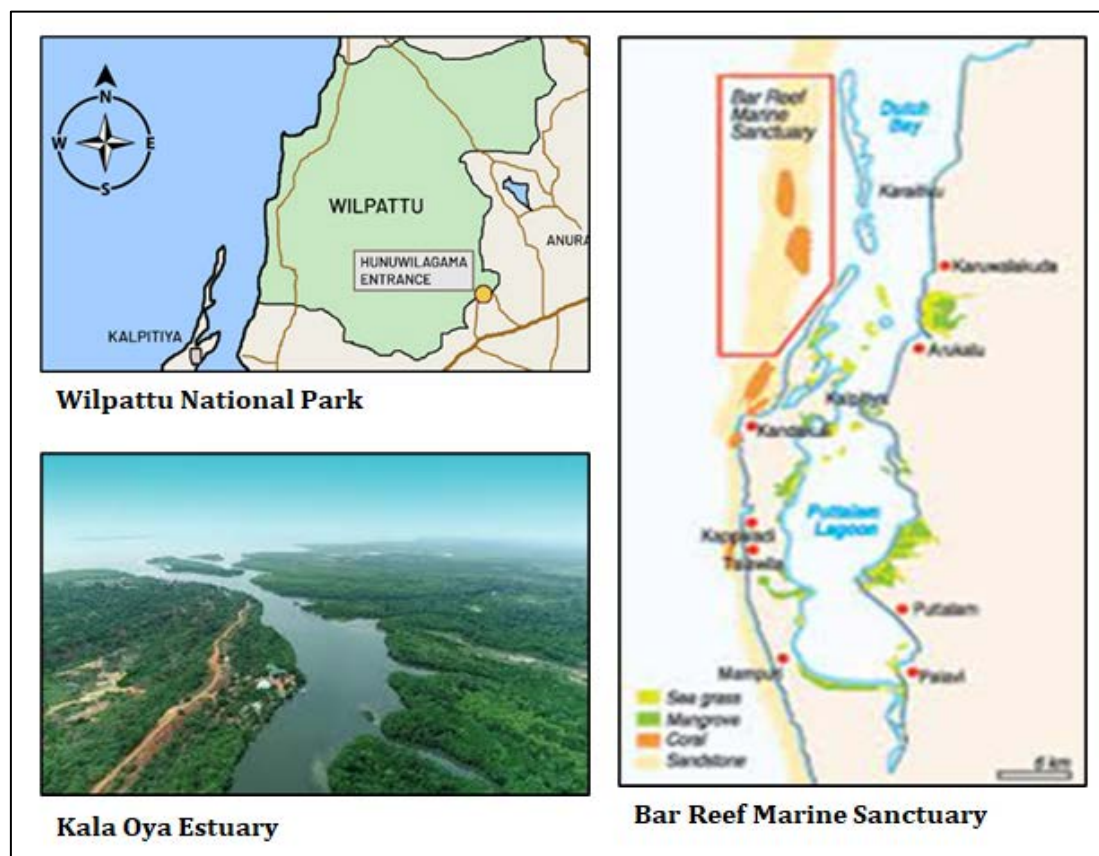


Figure 3.50: Protected Areas to the Project Area

3.3.8 Ecological Assessment of the Project Impacting Area

3.3.8.1 Land directly occupied by the turbine's foundation and tower

Each wind tower footprint was surveyed to assess its biological diversity, with a focus on floristic composition (plants). Surveys were conducted within a 200-meter radius around the planned site of each tower.



Table 3-17: Floristic Composition in WTG1

Tower 1	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting			
	The area is characterized by an open landscape dominated by grasses and scattered low-growing shrubs.	The area is characterized by an open landscape dominated by grasses and scattered low-growing shrubs.	Scattered water bodies with fringing mangroves occur within the wind turbine's rotational zone, particularly in the eastern area and along the western margin, where they merge with the shallow sea.
Florist Setting	<i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i> <i>Cyperus rotundus</i> <i>Cyperus arenarius</i>	<i>Indigofera oblongifolia</i> <i>Synostemon bacciformis</i> <i>Tephrosin purpurea</i> <i>Cyperus rotundus</i> <i>Hydrophylax maritime</i> <i>Cyperus arenarius</i>	<i>Avicennia marina</i> <i>Excoecaria agallocha</i> <i>Lumnitzera racemosa</i> <i>Azima tetracantha</i>
Ecological impact of the Project	The proposed activity does not require the removal of any trees or woody shrubs.	The proposed activity does not require the removal of any trees or woody shrubs.	The fringing mangroves will remain undisturbed, preserving existing coastal and lagoon-side vegetation and maintaining ecological integrity.



Table 3-18: Floristic Composition in WTG2

Tower 2	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	 The tidal wetland is characterized by scattered, stunted mangroves and shows signs of significant ecological degradation.	 The area is characterized by tidal channels and depressions, which play a vital role in its hydrological and ecological functions, yet it exhibits significant signs of ecological degradation.	 The rotating area extends to the lagoon in the east, which is characterized by fringing mangroves, while the western side is predominantly covered with coastal shrub vegetation.
Floristic setting	<i>Cyperus rotandus</i> <i>Eleocharis geniculata</i> <i>Avicennia marina</i>	<i>Cyperus rotandus</i> <i>Cyperus arenarius</i> <i>Eleocharis geniculata</i> <i>Avicennia marina</i> <i>Halosarcia indica</i>	<i>Avicennia marina</i> <i>Excoecaria agallocha</i> <i>Lumnitzera racemosa</i> <i>Opuntia dillenii</i>
Ecological impact of the Project	The landscape is severely degraded, with most woody vegetation, including mangrove species, largely destroyed.	The landscape is severely degraded, with most woody vegetation, including mangrove species, largely destroyed.	No clearance of fringing mangroves will occur, ensuring that the existing vegetation remains intact and ecological integrity is preserved.



Table 3-19: Floristic Composition in WTG3

Tower 3	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting			
	An open area of sandy substrate, partially covered with grass.	Vegetation dominated by thorny shrubs.	Mixed evergreen coastal forest interspersed with thorny shrub vegetation.
Floristic setting	<i>Cyperus rotundus</i> <i>Cyperus arenarius</i>	<i>Phoenix pusilla</i> <i>Tarennia asiatica</i> <i>Salvadora persica</i> <i>Dichrostachys cinerea</i> <i>Zanthoxylum asiaticum</i>	<i>Manilkara hexandra</i> <i>Syzygium cumini</i> <i>Salvadora persica</i> <i>Dichrostachys cinerea</i> <i>Acacia eburnea</i> <i>Bauhinia racemosa</i>
Ecological impact of the Project	The project does not require the removal of any trees or shrubs.	Shrub vegetation is planned to be cleared as part of the proposed activities.	The vegetation remains intact, ensuring that ecological integrity is preserved.

Table 3-20: Floristic Composition in WTG4

Tower 4	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	Open sandy patch.	Mixed thorny shrub.	Thorny shrub mixed dry evergreen forest.
			
Florist setting	<i>Cyperus rotundus</i> <i>Cyperus arenarius</i> <i>Trachys muricata</i>	<i>Pemphis acidula</i> <i>Premna obtusifolia</i> <i>Phoenix pusilla</i> <i>Opuntia dilleni</i> <i>Dodonaea viscosa</i>	<i>Manilkara hexandra</i> <i>Syzygium cumini</i> <i>Ficus beghalensis</i> <i>Dichrostachys cinerea</i>
Ecological impact of the Project	Open grassland with no woody vegetation, and therefore no tree or shrub clearance is required.	No trees will be cleared, and only shrub species will be removed. None of these shrubs are endemic or endangered; they are common species in the area.	The existing vegetation remains intact, ensuring the preservation of ecological integrity.



Table 3-21: Floristic Composition in WTG5

Tower 5	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	Dry, open landscapes with patches of woody vegetation dominated by low shrubs and stunted trees.	Dry, open landscapes with patches of woody vegetation dominated by low shrubs and stunted trees.	The rotating area in the west overlaps with the coast, while thorny shrubs and a mixed dry evergreen forest dominate the areas to the east.
			
Floristic setting	<i>Phoenix pusilla</i> <i>Dodonaea viscose</i> <i>Dichrostachys cinerea</i>	<i>Manilkara hexandra</i> <i>Pemphis acidula</i> <i>Premna obtusifolia</i> <i>Phoenix pusilla</i> <i>Dodonaea viscose</i>	<i>Manilkara hexandra</i> <i>Syzygium cumini</i> <i>Dichrostachys cinerea</i> <i>Catunaregam spinosa</i> <i>Toddalia asiatica</i>
Ecological impact of the Project	Thorny shrub species will be cleared; none are endemic or endangered, and all are commonly found in the area.	Thorny shrub and tree species will be cleared as required; none are endemic or endangered, and all are common in the area.	The existing vegetation will remain intact, thereby preserving the area's ecological integrity.



Table 3-22: Floristic Composition in WTG6

Tower 6	Base footprint	Service area (50 m radius)	Outer area (150 radius)
Habitat setting	The area is a seasonally inundated open landscape, characterized by salt marshes and grassy vegetation. 	Seasonally inundated open pasture landscapes with patches of woody vegetation, dominated by low shrub and stunted trees. 	The rotating area overlaps with the lagoon on the eastern side, where fringing mangroves line the lagoon edge. 
Florist setting	<i>Salicornia branchiate</i> <i>Cyperus arenarius</i> <i>Fimbristylis cymosa</i> <i>Eleocharis geniculata</i>	<i>Sesuvium portulacastrum</i> <i>Suaeda monoica</i> <i>Tecticornia indica</i> <i>Eleocharis geniculate</i> <i>Schoenoplectus pungens</i> <i>Phoenix pusilla</i>	<i>Avicennia marina</i> <i>Lumnitzer racemosa</i> <i>Excoecaria agallocha</i> <i>Rhizophora muronata</i>
Ecological impact of the Project	Thorny shrub species will be cleared; none are endemic or endangered, and all are common in the area.	Thorny shrub species will be cleared; none are endemic or endangered, and all are common species in the area.	The existing vegetation remains intact, ensuring the preservation of ecological integrity.



Table 3-23: Floristic Composition in WTG7

Tower 7	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	Open landscapes of sandy patch scattered with short shrubs. 	The vegetation consists of shrubs mixed with stunted trees and thorny shrub species. 	The rotating area overlaps with the sea on the western side, while the eastern side is covered with thorny shrubs mixed with stunted tree vegetation. 
Florist setting	<i>Phoenix pusilla</i> <i>Dodonaea viscosa</i>	<i>Phoenix pusilla</i> <i>Dodonaea viscosa</i> <i>Zanthoxylum asiaticum</i> <i>Dichrostachys cinerea</i>	<i>Dichrostachys cinerea</i> <i>Dodonaea viscosa</i> <i>Borassus flabellifer</i> <i>Spinifex littoreus</i>
Ecological impact of the Project	Shrub species, including a small number of <i>Phoenix pusilla</i> , will be cleared. None of these species are endemic or endangered; all are common in the area.	Shrub species, including thorny shrubs and a small number of <i>Phoenix pusilla</i> , will be cleared. None of these species are endemic or endangered; they are all common in the area.	The existing vegetation remains intact, ensuring the preservation of ecological integrity.



Table 3-24: Floristic Composition in WTG8

Tower 8	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open sandy area within a poorly managed, newly cultivated coconut plantation.	A relatively open, newly cultivated coconut plantation, where the vegetation consists of shrubs and pastures interspersed with coconut plants.	The rotating area overlaps with the lagoon on the southeast side, while the eastern side is covered with palmyra woodlands and thorny shrub vegetation scattered with trees from the west.
			
Florist setting	<i>Dodonaea viscosa</i> <i>Tephrosia purpurea</i>	<i>Dodonaea viscosa</i> <i>Zanthoxylum asiaticum</i> <i>Dichrostachys cinerea</i> <i>Imperata cylindrica</i>	<i>Dichrostachys cinerea</i> <i>Dodonaea viscosa</i> <i>Borassus flabellifer</i> <i>Phoenix pusilla</i> <i>Canthium coromandelicum</i> <i>Ziziphus mauritiana</i>
Ecological impact of the Project	Shrub species will be cleared; none are endemic or endangered, and all are common in the area.	Shrub species and a small number of coconut plants will be cleared. None of these species are endemic or endangered; all are common in the area.	The existing vegetation will remain undisturbed, thereby ensuring the preservation of ecological integrity.



Table 3-25: Floristic Composition in WTG9

Tower 9	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open area located within a thorny shrub forest.	Thorny shrub vegetation mixed with stunted trees.	The rotating area overlaps with the sea on the western side, while the eastern side is covered with thorny shrub vegetation interspersed with scattered trees.
			
Florist setting	<i>Phoenix pusilla</i> <i>Dichrostachys cinerea</i>	<i>Phoenix pusilla</i> <i>Dichrostachys cinerea</i> <i>Acacia eburnean</i> <i>Azima tetracantha</i>	<i>Dichrostachys cinerea</i> <i>Phoenix pusilla</i> <i>Zanthoxylum asiaticum</i> <i>Opuntia dillenii</i> <i>Calotropis gigantea</i>



Table 3-26: Floristic Composition in WTG10

Tower 10	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open sandy area located within a coastal shrub forest.	The area features relatively open patches of shrub forest, with ground vegetation nearly absent.	A coastal shrub forest, interspersed with canopy trees, covers nearly the entire area.
			
Florist setting	<i>Tephrosia purpurea</i>	<i>Dichrostachys cinerea</i> <i>Zanthoxylum asiaticum</i> <i>Salvadora persica</i>	<i>Dichrostachys cinerea</i> <i>Phoenix pusilla</i> <i>Manilkara hexandra</i> <i>Canthium coromandelicum</i> <i>Ziziphus mauritiana</i>
Ecological impact of the Project	Shrub species will be cleared; none are endemic or endangered, and all are common to the area.	Shrub and scattered trees will be cleared. None of these species are endemic or endangered; all are common in the area.	The existing vegetation will remain undisturbed, thereby ensuring the preservation of ecological integrity.



Table 3-27: Floristic Composition in WTG11

Tower 11	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open sandy area situated within a coastal shrub forest.	The area features relatively open patches of shrub forest, with sparse ground vegetation.	A coastal shrub forest, with interspersed canopy trees, covers almost the entire area.
			
Florist setting	<i>Cyperus rotundus</i> <i>Blumea oblique</i> <i>Launaea sarmentosa</i>	<i>Dichrostachys cinerea</i> <i>Zanthoxylum asiaticum</i> <i>Salvadora persica</i> <i>Acacia eburnean</i>	<i>Dichrostachys cinerea</i> <i>Phoenix pusilla</i> <i>Canthium coromandelicum</i> <i>Capparis sepiaria</i> <i>Borassus flabellifer</i>
Ecological impact of the Project	Sparsely spread ground vegetation will be cleared; none of these species are endemic or endangered, and all are common in the area.	Shrubs and scattered trees will be cleared. None of these species are endemic or endangered; all are common in the area.	The existing vegetation will remain undisturbed, thereby ensuring the preservation of ecological integrity.



Table 3-28: Floristic Composition in WTG12

Tower 12	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open sandy area situated within a scattered coastal shrub landscape.	The area features relatively open patches of shrubbery, with pasture-like ground vegetation.	Open coastal shrub vegetation, scattered throughout, dominates the entire area.
			
Florist setting	<i>Cyperus rotandus</i> <i>Cyperus arenarius</i> <i>Imperata cylindrica</i> <i>Launaea sarmentosa</i>	<i>Phoenix pusilla</i> <i>Cyperus rotandus</i> <i>Imperata cylindrica</i> <i>Dodonaea viscosa</i> <i>Phyllanthus reticulatus</i>	<i>Phoenix pusilla</i> <i>Dodonaea viscosa</i> <i>Imperata cylindrica</i> <i>Morinda coreia</i> <i>Phyllanthus reticulatus</i>
Ecological impact of the Project	Sparsely spread ground vegetation will be cleared; none of these species are endemic or endangered, and all are common in the area.	Shrubs and scattered trees will be cleared. None of these species are endemic or endangered; all are common in the area.	The existing vegetation will remain undisturbed, thereby ensuring the preservation of ecological integrity.



Table 3-29: Floristic Composition in WTG13

Tower 13	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open area situated within a sparsely vegetated coastal shrub landscape.	The area features relatively open patches of shrubbery mixed with short trees, and ground vegetation resembling pasture.	Open, scattered coastal shrub vegetation dominates the entire area.
			
Florist setting	<i>Cyperus rotandus</i> <i>Cyperus arenarius</i> <i>Launaea sarmentosa</i> <i>Tephrosin purpurea</i> <i>Indigofera oblongifolia</i>	<i>Phoenix pusilla</i> <i>Cyperus rotandus</i> <i>Imperata cylindrica</i> <i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i>	<i>Phoenix pusilla</i> <i>Imperata cylindrical</i> <i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i> <i>Calotropis gigantea</i>
Ecological impact of the Project	Sparsely spread ground vegetation will be cleared; none of these species are endemic or endangered, and all are common in the area.	Shrubs and scattered trees will be cleared. None of these species are endemic or endangered; all are common in the area.	The existing vegetation will remain undisturbed, thereby ensuring the preservation of ecological integrity.



Table 3-30: Floristic Composition in TW14

Tower 14	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open area located within a coconut-cultivated land, with ground vegetation mainly consisting of grasses and scattered low shrubs.	A coconut-cultivated land with ground vegetation mainly consisting of grasses and scattered low shrubs.	The rotational area overlaps with the lagoon and the fringing mangroves along the eastern lagoon coast.
			
Florist setting	<i>Cyperus rotundus</i> <i>Tephrosin purpurea</i> <i>Indigofera oblongifolia</i>	<i>Cocos nucifera</i> <i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i> <i>Calotropis gigantea</i>	<i>Cocos nucifera</i> <i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i> <i>Avicennia marina</i> <i>Rhizophora mucronata</i>
Ecological impact of the Project	Since there are no coconut trees within the area, only ground vegetation will be cleared; none of the species are endemic or endangered, and all are common in the area.	As the ground vegetation is minimally disruptive, only a few coconut trees will be felled.	The existing vegetation, including coconut trees and fringing mangroves, will remain undisturbed, thereby ensuring the preservation of ecological integrity.



Table 3-31: Floristic Composition in TW15

Tower 15	Base footprint	Service area (50 m radius)	Outer area (200 radius)
Habitat setting	An open area located within an old coconut-cultivated land, with ground vegetation mainly consisting of grasses and scattered low shrubs.	An old coconut-cultivated land with ground vegetation primarily consisting of grasses and scattered low shrubs and trees.	The rotational area overlaps with the sea on the western side and borders the vegetation along the beach zone.
			
Florist setting	<i>Cyperus rotandus</i> <i>Tephrosin purpurea</i> <i>Indigofera oblongifolia</i>	<i>Cocos nucifera</i> <i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i> <i>Phoenix pusilla</i> <i>Dodonaea viscosa</i>	<i>Cocos nucifera</i> <i>Indigofera oblongifolia</i> <i>Tephrosin purpurea</i> <i>Spinifex littoreus</i>
Ecological impact of the Project	Since there are no coconut trees within the area, only ground vegetation will be cleared; none of the species are endemic or endangered, and all are common in the area.	Shrubs and trees, mainly <i>Phoenix pusilla</i> and a few old coconut trees, will be felled.	The existing vegetation, including coconut trees and shore-based species, will remain undisturbed, thereby ensuring the preservation of ecological integrity.

3.3.9 Avifauna Movements

3.3.9.1 Birds inhabiting area within the proposed location

A total of 79 bird species were recorded during the survey period from April 22nd to 25th (Annexure III L), representing 38 families. The highest species richness was observed in the families *Ardeidae* and *Laridae* (9 species each), followed by *Scolopacidae* (8 species). This total accounts for approximately 15.5% of the country's known avifauna (508 species), including residents, migrants, and vagrants.

Out of the 79 species recorded, 56 are considered breeding residents on the island, although their breeding status within the project area remains unknown. The remaining



23 species are migrants. Previous studies conducted in the Kalpitiya Peninsula reported a total of 132 species, including 32 migrant species and 4 considered vagrants. Ten species were recorded as having both resident and migrant populations on the island (Kaluthota and Kotagama, 2009).

Due to limitations related to weather conditions and the short duration of the survey, it was not possible to distinguish separate populations or confirm the breeding status of resident species in the area. Further surveys during the breeding season are necessary to obtain comprehensive data on breeding populations.

3.3.9.2 Abundance of bird species

An index was developed to estimate species abundance based on the frequency of observations during this late stage of the migratory period. Species observed in more than 65% of the survey instances were categorized as **very common**. Accordingly, common species (45-65%), uncommon species (25-45 %), rare species (15 – 25%) and very rare species (0-15%) were identified. It should be noted that this classification (**Table 3-32**) reflects local abundance in the study area and does not necessarily represent the national abundance status of the species. Based on this index, only forty four were classified as either very common or common in the study area.

Table 3-32: Commonness of the species found in the project area

Commonness	Number of Species
Very Common	15
Common	29
Uncommon	03
Rare	18
Very Rare	14

Figure 3.51 illustrates the abundance of bird species based on their frequency of observation at each turbine location. The data indicate that Turbine 1 (T₁) supports a significantly higher abundance of birds compared to the other turbine plots. This suggests that T₁ may be located in or near a key foraging or resting habitat, making it particularly important from an avifaunal perspective.



3.3.9.3 Bird Group

The list of species identified through the surveys includes 10 species classified as waterbirds (seabirds and waterfowl), 3 species of raptors, 24 species of waders, and 42 species belonging to passerines and other bird families. These avian groups exhibit distinct behaviors and habitat preferences, which should be carefully considered when evaluating the potential impacts of the proposed wind energy development.

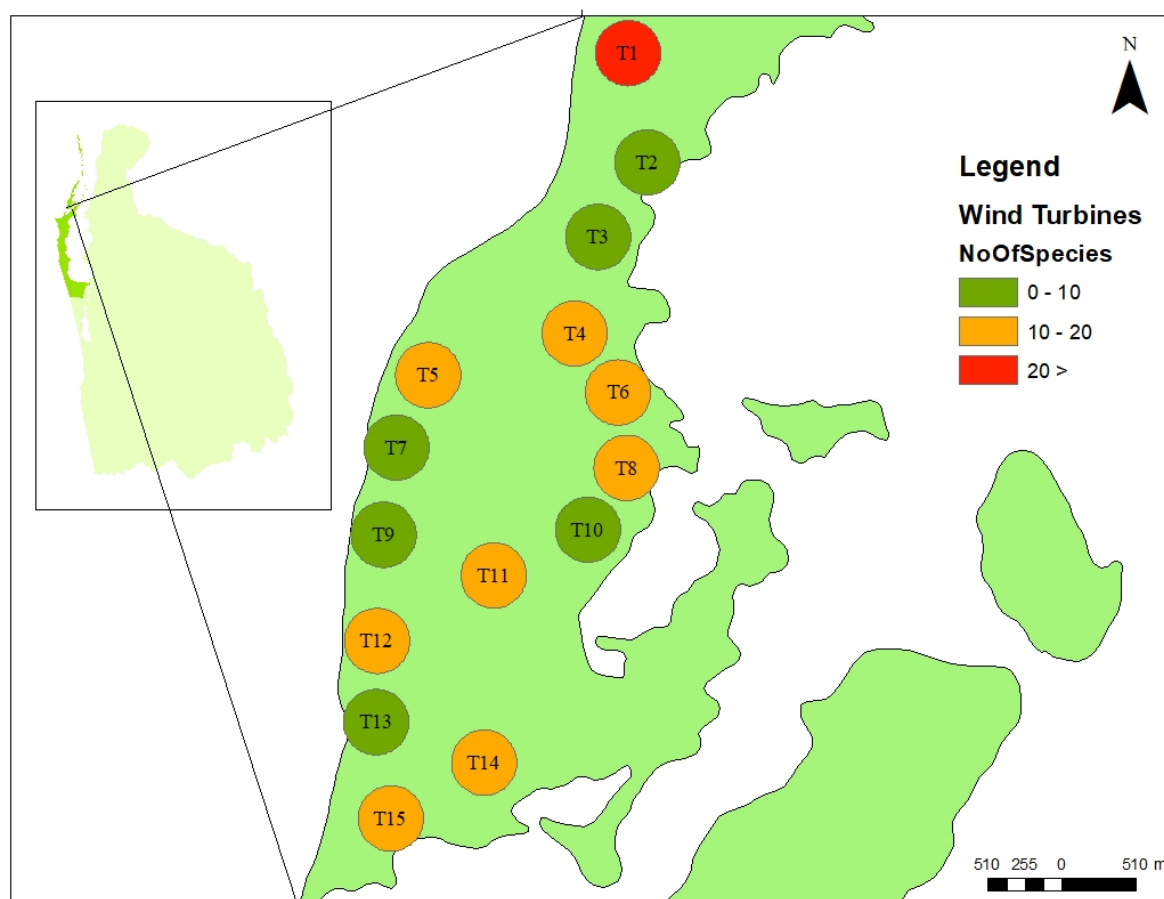


Figure 3.51: The abundance of bird species based on their frequency of observation at each turbine Location

3.3.9.4 Water birds (Including seabirds, waterfowls and waders)

Seabirds spend much of their lives in marine environments, typically venturing no farther inland than the shoreline. However, during the survey period, almost all seabird species observed—primarily terns and gulls—exhibited high mobility between the sea and the lagoon. This group was identified as the highest risk group in relation to the proposed wind energy development. Given that the Puttalam Lagoon provides critical habitat for aggregating terns and gulls, it is essential to identify and map their flight paths between the sea and the lagoon.



Waders are another group of aggregating birds that exhibit high mobility between suitable habitats. Shallow coastal water bodies serve as ideal feeding grounds for migratory waders. Given that the Kalpitiya Peninsula is rich in such coastal habitats—including the Puttalam Lagoon Important Bird Area (IBA)—a high level of wader movement is expected in the region. All wader species identified during the survey, except for lapwings and thick-knees, demonstrated this high habitat mobility.

However, during the data collection period, fewer than 60 individual waders were recorded within the proposed project area. Similarly, previous studies conducted during the migratory season also reported no large aggregations of waders (i.e., over 100 individuals) along the coast (Kaluthota and Kotagama, 2009).

3.3.9.5 Raptors

Raptors are high potential group of birds to have collision risk with the wind turbine and therefore extensively studied all over the world. Previous studies in Puttlam peninsula recorded 16 species of raptors. There were 4 species recorded in this survey and only the Brahminy Kite was very common while others are rare. However, all these species move through all the vertical layers, especially the space 30m above ground. The habit raptors have of perching on tall structures make them especially vulnerable to Collision with wind turbine blades.

3.3.9.6 Other birds

Most of the passerine and other bird species recorded in the proposed project area are resident breeders, with only a few migratory species observed during the survey. Notably, Ashy-headed Swallows (*Hirundo dumicola*) and Asian Palm Swifts (*Cypsiurus balasiensis*) were among the migratory or seasonally mobile species identified. These species are of particular concern due to their high abundance and consistent use of the upper vertical strata often flying at heights that intersect with the rotor-swept zone of wind turbines. As such, they are considered to be at elevated risk of collision.

Among the resident species, the Common Myna (*Acridotheres tristis*) and House Crow (*Corvus splendens*) were found to be very common and widespread across the area. Indian Rollers (*Coracias benghalensis*) and Rock Pigeons (*Columba livia*) were also frequently recorded and are known for their tendency to fly and forage at higher altitudes. These behavioral patterns increase



their potential exposure to turbine blades, making them relevant species to monitor in assessing the impact of the proposed wind energy development.

3.3.9.7 Species of Special Conservation Concern

There are 6 species recorded as globally near threatened species. Three are endangered, 05 species are Critically endangered while one species namely Little Tern (*Sternula albifrons*) considered as Vulnerable species (**Figure 3.52**). Kentish Plover and Grey Francolin reported as Endangered while Ashy Wood swallow and Great Cormorant considered as near threatened at the national level. Among these species the Kentish Plover was the only species commonly recorded during the study period (**Annexure III L**). No restricted range species (Endemic) recorded in the proposed area.



Figure 3.52: Threatened avifaunal species recorded within the proposed area

3.3.9.8 Species Richness and abundance in different turbine points

Heavy rains and cloudy conditions were experienced during the survey period; therefore, the recorded number of species and individual birds may not fully represent the actual avifaunal diversity in the area. **Figure 3.53** illustrates species richness and abundance at each turbine plot (T). According to the figure, the most important location is Turbine 1 (T₁), while Turbines 2 (T₂), 3 (T₃), and 10 (T₁₀) appear to be the least significant in terms of bird activity. Near T₁, a wading area was observed, along with the presence of gulls and terns along the coast (**Figure 3.53**).

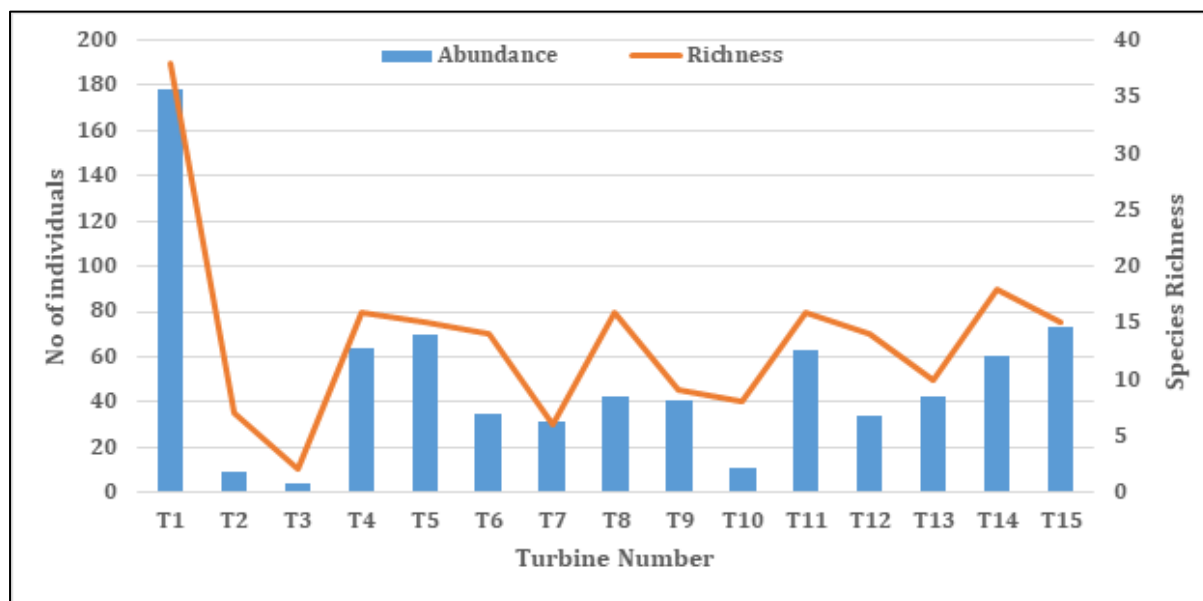


Figure 3.53: Species richness and abundance in different turbine points

3.3.9.9 Seasonal changes of the diversity and abundance

There are no available data specifically for the Kalpitiya area, making it difficult to determine seasonal changes in avifaunal composition. However, based on literature and eBird data from the Puttalam Peninsula, species richness fluctuates throughout the year. The number of bird species increases significantly during September and October, then continues to fluctuate until February (Figure 3.54). After February, species numbers decline. This increase is driven by two main factors: first, the arrival of migratory species in September and October; second, the influx of resident species due to favorable conditions, particularly the reduction in wind speed in the area.

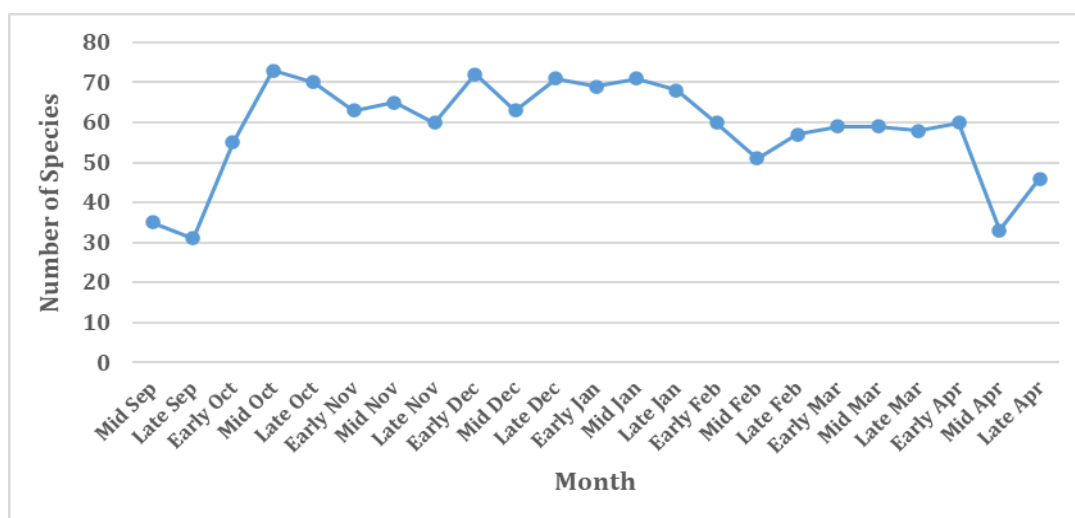


Figure 3.54: Changes in avian diversity during the migratory period (Kaluthota and Kotagama 2009)



Figure 3.54 and **Figure 3.55** presents the migratory bird species recorded in the Kalpitiya area, based on available eBird data. However, due to the limited quantity and temporal coverage of these records, it is difficult to identify or illustrate clear seasonal patterns in species occurrence. No comprehensive, peer-reviewed studies have been published specifically on the avifauna of Kalpitiya, apart from the data generated through our current research efforts. This lack of scientific documentation highlights a critical knowledge gap regarding bird migration patterns, habitat use, and seasonal dynamics in the region.

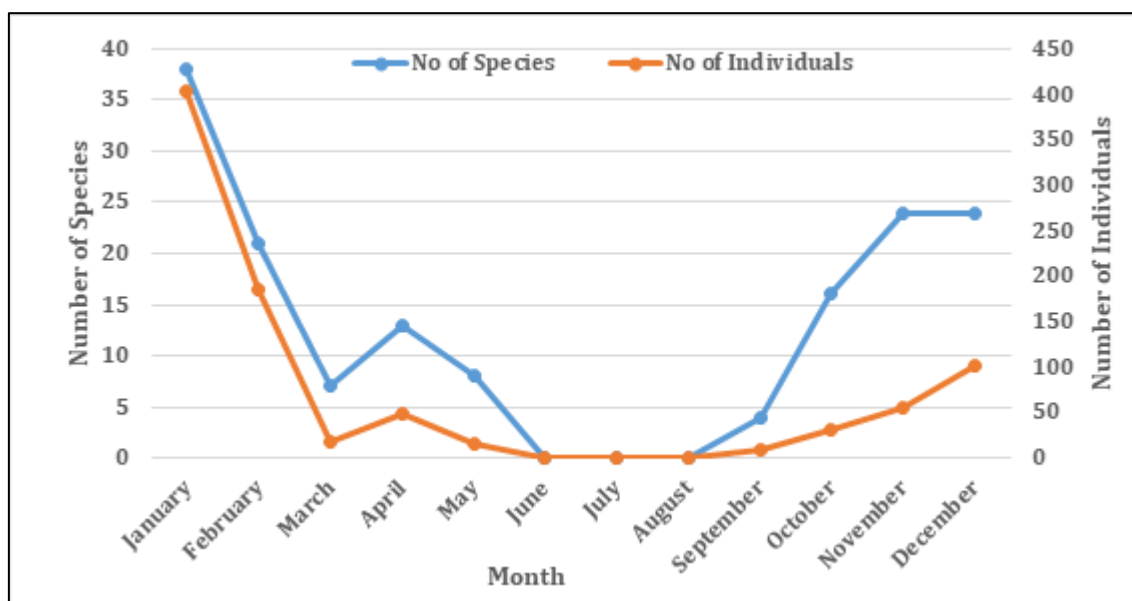


Figure 3.55: Monthly variation of migratory birds in Kalpitiya area (eBird data 2024)

3.3.9.10 Diurnal change of the bird composition

Diurnal change of bird composition was observed during the study period (Figure 3.56). Based on the results from point counts the number of birds, and species were higher in the mornings. A considerable drop of both the average number of species and number of individuals were observed during mid-day and the evenings. Hence, it can be concluded that risk of birds colliding with the wind towers is higher in the mornings. The same patterns observed in previous records as well.

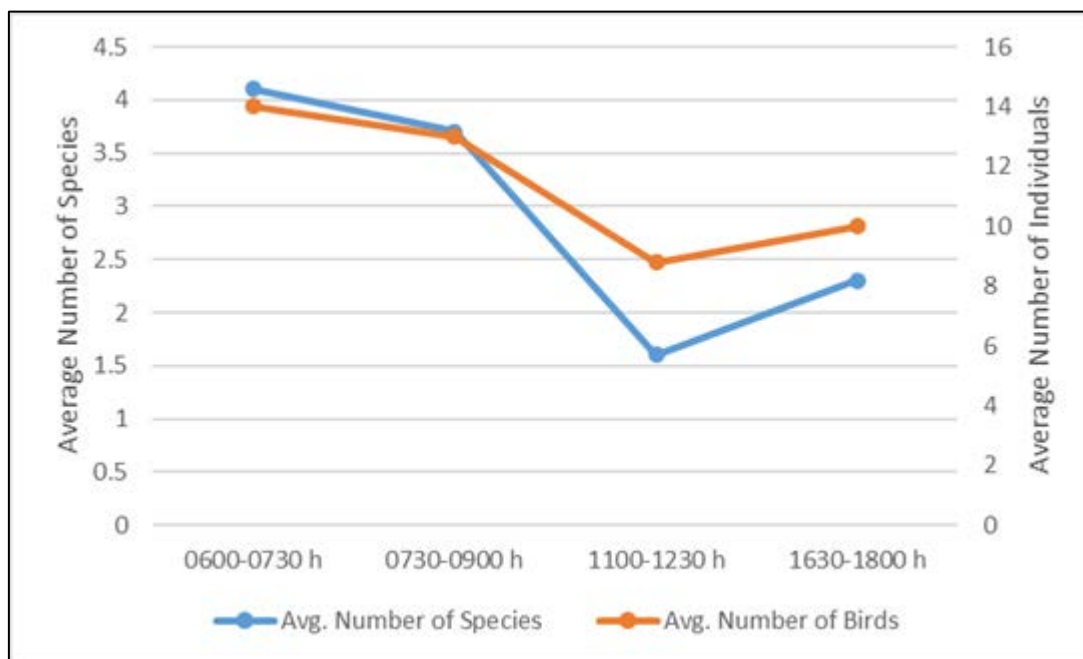


Figure 3.56: Diurnal variation of species composition

3.3.9.11 Vertical stratification of bird movements

Based on observations of 761 bird movements recorded during April 2025, the majority (78.84%) occurred below the rotor-swept zone (Level 1), posing minimal risk of collision (**Figure 3.57**). Only a small proportion (3.81%) of bird movements were observed above the height of the rotors, further reducing collision concern at higher altitudes. However, 17.35% of bird movements occurred within the critical vertical stratum—directly within the rotor-swept zone—during the migratory period. This represents the maximum potential risk of bird collisions with turbine blades. It is important to note that actual collision risk depends on several additional factors, including the angle of bird flight in relation to rotor orientation, wind speed, wind direction, and bird avoidance behavior. Comparable patterns of vertical bird distribution were also recorded during a similar survey conducted in 2009 in Puttalam Peninsula, suggesting consistency in movement patterns across years.

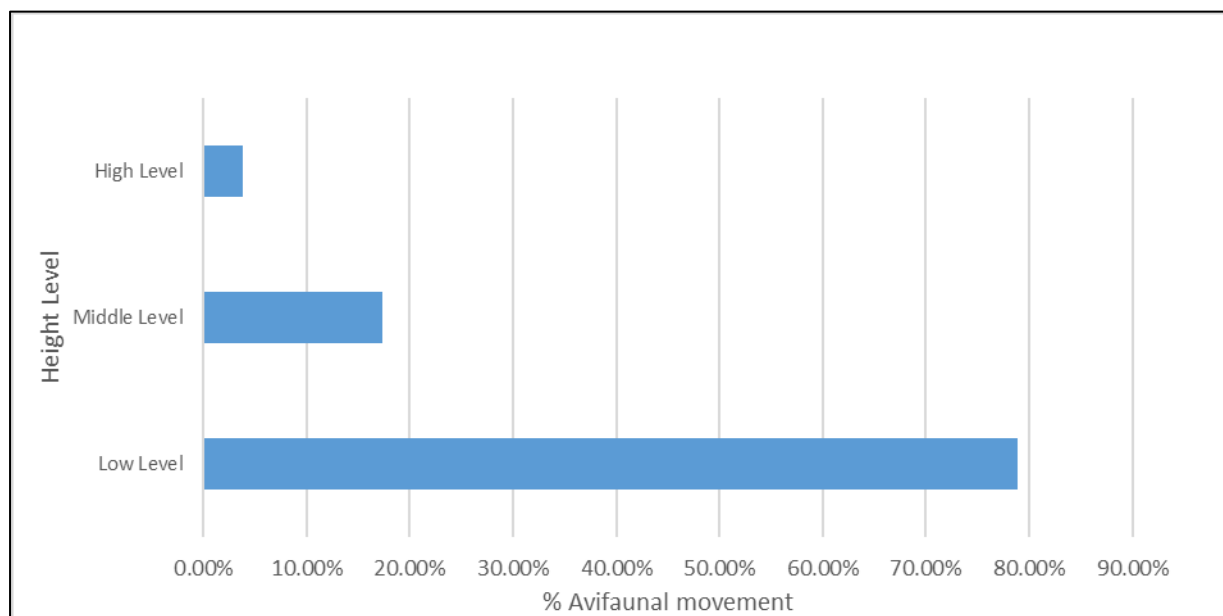
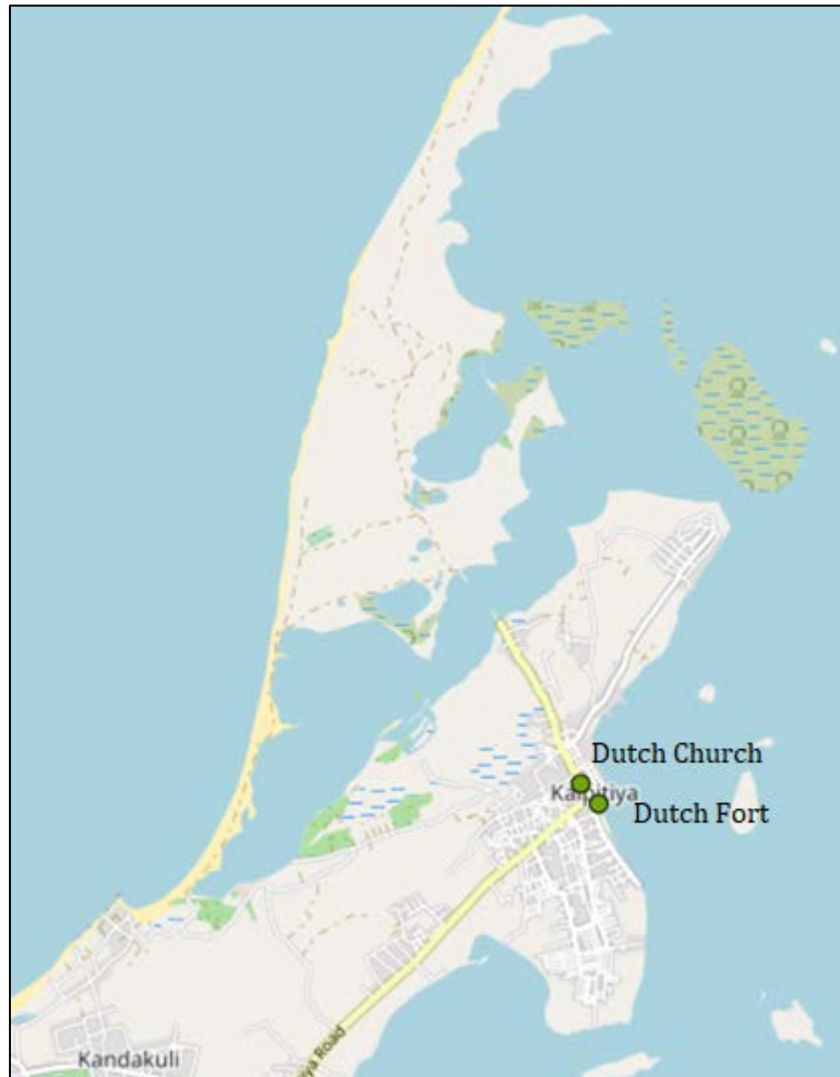


Figure 3.57: Vertical stratification of bird movements as a percentage

During the survey period, movements of certain species particularly terns and gulls from the sea toward the lagoon were observed. These movements indicate possible foraging or roosting behavior across the coastal-lagoon interface. However, due to the short duration of the study and unfavorable weather conditions, detailed identification of flight paths and quantification of movement density were not carried out. As a result, further long-term observations under stable weather conditions are recommended to accurately assess flight patterns and potential risks, especially in relation to turbine placement.

3.4 Historical and Archaeological Importance

Dutch Bay GN division where the project is located does not fall within any of the Archaeology zone or reserves declared under any of the government gazettes according to the Archaeology Clearance certificate decision support system of Archaeology Department's official website. Nearest protected monuments, Dutch fort and Dutch church are positioned beyond 2 miles away from the project site in Kalpitiya (**Figure 3.58**).



Source: <https://archaeologysl.maps.arcgis.co>

Figure 3.58: Nearest protected monuments to project location

Mohonthuwarema St Antony's Church and Muhathuvaram Holy Cross Church in Keerimundalam village where annual feast is celebrated in the second week of September every year are historical churches in the area with cultural and religious significance, while, Sri Veal Murugan Hindu Temple and Muhathuvaram Jumma Masjid mosque are other religious places in Dutch Bay Island (Figure 3.59).



Figure 3.59: Religious places in Dutch Bay Island



3.5 Socio-economic Environment

3.5.1 Delineation of the Social Area of Influence

According to the Terms of Reference (ToR) for the Environmental Impact Assessment (EIA), issued by the CC&CRMD, no specific Social Area of Influence (AoI) has been defined. However, considering the project footprint, proposed activities, and associated construction and operational procedures, the Social AoI is delineated to include the Dutch Bay Grama Niladhari Division (GND) and the immediate coastal environment.

While the project is located within the Dutch Bay GND, the adjacent Palliyawatta GND to the north and Kudawa-Kandakuliya GND to the south have limited land-based connectivity due to the lack of proper road infrastructure. As a result, social and economic linkages between these communities are minimal, and no significant impact transmission channels are evident. Therefore, in the assessment of the socio-cultural impacts of the proposed wind power project, these GNDs were only partially included within the scope of analysis.

3.5.2 Present Land use of the study area and zoning

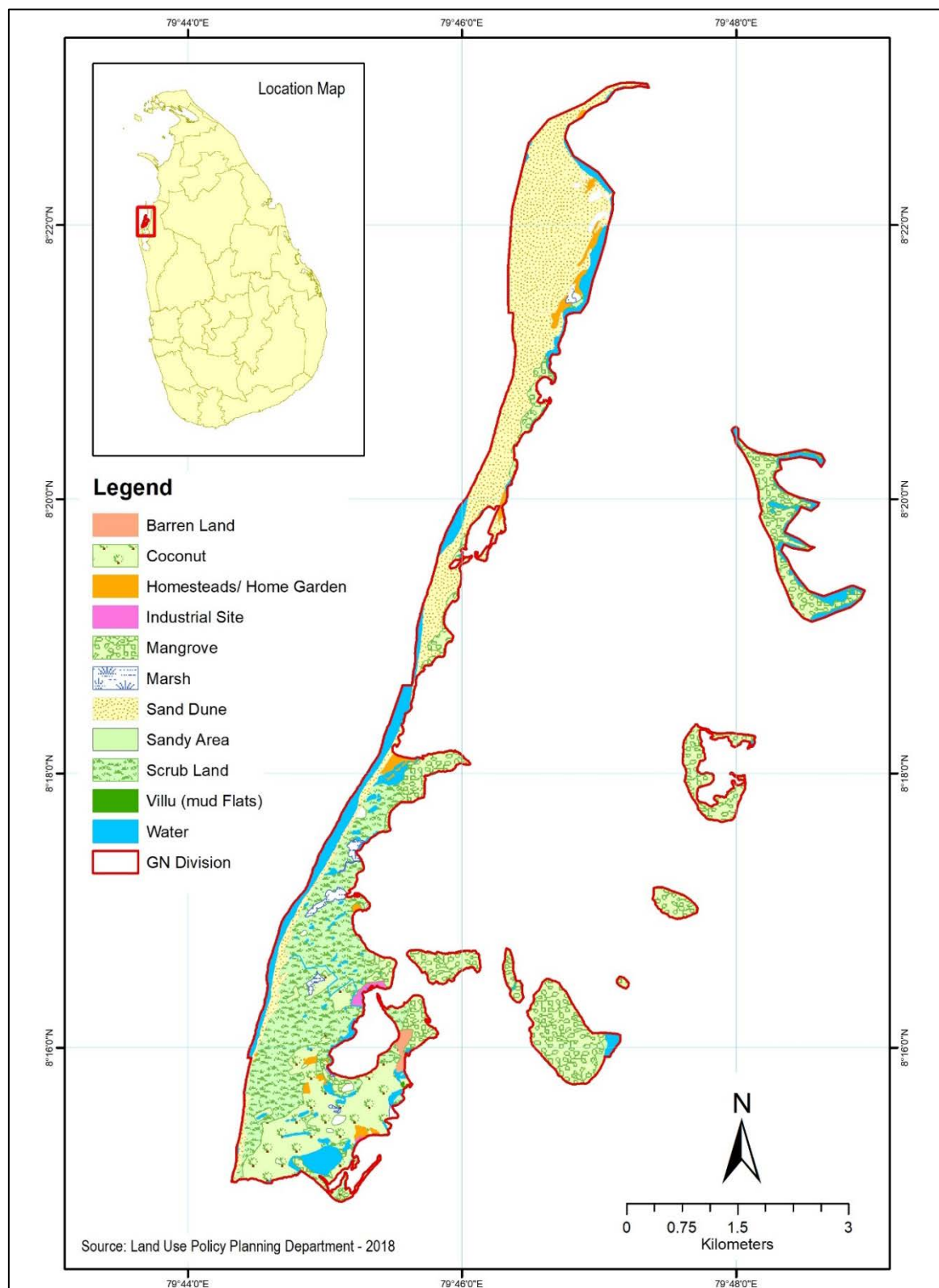
The land use pattern within the social Area of Influence (AoI) of the project exhibits a diverse range of coastal ecosystems. The dominant land cover is sand dunes, occupying approximately 439.6 hectares, primarily located along the coastal belt. This is followed by mangrove forests covering 349.7 hectares, scrubland (334.1 ha), open sea (175.6 ha), and coconut plantations (174.3 ha). Home gardens and marshlands account for 31.7 hectares and 16.0 hectares, respectively. The distribution of these land use types within the Dutch Bay Grama Niladhari Division (GND) is depicted in Figure 3.60 and detailed in Table 3-33.

According to the Kalpitiya Development Plan 2019–2030, prepared by the Urban Development Authority (2019), the Dutch Bay area has been designated as a “Beach-based tourism development zone” (p. 284), highlighting its potential for tourism-driven economic growth. In addition to tourism, the plan also emphasizes the importance of renewable energy development, particularly wind power. It states: “The wind power generation too needs to be given a significant place in the development process as it can play an important role in power generation in the national context. Under these circumstances, this development plan needs to take into consideration the physical setting of the peninsula and provide guidance for the development process towards the envisioned future” (Kalpitiya Development Plan 2021–2030, Urban Development Authority, p. 80).



This statement highlights the dual focus of the development strategy: promoting tourism while also leveraging the region's natural resources for sustainable energy production. Furthermore, the Renewable Energy Resource Development Plan 2021–2026 has specifically identified areas within the Kalpitiya Peninsula, including the proposed project site, as suitable for wind power development, reinforcing the strategic importance of integrating renewable energy initiatives into the broader development framework of the region.

Additionally, the Sri Lanka National Coastal Zone and Coastal Resource Management Plan 2024 outlines specific regulatory measures for coastal development. It stipulates a mandatory 60-meter structure-free coastal setback along the stretch from Uchchamunai to Mohonthuwarema (northern boundary of Mohonthuwarema Spit). Moreover, the zone extending from the Mohonthuwarema Spit northern boundary to the Mohonthuwarema/Kandakuliya Spit southern boundary is designated as a conservation area and strictly regulated as a no-build zone.



Source: Indika, W.A.K., Matharaarachchi, H., de Silva, D.Y.Y. (2022). Kalpitiya Socioeconomic and Ecological Profile 2022. Environmental Foundation (Guarantee) Ltd

Figure 3.60: Land Use Pattern of the Dutch Bay GND



Table 3-33: Land Use Pattern of Dutch Bay GND

Land Use	Area (ha)	Percentage (%)
Sand Dunes	439.6	27.9
Mangroves	349.7	22.2
Scrub Lands	334.1	21.2
Sea	175.6	11.2
Coconut	174.3	11.1
Home Garden	31.7	2.0
Marsh	16.0	1.0
Odaya	15.6	1.0
Lagoon	9.5	0.6
Water Hole	8.6	0.5
Barren Land	6.9	0.4
Sandy Areas	5.6	0.4
Industrial Site	5.6	0.4
Canals	1.0	0.1
Villu	0.3	0.0
Total	1574.1	100

Source: Indika, W.A.K., Matharaarachchi, H., de Silva, D.Y.Y. (2022). *Kalpitiya Socioeconomic and Ecological Profile 2022*. Environmental Foundation (Guarantee) Ltd

3.5.3 Population distribution and nature of households in the area

This section outlines the socio-economic characteristics of the project's Area of Influence (AOI), which is primarily located within the Dutch Bay Grama Niladhari Division (GND), with relevant information also included from the neighboring GNDs of Kandakuliya Kudawa and Palliyawatta. It covers key aspects such as the socio-demographic profile of the population, livelihood activities, housing and land ownership patterns, household amenities and assets, energy usage, public infrastructure, and access to health and educational services. Additionally, prevailing socio-economic and environmental challenges, such as poverty and gender-related issues, are also discussed.

To understand the socio-economic context of the Project Area of Influence, primary data was collected through Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), and a sample social survey conducted in the Dutch Bay GND, covering the villages of Mohottuwarama, Penapidugama, and Kirimundal. In addition, secondary information was collected from the District Office of the Department of Fisheries and Aquatic Resources Development in Puttalam, as well as through a review of relevant literature.



3.5.3.1 Demographic Profile

The 2023 Resource Profile of the Kalpitiya Divisional Secretariat indicates that the Kalpitiya DSD consists of 31 Grama Niladhari Divisions (GNDs), comprising 28,503 households and a total population of 108,066.

The Dutch Bay GND (633), where the project is to be located, is diverse in socio-economic and environmental characteristics. The GNDs of Palliyawatta (629C) and Kudawa-Kandakuliya (633A) were located north and the south of the Dutch Bay GND. The land extent of the GND is approximately 15 sq.km and There are 345 households in the Dutch Bay GND with a total population of 1683. The female population is slightly higher than the male population. The population density within the GND is 112 per sq.km. The average household size in the GND has been reported as 4. The dominant ethnic group is Tamil at 63%, and 80% of the population follows Catholicism. The gender, ethnic, and religious-wise population distribution in the Dutch Bay GND is depicted in **Table 3-34**. The population distribution in the adjacent GNDs of Palliyawatta and Kudawa-Kandakuliya is also depicted in Table 3-35 and **Table 3-36**. It is important to highlight the notable discrepancy between the official population figures recorded for the Dutch Bay Grama Niladhari Division (GND) and the actual number of residents physically present in the area. The current resident population is significantly lower than the registered population. This discrepancy has arisen primarily because many individuals, though officially registered under the Dutch Bay GND, reside in the Kalpitiya mainland, where better access to social infrastructure is available. This trend is particularly evident in Kirimundal village, where the gap between registered and resident populations is most noticeable.

Table 3-34: Gender, Ethnic, and Religious-wise population Distribution – Dutch Bay GND 633 (Land Area Sq.km. 15)

Gender-wise Population Distribution			
Households	Total Population	Male	Female
345	1683	707 (58)	976(42)
Ethnic -wise Population Distribution			
Sinhala	Tamil	Moor	Other
396 (23%)	1055(63%)	232(14%)	0
Religious-wise Population Distribution*			
Buddhist	Hindu	Catholic	Islamic
5	29	1106	285
Population density			
Sq. km 1 = 112			

Source: Divisional Resource Profile - Kalpitiya 2023



Table 3-35: Palliyawatta GND 633 A (Land Area 6.1 Sq.km)

Gender-wise Population Distribution			
Households	Total Population	Male	Female
27	88	41(47%)	47 (53%)
Ethnic -wise Population Distribution			
Sinhala	Tamil	Moor	Other
62	26	00	
Religious-wise Population Distribution			
Buddhist	Hindu	Catholic	Islamic
3	3	82	0
Population density			
Sq. km. 1= 77			

Source: Divisional Resource Profile - Kalpitiya 2023

Table 3-36: Kandakuliya-Kudawa GND 629 C (Land Area sq. km 1.8)

Gender-wise Population Distribution			
Households	Total Population	Male	Female
165	627	295	332
Ethnic -wise Population Distribution			
Sinhala	Tamil	Moor	Other
627			
Religious-wise Population Distribution			
Buddhist	Hindu	Catholic	Islamic
Population density			
Sq.km. 1= 313			

Source: Divisional Resource Profile - Kalpitiya 2023

3.5.4 Socio - economic status of the study area and Principle economic Activities

According to the Divisional Resource Profile of 2021, the total population within the age range of 15 years and above in the Dutch Bay Grama Niladhari Division (GND) is approximately 1,221 individuals. Among them, 267 individuals, or 21.2%, have been classified as economically inactive, indicating they are neither employed nor actively seeking employment. The economically active population, or the labour force, is estimated at 954 individuals, representing around 78% of the working-age population. Of this labour force, 317 individuals, equivalent to 33.2%, have been identified as unemployed. This reflects a notably high unemployment rate



within the division. Comparatively, the unemployment rate in Dutch Bay GND is higher than that of the neighbouring GNDs, with Palliyawatta reporting a rate of 20% and Kudawa-Kandakuliya at 22%. This suggests a relatively greater economic vulnerability or lack of employment opportunities in Dutch Bay. Further details regarding the current composition of the workforce are provided in following **Table 3-37**, while the age-wise distribution of unemployment in the GND is presented in **Table 3-38**

Table 3-37: Workforce Distribution in the Dutch Bay GND 2021

Total Population (Age between 15-60+)	Economically inactive population	Labor Force	Employment	Unemployment
1221	267 (21.8%)	954((78.13%)	637 (66.7%)	317 (33.2%)

(Source: Divisional Resource Profile - Kalpitiya 2021)

Table 3-38: Unemployment According to the Age Group

Age Group (Years)	Unemployment Population		
	Female	Male	Total
15-18	53	34	87
19-30	38	19	57
31-40	23	17	40
41-60	37	19	56
60 +	37	40	77
Total	188	129	317

(Source: Divisional Resource Profile - Kalpitiya 2021)

Household Income

The primary source of income for most households within the Social Area of Influence (AoI) is fishing. A sample social survey indicates that 94% of households in Moththuwarema and Kirimundal villages rely on fishing as their principal livelihood. Additionally, 85% of these households receive government social safety net through the *Aswesuma* programme. This trend is further corroborated by the Divisional Resource Profile (2021), which reports that in the Dutch Bay Grama Niladhari Division (GND), 229 families (60%) earn an average monthly income of less than LKR 16,000, below the national poverty line for March 2024, which is LKR 16,619. These figures highlight the significant income disparities and the prevalence of poverty in the Dutch Bay area. **Table 3-39** presents the average monthly income of households in Dutch Bay GND.



Table 3-39: Household Monthly Income Distribution

Household Monthly Income (LKR)	Number of Households	Percentage (%)
0 - 3,999	65	18%
4,000 - 7,999	63	17%
8,000 - 11,999	43	12%
12,000 - 15,999	49	13%
16,000 - 19,999	44	12%
20,000 - 23,999	47	13%
24,000 - 27,999	23	6%
28,000 - 31,999	17	5%
Above 32,000	14	4%
Total	365	100%

3.5.4.1 Livelihood of settlers (dependency of type of agriculture and other livelihoods)

There are no industries or commercial-level agriculture beside small and medium-level coconut plantations. According to the findings of the social survey conducted within the project's Area of Influence (AOI), it was revealed that a significant majority, 94.2% of households in the villages of Mohonthuwarama and Kirimundal are primarily engaged in fishing or fisheries-related activities as their main source of livelihood. This underscores the critical dependence of local communities on the fisheries sector for economic sustenance. In addition to fisheries, a smaller proportion of residents are involved in alternative income-generating activities, including small-scale retail businesses, employment overseas (foreign employment), and casual labour in both agricultural and non-agricultural sectors. The detailed employment distribution by type of economic activity is presented in Table 3-40, providing a clearer understanding of the local livelihood dynamics.



Table 3-40: Employment Distribution by type of the Dutch Bay GND

Type of Employment	Number
Executive management	1
Foreign employment	7
Business	22
Self-employment	72
Labourers	63
Plantation owners	105
Others	272

Source: Divisional Resource Profile - Kalpitiya 2021

Agriculture and Coconut Cultivation

Apart from homesteads and coconut cultivation, there are no significant agricultural activities present in the Dutch Bay Grama Niladhari Division (GND), where the proposed project is to be implemented. In addition to engaging in fisheries, some households in the area earn supplementary income through the cultivation of coconut, which remains the primary form of agriculture practiced within the GND. According to the Divisional Resource Profile of Kalpitiya (2021), a total of 74 landowners own and manage approximately 131 acres of land within the Dutch Bay GND. The extent of land parcels and the number of owners is given in Table 3-41.

Table 3-41: Extent and Ownership of Coconut Cultivation

Extent of land	Number of owners	Total Extent (acres)
1/4 – 1/2 acres or 15-30 trees	24	6
1;2 -1 acre or 31-65 trees	13	1
1-5 acres	22	54
5 acres and above	15	70
Total	74	131

Source: Divisional Resource Profile - Kalpitiya 2021

3.5.4.2 Role of women and Gender Issues

In the project AOI, women play a vital yet often limited role in income-generating activities. Their involvement primarily centers around small-scale domestic livestock farming, including the rearing of goats, pigs, poultry, and cattle. Additionally, many women engage in dry fish processing



at the household level, collecting oysters from the lagoon and selling oyster shells (to use for poultry feed) to support their families economically. Another key contribution comes through the maintenance of home gardens, which typically include coconut trees and other fruit-bearing plants such as mangoes, providing both food and supplementary income.

Focus Group Discussions (FGDs) and KII held with women in the communities of *Mohonthuwarema* and *Kirimundal* revealed several pressing gender-related challenges. Participants highlighted the lack of access to maternity services and general healthcare, as well as inadequate transportation and limited access to clean drinking water. These issues significantly hinder their daily lives and overall well-being, underscoring the need for targeted interventions that address both economic empowerment and basic infrastructure and service provision for women in these areas.

3.5.4.3 Fishing Activities along the coastal line and Dutch bay/lagoon of the project area and existence of fishing vessels, landing sites

Both marine and lagoon fisheries serve as the primary sources of livelihood in the Puttalam Fisheries District. To facilitate effective fisheries management and administration, the district is subdivided into eight Fisheries Inspector Divisions (FIDs): Vanatavillu, Puttalam, Mangalaeliya, Baththuluoya, Palakuda, Kandakuliya, Kalpitiya Mainland, and Kalpitiya Island. The proposed project site, located in Dutch Bay Grama Niladhari Division (GND), falls within the southern part of the Kalpitiya Islands FI Division, (one of the eight FI divisions), as illustrated in **Figure 3.61**.

According to the most recent data from the Puttalam Fisheries District Office (2024), a total of 3,197 active fishers are currently engaged in fishing activities within the Kalpitiya Mainland and Island FI Division. Of these, 1161 active fishers are based in the Kalpitiya Islands, highlighting the area's significant role in supporting local fisheries-based livelihoods. The details of the active fishers and existing fishing fleet in the FI Division are given in **Table 3-42** and **Table 3-43**.

According to the Divisional Resource Profile, Kalpitiya (2021), there are a total of 365 fishing households residing in the Dutch Bay Grama Niladhari Division (GND). The area supports a fisheries sector, with 6 landing sites and 16 designated beach seine sites. Additionally, 160 operational fishing crafts are contributing to the local fishing economy. The community is further organized through the presence of three fisheries societies (*Ekamuthu Fisheries Society- 52 members, Mohuththuwarema, Velan Kanni Matha -Uchchamunai - 465 members and Kirimundal*), which play a key role in supporting the livelihoods of local fishers, as detailed in **Table 3-44**.



Table 3-42: Active Fishers and Fishing Population

Kalpitiya mainland and Island FI Division	Number of Active Fishermen	Fisher Population
Kalpitiya mainland	2036	4454
Islands (including Dutch Bay)	1161	2568

Source: Assistant Director, DFAR, Puttalam 2024

Table 3-43: Details of the Fishing Fleet in the Kalpitiya Mainland and Island FI Divisions

	Kalpitiya mainland	Islands
OFRP - Outboard engine Fiberglass Reinforced Plastic Boats	640	395
MTRB (Motorized Traditional Boats)	48	0
NTRB (Non-motorized Traditional Boats)	196	40

Source: Assistant Director, DFAR, Puttalam 2024

Table 3-44: Fisher households, landing sites, beach seine sites, and number of crafts in the Dutch Bay GND

Length of the coastline (km.)	Number of Fisher household	Number of landing sites	Number of Fishing crafts	Number of beach seine sites	Number of Fisheries societies
27	365	6	160	13	3

Source: Divisional Resource Profile 2021, Kalpitiya

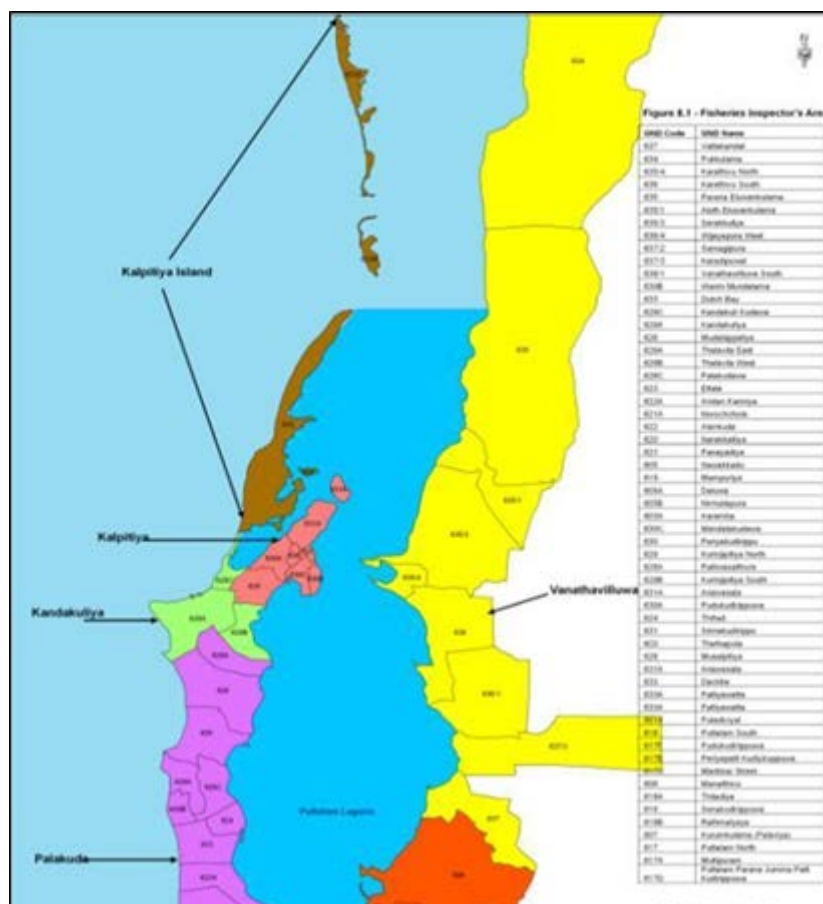


Figure 3.61: Proposed Project Site Details

3.5.4.4 Presence of Fish Farms, aquaculture activities etc.

As part of the field investigations, interviews were carried out with the Assistant Director of Fisheries, the Presidents of the *Ekamuthu* Fisheries Society in *Mohonthuwarema* and the *Velan Kanni* Matha Fisheries Society in *Uchchamunai*. According to the information gathered during these discussions, there were no existing records or evidence indicating the presence of fish farms or aquaculture activities within the project's Area of Influence (AOI).

3.5.4.5 Details of tourism related activities including tourism structure

Tourism-related activities in the Dutch Bay area, which lies in close proximity to the proposed project footprint, are primarily driven by the area's rich natural, cultural, and historical attractions. Kalpitiya, in particular, is known for its unique blend of ecological and heritage assets. Key attractions in the region include the Bar Reef Marine Sanctuary, one of the largest and most pristine coral reef ecosystems in Sri Lanka as well as diverse marine life such as dolphins, whales, and the elusive dugong. Additionally, the region is home to ecologically significant mangrove



communities, as well as several notable archaeological and religious monuments, including the Dutch Fort, Dutch Church, and the revered St. Anne's Church.

These environmental and cultural resources, along with the significant coastal and lagoon landscapes, support a range of tourism and recreational activities that are popular among both domestic and international visitors. Among the most prominent of these activities are diving, snorkeling, kitesurfing, horse riding, and dolphin and whale watching.

Currently, most accommodation facilities catering to tourists are concentrated in the Kandakuliya–Kudawa Grama Niladhari Division (GND) and several other GNDs located outside the Dutch Bay GND. Although the Kalpitiya Tourism Development initiative had previously introduced tourism infrastructure within the Dutch Bay area, these facilities have since been severely damaged or abandoned due to severe sea erosion, as illustrated in Figure 3.62.

Despite the lack of operational accommodation facilities in Dutch Bay, Palliyawatta, Uchchamunai, and the surrounding islands, various tourism-related recreational activities, such as kitesurfing, horse riding, diving, and snorkeling, continue to take place outside the immediate project area. The main kitesurfing season runs from May to September, characterized by strong winds ranging from 20 to 30 knots and attracting experienced kitesurfers. A secondary, lower season occurs from November to April, with more moderate winds of 15 to 20 knots.

Kitesurfing activities are primarily concentrated in popular locations such as Magic Island, Vella Island, Dream Spot, Kalpitiya Lagoon, Donkey Point, Kappaladi Lagoon, and Puttalam Lagoon. With the exception of Kalpitiya Lagoon, all these sites are situated well outside the project footprint in Dutch Bay.



Horse riding at Kirimundal



Horse riding at Kirimundal



*Bay Watch Hotel at Dutch Bay- abandoned
due to Sea Erosion*



Sand Hotel – Abandoned due to Sea Erosion

Figure 3.62: Abandoned tourism Development Facilities



3.5.5 Existing and proposed Recreational Activities (area including the Kite surfing, (identified kite surfing area should be submitted with consultation of UDA and SLTDA))

The existing recreational activities within the Project Area of Influence (AOI) and its surrounding areas have been described in Section 3.5.4.5. The most prominent activities include diving, snorkelling, kitesurfing, horse riding, and dolphin and whale watching, both in the nearshore waters and within the lagoons.

Consultations with the Urban Development Authority (UDA) and the Sri Lanka Tourism Development Authority (SLTDA) revealed that a new tourism development plan is currently being prepared for the Kalpitiya area. This plan aims to align with existing policies, development strategies, and market trends, while also incorporating social and environmental considerations. However, the plan is still under development and has not yet been finalized or officially gazetted.

In addition to the tourism development plan, efforts are underway to develop a zoning plan specifically for kitesurfing in the Kalpitiya area. This zoning plan is currently in draft form and also awaits finalization and gazetting. The zones proposed for designation as kitesurfing areas are illustrated in the following **Figure 3.63**.

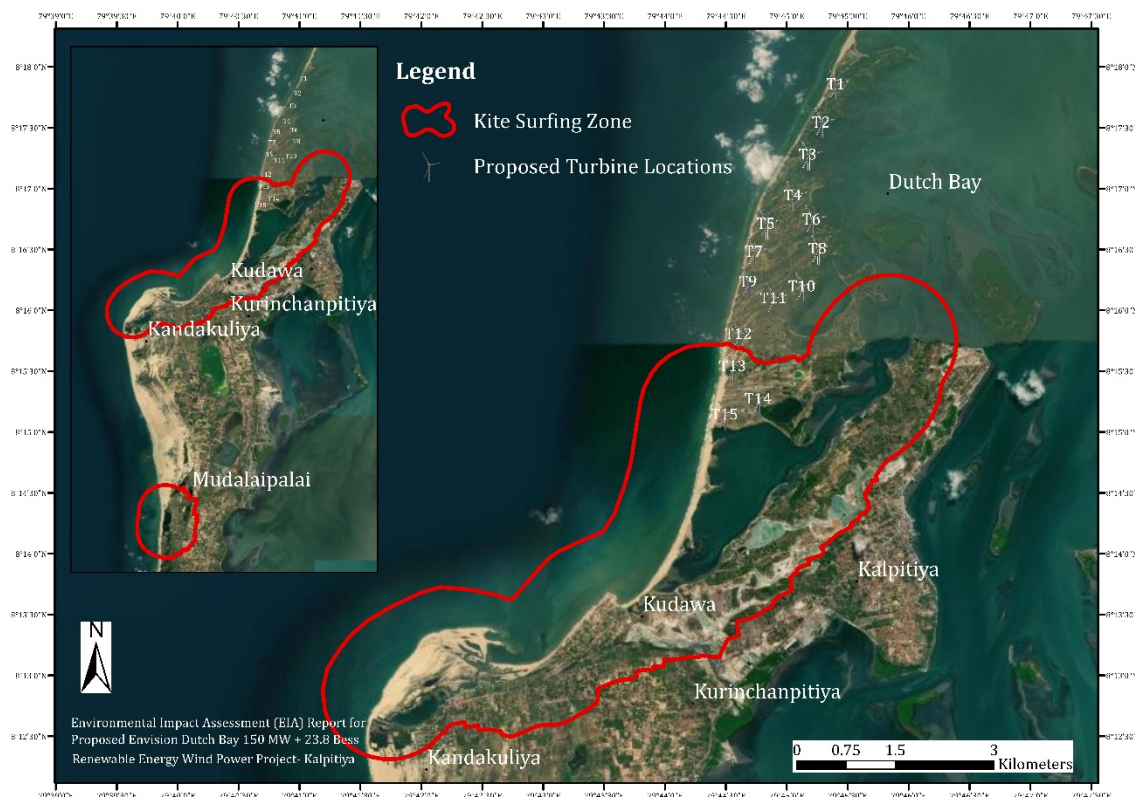


Figure 3.63: Zones proposed for designation as kitesurfing areas



3.5.6 Cultural, historical, religious, protected resources and archaeological aspects / considerations

According to Premathilake (1984), the region's cultural development has been influenced by trade relations dating back to prehistoric times. Among the artifacts discovered are coins and bronze sculptures from the Old Dutch Bay at Kalpitiya. These trade connections, along with fishing activities, attracted various ethnic groups to the area, particularly Moors and Tamils, in addition to the Sinhalese. This diverse settlement is reflected in the presence of mosques and Kovels in the region. Over time, the area was also shaped by the occupations of Western colonial powers. The Portuguese arrived in 1544 A.D., followed by the Dutch in 1644 A.D., and later the British in 1795. Within this historical context, several religiously significant and recent origin sites can be found in the Dutch Bay Grama Niladhari Division (GND), including:

- St. Mary's Church) built in 1950, is located approximately 200 meters from the beach in Palugahaturai
- St. Anthony's Church, built in 1934, is situated in the village of Mohonthuwarema
- Holy Cross Church (Avey Maria Church), located in Kirimundal
- Mari Amman Hidu Kovil- Mohonthuwarema
- Masjud Rahuman Mosque - Mohonthuwarema
- Masjith Mosque - Mohonthuwarema

3.5.7 Availability of infrastructure facilities (roads, power, water supply, communications, power supply, housing and sanitation health care, hospital, schools etc.

Infrastructure and service delivery in both the Dutch Bay Grama Niladhari Division (GND) and Palliyawatta are relatively poor. According to findings from the social survey, there is no access to safe drinking water in the villages of Mohonthuwarema and Kirimundal. Residents are compelled to purchase drinking water from external sources, as the available well water is saline and unsuitable for consumption. Further, the existing community water supply program operated by the villages in Mohonthuwarema provides water to 70 households, but it is currently used only for domestic washing purposes. The survey also revealed that while most houses are connected to the national electricity grid or rely on solar power, the overall availability of basic household amenities remains limited. These conditions are summarized in Table 3-45.

Dutch Bay GND lacks essential healthcare infrastructure, including health centers, hospitals, and maternity services provided by the public or private sectors. The hospital that is nearest to the project site is located at Kalpitiya city center.



The road network within the area is narrow, unpaved, and in poor condition, with only a short stretch of concrete road (less than 1 km) located inland. During the rainy season, many roads become flooded, significantly hampering accessibility. Public transportation is non-existent in both Dutch Bay GND and the interior areas of the Kalpitiya Peninsula.

Education facilities are also limited. The two local government schools, Penapidugama Vidyalaya and Mohonthuwarama Muslim School operate with inadequate resources and staff, offering classes only up to Grade 9. For Advanced Level education, students must travel to schools in the Kalpitiya Town area. However, the absence of a bridge between Mohonthuwarama and Thoradiya forces schoolchildren and community members to rely on unsafe boat services to cross the water.

Besides a handful of small retail shops, there are no marketplaces or other essential amenities in the GND, highlighting the pressing need for improved infrastructure and services in the area.

Table 3-45: Availability of amenities for the houses

Total Number of Houses	283		Percentage	
Amenities	Available	Not available	Available	Not available
Latrines	252	31	89%	11%
Electricity	120	163	42%	58%
Water	40	243	14%	86%
Telephone(fixed)	171	112	60%	40%
Internet	10	273	4%	96%

Source: Resource Profile Kalpitiya DSD 2021

Housing conditions in the Dutch Bay area are generally poor. Approximately 54% of the houses are temporary structures built with materials such as clay, wooden planks, and cadjan. In terms of ownership, only 76% of residents own their homes, while 22% do not have secure or formal ownership. Further details regarding the types of housing and ownership status are presented in Table 3-46 and Table 3-47, respectively.

Table 3-46: Nature of Ownership of Houses

Total households	Number of HH with own house	Number of HH With Rent /Leased houses	Number of HH without House
365	278	5	82
Percentage	76%	1%	22%



Table 3-47: Nature of Houses in the Dutch Bay GND

Permanent Houses (bricks, Cement, tile)			Semi-permanent (Clay, planks, Cadjan)	Temporary houses (shanty)	Other	Total
Total	Completed	Half completed				
59	32	27	71	153		283
Percentage	11%	10%	25%	54%	0%	100%

3.5.8 Existing security points /camps

There are two security points run by the SL Navy located in the project area, close to the abandoned Bay Watch Hotel and the entrance point of the Kudawa- Mohonthuwarama road near the coastal information center managed by CC&CRMD. These security points operate as observation points to control smuggling and maintain national security.

3.5.9 Existing environmental problems and issues or any social cultural conflicts that may have bearing on the projects

Based on stakeholder consultations carried out by the social team through Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), and a comprehensive social survey, several key environmental and social issues have been identified. These issues are of significant concern and could potentially impact the successful implementation of the project. They should therefore be carefully addressed during the planning and execution phases:

Coastal Erosion: Severe coastal erosion has been observed in the area along the coastline that overlaps with the project footprint. This ongoing environmental challenge poses a threat to both natural ecosystems and the structural stability of any coastal development planned under the project.

Infrastructure and Accessibility: The main access road that cuts through the project site is currently in poor condition, being narrow and composed primarily of sand. This limits transportation capacity and could hinder the efficient delivery of materials and equipment necessary for the project. Upgrading this road is essential to ensure smooth logistical operations.

Community Perceptions and Stakeholder Dynamics: The majority of residents in the villages of Mohonthuwarama and Kirimundal view the proposed project as a generally positive initiative that could bring tangible benefits to the community. However, there is also a negative perception among particularly among trainers of kite surfers due to negative impacts such as gusty effects.



3.5.10 Community Awareness about the proposed project

During the social survey conducted in the villages of Mohonthuwarama and Kirimundal, efforts were made to evaluate both the general perception of the proposed wind power project and the extent of stakeholder awareness regarding its implementation. The findings indicated that approximately 71% of the surveyed households were aware of the proposed project. Most of this information had been disseminated through community meetings and local fisheries societies, which served as key channels for communication.

However, despite this relatively high level of basic awareness, many respondents expressed a lack of detailed knowledge about the specific measures and methodologies the project proponents intended to employ during both the construction and operational phases. This suggests the requirement for meaningful stakeholder engagement, particularly in terms of communicating technical details and project timelines to the communities (Figure 3.64).



Figure 3.64: Social Awareness Program

The following proofs & details related to the conducted stakeholder awareness programs, stakeholder interviews, etc. are included in **Annexure III P**.

- List of Stakeholders Interviewed - Stakeholder Meetings Conducted for the SIA Preparation Phase
- List of Attendees for meetings
- List of respondents engaged for social surveys & List of Stakeholders interviewed
- Minutes of the Meeting with Key Informants
- Photographs taken during social awareness programs



4 ANTICIPATED ENVIRONMENTAL IMPACTS OF PROPOSED PROJECT

Environmental assessment is a systematic method of appraising the environmental effects of development on the proposed location and surrounding environment and recommending mitigation measures to reduce and eliminate those environmental effects. The potential environmental impacts of the proposed infrastructure components are presented in this section.

4.1 Categorization and Method for Identification of Environmental Impacts

Prediction of potential impacts is the most important step during the EIA process. Potential environmental impacts were identified through a systematic process. The activities associated with the project stages have been considered with respect to their potential to interact with existing environment resources or receptors. This effort builds on the identification step in scoping, whereby the potential interactions between the project and the baseline environment were identified. From these interactions, the potential impacts on the various resources or receptors were identified and described to the extent possible. The evaluation of baseline data provides crucial information for the process of evaluating and describing how the project could affect the project environment.

Superimposition of predicted impacts over baseline environmental conditions gives the ultimate environmental scenario. Impacts can be either adverse⁸ (negative) or beneficial⁹ (positive) based on the existing characteristics and sensitivity of receivers, resources and the areas to be affected.

⁸ An impact that is considered to represent an adverse change from the baseline or introduces a new undesirable factor

⁹ An impact that is considered to represent an improvement on the baseline or introduces a positive change



4.1.1 Identifications of Potential Impacts

The nature and types of potential impacts addressed in this EIA are defined below.

Impacts	Description
Negative impacts	Potential impacts that are detrimental to the impacted medium
Positive impacts	Potential impacts that are beneficial to the impacted medium
Direct impact	Any change to the environment, surrounding communities, or habitats
Indirect impact	Environmental impacts that do not result directly (immediately) from the project, occur at some distance from the project area, or occur from a chain of cause and effect from the project
Cumulative impacts	Each project activity may have an insignificant effect on its own, but the combined effects may be considerable.
Receptor	The attributes of the environment (e.g., air, water, soil, sediment, etc.), which may be potentially impacted by the project activities.
Event-related impacts	Potential impacts stemming from a specific unintentional event such as an oil spill, fire, etc.

4.1.2 Anticipated Impact Prediction Methodologies

Predicting the magnitude of the development's likely impacts and evaluating their significance is at the core of the environmental impact assessment process. The prediction was made based on professional judgment, adequate reasoning, and a supportive environmental baseline for the project area. The predictions were described in qualitative terms.

4.1.3 Considerations in Anticipated Impact Prediction

4.1.3.1 Magnitude or Intensity of impact

This is defined by the severity of each potential impact and indicates whether the impact is irreversible or reversible and the estimated potential rate of recovery. The impact resulting in the destruction or serious impairment of a social or environmental feature or system, or deterioration of the economic, social or cultural well-being of a large number of people will be assessed under this. Further, the possibility of managing the impacts (to avoid or mitigate) using relatively uncomplicated and accepted measures, or the requirement of a detailed study to understand if the impacts can be managed and which management measures are needed also considered under this.



4.1.3.2 Extent (Location) of impact

The spatial extent or zone of influence, of the impact should always be determined. An impact can be site-specific or limited to the project area; a locally occurring impact within the locality of the project; or a regional impact that may extend beyond the local area. If the site is located in sensitive areas (density densely populated areas, near critical habitat, near educational institutions, homes, hospitals, and protected areas) high weightage should be given to the impacts.

4.1.3.3 Duration of impact

Environmental impacts have a temporal dimension, which was considered in the EIA study. Impacts arising at different phases of the project cycle were also considered. An impact that is generated during the construction period is considered short-term. An impact that generally lasts less than 5 years after project completion may be classified as medium-term. An impact that continues for 10 years may be defined as long-term, and impacts that last beyond 10 years are considered multiyear or permanent.

4.1.4 Integrated Impact Assessment Methodology

The impacts on the environment that can be caused by the project interventions were predicted, assessed, and separated into two distinct stages of the project life cycle. (a) Stage 1: Site development and construction; (b) Stage 2: Operation and maintenance.

Relative qualitative classification of these impacts as 'Positive', 'No', 'Low', 'Moderate' 'High' and 'Severe' were made mainly based on criteria such as magnitude, spatial extent, time period within the day and duration, reversibility as well as the existing characteristics and sensitivity of receivers, resources and the areas to be affected. The referencing environmental criteria such as standards and associated threshold values, professional opinion where applicable, etc. were also considered for the classification of impacts.

The aim of impact prediction and assessment is to ensure that potential environment risks are foreseen and avoided at an early stage in the planning cycle so as to prevent issues.

There are several methods to analyse impacts ranging from simple qualitative analysis to detailed quantitative survey or modelling. The risk assessment is prepared based on the guidance on evaluating and managing the risk of the AS/NZS ISO 31000:2009 Risk management – Principles and guidelines (Standards Australia 2009).



4.1.5 Likelihood and Consequence

Each environmental risk was given a rating in terms of likelihood (how likely is that event/issue will occur) and consequence (what will be the consequence/result if this issue does occur) using the criteria in the tables below (Table 4.1 and Table 4.2).

Table 4.1: Likelihood of the Environmental Risk

Likelihood/ Probability Level (how likely is it that this event/issue will occur)	
Highly likely	Is expected to occur in most circumstances
Likely	Will probably occur during the life of the project
Possible	Might occur during the life of the project
Unlikely	Could occur but considered unlikely or doubtful
Rare/Slight	May occur in exceptional circumstances

Table 4.2: Consequences of the Environmental Risk

Consequences (what will be the consequence/result if this issue does occur)	
Minor	Minor incident of environmental damage that can be reversed
Medium	Isolated but substantial instances of environmental damage that could be reversed with intensive efforts
High	Substantial instances of environmental damage that could be reversed with intensive efforts
Major	Major loss of environmental amenity and a real danger of continuing
Critical	Severe widespread loss of environmental amenity and irrecoverable environmental damage

4.1.6 Significance of the Impact

The significance of the impact (risk rating) refers to the value or amount of the impact and is assessed by assigning a likelihood rating and a consequence rating. It is determined whether the impact (especially for the negative impacts) is 'Low', 'Medium', 'High' or 'Severe' using the rating table below.



		Consequences				
Likelihood		Minor	Medium	High	Major	Critical
	Highly Likely	Moderate	High	High	Severe	Severe
	Likely	Low	Moderate	High	Severe	Severe
	Possible	Low	Moderate	Moderate	High	High
	Unlikely	Low	Low	Moderate	Moderate	High
	Rare/slight	Low	Low	Low	Low	Moderate

The risk rating can be used as a guide to the amount of time and resources that will be required to manage each impact. Impacts with 'low' risk ratings could be neglected without providing any mitigating measures while other ratings required special mitigating measures to avoid/minimize its impact. The impacts with a 'moderate' risk rating will usually require significantly less management than 'high' and 'severe' risks.

In this section, based on the potential impacts identified during the EIA process, a risk assessment was carried out for each potential impact associated with the proposed development activities.

4.2 Anticipated Environmental Impacts

The proposed development activities have been examined in detail to identify the construction methods, technology, and other factors that would determine the potential impact of the various activities on the environment. These activities span from pre-construction to decommissioning, and each phase can have distinct environmental, social, economic, and technical impacts.

Following development activities are taken into consideration while preparing the potential impacts and corresponding risk ratings;

- Ground formation (land clearing and grading) and compaction
- Construction of wind towers (foundation, turbine transport and assembly)
- Construction of access roads.
- Grid infrastructure and other installations / infrastructure facilities
- Operation phase of the wind farm



The impact identification, assessment and proposed mitigating measures for the identified activities during the construction and operation phases of the project are described in below sections.

4.3 Anticipated Impacts on Coastal Environment

4.3.1 Anticipated Impacts to the Beach, Shoreline and Coastal Vegetation

Main structures to be constructed on the shoreline on the open sea would be the temporary jetty during the construction period and segments of the proposed road along the shoreline. Since the temporary jetty to be construct with standardized floating box modules, minor disturbances to the existing beach, shoreline and coastal vegetation could be expected.

Reinforced roadway segments to be constructed would be positively affect the stability of the shoreline from further erosion prevailing in the coastal stretch of the Island.

4.3.2 Anticipated Impacts to the Dutch bay/lagoon environment

All the project activities will be confined within the land area and hence no disturbance will be on the lagoon side of the Dutch Bay Island during both construction and operational period of the proposed Wind Power Project. Zero discharge of wastewater or oils into the lagoon will be ensured to maintain the water quality without any negative impact. No mitigation measures are required.

4.3.3 Anticipated Impacts on stability and seasonal dynamics of land strip of Kalpitiya Peninsula

Main project footprint of the proposed Envision Dutch Bay 150 MW + 23.8 Bess Renewable Energy Wind Power Project is confined to the Dutch bay Island which is located north of Kalpitiya Peninsula. The sediment transport on the North-western coastline around Kalpitiya is towards north and any disturbance that arrest the sediment transport will only affect the shoreline. Although the main construction of the wind towers and the substation are located within the land area, a temporary jetty construction is sited on the beach to facilitate the unloading operations. However, the impacts to the shoreline and the sediment transport pattern of the area are mitigated through design by introducing the floating jetty for the operation. There would be minor disturbance to the shoreline during the construction of temporary jetty, but it will not create a barrier for the existing hydrodynamics and the sediment transport pattern of the area.



And hence it would not affect the stability and seasonal dynamics of land strip of Kalpitiya Peninsula.

There will not be any effect on the area during the operational period. However, having noticed many uncertainties associated, it is always recommended to monitor the shoreline very closely for any adverse impacts due to longshore sediment movement and resulting coastal erosion and then, take appropriate remedial measures. Beach profile surveys before and after construction of the jetty is proposed along the project site to visualize the real behaviour.

4.4 Anticipated Noise and vibration impacts

4.4.1 Expected noise and vibration level during the construction period

The major noise-generating sources at the proposed site during the construction phase are vehicular traffic, the use of construction equipment and heavy machinery. Noise levels generated by construction equipment vary significantly depending on the type and condition of equipment, the operation method and schedule and the site of the activity. Construction activities at site are expected to produce noise levels in the range of 75–110 dB (A) which is generally higher than the permissible noise levels. Therefore, these activities may have an impact on the immediate surrounding of the project area, with most work carried out during daytime. However, the noise generated during the construction works is temporary and will be intermittent. The adverse impact will only be short-term, highly likely in occurrence and moderate in consequence, hence “**Moderate**” in significance.

Operation of heavy machinery, deep excavations, compaction, and concrete construction vibration etc. can cause high vibration. There will be temporary inconvenience to some of the noise receptors, depending on the construction activities.

The proposed measures to mitigate the impacts are presented in Chapter 5.

4.4.2 Predicted noise levels distribution patterns during the operation period.

Wind turbines produce noise through several different mechanisms, which can be roughly grouped into mechanical and aerodynamic sources. Modern wind turbines, including models in the **10 MW** range such as those manufactured by Envision Energy, typically generate broadband aerodynamic noise due to blade–air interaction and mechanical components. Published data and benchmarked turbine models suggest the following sound pressure levels (SPL):

- ~113 dB(A) at the nacelle (source level)



- ~60 dB(A) at 100 meters distance from the turbine
- ~55 dB(A) at 300 meters
- ~45 dB(A) at 500–750 meters, depending on terrain and atmospheric conditions

The topography of the Dutch Bay area, flat with sparse residential clusters, supports natural attenuation of turbine noise.

Arrays of turbines may cause overlapping zones of noise exposure; however, the additive nature of SPL means doubling turbine count increases total SPL by approximately 3 dB. Cumulative effects of multiple turbines operating simultaneously have been modelled based on international practice.

Predicted noise level distribution pattern during the operation of wind turbines are illustrated in the Figure 4.1 below.

According to the Sri Lankan Noise Level Standards, project area falls in the low noise zone, in which permissible noise levels are 55 dB LA_{eq} and 45 dB dB LA_{eq} during day time and night time respectively. The allowed increase of noise level is +3 dB(A) with reference to Measured Background Noise level.

It is observed that the values beyond 300 m circle are within permissible exposure levels set by both the Sri Lankan Standards and the Standards given by World Health Organization (WHO) for community noise during the day (55 dB LA_{eq}) and night (45 dB LA_{eq}). The noise generation within the first 100 m seem to be bit elevated from the permissible levels. However, only there is hardly any sensitive receptors are available within the 100m buffer zone. Refer Table 3-12. Only Dutch Bay resorts which is currently abandoned is located near to 100m buffer zone. Almost all the other sensitive receptor are located beyond 160 m and falls within the permissible levels.

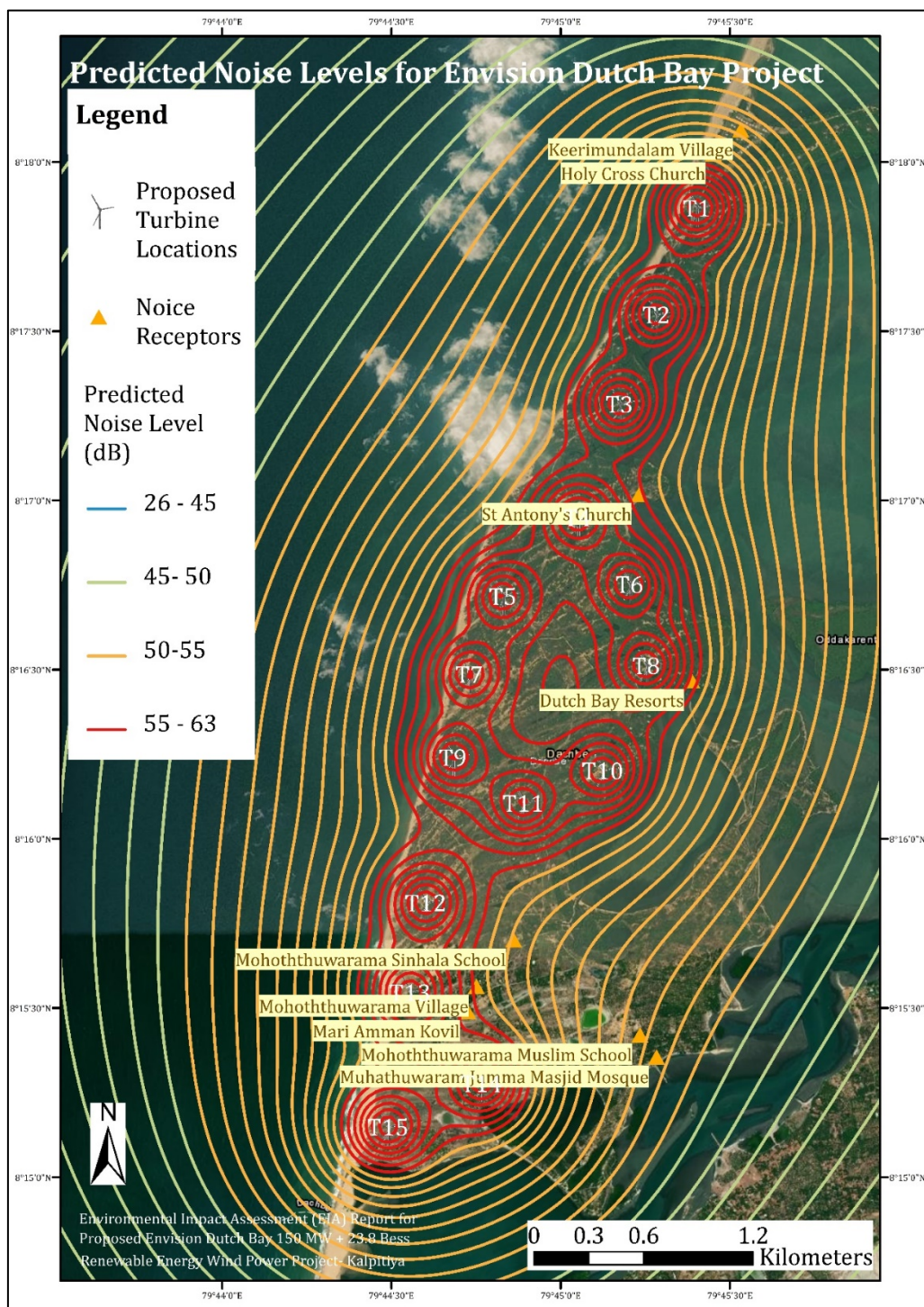


Figure 4.1: Predicted noise levels distribution patterns during the operation period

During project operation, there may be minor noise generations from vehicle movements and machinery operation around the site for maintenance and repair purposes.



4.4.3 Anticipated Impact on dwellers, fishing community, fishing activities, fish harvest and security point due to noise during operation period

Few noise receptors were identified in the area as Dutch Bay Resort (located 100m away from T8), Mohanthuwarama Village (located 160m away from T13) and all others located more than 300m away from its closet tower location (refer Table 3-12). The predicted noise levels from the turbine is 60 dB(A) at 100 meters and 55 dB(A) at 300 meters would not create any significant impact during the wind farm operation.

Noise generated by onshore wind turbines would be outcast by the sound of the wave action and wind in the ocean. Hence quality of the working environment for fishing activities carried out in the ocean would not be affected significantly. Although there are several studies carried out about the noise generated from offshore wind turbines on fish, there is a knowledge gap of scientific literature on the direct impact of onshore wind turbine generated noise on marine organisms including fish.

No special mitigation measures are required to mitigate the noise generation during the operation phase. However, an immediate complaint handling will be done by implementing proper noise mitigation action whenever there is a complaint.

4.5 Anticipated Ecological Impacts

The proposed project site is located within a narrow coastal strip along Dutch Bay on the Kalpitiya Peninsula, bordered by the sea to the west and the Puttalam Lagoon to the east. Despite its limited size, the site includes a diverse mix of habitats—natural wetlands, terrestrial ecosystems, and managed landscapes. While core construction will take place in inland zones, certain activities such as land clearance, infrastructure development, and temporary storage may occur near or within sensitive habitat areas. These interventions could potentially disrupt or degrade ecological features, especially in areas where habitat connectivity and biodiversity are already under pressure from human activity.

The wind farm is sited in a sparsely populated area, deliberately chosen to avoid tourist zones, fishing settlements, and known resident or migratory wildlife habitats. Potential future urban development was also taken into account during site selection. Ecologically sensitive areas—including mangrove stands, coastal forests, salt marshes, inundated zones, pastures, and palmyra woodlands—were specifically excluded to minimize habitat disturbance. Where feasible, turbine foundations are located on elevated sandy terrain or previously disturbed areas to reduce environmental risk.



At each turbine site, approximately 0.2 ha of vegetation, including trees and shrubs, will be cleared for construction. However, the remaining area will be preserved to maintain natural habitat. Turbines are variably spaced to retain and connect vegetation corridors, helping to ensure that large sections of the landscape remain ecologically functional.

Following construction, natural regeneration of vegetation is anticipated. Landscaping will prioritize native plant species characteristic of the region to support habitat restoration and enhance both structural and floristic diversity. These efforts aim to improve habitat quality for local and migratory wildlife—including birds, small mammals, reptiles, and amphibians—while maintaining the ecological integrity of the interconnected land, lagoon, and marine environments.

Ecological assessments have identified four primary areas of potential impact associated with project implementation: habitat alteration, vegetation loss, disruption to fauna, and disturbances to surrounding sensitive and protected areas within Dutch Bay on the Kalpitiya Peninsula.

Table 4-3 summarizes the project's key potential impacts on the terrestrial and aquatic environments—both positive and negative, direct and indirect—affecting biological resources during the planning, construction, and operational phases.



Table 4-3: Potential Impact Matrix – Anticipated Biological Environmental Impacts of the Project Presented in the Leopold Matrix

Environmental attribute	Potential impacts	Nature of impact	Magnitude of impacts			Mitigation measures	Implementation & Monitoring
			L	M	H		
Aquatic Ecology	No significant impact on aquatic ecology	Direct/Local/reversible	X			- Implement effective waste management practices to prevent contamination of nearby water bodies, including the lagoon, by soil, construction debris, and other waste materials. During piling, drilling, and land clearing activities, appropriate precautions must be taken to minimize environmental impacts. This includes the implementation of erosion control measures, sediment containment systems, and pollution prevention protocols. - Construction activities should be scheduled during the dry period to reduce the risk of soil erosion, sediment runoff, and water pollution, thereby minimizing environmental impacts.	Before and during construction phase
Terrestrial Ecology	Loss of vegetation	Direct/Local/irreversible		X		- The selection of sites for wind turbine installation, storage areas, and access routes should primarily be confined to the zone between the landward coastal setback (Zone) of 300 meters from the high tide line and 100 meters from the lagoon shoreline. To minimize environmental impact, these locations should, as far as possible, be limited to open sandy terrain or previously disturbed lands. - Wherever feasible, existing truck roads should be utilized or upgraded to serve as access routes. This approach reduces the need for new land clearance and minimizes additional disturbance to natural habitats.	Before the construction phase



Environmental attribute	Potential impacts	Nature of impact	Magnitude of impacts			Mitigation measures	Implementation & Monitoring
			L	M	H		
Terrestrial Fauna	Disturbance to the local fauna during construction	Direct/Local/reversible	X			Although wildlife diversity and population density in the project area are relatively low, preferred habitat areas must be pre-identified and carefully avoided when selecting specific locations for wind turbines, storage facilities, and access routes, in order to further minimize ecological impacts.	Before and during construction phase
	Disturbance to the local fauna during operation	Direct/Local/reversible	X			Regular monitoring should be conducted to detect animal occurrences by comparing current observations with established baseline data. Any animal carcasses—particularly those resulting from hunting or bird collisions during wind park operations—must be documented and analysed to assess potential impacts and inform adaptive mitigation measures.	During operation phase
Lagoon/ coastal sea area	Significant impacts if works are not undertaken carefully	Direct/local/irreversible		X		Disturbance in areas affected by wind turbine construction should be minimized, with particular attention to coastal zones, sand dunes, and scrublands. The removal or destruction of sand dune vegetation and surrounding scrubland habitats must be avoided. Additionally, coastal stretches that are vulnerable to sea erosion should not be selected for development activities, in order to preserve natural shoreline stability and maintain ecological integrity.	During construction phase



4.5.1 Anticipated Impacts to Habitat

Habitat impacts from the proposed wind farm development may manifest in several interrelated forms, potentially affecting the ecological integrity of the project area and its surroundings. These impacts stem primarily from land conversion and construction activities and can have cascading effects on biodiversity, ecosystem services, and landscape connectivity.

Habitat impact can be in different ways

- Loss and alteration of natural habitats such as coastal scrublands and grasslands due to land clearing for access roads, turbine pads, and temporary storage areas.
- Habitat fragmentation, potentially disrupting wildlife movement corridors and ecological connectivity
- Edge effects, introducing microclimatic changes can promote invasive species colonization along disturbed boundaries.

4.5.1.1 Loss and alteration of natural habitats

One of the most immediate and visible ecological impacts is the removal and modification of natural habitats, particularly coastal scrublands, grasslands, and transitional zones between terrestrial and wetland ecosystems. These habitats are often characterized by specialized and adapted plant communities that support a variety of faunal species, including birds, reptiles, small mammals, and invertebrates. Vegetation clearance to accommodate turbine foundations, access roads, temporary laydown areas, and associated infrastructure leads to permanent or semi-permanent habitat loss, thereby reducing the overall habitat availability in the area. The removal of native vegetation also alters soil structure, microclimate, and hydrology, which may compromise habitat regeneration and ecological resilience over time.

4.5.1.2 Habitat fragmentation

The construction of linear infrastructure—such as roads and service corridors—introduces fragmentation within previously contiguous habitat zones. It may also lead to genetic isolation of small populations, reducing their long-term viability. Fragmentation disrupts wildlife movement patterns, especially for species that rely on large, connected territories for foraging, nesting, or breeding. In particular, species with limited dispersal capacity or specialized habitat requirements are at heightened risk. Fragmentation may also interrupt ecological processes such as seed dispersal, pollination, and predator-prey dynamics, with broader implications for local biodiversity.



However, the road infrastructure will be developed mainly by improving the existing roads, hence additional fragmentation will not occur. Only few new roads will be introduced to connect the turbine location to the road network, but the area will be insignificant.

4.5.1.3 Edge effects

The creation of new habitat edges (transitional zones between disturbed and undisturbed areas) leads to changes in microclimatic conditions, including increased exposure to sunlight, wind, and temperature fluctuations. These edge conditions often favour the establishment of opportunistic or invasive species, which may outcompete native flora and alter habitat composition. In the long term, edge effects can contribute to habitat degradation, loss of ecological function, and reduced suitability for native fauna. Additionally, some species may avoid edge zones due to higher predation risk or decreased shelter availability, further shrinking the effective habitat area.

4.5.2 Anticipated Impact on Vegetation

The commencement of the project is expected to result in both direct and indirect impacts on the existing vegetation within the project area. The landscape is currently characterized by a mosaic of natural and semi-natural plant communities, including mangroves, salt-tolerant shrubs, grasses, and scattered terrestrial flora, particularly remnants of thorny shrub (mixed dry evergreen forest) all of which contribute significantly to local biodiversity and ecological balance.

Key anticipated impacts include:

- Direct removal of native vegetation during site preparation and construction, particularly affecting dune flora and other salt-tolerant coastal plant communities.
- Soil compaction and erosion resulting from heavy machinery and construction activities, which may hinder natural vegetation regrowth and reduce overall habitat quality.
- Introduction and spread of invasive plant species along disturbed areas, especially where construction equipment and materials are transported from external sites or where soil is exposed and vulnerable to wind dispersal, potentially outcompeting native flora.

4.5.2.1 Direct removal of native vegetation

During the site preparation and construction phases, large areas of native vegetation are likely to be cleared to accommodate infrastructure development, access roads, and staging areas. This clearance will particularly affect dune flora and salt-tolerant coastal plant communities, which are often adapted to specific microhabitats and are slow to recover once disturbed. These plant communities—adapted to high salinity, sandy soils, and strong coastal winds—play a critical role



in stabilizing the shoreline and supporting niche fauna. Their removal not only causes immediate biodiversity loss but may also trigger long-term changes in ecological structure and function, especially if sensitive or endemic species are impacted.

4.5.2.2 Introduction and spread of invasive plant species

Construction activities disturb the soil and create open niches that are highly susceptible to colonization by invasive plant species. These species may be introduced unintentionally via machinery, construction materials, or workforce movement between sites. Additionally, exposed soils are more vulnerable to wind-dispersed invasive species. Invasives such as *Prosopis juliflora* and *Opuntia dillenii* are particularly aggressive and can rapidly outcompete native vegetation by monopolizing essential resources like light, water, and nutrients. Once established, these species can alter natural fire regimes, soil chemistry, and hydrological patterns, further inhibiting native plant regeneration. Their spread significantly reduces plant diversity, disrupts ecological interactions, and often requires long-term and resource-intensive management for control or eradication.

4.5.3 Anticipated Impact on Fauna

Project activities are expected to impact local fauna both directly and indirectly. The project site and its surrounding areas, such as the coastal habitats of Dutch Bay and remnants of dry evergreen forest support a diverse range of terrestrial, avian, and aquatic species, some of which may be migratory or of conservation concern. Habitat alteration and fragmentation, human disturbance, noise, and pollution are among the primary drivers of faunal displacement or population decline during the various phases of the project.

Key anticipated impacts on fauna include:

- Habitat loss resulting from land clearing, vegetation removal, and infrastructure development, disrupting shelter, nesting, and foraging grounds for numerous species.
- Disturbance and displacement caused by noise, vibrations, and increased human activity, particularly affecting sensitive or nocturnal species.
- Increased mortality risks for small mammals, reptiles, amphibians, and invertebrates due to machinery movement, vehicular traffic, and open excavations.
- Disruption to avian populations, especially migratory birds and waders that depend on coastal and wetland habitats for feeding and roosting.



4.5.3.1 Habitat loss

The clearance of vegetation and modification of the natural landscape for construction purposes will result in the loss of critical habitats used by various faunal species. This includes nesting sites for birds, cover for small mammals and reptiles, and foraging areas for herbivores and pollinators.

4.5.3.2 Disturbance during Construction period

Construction-related noise, human presence, and light pollution can stress or displace wildlife, especially more sensitive or territorial species. Disturbances may alter feeding, mating, or migratory behaviors and can lead to permanent abandonment of traditional habitat areas. Ground-nesting birds and burrowing animals are particularly at risk during breeding seasons, while nocturnal species may suffer from light pollution and disrupted activity patterns.

4.5.3.3 Faunal mortality

The movement of heavy machinery, vehicles, and equipment across the site increases the risk of direct injury or mortality to slow-moving or burrowing animals, such as amphibians, reptiles, and small mammals. Excavations and open pits can trap animals; while fencing or barriers may obstruct safe movement, further raising mortality risks.

4.5.3.4 Water Pollution and habitat degradation

Sediment runoff from disturbed soil, accidental spills of fuel or construction chemicals, and improper waste management can severely impact aquatic ecosystems. Polluted waters can reduce oxygen levels, introduce toxins, and alter the composition of aquatic food webs, affecting fish, amphibians, and aquatic invertebrates. These changes can ripple through the ecosystem, affecting predators and dependent terrestrial species as well.

4.5.4 Anticipated Impact on surrounding sensitive and protected areas

The project areas close to the project sites were identified as Wilpattu National Park, Bar Reef Marine Sanctuary (BRMS) and Kala Oya Estuary. Among them, the Kala Oya estuary is 7 km away from the northern end of the project area while the Wilpattu National Park is located around 7.5 km from the northern end of the project area.

The defined South-eastern BRMS boundary falls very closer to the project site. However, all project components, including turbines, jetty, and access roads, are located outside the



designated boundaries of the Bar Reef Marine Sanctuary. This has been confirmed through spatial overlays using CZMP 2024 GIS data and coastal conservation clearance boundaries.

To the east, the project adjoins the Dutch Bay section of the Puttalam Lagoon, an estuarine ecosystem comprising extensive mangrove forests, intertidal mudflats, and seagrass meadows. These habitats serve as critical breeding, feeding, and nursery grounds for various species of fish, crustaceans, and birds, including migratory waterfowl and other species of conservation importance.

While the proposed wind farm is not located within any officially designated protected area, its proximity to these sensitive coastal and marine ecosystems raises legitimate concerns about potential indirect impacts, such as:

- Increased sedimentation and surface runoff during site clearance and construction, potentially degrading marine habitats and water quality in Bar Reef and Dutch Bay.
- Reduction in buffer habitat quality, which may affect species that depend on both terrestrial and marine ecosystems for different life stages.

Given these sensitivities, it is essential that project planning includes rigorous environmental safeguards, such as adequate buffer zones, erosion control measures, and biodiversity monitoring protocols. This will help ensure that the ecological integrity of these surrounding natural areas is preserved and that long-term impacts on protected and sensitive ecosystems are minimized.

4.5.5 Anticipated Impacts to Migratory Birds

The main potential impacts of wind farms on birds include the risk of collision with turbine blades, the direct loss of breeding or foraging habitats due to land clearance, and the indirect loss or degradation of habitat caused by disturbance. This disturbance may be temporary, such as during the construction phase, or long-term and persistent, resulting from the operation of turbines. Operational disturbances can alter bird behavior, leading to displacement from otherwise suitable habitats, changes in flight patterns, or reduced breeding success in nearby areas (**Table 4-4**).

The construction activities would involve the usage of approximately 0.7 ha area around the locations of each wind turbine, in view of the large scale of turbines and associated tower structures and the equipment/machinery to be used for construction. Clearing of vegetation, excavating, filling, compacting, levelling and other related activities are envisaged during construction phase. A possibility thus exists for soil erosion, siltation, blockage/restriction of



natural drainage patterns leading to soil degradation and water logging/localized flooding in the area.

Table 4-4: Impacts of Wind Power establishment on Migratory Birds in proposed area

No	Phase	Main effects	Impacts on avifauna
1	Construction	Habitat Disturbance and Loss	- Clearing of vegetation and land modification can destroy temporary roosting or feeding sites. - Wetlands and coastal habitats near development areas may be degraded or fragmented.
		Noise and Human Activity	- Construction noise and increased human presence can cause migratory birds to avoid the area. - Potential disruption of migratory stopover behavior due to stress or disturbance.
		Light Pollution	Use of artificial lighting during night-time construction may disorient nocturnal migrants and increase collision risk.
		Pollution and Waste	Improper waste disposal and machinery leaks can contaminate nearby habitats, affecting food resources (e.g., invertebrates in mudflats).
2	Operational	Collision Risk	- Migratory birds, particularly those flying at rotor-swept altitudes, are at risk of fatal collisions with turbine blades. - Species most vulnerable: large-bodied, slow-flying birds (e.g., raptors), and birds flying in flocks at night.
		Barrier Effect	- Wind farms may act as barriers to flight paths, forcing birds to detour, increasing energy expenditure during migration. - Disruption of traditional migratory routes and stopover sites.
		Displacement and Avoidance	- Some species may avoid turbine areas due to noise, movement, or habitat changes, leading to functional habitat loss. - Long-term displacement can reduce access to feeding / resting grounds.
		Cumulative Impacts	When combined with other developments (roads, power lines, deforestation), wind farms contribute to broader landscape-level threats to bird populations.

According to existing data, the highest number of migratory birds was recorded around Turbine 1, which is located in an area characterized by semi-aquatic habitats and lies in close proximity to the coastal belt. This coastal zone functions as an important feeding ground for several species,



including the Sanderling (*Calidris alba*) and the Brown-headed Gull (*Chroicocephalus brunnicephalus*). Additionally, the area surrounding Turbine 1 has recorded the presence of several conservation-sensitive species, such as the Near Threatened Ruddy Turnstone (*Arenaria interpres*) and Grey Francolin (*Francolinus pondicerianus*), the nationally Endangered Kentish Plover (*Charadrius alexandrinus*), and the Yellow Bittern (*Ixobrychus sinensis*). These observations highlight the ecological significance and sensitivity of the habitat associated with Turbine 1. (Table 4-5).

Table 4-5: Impact Levels of Wind Turbines on avifaunal species and Sensitive Habitats

Category	Turbine Number
Low Impact	T ₂ T ₃ T ₇ T ₉ T ₁₀ T ₁₃
Moderate Impact	T ₄ T ₅ T ₆ T ₈ T ₁₁ T ₁₂ T ₁₄ T ₁₅
High Impact	T ₁

Given the observed species richness, abundance, and ecological sensitivity of habitats in the vicinity of different turbines, the turbine sites can be tentatively categorized based on their potential ecological impact. However, to accurately assess the long-term effects of turbine operation on migratory birds, it is essential to conduct continuous, year-round monitoring. Furthermore, mapping the migratory routes of these bird species is crucial to understanding collision risk and habitat usage throughout the migration cycle.

4.6 Anticipated Hydrological and Water Quality Impacts

4.6.1 Anticipated impacts on natural drainage pattern (flow pattern) of the area and surface water bodies

4.6.1.1 Impacts during the Construction Phase

Each tower site is managed by adopting a proper site management plan. The excavation material, material stockpiling, storage of disposal material will be kept without interfering with natural flow patterns of the area. There are no other significant disturbances to the drainage pattern in the area during the construction phase as the tower locations are carefully selected by avoiding such natural drainage paths in the area. Therefore, the impacts due to changes in drainage patterns can be rated as 'Low' and no special mitigation measures are required.



4.6.1.2 Operational Phase

During the operation of the project, no turbine will intercept any water body such as Odais and Thonas (A coastal stream) etc., hence there will not be impacts on the waterbodies. As the turbine access roads and footings have been hydrologically designed by checking the backwater impact on the flood model using access road culverts and the access road levels will be optimized during the detailed design after taking levels there cannot be any significant hydrological impact. Required return periods for design will be decided during detail design stage.

Therefore, the impact on the existing drainage patterns can be rated as 'Low' during the operational phase as well. No special mitigation measures are required.

4.6.2 Anticipated impacts on groundwater table and availability of groundwater

The construction and operation of a wind farm can have some impacts on the **groundwater table** and **groundwater availability**, depending on the methodology adopted.

4.6.2.1 Construction Phase

Excavation and Drilling can cause impacts as the excavation for the foundation of wind turbines will required to reach up to 3m depth. This may disturb shallow aquifers, particularly if the site is located in an area with high water tables.

Dewatering might be needed temporarily, lowering the water table locally.

Use of Water for Construction, for purposes such as dust suppression, and cleaning will be obtained outside of the project area via water bowsers. Further, the concrete mixing would take place outside the project area and hence no direct impact from it. .

4.6.2.2 During Operation Phase

Reduced Recharge Area and land use changes can cause impacts on the groundwater table and availability in case if the area reduction is significant. However, the proposed project envisages minimal area (very low <2% compared to the entire project area) for footprint of turbines, substations and service roads in which creates minimum impacts.

Long-Term Groundwater Use: in most wind farms, ongoing water use is minimal. However, maintenance activities in some regions may involve periodic water use (e.g., for cleaning



equipment), which could affect local availability in arid areas. However, there is no plan for usage of groundwater in long-term basis. Hence the impact to the groundwater would be insignificant.

4.6.3 Anticipated Impacts on surface and ground water quality (during the construction and operation)

During Construction Phase

A reduction in the water quality of the nearshore water could be expected in terms of suspended solids during the following activities;

- wash-out of sediments from the fill material
- placing of core material and rock armours

However, the adverse impact is likely to be localized and temporary, occurring only during the construction period. Moreover, any adverse impact on water quality is not expected to create any additional impact on ecology. The adverse impact is expected to be short-term, high in occurrence and low in consequence and hence '**Moderate**' in significance.

During the construction stage, the following activities may also present a risk to water quality;

- Wastewater generated by site activities being discharged to sea
- The solid waste generated on-site
- Fuel spillage from machinery that will be used on-site

Any adverse impact on water quality due to the above activities is expected to be in the short term only (only limited to the construction period). Hence impact on water quality will be moderately likely in occurrence, but moderate to low in consequences. Hence the significance level is taken as '**Moderate**'. The proposed measures to mitigate the impacts are presented in Chapter 5.

During Operation Phase

No water pollution sources could be identified during the operational phase of the project and hence special mitigation measures are not required.

4.6.4 Anticipated Impacts on stability and seasonal dynamics of sand dunes / coastal features

The proposed wind turbine constructions are located away from the existing dunes. Only the temporary jetty construction will disturb the beach and the associated coastal features during the construction. This disturbance is temporary and could be easily re profiled after the placing of the



jetty. Since the is proposed as a floating structure, it will not create any disturbances to the existing hydrodynamic nature of the area. Therefore, the anticipated impacts on seasonal dynamics of land strip would be negligible.

4.6.5 Anticipated Impacts due to disturbance to the reservation and bank stability along the Dutch bay / lagoon or other water bodies located within the project site frontage (if any)

The proposed project does not envisage impacts/disturbances on the reservations and banks along the Dutch bay/ lagoon or other water bodies.

4.6.6 Anticipated Impacts due to Wastewater / Sewage disposal

Wastewater/sewage would be generated by the presence of workers on site. The contractor will be providing temporary toilet facilities for his personnel, which will have to be established in accordance with prevailing norms for such facilities to prevent contamination of the environment during the construction phase. No special mitigation measures are required.

4.6.7 Impacts of waste oil spills on the environment during construction and Operation period including barge movement, oil changing of equipment

Accidental spills, Leaks of fuels, lubricants, or concrete additives can contaminate surface water bodies, soil and shallow groundwater if not managed properly.

Servicing of all machinery will be done at off-site service stations. No servicing facilities will be located within the project site. Any accidental spills or environmental releases if they occur will be attended immediately complying to Safety Data Sheets (SDS) information.

4.7 Anticipated Impacts due to Waste Disposal

4.7.1 Anticipated Impacts due to Solid Waste Disposal

Construction Phase

Accumulation of construction waste may cause health impacts to the communities in the vicinity. If construction waste is not properly managed, washing out waste to sea, blockage of natural drainage paths and spreading of dust may lead to serious environmental pollution. Disposal of hazardous waste may cause adverse impacts to communities as well as natural ecosystems. In addition, the waste generated in the labour camps if any should be properly disposed of as



proposed in the migratory measures and disposal methods of solid waste are proposed in the solid waste management plan.

Waste disposed of in watercourses and beaches could lead to water pollution and cause threats to marine and aquatic life.

With all the above mentions, the anticipated impacts of solid waste disposal during the construction phase of the project can be rated as 'Moderate'.

The proposed measures to mitigate the impacts are presented in Chapter 6.

Operational Phase

Solid waste will be generated during the operation phase of the project will not occur at any wind tower locations, but will be limited to the substation area. These wastes, if not managed properly, may lead to soil contamination, surface water and groundwater pollution and may also generate odour as well as other negative visual.

The anticipated impacts of solid waste disposal during the operational phase of the project can be rated as 'Moderate'.

To mitigate the impacts, measures have been proposed in Chapter 5.

4.7.2 Anticipated Impact due to haphazard disposal

Haphazard disposal of solid and liquid waste, oils and grease, waste construction material, spent concrete mix etc. near sensitive habitats such as near Odais, freshwater pools etc. could pollute the water resources, damaging the ecosystem quality. This could be avoided with proper site management practices,

4.7.3 Anticipated Impacts of Siltation

Siltation impacts are not clearly expected / identified in the project.

4.8 Anticipated Socio-economic Impacts

This section outlines the project-related activities that may impact social receptors. Social receptors include people, places, assets, and structures that could be affected positively or negatively, directly or indirectly, by the Project. These receptors are identified through a combination of desk-based research and field studies, with consideration given to the potential impacts of the Project.



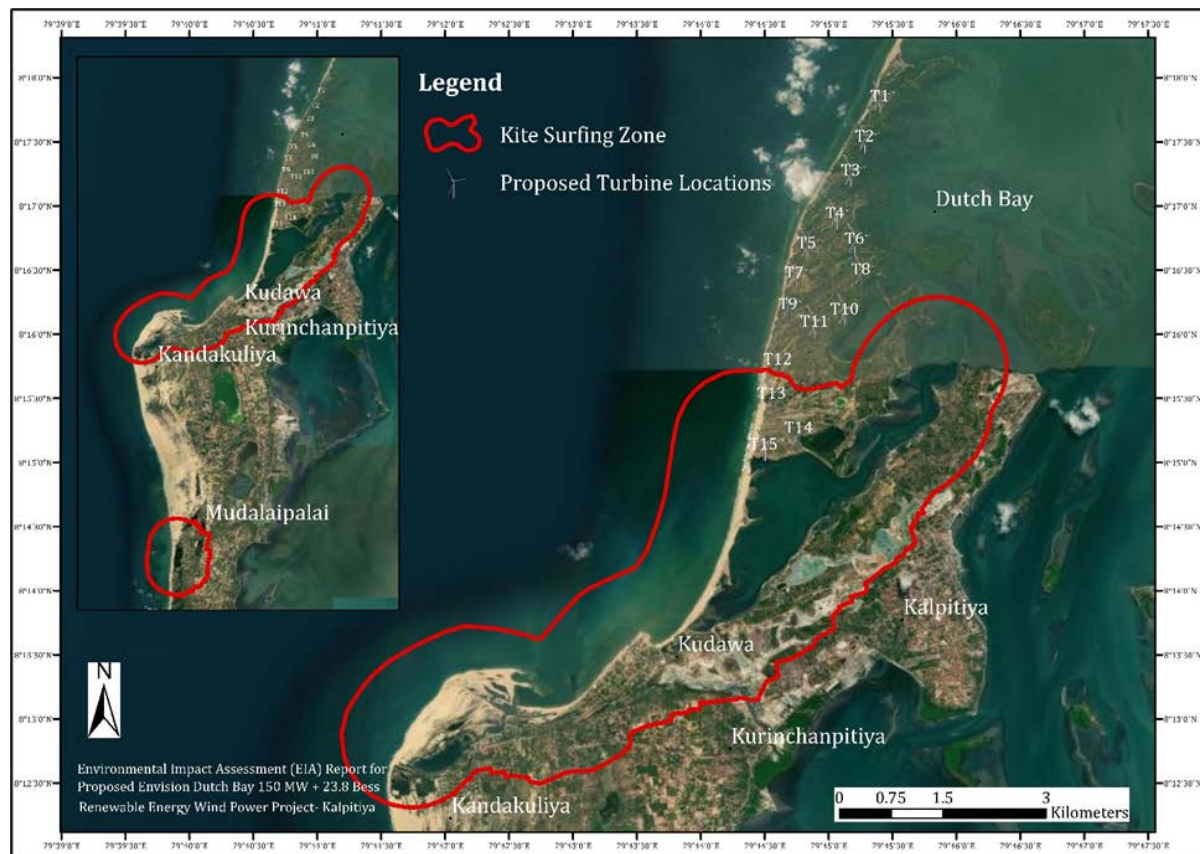
4.8.1 Anticipated Impact on land use and zoning

The Kalpitiya Development Plan 2019–2030, prepared by the Urban Development Authority (2019), and the Renewable Energy Resource Development Plan 2021–2026 by the Sustainable Energy Authority of Sri Lanka, have both identified the Kalpitiya Peninsula, including the Dutch Bay area, as suitable for wind power generation. Although the Kalpitiya Development Plan designated the Dutch Bay area as a “Beach-based Tourism Development Zone,” aiming to promote tourism-driven economic growth, this potential has not been fully realized due to significant environmental challenges. In this context, the proposed project is not expected to conflict with the local zoning regulations, nor will it alter the existing landscape or land use patterns in the project area.

According to discussions with the Assistant Director of the Urban Development Authority in Puttalam and the Resort Officer of the Sri Lanka Tourism Development Authority, it was revealed that the Dutch Bay area has been identified and designated as a tourism development zone under the new tourism development plan, which is currently being prepared but has not yet been gazetted.

Similarly, the Urban Development Authority (UDA), in collaboration with the Sri Lanka Tourism Development Authority (SLTDA), has been involved in preparing a zoning plan for kite surfing in the Kalpitiya area, as illustrated in the following figure. According to the draft zoning plan, a small portion of land overlaps with the locations identified for three wind turbines (T13, T14 and T15) as indicated in **Figure 4.2**.

However, this zoning plan has not yet been finalized or gazetted.



Source: Urban Development Authority(2025)

Figure 4.2: Draft Zoning Area for Kite surfing

4.8.1.1 Impact on Land Use

In the Dutch Bay Grama Niladhari Division (GND), only 2% of the total land area is currently used for home gardens, while coconut cultivation occupies approximately 11%, accounting for around 206 hectares. The proposed project, which involves the installation of 15 wind turbines along with the construction of access roads, improvement of the main access route, stockpiling areas, labour camps, and other auxiliary infrastructure, is estimated to require approximately 13 hectares of land.

Out of this, the permanent land take for project infrastructure, such as turbine foundations. Structure structure-free buffer area and permanent access roads, is expected to be approximately 9.7 hectares. The actual land requirement for different project interventions is presented in **Table 4-6**. This represents only a small percentage of the total land area within the GND and will primarily be drawn from land currently covered with scrub vegetation. Therefore, no significant impacts are anticipated on key land use categories such as mangroves, home gardens, or coconut plantations. The proposed locations of the wind turbines and approach roads are illustrated in **Figure 2.18**.



Table 4-6: Actual Land Requirement for different project interventions

Wind Turbine area		Total (ha)	Approach roads(ha)		
Foundation area for 15 Nos.	Structure-free Buffer area for 15 Nos.		Approach (14.5km x 4m) Roads	Main access road improvement	Other auxiliary facilities
0.66	2.94		9.02		

The land designated for the project, intended for activities such as the construction and installation of wind power turbines, the development of access roads, and the establishment of stockpiling areas, has either been purchased or leased. This land is currently unoccupied, lacking any existing structures, agricultural cultivation, or valuable tree cover, aside from minor scrub vegetation. As a result, the sensitivity of environmental and human receptors in this area is considered low, and any potential impacts from the project are expected to be negligible.

4.8.1.2 Anticipated Impacts on existing infrastructure facilities, properties, and crops within the study area due to the construction and operation of the project

The project area is situated within the Dutch Bay Grama Niladhari Division (GND), one of the administrative divisions under the Kalpitiya Divisional Secretariat Division (DSD). This region remains underdeveloped, with minimal public infrastructure and limited access to essential services such as proper road networks, reliable water supply, and other communal amenities. Notably, the only existing facilities include two schools located in Mohonthuwarama and a basic electricity supply provided via the national grid.

An existing unpaved sandy road that runs through the villages of Mohonthuwarama and Kirimundal, extending to Uchchamunai, is proposed to be used for transporting construction materials to the project site. Minor upgrades to this road will be undertaken to facilitate transportation. Given the current use and condition of the road, its sensitivity as a receptor is considered low, and any anticipated negative impacts during the construction phase are expected to be minimal and temporary. These impacts may include:

- Safety concerns for school children using the same road
- Temporary disruptions to the free flow of local vehicular traffic
- Reduced accessibility and mobility during the rainy season due to the movement of heavy vehicles



In addition to using the existing road network, the project includes the enhancement of the main access road, which will involve widening, culvert construction, and gravel surfacing as appropriate to improve its durability and usability. Furthermore, the project plans to introduce a motor ferry service to transport construction materials from Kalpitiya to the site via Thoridiya, using the Mohonthuwarama jetty to connect to the main access road. Upon completion of the construction phase, the motor ferry will be handed over to the Local Authority for use as a community transport service. **These infrastructure improvements will not only support the immediate logistical needs of the project but also deliver lasting benefits to the local population by enhancing overall accessibility, connectivity, and transportation efficiency in the area.**

In terms of land use, the Dutch Bay Grama Niladhari Division (GND) has a relatively low housing density, with approximately 19 housing units per square kilometer. The built-up areas are primarily concentrated in the villages of Penapidugamam and the Muslim village in Mohonthuwarama. The proposed locations for the installation of wind turbines are situated at a considerable distance from these settlements. As a result, **no direct impact on existing housing structures is anticipated.**

4.8.1.3 Anticipated Impacts on livelihood of the neighbouring communities

The primary livelihoods of the neighboring communities—particularly those residing in the villages of Kirimundal, Palliyawatta, and Uchchamunai to the north of the project site, and Kudawa and Kandakuliya to the south—are based on fishing, tourism, and related recreational service provisions. During the construction phase of the project, materials are expected to be transported via a sea route from Kandakuliya to the project site. This may result in minor, short-term disruptions to individuals involved in nearshore fishing and recreational activities, such as dolphin watching. However, the **anticipated impact is considered low in magnitude and temporary in duration.**

Furthermore, since most residents of Uchchamunai and Palliyawatta villages primarily rely on sea routes for transportation, the occasional use of land routes for material transport is not expected to significantly affect their livelihoods. **The overall impact on the socio-economic activities of these coastal communities during the construction phase is, therefore, expected to be minimal.**



4.8.1.4 Any relocation of families, if so, relocation sites facilities to be provided etc.

The wind turbines, along with associated auxiliary facilities and approach roads under the project, will be located on scrub land, barren land, or land with minimal vegetation, such as a few coconut trees. These lands have been or will be acquired through purchase or lease agreements by the project proponents. Currently, there are no houses, buildings, or other permanent or temporary structures present on these lands. Accordingly, no land acquisition from private owners is envisaged for the project's purposes, and there will be no requirement for either voluntary or involuntary resettlement. As a result, **no physical displacement of people is anticipated.**

4.8.1.5 Anticipated Impacts on fishery/fishery farms/fishing community/ongoing aquaculture activities

The siting of wind turbines and associated auxiliary facilities, including the construction and upgrading of approach roads, will be carefully planned to avoid areas with fishery-related activities. Additionally, all project-related structures will be located with a setback of at least 60 meters from the permanent vegetation line along the beachfront, as stipulated in the Sri Lanka Coastal Zone and Coastal Resources Management Plan 2024 (CZ&CRMP 2024).

Existing fishing activities, including access to landing sites, beach seine (Madel) sites, and fishing camps (Vadis), along the coastline within the Project Area of Influence (AOI) will not be disrupted. It is noted that six beach seine sites are currently operating within the project AOI. These traditional fishing operations occur seasonally, from September to May each year.

According to information provided by the beach seine operators, each operation requires approximately 1–1.5 kilometres of nearshore sea area free of disturbances to successfully deploy and retrieve nets. The operations typically utilize tractors and mechanical winches to pull in large nets, supported by a workforce ranging from 20 to 60 individuals, depending on the scale of the operation.

As a key social receptor, the beach seine operations within the AOI are classified as having low to moderate sensitivity. Potential temporary impacts may arise during the construction period, primarily due to transportation and unloading of construction materials via the sea route using barges.

Additionally, a temporary pier or jetty is planned to be constructed near the Bay Watch Hotel and the Navy security point to facilitate the unloading of materials transported via the sea route to the project sites. This facility will be used primarily during the construction phase to streamline



the delivery of heavy equipment and materials. As a result of these activities, **there is a potential for low to moderate levels of temporary impact on nearby fishing operations**. These impacts may include restricted access to certain areas, temporary congestion near landing sites, and minor disruptions to traditional fishing patterns, particularly during periods of high construction activity.

Furthermore, throughout the construction phase, the project will continuously monitor the situation and assess any unanticipated temporary effects on fishing activities, particularly regarding access to landing sites and fishing camps. Where such impacts are identified, additional mitigation measures, including timely compensation for affected parties, will be put in place to address any disruptions. This adaptive management approach will ensure that the livelihoods of local fishing communities are respected and preserved during project implementation.

4.8.1.6 Anticipated Impacts on fisheries resources in the area during the operation of the project

According to the findings of the social survey and interviews conducted with beach seine operators within the Project Area of Influence (AOI), it was observed that the majority of respondents do not anticipate any significant impacts on fishery resources as a result of the operation of wind turbines in the coastal areas. The fishermen expressed confidence that the functioning of the turbines would not interfere with fish stocks, fishing grounds, or their traditional fishing practices.

However, some members of the fishing community raised concerns regarding potential sound emissions from the wind turbines. While they did not foresee direct impacts on fishery resources, there was uneasiness about the possible disturbance caused by turbine noise, both to marine life and to the quality of their working environment. Refer section 4.4 for this perspective.

4.8.1.7 Anticipated Impacts on Palmyrah trees, coconut cultivation/plantation due to the erection of wind towers and transmission lines

Palmyrah and coconut trees are sparsely distributed within the project area, with only a limited number observed in the buffer zones surrounding the proposed turbine locations. It is anticipated that these trees will not need to be removed during the construction phase. Careful planning and alignment of approach roads and construction activities will be undertaken to avoid disturbing or removing Palmyrah and coconut trees present in these buffer areas. If the removal of any trees becomes unavoidable, an equivalent number of trees will be replanted as a mitigation measure.



to offset the environmental impact. **Refer Section 3.3.8 Ecological Assessment of the Project Impacting Area for further details.**

4.8.1.8 Anticipated Impacts on agricultural activities in the project area

Agricultural activities within the Dutch Bay Grama Niladhari Division (GND) are primarily limited to small-scale home gardens. No commercial-scale agricultural operations are currently taking place within the project's Area of Influence (AOI), with the exception of coconut cultivation, as detailed in the preceding section.

For the development of the project, a few parcels of land have been either purchased or leased to accommodate the installation of wind turbines, the construction of access roads, and other supporting infrastructure. These selected lands contain a limited number of coconut trees. However, the area required for the construction of the wind turbine foundations is relatively small. As such, **no significant adverse impacts on the existing coconut plantations are anticipated.** Efforts will also be made to minimize any disturbance to the surrounding vegetation during the construction phase.

4.8.1.9 Anticipated Impacts on tourism/tourism activities in the area, including kite surfing etc.

The direct impact of the project on tourism and tourism-related activities within the Area of Influence (AOI) is considered negligible, based on two key factors. Firstly, the active tourism infrastructure including accommodation and recreational facilities is primarily concentrated in Kudawa, Kandakuliya, and other locations outside the immediate project area. As a result, the core zones of tourism activity remain unaffected by project operations. Secondly, within the Dutch Bay area itself, the only two previously existing tourism-related facilities, the Bay Watch Hotel and the Sand Hotel, have been significantly damaged and abandoned, primarily due to sea erosion and other natural factors. Consequently, no functional tourism establishments currently exist within the project AOI.

However, a minor visual impact may be experienced by tourism facilities located on the Kalpitiya mainland, outside the Dutch Bay area, particularly in areas with direct line-of-sight to the project site (T14 and T15). These visual effects are expected to be minimal and are not anticipated to substantially affect the overall attractiveness of the region for tourists.



4.8.1.10 Impacts on access to the beach and Dutch Bay/lagoon

Lateral and vertical public access to and along the beach or the lagoon wedge will remain unobstructed throughout the project lifecycle. The siting of wind turbines, stockpiling of construction materials, transportation activities, and all other project-related operations will be carefully planned and managed to ensure they do not disrupt or hinder public access. Furthermore, the turbine locations will not be enclosed by fencing, thereby maintaining open access in the surrounding areas. **All turbine installations will respect a minimum coastal setback of 60 meters from the shoreline and will ensure an adequate reservation area is maintained along the lagoon, in compliance with set-back standards and planning regulations.**

4.8.1.11 Anticipated effects from Shadow flickering

Shadow-flicker from wind turbines is defined as alternating changes in light intensity caused by rotating blades casting shadows on the ground and stationary objects such as a window at a dwelling. Shadow flicker at a receptor may occur, only when the following four conditions are met:

- The sun is shining with no cloud cover present,
- the turbine is operating,
- the turbine blades are positioned on a line between the sun and the receptor,
- the receptor is close enough to the turbine to distinguish the shadow created by the blades.

However, obstacles such as terrain, trees, or buildings between the wind turbine and a potential shadow-flicker receptor significantly reduce or eliminate shadow-flicker effects. There are no houses located at least 300 m from all turbine locations in the proposed project. Therefore, **the impacts of shadow flicker are not envisaged**. Shadow flicker is very localised. In addition, the sensitive communities are located bit far from the turbine locations.

4.8.1.12 Anticipated Impacts on Blade Accidents / Fall Hazards etc.

Blade fall hazards would occur only due to improper maintenance or due to design failures. These kinds of impacts can be prevented with proper care during design and maintenance.

However, the wind turbines have been located reasonably away from the permanent settlements to avoid such issues. The hazards that could occur during the construction period and the defect liability period will be covered by the insurance of the contractor under the civil work contract.



The ENVISION will purchase an insurance policy directly or indirectly through the operation and maintenance contract to cover these hazards.

4.8.1.13 Anticipated Impacts on scenic beauty and visual environment

During the Focus Group Discussions (FGDs) conducted with stakeholders involved in tourism and kitesurfing operations in the Kudawa and Kandakuliya coastal areas, concerns were raised regarding the potential visual impacts of wind turbines on the surrounding environment and scenic beauty. These impacts largely depend on factors such as the siting location of the turbines, their colour, height, and the number installed in a given area. Stakeholders expressed particular concern about the visual impacts on tourist establishments (Kite Land, Arasi Resort, Vila Sethawadi Hotel, De Silva Kite Surfing School and Hotel, and the Blue Lagoon Resort) located along the eastern side of the Kudawa lagoon, which is a popular area for kitesurfing. Currently, a significant number of wind turbines are already situated on the land located eastern side of the lagoon and are prominently visible from tourist establishments within the surrounding Grama Niladhari Divisions (GNDs), including Kudawa-Kandakuliya and Kudawa. To address these concerns, it is anticipated that careful planning will be undertaken to optimize the siting and layout of future wind turbine installations. This will include strategies to minimize adverse visual and aesthetic impacts, to preserve the natural scenic value of the area.

4.8.1.14 Employment generation during construction and operation

According to the findings of the social survey and focus group discussions (FGDs), one of the key expectations expressed by the local community is the creation of employment opportunities as a direct benefit of the proposed wind power project. This expectation is grounded in the area's socioeconomic context, particularly in the Dutch Bay Grama Niladhari Division (GND), which currently faces a high unemployment rate of approximately 33.2%.

In practical terms, the project is expected to generate both direct and indirect employment opportunities during the construction phase, which is projected to last between 12 to 18 months, as well as throughout the operational phase, which may extend up to 22 years. Employment will be available across a range of job categories, including both skilled (professional and technical roles) and unskilled labour positions.

However, considering the prevailing educational and skills levels within the Dutch Bay GND, it is anticipated that a greater share of local employment will be in unskilled job categories



4.8.1.15 Anticipated any impact on the proposed or planned government /private sector development activities in the vicinity

According to consultations conducted with the Urban Development Authority (UDA), the Sri Lanka Tourism Development Authority (SLTDA), and the Acting Divisional Secretary of the Kalpitiya Divisional Secretariat Division (DSD), it was confirmed that there are no ongoing or planned government or private sector development activities within the immediate footprint of the proposed project area. Furthermore, the Resort Officer (Kalpitiya) of the SLTDA indicated that eight out of the twelve islands located within the lagoon area have been leased out for tourism development. However, to date, only one of these projects has commenced construction activities, specifically the development of water bungalows on one of the islands. These islands are situated at a considerable distance from the proposed project site. As such, **only a minor visual impact from the wind turbines is anticipated on these tourism developments**, with no direct interference or overlap with their operations.

4.8.1.16 Anticipated Impacts due to transportation of material and equipment, and an increase in traffic volumes during the construction period

The transportation of materials, equipment, and labor required for the project presents unique logistical challenges due to the remote location of the proposed wind turbine siting plots. Unlike typical development sites, these areas lack adequate land-based transportation infrastructure, as illustrated in the following figures. As a result, a combination of land and sea routes must be utilized during the construction phase to ensure the timely delivery of essential components. Key materials and equipment needed for the construction of approach roads, turbine foundations, and temporary labor camps include cement, sand, gravel, rock aggregates, steel, and heavy machinery such as bulldozers, graders, excavators, front-end loaders, compactors, and dump trucks. Additionally, the transport of oversized components, including large wind turbine parts and blades, will require special logistical arrangements.

Given the site conditions and the surrounding environment, it is planned to use both land and sea transport routes. When utilizing the sea route, oversized turbine parts, large blades, and heavy equipment will be transported via barges. This approach is expected to result in minor to moderate impacts on six active beach seine fishing operations along the Dutch Bay Fisheries Inspector Division (FID) coastline, as well as on nearshore fishing activities.

Furthermore, temporary and minor to moderate impacts are anticipated for tour boat operators engaged in dolphin and whale watching in the Bar Reef area and its vicinity. Additionally, minor disruptions may occur at fishing and beach seine operation sites during the unloading of



materials at the designated beach area near the Bay Watch Hotel. The anticipated impact needed to be considered carefully, because transportation of materials needs to be conducted during the non-monsoon period where the fishing and recreational activities also largely conducted. On land, it is expected that vehicular movement along the sandy road passing through Mohoththwarama to Kirimundal will be temporarily disrupted during the transport of materials and equipment throughout the construction period.

The anticipated impacts must be carefully evaluated, as the transportation of materials is scheduled to take place during the non-monsoon period, which coincides with the peak season for both fishing and recreational activities in the area. This overlap raises the potential for increased conflicts between project logistics and the livelihoods of local fishing communities, as well as tourism operators who rely on uninterrupted access to marine and coastal resources.



Figure 4.3: Sandy Road traverses through Mohoththuwarama village

4.8.1.17 Anticipated Impacts on roads, buildings, and other properties during the transportation of material, equipment during the construction period

The materials and equipment required for the construction of approach roads, turbine foundations, and temporary labor camps include cement, sand, gravel, rock aggregates, steel, and heavy machinery such as bulldozers, graders, excavators, front-end loaders, compactors, and dump trucks. To facilitate the delivery of these materials to the designated turbine construction plots, the existing road network will be utilized, with planned improvements including road widening and the construction of culverts.

As the majority of the existing road network is unpaved and primarily composed of sandy terrain, except for a short concrete segment of less than 1 km within the Penapidugama interior, no



significant negative impacts on the road infrastructure are anticipated. On the contrary, the proposed upgrades, including widening and culvert installation, are expected to result in positive impacts by enhancing local road conditions and improving accessibility for both project-related and community use.

The buildings located on both sides of the main access road consist primarily of single-story dwellings, both permanent and semi-permanent, concentrated in the villages of Mohoththuwrama and Kirimundal. Additionally, a Hindu Kovil, a church, fishing camps (Vadis), and Cadjan huts constructed near the kite surfing training site are also situated close to the access road. The movement of heavy vehicles and the transportation of materials along this road may **result in minor to moderate impacts, such as structural damage caused by vibrations.**

Furthermore, the transportation of materials and equipment by motor ferry from Thoraidiya to Mohonthuwarama, via the lagoon and using the existing jetty, may pose certain negative impacts. One concern is the potential disruption to schoolchildren who rely on the current boat services to travel to schools in Kalpitiya, as increased ferry traffic could interfere with their daily commute and compromise safety. In addition, the frequent movement of heavy vehicles along the adjacent roads may lead to adverse effects on the surrounding built environment, particularly the existing buildings located on both sides of the road, which may be vulnerable to structural damage caused by sustained vibrations and increased vehicular load.

4.8.1.18 Anticipated Impacts due to waste and hazardous waste

Generation of Solid wastes during the Construction Phase

Accumulation of construction waste may cause health impacts to the communities in the vicinity. If construction waste is not properly managed, washing out waste to sea, blockage of natural drainage paths and spreading of dust may lead to serious environmental pollution.

Disposal of hazardous waste may cause adverse impacts to communities as well as natural ecosystems. Waste disposed of in watercourses and beaches could lead to water pollution and cause threats to marine and aquatic life.

In addition, the waste generated in the labour camps if any should be properly disposed of as proposed in the migratory measures and disposal methods of solid waste are proposed in the solid waste management plan.

With all the above mentions, the anticipated impacts of solid waste disposal during the construction phase of the project can be rated as 'Moderate'.



The proposed measures to mitigate the impacts are presented in Chapter 5.

There are no waste generation sources during the operational phase of the project.

4.8.1.19 Public and occupational health/safety

Wind power plants provide clean, renewable electricity to communities, reducing their reliance on fossil fuels. Like any other kind of construction project, the project will pose some health and safety risks to both the public and the people doing the installation and maintenance of the turbines.

During the construction stage, employees may be exposed to a range of occupational health and safety concerns, including threats associated with heavy machinery and equipment, electrical hazards, and falls from great heights. The project will implement strict health and safety regulations, provide staff with the required training and personal protective equipment, and ensure that all machinery and equipment are maintained in excellent working order.

The project team has also ensured that the neighborhood will not face any health or safety risks from the infrastructure surrounding the wind turbines. The placements of the turbines were chosen to prevent or reduce the possibility of danger or damage from debris or turbine failure.

One of the key concerns is noise pollution because it can have a negative impact on both mental and physical health. The wind turbines have been located safely away from populated and settled regions to reduce noise pollution.

The construction of wind power projects can have substantial effects on the local labour market and workforce, given the lack of trained and unskilled workers available in that project location. The project area's public health would suffer from the influx of workforce, particularly communicable diseases.

4.8.1.20 Anticipated Impacts on security activities in the area

The project itself will not introduce new security systems or structures and will fully adhere to local and national security policies and regulations in the area. There are no plans for fencing or other security-related constructions associated with the project.

However, it is important to mention that wind turbines have the potential to impact radar systems, regardless of their type. Previous studies have demonstrated that wind projects can affect various types of radar systems, including military, commercial aviation, weather, and border surveillance radar.



To address this concern, the developer has taken proactive steps by consulting with the local security forces. Because of these consultations, the turbine locations have been adjusted to mitigate any potential disturbances to radar operations. This demonstrates the project's commitment to ensuring that security systems in the area continue to function effectively while harnessing wind energy for sustainable power generation.

4.9 Anticipated Impacts on navigation of air crafts

In general, wind turbine's physical size, in particular their height, wind farms can have an effect on aviation. Additionally, rotating wind turbine blades may have an impact on certain aviation operations, particularly those involving radar. Low-flying general aviation and military aircraft need to avoid wind turbines, resulting in the channeling and concentration of traffic into narrow corridors. If inappropriately lit or marked, wind turbines can pose a hazard to low-level aircraft, particularly at night.

As the project area does not fall directly within an air traffic route, there will not be any adverse effects on aircraft navigation. Since some adjacent routes are possible, ENVISION will obtain necessary principal approvals from Civil Aviation Authority as appropriately if required.

4.10 Anticipated Impacts on land excavation for transmission lines

The EIA related to the transmission lines will be done after CEB approves the proposal.

4.11 Anticipated Historical and archaeological impacts

The proposed project does not create impacts on any historical and archaeological sites.

4.12 Anticipated any other impacts not listed here but may be significant in view of the project proponent

EIA related to the proposed Transmission line will be carried out as a separate study after CEB approves the proposal.



4.13 Anticipated Risk of oil spills

The risk of oil spills or hydrocarbon leakage during marine transportation is considered minimal, as no refuelling activities will occur at the temporary jetty. All transport barges will be pre-fuelled at the Colombo Harbour and will return there for refuelling under port-controlled protocols. Nevertheless, comprehensive safeguards will be in place as a preventive measure. All barges will be required to carry marine-grade oil containment booms, absorbent pads, and emergency spill kits on board. The jetty will maintain a standby oil-leakage pool and sediment trap around its structural perimeter. In the unlikely event of a spill, a pre-approved Oil Spill Response Plan (OSRP) will be activated, which includes immediate containment using floating barriers, deployment of sorbents, and coordinated removal using skimmers or vacuum units. CC&CRMD guidelines and IMO MARPOL Annex I recommendations for coastal oil pollution prevention will be strictly followed, with incident reporting protocols integrated into the Environmental Management Plan (EMP) and project-specific Emergency Response Plan (ERP).



5 PROPOSED MITIGATION MEASURES

5.1 Impact Mitigation and Impact Management

Once the significant negative potential impacts (moderate and major) were characterized, the next step was the development of mitigation, enhancement, and monitoring measures. Under the mitigation hierarchy, preference is given first to measures in the following sequence:

- Avoid at source, reduce at source—avoiding or reducing negative potential impacts from the project through the design of the project or through engineering control to prevent unplanned events;
- Abate on site: add something to the design to abate the potential impact (e.g., stockpiling arrangement, pollution control equipment, sea traffic controls, etc.);
- Abate at the receptor: if a negative potential impact cannot be abated on site, then control measures can be implemented off-site (e.g., rocks and boulders to be put in place will be cleaned and washed prior to placing them in the sea to prevent spreading invasive plant species).
- Repair or remedy: some potentially significant impacts involve the potential for unavoidable damage to a resource (e.g., oil spill, water quality deterioration, etc.), and these potential impacts can be addressed through repair, restoration, or reinstatement measures if they occur;
- Compensate or offset: where none of the above approaches are practicable, compensate for potential project impacts by, for example, replacing loss or damage at another location (e.g., restoration of dune vegetation).

For positive potential impacts or project benefits, the EIA team identified opportunities to enhance potential project benefits. In developing the mitigation and enhancement measures, the following points were considered:

- The effectiveness of each potential mitigation measure was assessed and potential synergies identified (i.e., where one measure mitigates several potential impacts, e.g., sediment control);
- The level of impact reduction or benefit enhancement that results from implementing each potential mitigation measure was determined, e.g., by employing hybrid structures;
- The preferred measures were selected based on the previously described analysis, e.g., soft structures;



- All mitigation and benefit enhancement alternatives were reviewed to:
 - Determine if there are potential conflicts or adverse cumulative effects;
 - Determine if there are potential implications for other project parameters, such as safety, security, livelihood, or stakeholder relationships;
 - Identify potential synergies.
- The plan also identified the monitoring to be conducted during the site development, construction and operation stages of the project to provide the data needed to assess:
 - Actual project impacts against those evaluated in the impact assessment;
 - The effectiveness of mitigation and enhancement measures being implemented
- When developing mitigation, enhancement, and monitoring measures, the plan clearly distinguishes:
 - The measures implemented to mitigate potential negative impacts resulting from the project
 - The measures taken to address potential multi-factorial indirect effects
 - The measures taken to enhance the potential positive project impact

5.2 Proposed Mitigation measures for the Identified Impacts on Coastal Environment

As it is described in Section 4.3 , the project activities would not create considerable impact on the coastal environment. However, it is suggested to prevent the **potential erosion / accretion** during the construction phase, by following the below measures into consideration,

- provision of sediment traps
- construction activities to be carried out during the dry season and calm weather
- original beach profile be restored after construction activities

Further, in order to prevent Degradation of Sea Water Quality following measures are to be taken.

- Avoiding spilling overboard during loading, transit and unloading construction materials
- provision of sediment traps
- Any accidental spills or environmental releases that occur should be attended immediately



It is recommended that proper construction practices (during roads construction) in the form of the usage of minimal areas required for constructions, provision of drainage systems-with silt traps where necessary, non-obstruction of existing drainage paths/flow in water bodies-with culverts provided where necessary-and restoration of the affected area as much as possible after the construction stage need to be adopted as impact mitigation measures.

It is also recommended that these measures be put in place prior to the rainy season and regular monitoring and maintenance activities be carried out during the construction stage. Facilitating the regrowth of vegetation cover in the areas cleared for construction purposes is also recommended to minimize soil erosion. Moreover, should conduct beach profile monitoring.

5.3 Proposed Mitigatory Measures- Noise and Vibration

Site noise management measures during construction period

The identified potential noise impacts could be easily mitigated by applying the following mitigatory measures.

- Noise-generating activities during the night time will create annoyance in the neighbourhood. Hence, construction activities should be restricted to daytime (between 7 am and 5 pm), especially for the noise-generating activities and use of noise-generating machinery. Thus, during night hours there will not be a nuisance to the people within the area due to noise-generating activities.
- The contractor will have to ensure that equipment and machinery deployed for the works and vehicles used for the transportation of materials at the site confirm statutory noise levels.
- Planning of alternative routes will be done to minimize traffic during the construction period (e.g. diverting traffic away from noise-sensitive areas using existing alternative/bypass roads)
- On the construction site, workers who may be exposed to a high level of noise for a long period of time must be provided with appropriate personal protective equipment as per the Occupational Health and Safety Act.

The identified potential vibration impacts could be easily mitigated by applying the following mitigatory measures.



- As much as practicable, construction methods that produce minimal vibration will be adopted, especially in sites adjacent to residential and institutional areas where possible damage to the structures due to vibration may occur.
- Pre-crack surveys will be carried out for any identified sensitive receptors in the area before construction work starts and after the work is complete. In case of building cracks due to project induced vibration, the affected party will be immediately informed, and the damage repaired at their own cost with reimbursement of the cost, or the damage will be compensated without delay.

5.4 Proposed Mitigatory Measures for Identified Ecological Impacts

To minimize the adverse effects of project activities on natural habitats—including coastal scrublands, grasslands, and associated ecosystems—the following mitigation measures are recommended:

5.4.1 Proposed Minimization measures for habitat loss

Minimizing habitat loss is a primary goal during project design and execution. This involves careful planning to avoid ecologically sensitive areas and reduce the overall footprint of development. Key actions include:

- Avoidance of high-value habitats such as native coastal scrublands, grasslands, mangroves, and seasonal wetlands such as salt marshes through detailed baseline ecological surveys and geospatial mapping.
- Optimizing infrastructure placement by locating turbines, access roads, substations, and storage facilities in already degraded or previously cleared lands, thereby preserving intact ecosystems.
- Reducing construction footprints through compact or linear site layouts, multi-use corridors and limiting the width of access paths.
- Implementing a no-go zone policy around ecologically sensitive patches identified during Environmental Impact Assessments (EIAs), especially areas of sensitive habitats or that host rare or endangered species.

Habitat fragmentation management

Fragmentation of habitats can severely disrupt ecological connectivity and wildlife movement. Mitigation focuses on maintaining continuity and minimizing barriers. Actions include:



- Prepare Landscape Conservation Plan is to conserve and restore ecological integrity within and around the project area, Coordinate with landscape-scale conservation plans to align turbine siting and access paths with existing ecological networks or green belts.
- Stagger construction activities to reduce the extent of active disturbance at any one time, thereby allowing wildlife to adapt and move through less disturbed zones.

Edge effect reduction

The creation of edges between disturbed and undisturbed areas introduces changes in temperature, humidity, light penetration, and wind exposure, often leading to habitat degradation. To mitigate edge effects:

- Maintain buffer strips of natural vegetation along all cleared boundaries to serve as transitional zones that reduce abrupt changes in microclimate and protect core habitats.
- Use native vegetation for restoration of cleared margins to enhance structural continuity and ecological function, promoting the return of native fauna and flora.
- .
- To address edge effects and potential microclimate shifts, turbines bordering native habitats will be buffered by replanting strips of native shrub and coastal palm belts.

Invasive species prevention and reduction

Disturbed soils, construction materials, and machinery can introduce invasive species that outcompete native vegetation and alter ecosystem dynamics. Preventative strategies include:

- Strict sanitation protocols for construction machinery and vehicles entering and exiting the site, including pressure-washing equipment and checking plant materials.
- Material sourcing protocols to ensure soil, gravel, or fill is free from invasive propagules or seeds.
- Training construction personnel and site managers to recognize invasive species and report sightings for prompt action.

Invasive species control will be executed via strictly manual and mechanical methods – including removal of *Prosopis juliflora* and other scrub – without the use of herbicides. A quarterly re-clearing regime will be enforced during the first 3 years.



Post-construction rehabilitation

After construction is complete, disturbed areas must be actively restored to recover ecological function and biodiversity. This step is critical to ensuring long-term sustainability:

- Replant native vegetation suited to local soil and climatic conditions, with a focus on species that were originally present or that support local wildlife.
- Establish monitoring programs with clear indicators for vegetation cover, species diversity, and soil health, and adjust management actions based on performance outcomes.
- Engage local communities or conservation groups in restoration efforts to build stewardship and enhance local ecological knowledge.

5.4.2 Proposed Mitigation measures for impacts on vegetation

The construction and operation of the project pose several threats to local vegetation, including direct removal, soil degradation, and the spread of invasive species. The following detailed mitigation strategies are proposed to minimize these impacts and ensure long-term ecological sustainability of the site.

5.4.2.1 Mitigating direct removal of native vegetation

To address the potential loss of native flora, especially salt-tolerant dune vegetation and coastal plant communities the following measures are proposed:

- Adjust project design and site layout through micro-siting techniques to avoid high-conservation-value vegetation areas wherever possible. Prioritize placing infrastructure (e.g., turbine pads, access roads, substations) in already degraded or low-biodiversity zones.
- Define and demarcate conservation zones and vegetation buffers clearly in the field using signage and fencing to prevent accidental clearing by construction crews.
- Apply a phased clearing strategy, only removing vegetation in areas immediately needed for active construction. This reduces unnecessary loss and gives other parts of the site time to recover or be restored earlier.
- Promote on-site translocation of rare or slow-growing plant species, where feasible, from high-impact zones to safe, managed receptor sites under ecological supervision.



5.4.2.2 Preventing and managing invasive plant species

Disturbed land is highly susceptible to colonization by invasive alien species, which can outcompete native flora and degrade habitats. To minimize this risk:

- Enforce strict biosecurity protocols requiring all machinery, vehicles, and materials entering the site to be cleaned and inspected to prevent the introduction of invasive plant seeds or propagules.
- Establish an invasive species monitoring program, beginning at the start of construction and continuing into the operational phase, to detect and respond to early invasions rapidly.
- Employ only manual or mechanical control methods (e.g., uprooting, cutting, or slashing) for the management of invasive plant species. The use of chemical herbicides or pesticides shall be strictly prohibited under all circumstances, to prevent contamination of soil, water bodies, and adjacent sensitive ecosystems.
- Encourage dense and early revegetation of disturbed areas using fast-growing native groundcovers and grasses that can act as a natural barrier against invasives.
- Train site personnel, contractors, and community workers in the identification of priority invasive species (e.g., *Prosopis juliflora*, *Opuntia dillenii*) and in basic removal methods.
- Encourage replanting with native species in disturbed areas as a natural defense against invasive colonization, while simultaneously restoring habitats for dependent fauna.
-

Together, these mitigation strategies aim to ensure that vegetation loss is minimized, ecological balance is maintained, and the landscape's resilience to long-term disturbance is strengthened. Active ecological oversight and periodic monitoring will be essential to ensure these measures are implemented effectively and adaptively over time.

5.4.3 Proposed Mitigation Measures for Impacts on Fauna

Project development and construction activities have the potential to adversely affect terrestrial and aquatic fauna through habitat loss, noise, pollution, light disturbance, and direct physical harm. To minimize such impacts, the following comprehensive and practical mitigation measures are proposed:



5.4.3.1 Minimize habitat loss of fauna

To preserve the integrity of faunal habitats and maintain ecological connectivity across the landscape:

- Preserve natural vegetation buffers along natural drainage paths, mangroves, forest margins, and known animal routes. These green corridors act as both habitat and movement pathways for many species, especially small mammals, reptiles, and amphibians.
- Schedule construction activities outside of critical biological periods, such as breeding or nesting seasons for key faunal groups (e.g., migratory birds, turtles, and amphibians). Timing restrictions can significantly reduce reproductive disruption and stress on wildlife.
- Designate and enforce “no-go” zones within or adjacent to the project area, based on ecological survey data. These zones should be physically demarcated, mapped in construction plans, and communicated to all personnel.

5.4.3.2 Proposed Mitigation Measures for Disturbances during construction

To prevent stress or displacement of sensitive wildlife:

- Use low-noise construction equipment and install noise suppressors or silencers on heavy machinery. Activities should be restricted to daylight hours to minimize disruption to nocturnal species.
- Limit worker access and movement in high-sensitivity ecological areas, especially during dawn and dusk when many species are most active. Human presence should be kept to a minimum in riparian zones, nesting sites, and roosting areas.
- Conduct mandatory environmental awareness and wildlife sensitivity training for all site workers and contractors. Training should emphasize the importance of minimizing disturbance and reporting wildlife sightings or incidents.

To reduce disorientation and disruption to nocturnal and migratory species:

- Avoid or limit artificial lighting during night-time, particularly near water bodies, forest edges, and known bird roosts or nesting sites. Bright lights can interfere with natural behaviors such as navigation, foraging, and predator avoidance.
- Use low-intensity, downward-shielded lighting in areas where illumination is required for operational safety or site security. This measure helps minimize light spill into



surrounding habitats, reducing disturbance to nocturnal wildlife and maintaining the natural behavior patterns of sensitive species.

5.4.3.3 Prevent direct wildlife mortality

To avoid accidental injury or death of fauna during site preparation and construction:

- Inspect all work areas before initiation of clearing or excavation, and safely relocate any animals discovered. A trained wildlife handler or environmental officer should supervise this process.
- Cover or install escape ramps in open pits, trenches, or storage tanks to prevent entrapment of small animals. Regular checks should be conducted, especially overnight.
- Implement a low-speed limit within the construction zone and post wildlife crossing signage to alert drivers and reduce vehicle-wildlife collisions.

5.4.3.4 Maintain water quality and aquatic habitat integrity

To protect aquatic species and prevent downstream ecological degradation:

- Implement erosion and sediment control measures, such as silt fences, sediment traps, and vegetative buffer strips, especially near drainage lines, lagoons, or estuarine habitats.
- Ensure proper management of solid and liquid waste on-site, including designated disposal and spill containment areas, to prevent pollutants from entering nearby wetlands, lagoon or marine systems.
- Monitor water quality regularly in adjacent aquatic habitats for signs of turbidity, chemical and oil contamination, or eutrophication that could affect fish, invertebrates, and amphibians.

By incorporating these fauna-sensitive mitigation strategies into all stages of project planning and execution, adverse effects on local and migratory species can be significantly minimized. Regular ecological monitoring, adaptive management, and enforcement of mitigation commitments are critical to ensuring long-term biodiversity conservation in and around the project area.



5.4.4 Proposed Mitigatory measures to reducing bird mortality at wind energy facilities

Based on global studies, several methods have been identified to reduce bird collisions with wind turbines. One of the key approaches to addressing this issue is conducting a thorough risk assessment. The primary objective of such an assessment is to carry out cumulative collision risk modeling for the proposed projects. This will assist the EIA consultant in preparing the ornithological component of the Environmental Impact Assessment (EIA). The assessment is intended to support in evaluating potential impacts on bird populations associated with both the proposed power evacuation infrastructure and the broader large-scale wind power developments.

As recommended by the project proponent, an avifauna detection and control mechanism has been proposed for implementation. A radar-based automated early warning system will be installed to mitigate bird collisions across the entire wind farm area. This system utilizes a multi-sensor bird detection setup, which triggers curtailment actions at both the plant and turbine levels through integration with the Supervisory Control and Data Acquisition (SCADA) software of the Wind Turbine Generators (WTGs). Based on our observations along the Dutch Bay area, this system can be installed in key turbine locations following a detailed feasibility study to be conducted during the migratory season, ensuring minimal impact on bird movement and enhancing conservation efforts.

Construction activities for the wind farm will be prohibited during the peak wintering season for waterfowl, which will help to significantly reduce the number of birds at risk of disturbance during this sensitive period. Shutdown or curtail operations during peak migratory periods (October–March). Try to avoid breeding period of native birds from May to August. However, during the operational phase, ongoing disturbance may still occur and has the potential to displace birds, particularly those utilizing the beach and other coastal habitats. Such displacement could lead to reduced access to important feeding or roosting areas, thereby impacting local and migratory bird populations over time. Use bird-friendly lighting to reduce disorientation.



5.5 Proposed Mitigatory Measures for Hydrological and Water Quality impacts

5.5.1 Mitigation measures for disturbances for the natural drainage pattern

5.5.1.1 Drainage Management Plan for access Roads

Drainage Management Plan will be developed for the construction phase of the access roads and road design processes will ensure proper conveyance of flow paths across the roads with necessary hydraulic structures even during the operational stage. The access road designs will consider the existing natural flow paths across the roads if any and ensure the safe conveyance without backwatering through designs.

5.5.1.2 Drainage Management Plan for wind tower sites

No drainage management plans are required for the wind tower sites during operation, however, considerations should be paid during the construction period.

5.5.2 Mitigating Measures for the Anticipated Water Quality Impacts

All of the impacts identified above can either be eliminated or mitigated to minimize the negative effect of water quality deterioration on the environment. The following mitigation measures are designed to be practical and cost-effective in achieving compliance with water quality standards and should be undertaken by the construction contractor during the construction period to mitigate the potential impacts;

- The construction activities will have to be planned in such a way to minimize sediments from fill material to be carried away into the sea/lagoon water body, the stretch of the water area at the site would be fenced with a silt curtains to limit the spread of sediments away from the working area. This measure will provide better control over sediment propagation and limit the impact of a change in sediment level in the lagoon near shore.
- Cover stockpiles of soil and sand to minimize exposure to wind and water.
- Scheduling construction to minimize soil exposure during the wet season or to limit exposure times overall
- Creating temporary turbidity barriers along the shoreline to prevent erosion during onshore construction to prevent or lessen the spread of fine material



- Maintenance of the vehicle should not be carried out on-site but in a proper workshop located far away from the lagoon water body. Refuelling of the equipment should be carefully planned such that no fuel is spilt on the ground, into the lagoon.
- The contractor should provide an emergency preparedness plan for approval by the Engineer in case of a spill of fuel or other fluids from equipment/machinery used or accidents that may lead to fuel spillage of fluids from equipment/machinery on land or in the lagoon.

5.6 Mitigatory Measures proposed to avoid/ minimize sociological and cultural impacts

The proposed mitigation measures to avoid or minimize social and cultural impacts include both embedded strategies and additional interventions. In accordance with Good International Industry Practice (GIIP), the following measures have been embedded into the project design and align with the Social Impact Assessment (SIA):

Avoidance of land acquisition

To minimize displacement and resettlement impacts, the project prioritizes the use of uninhabited or barren land that can be either purchased or leased, thereby avoiding disruption to local communities.

Support for fisheries communities

The project will collaborate with local fisheries communities, particularly beach seine operators within the Area of Influence (AoI), to:

- Schedule transportation activities during off-peak hours to reduce conflicts. Establish designated unloading zones away from key fishing and tourism access points.
- Ensure clear and continuous communication to minimize disruptions.

Stakeholder Engagement Plan (SEP) and Grievance Redress Mechanism (GRM)

The project will develop and implement a SEP and GRM with the following objectives:

- To engage and communicate with affected communities and stakeholders, especially regarding fisheries and recreational activity-related impacts.
- To identify and respond to concerns raised by community members and tourism-related business operators about disturbances caused by wind turbine construction (e.g., increased noise, dust, vibration, visual impacts), construction traffic, accidents or damages, workforce influx, unmet employment expectations, and security arrangements.



- To provide a publicly accessible grievance mechanism through which stakeholders can raise and resolve project-related concerns.

Compensation for temporary disturbances

Affected parties will be compensated for temporary disturbances or accidental damage to property directly attributable to the project, based on evidence submitted through the GRM.

Local Employment Plan

The project will develop a Local Employment Plan that ensures transparency in recruitment. This includes:

- Posting job opportunities in centrally located, easily accessible public areas.
- Communicating eligibility criteria and the hiring process to local communities.

5.6.1 Mitigation of Construction Phase Social Impacts

5.6.1.1 Mitigation measures to minimize/avoid Impacts on existing infrastructure facilities, properties, and crops within the study area due to the construction and operation of the project

In addition to the embedded mitigation measures incorporated during the project design phase, supplementary measures will be implemented to avoid or minimize adverse impacts on existing infrastructure, properties, and crops within the social Area of Influence (AoI) during both the construction and operational phases of the project. These additional mitigation measures include the following:

- **Minimizing potential impacts on utility infrastructure:** The project will take steps to prevent or reduce any potential damage to public utility lines, including community water supply systems, electricity lines, and access roads. Special attention will be paid to the protection of fish landing sites and ensuring continued public access to key infrastructure such as the Thoradiya–Mohonthuwaramaa jetty.
- **Assessment and coordination:** Before construction activities, a detailed assessment will be conducted to evaluate the likelihood and extent of any damage or the need for temporary relocation of utility lines and services. This process will be carried out in close coordination with the relevant government agencies, utility providers, and community stakeholders to ensure that appropriate precautionary measures are identified and implemented in a timely manner.



- **Compensation for unavoidable impacts:** In cases where impacts to infrastructure, private property, or agricultural crops are unavoidable, the project will provide fair and timely compensation to affected parties. This will apply to both government-owned and privately owned assets, ensuring that those impacted are adequately compensated in line with national regulations and project policies.

5.6.1.2 Mitigation measures to minimize/avoid Impact on fishing, fish farms, fishing communities, and ongoing aquaculture activities

Although aquaculture activities and fish farms are not located within the project's social Area of Influence (AoI), no direct or indirect impacts on these operations are anticipated during either the construction or operational phases of the proposed wind power project. However, minor to moderate impacts on local fishing activities, particularly those involving beach seine operations may occur during the transportation of materials via sea routes. To minimize or avoid such potential impacts, the following mitigation measures will be implemented:

- **Consultation-Based Site Selection:** Loading and unloading points for sea-based material transport will be determined through meaningful consultations with relevant fisheries societies and beach seine operators to ensure minimal disruption.
- **Transportation Scheduling:** Material transportation schedules will be planned to avoid beach seine operation times as much as possible, thereby reducing potential conflicts.
- **Safe Navigation Practices:** Barges transporting materials will maintain a safe and adequate distance from active beach seine operations to prevent interference.
- **Monitoring:** Throughout the construction phase, the project will continuously monitor the situation and assess any unanticipated temporary effects on fishing activities, particularly regarding access to landing sites and fishing camps
- **Grievance Redress Mechanism (GRM):** Prior to the start of the project, a cost-effective and accessible GRM will be established to address any concerns or complaints from affected communities, helping to mitigate unanticipated impacts. (Please refer to the GRM section of this report for further details.)

To mitigate potential impacts on fishing activities, several embedded mitigation measures will be implemented. These include the preparation and adoption of carefully planned transportation schedules and the maintenance of appropriate buffer distances to avoid any disturbances to ongoing beach seine operations. Transport activities, particularly those involving the use of



barges and the temporary pier, will be timed to minimize overlap with peak fishing periods and critical fishing operations.

5.6.2 Mitigation measures to minimize/avoid Impacts on recreational activities in the area, including kite surfing

Among the popular kite surfing locations in the Kalpitiya Divisional Secretariat Division (DSD)—including Magic Island, Vella Island, Dream Spot, Kalpitiya Lagoon, Donkey Point, Kappaladi Lagoon, and Puttalam Lagoon—only Kalpitiya Lagoon (Kudawa) is situated in close proximity to the proposed project footprint. All other kite surfing sites are located at a considerable distance from the project area, and therefore, no significant impacts are anticipated due to the proposed wind power project.

According to the current project layout, only turbines T14 and T15 are positioned approximately 1 km from the Kalpitiya Lagoon kite surfing zone. During stakeholder consultations with kite surfing operators, some expressed concerns about potential indirect impacts from the project, specifically the possibility of a "gusty effect"—a phenomenon where wind flow becomes irregular and turbulent, rather than smooth and consistent, potentially affecting kite surfing conditions.

However, given that all turbine sites, including T14 and T15, are located at a reasonable distance from kite surfing areas, the likelihood of such an effect occurring is considered very low and remains to be verified through technical analysis. To address these concerns proactively, a comprehensive wind monitoring program will be initiated prior to the installation of turbines T14 and T15. This program will assess any changes in wind patterns and their potential effects on kite surfing conditions. Final turbine siting decisions will be made in consultation with kite surfing operators to ensure that any adverse impacts are avoided or minimized.

5.6.3 Mitigation measures to minimize/avoid Impacts on access to the beach and to the Dutch Bay lagoon

Public access, both vertical (from inland areas to the shoreline) and lateral (along the beach and lagoon wedge), will remain unrestricted and undisturbed throughout the project. Access to common properties in the area will also be preserved. Furthermore, land parcels where wind turbines are installed will not be enclosed by fences, ensuring open access to these areas. The public will be granted the right to use the upgraded main access road as well as the approach roads that will be constructed as part of the project. These improvements are intended to benefit not only the project but also the local community by enhancing overall connectivity.



Additionally, the motor ferry that will be used to transport construction materials from *Thoradiya* to *Mohonthuwaremma* during the implementation phase will be transferred to the local authority upon completion of the project. This ferry will subsequently be made available for public use, contributing to improved local transportation infrastructure.

5.6.4 Mitigation measures to minimize/avoid Impacts due to transportation of material and equipment, and an increase in traffic volumes during the construction period

Although a significant increase in traffic volume on the main access road is not anticipated during the construction phase, any temporary disruptions to the free movement of vehicles or pedestrians, particularly during the transportation of heavy equipment and materials, will be mitigated or avoided. Measures to be adopted include:

- Issuing advance public notices with transportation schedules;
- Deploying pilot vehicles to alert the public to incoming convoys;
- Establishing temporary bypass routes at key locations to facilitate uninterrupted traffic flow.
- Transportation activities will be scheduled to avoid peak times, including the start and end times of school sessions, as well as during religious services at the church and mosque.
- One-way vehicular movement along narrow or constrained sections of the access routes will be carefully managed by deploying trained flagmen equipped with "Stop" and "Go" signboards. These personnel will coordinate traffic flow to ensure the safe and orderly passage of vehicles, minimize delays, and prevent congestion or accidents during material transportation.

These proactive strategies are designed to minimize inconvenience to local communities and ensure safety and accessibility throughout the construction period.

Given the high level of activity during this period, special attention must be paid to minimizing disruptions. This could include scheduling transportation activities during off-peak hours, establishing designated unloading zones away from key fishing and tourist access points, and maintaining clear communication with stakeholders such as fishermen, beach seine operators, and tour boat service providers. Additionally, implementing a coordinated transport management plan and conducting prior consultations with affected groups will help to ensure that the project proceeds with minimal social and economic disruption.



5.6.5 Mitigation measures to minimize/avoid Impacts on roads, buildings, and other properties during the transportation of material, equipment during the construction period

Any potential impacts resulting from the transportation of construction materials on residential dwellings and other structures—such as public buildings and facilities in the jetty areas at Thoradiya and Mohonthuwaramaa, as well as along both sides of the access roads and loading/unloading points—will be carefully minimized. The appointed contractor will bear full responsibility for rehabilitating and restoring any damaged private properties or public infrastructure to their original or improved condition. To ensure timely restoration, the project proponent will oversee the implementation of these measures and will secure the necessary financial commitments in advance to guarantee prompt action.

5.6.6 Mitigation measures to minimize/avoid Impacts on waste and hazardous waste

Construction site solid waste management measures

- Each profession shall be responsible for the implementation of the relevant environmental management infrastructure in the construction organization design, and the solid waste storage facilities for different purposes shall be clearly marked according to the requirements, such as: recyclable waste storage, non-recyclable waste storage, and construction waste.
- Each profession shall strictly implement the relevant rules and regulations of the project department, and do a good job in the management of safe and civilized construction in accordance with the requirements of the safe and civilized construction management methods and the specified areas of responsibility. The construction site garbage and slag should be cleaned out of the site in time, so that the site garbage is cleaned out on a day-to-day basis
- Recyclable solid waste shall be recycled and disposed of regularly; non-recyclable solid waste shall be managed uniformly in accordance with the management program and measures formulated by the profession;
- For the earth excavated on site, pile up according to the disposal site designated by the general contractor. In the construction measures, a reasonable excavation program shall be considered to reduce the generation of abandoned soil and backfill on site as far as possible. The retained backfill will be transported to the planned temporary disposal site for stockpiling.



- The Safety Department is responsible for contacting with the general contractor or the competent unit to determine the location for storage and disposal of solid waste at the construction site, and is responsible for the removal and disposal of solid waste at the construction site.

Construction site hazardous waste management measures

- The waste hazardous chemicals shall be handed over to the material supplier for unified and centralized disposal or other effective measures stipulated by the government environmental protection department for treatment, recycling and unified treatment of waste materials as far as possible.
- Before the construction, each profession should formulate the specific management plan and measures for the hazardous waste products and important environmental factors listed in the list of the construction process of each profession, make technical briefing for the operators, make a good record of the briefing, and be responsible for the organization, implementation and management.
- Reasonably set up storage facilities for hazardous waste products in each specialty and mark them.
- For on-site storage of oil and paint, special oil and paint storage should be set up, other substances are strictly prohibited to be placed in the storage, and the floor and wall of the storage room should be treated with special anti-seepage treatment. The storage, use custody should be handled by specialized personnel to prevent oil from running, risking, dripping, leaking and polluting water bodies.
- Each profession should recycle the waste oil and paint; it is responsible for the unified storage and management of paint, paint drums, oil-containing cotton yarn, waste oil and other dangerous waste products within the scope of this profession.
- The Safety Department is responsible for contacting units with processing qualification and capability, signing agreements and entrusting them to dispose of hazardous waste generated on site in a unified manner.

5.6.7 Mitigation measures to minimize/avoid Impacts on occupational health/safety

The identified potential impacts could be easily mitigated by applying the following mitigatory measures.

- All precautions must be taken as per the Occupational Health and Safety Act as well as other precautionary measures required by the authorities. Appropriate traffic management



schemes should be developed during the construction phase to ensure the safety of road users, including pedestrians who transit at the site.

- Appropriate personal protective equipment must also be provided to workers on site.
- Furthermore, the Contractor will have to comply with all requirements of the Occupational Health and Safety Act and other legislation with respect to health and safety on site.

5.6.8 Proposed Mitigation Measures for Impacts on Surrounding Sensitive and Protected Areas

The proximity of the project site to Bar Reef Marine Sanctuary and necessitates a suite of robust and proactive environmental safeguards. These protected and sensitive areas support critical biodiversity, including coral reefs, mangroves, estuarine ecosystems, and migratory bird habitats. Mitigation measures must be both precautionary and adaptive, addressing direct, indirect, and cumulative risks.

The identified potential impacts could be easily mitigated by applying the following mitigatory measures.

To safeguard downstream aquatic ecosystems and maintain water quality:

- The construction activities will have to be planned in such a way to minimize sediments release into the water bodies attached to these sensitive areas by fenced with a silt curtain to limit the spread of sediments away from the working area. This measure will provide better control over sediment propagation and limit the impact of a change in sediment level in the water body.
- Install sediment control structures such as silt fences, bunds, and sedimentation basins at all runoff points, especially where drainage leads to Dutch Bay or the Kala Oya system.
- Enforce strict solid and liquid waste management protocols, including separation, containment, and proper disposal through licensed service providers.
- Prohibit use and storage of chemical herbicides, pesticides, fuel, and lubricants near drainage paths, watercourses, or within buffer areas.
- Include spill containment kits and emergency protocols on-site to deal with accidental leaks or chemical spills.

To minimize disruption to nearby fauna:

- Limit heavy machinery operation to standard daylight working hours and conduct regular maintenance to reduce exhaust emissions and noise levels.



- Apply dust suppression methods, such as regular water spraying on access roads and open areas, especially during dry and windy periods.
- Avoid installing bright or continuous lighting near marine and lagoon edges, particularly Bar Reef and Dutch Bay to avoid disorientation or behavioral disruption of nocturnal and migratory species.
- Use low-intensity, downward-shielded, and motion-activated lights only where necessary for safety and site security.

To support uninterrupted wildlife movement and maintain regional biodiversity:

- Avoid infrastructure placement that could fragment habitats or obstruct terrestrial or aquatic movement paths.
- Retain vegetated corridors and minimize linear clearings that could lead to isolation of wildlife populations.

To ensure long-term ecological protection through informed project governance:

- Monitor key indicators including water quality, turbidity, invasive species presence, and wildlife activity near protected zones.
- Establish a grievance redress and incident response system to capture community reports, wildlife incidents, or environmental breaches in real time.

By integrating, these detailed and ecologically sound mitigation measures, the project can substantially reduce its potential impacts on adjacent protected and sensitive habitats. This approach aligns with national environmental legislation and international conservation obligations, ensuring that development progresses in harmony with biodiversity conservation and ecosystem health.

5.7 Proposed Mitigation measures for shipment unloading and transportation of oversized turbine parts.

Wind turbine equipment is heavy and over-dimensional cargo, which must be handled at the location of unloading and transported in extra-large trailers in internal proposed roads. The land transport vehicles usually need wider roads with minimum transport bottlenecks such as sharp bends, bridges and culverts, and selection of wind turbine size must be carefully assessed for these limitations. A 10 MW wind turbine has critical dimensions of turbine equipment for the transport and installation arrangement of the proposed machinery.



5.8 Proposed Mitigatory Measures for Impacts due to Disposal of Debris and Waste

When construction commences, ground materials of each of the wind turbine localities will have to be cleared. When making foundation structures at each wind turbine location, either piling will have to be done or suitable foundation method will have to be applied under the supervision of experienced structural engineer, based on the N values at depth based on the drilling data. Based on the N values of drilling data, either piling or suitable foundation method should be applied between 13 and 15m depth levels of all wind turbine locations. Some of these excavated materials (sandy materials) could be used for filling purpose of each of the wind turbine location to certain extent, after constructing the wind turbine structures. The other excavated materials, such as clay, sandy clay, silt and silty clay materials, will be disposed systematically within the identified location. Approval will be obtained from the Kalpitiya Pradeshiya Sabah/ Divisional Secretary of Kalpitiya/Puttalam for the disposal of debris in systematic way within the identified location.

After erecting turbine structures, its surrounding area will be systematically filled with suitable soil materials (preferably Red Beds) and its slopes will be maintained in 1:3 ratio. The soil filling process will be undertaken considering the elevation of the surrounding area to avoid water stagnation during rainy period in and around each turbine structure. Then silt traps will be erected at each location by considering the drainage path at each locality. The drainage path will be maintained in order to drain water properly during rainy periods without stagnation.

5.9 Mitigation Commitments from Agency Clearances

This Section compiles the mitigation measures and commitments required by various line agencies as conditions of their **in-principle clearances** for the project. The actions are grouped by thematic area and further organized by project phase (pre-construction vs. construction implementation). Each measure indicates the responsible *agency*, the clearance *condition* imposed, and the *mitigation action* the project will undertake in response. The proposed mitigation measures are tabulated in **Table 5.1** and Table 5.2.



Table 5.1: Summary of proposed mitigation commitments for line agency clearances – Pre Construction Phase

Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
1.0 Biodiversity and Ecosystems	Palmyrah Development Board	Palmyrah Tree Inventory	Conduct a comprehensive survey of all palmyrah palm stands within or adjacent to turbine sites, access roads, the jetty, and trenching areas.	Prior to construction, the project will complete this GPS-tagged palmyrah inventory (with photographs) and use it to inform micro-siting – turbines, roads and structures will be positioned to avoid removing mature palmyrah trees wherever feasible. Stands of palmyrah near work areas will be clearly demarcated and protected with buffer zones or temporary fencing before any site clearing.
		Permits and Replanting for Tree Removal	If any palmyrah trees absolutely must be felled for the project, it must only occur after obtaining a palmyrah tree felling permit from the Palmyrah Board. Any removed palmyrah must be compensated by replanting at least 10 new palmyrah saplings for each tree cut . A detailed replanting plan (species, planting locations in the Kalpitiya area, and maintenance duties) is to be submitted with the prior to construction.	The project will secure Palmyrah Board felling permits in advance for any unavoidable tree removals and prepare a palmyrah replanting program at a 1:10 ratio , to be implemented locally. The replanting plan, including long-term maintenance commitments, will be provided at the time of construction and cleared with the Board.
	Department of Wildlife Conservation	Avifauna Integration Study	Incorporate the findings of the one-year avifaunal (bird) survey for this project . A standalone report on the bird study (methods, seasonal observations, species list, and mitigation proposals) must be submitted to DWC (copied to CC&CRMD) for review.	The project will complete a 12-month avian survey and will fully integrate its results into the prior to construction , including a separate technical report to be provided to the DWC. This ensures that bird migration patterns and sensitive habitats identified are acknowledged and addressed in project planning.
		Wildlife-Sensitive Design	The final wind turbine layout must avoid critical bird areas such as migration corridors, roosting and nesting sites identified by the avifauna study. Any required turbine lighting should be minimized to aviation safety essentials and use bird-friendly features (downward-shielded, radar-activated beacons) to avoid attracting birds.	The project will adjust micro-siting of turbines and infrastructure to steer clear of important bird habitats flagged by DWC. Turbine obstacle lights will be configured to the lowest safe intensity and employ shielding and radar activation so that they illuminate only when needed for aircraft, thereby reducing sky glow and bird attraction.
		Habitat Offset Plan	Any unavoidable disturbance to natural habitats must be offset through habitat	The project commits to compensate for habitat loss by implementing DWC-approved habitat restoration measures . A habitat management plan will be developed in the pre-construction stage,



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
			restoration or reforestation programs approved by DWC. A habitat management plan detailing these offsets (with timelines and responsible parties) should accompany the construction plan.	identifying areas where native vegetation will be replanted or degraded habitats improved to offset construction impacts. This plan will specify the schedule of restoration activities and long-term monitoring of their success.
2.0 Marine/Coastal and Fisheries Impacts	Coast Conservation & Coastal Management (via UDA/SLSEA)	Coastal Setback Buffer	All project structures must observe a minimum 60-meter setback from the permanent coastal vegetation line , on both the Dutch Bay side and the open sea side.	The site plan will be finalized to maintain at least a 60 m buffer zone between all project works and the shoreline vegetation line , in full compliance with Coast Conservation Department requirements. This coastal buffer will protect dune ecosystems and guard against shoreline erosion. The layout drawings in the construction plan will clearly indicate these coastal reservations. Additionally, project engineers will incorporate any setback guidelines for dune areas – ensuring turbines or roads are sited away from sensitive sand dunes as directed by regulators.
	Palmyrah Development Board	Dune Area Restrictions	Construction activities must respect defined setbacks for dune vegetation , and heavy machinery is not permitted to traverse dune areas.	The project will avoid any direct disturbance to sand dunes along the coast. Construction equipment access will be planned to use existing paths or cleared routes away from dunes, and no vehicle traffic will be allowed on beach/dune vegetation . Sensitive dune zones will be marked as no-go areas on construction maps, preserving these coastal features from compaction or degradation.
	Department of Fisheries & Aquatic Resources	Barge Operations Plan	Provide a detailed jetty and barge logistics plan that aligns with local fishing seasons and monsoon cycles. The plan must specify barge navigation routes, loading/unloading points, schedules, and navigational safety measures (lights, markers), and ensure no waste, ballast or bilge water is discharged during marine transport. DFAR may require adjusting the timing of marine transports to minimize interference with fishing activity.	Before construction, the project will develop a comprehensive marine logistics plan for moving equipment via Dutch Bay , created in consultation with DFAR and local experts. This plan will schedule all barge trips around peak fishing periods (e.g. avoiding monsoon fishing seasons and daily peak fishing hours). Designated navigation channels and mooring points will be chosen to avoid fishing grounds, with proper lighting and markers installed on barges for safety. The plan will include a strict “ no-discharge ” policy – barring any dumping of waste or oily bilge water from vessels – to protect fisheries and water quality. The Final construction plan will append this barge logistics plan, and DFAR will review and approve the scheduling; the project is prepared to shift schedules per DFAR’s recommendations to accommodate fishermen.
	National Aquaculture Development Authority	Navigational Management Aquaculture for	All construction-related boat/barge movements must be timed to avoid peak feeding and harvest periods of aquaculture farms in Dutch Bay , and waterways serving fish farms must remain navigable at all times . NAQDA requires a Navigational Management	The project’s marine logistics plan (above) will also incorporate aquaculture schedules by consulting NAQDA and local farm operators on their critical feeding/harvest times. Barge routes and timing will be adjusted so as not to disrupt access to aquaculture pens, and no work will block the narrow channels and canals that fish farmers use daily. This plan will be documented in the construction plan and shared with NAQDA for concurrence.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
2.0 Marine/Coastal and Fisheries Impacts			Plan in the construction plan addressing these constraints.	
	Central Environmental Authority	Marine Spill Contingency Planning	Prepare an oil-spill contingency plan covering fuel transport and handling for marine works (e.g. barges, cranes). The plan must detail spill prevention measures, on-site response equipment, notification protocols, and coordination with the Marine Environment Protection Authority (MEPA), including obtaining MEPA's approval for the plan.	An emergency oil/chemical spill response plan will be developed and integrated into the project's Environmental Management Plan before construction begins. This plan will enumerate preventive steps (such as double-walled fuel tanks, drip trays during refueling), a list of response equipment to be stationed at site (oil absorbent booms, skimmers, etc.), and clear communication lines to immediately involve MEPA in case of any spill. The project will seek MEPA's endorsement of the spill response plan and include that approval reference in the construction plan, ensuring regulatory buy-in on marine pollution countermeasures.
	Sri Lanka Navy	Secure Marine Route Clearance:	Submit detailed plans of barge routes, temporary anchorage and mooring sites for transporting project components by sea. These plans must show all navigation channels and water depths and confirm that planned routes do not encroach into naval operational areas or restricted zones . Any anchoring within controlled Navy zones requires prior written permission.	The project will coordinate proposed marine transport routes with the Sri Lanka Navy in advance , mapping out barge paths that steer well clear of naval exercise areas, defense installations, or restricted waters. If the preferred route necessitates passing near a naval reservation, the project will obtain written Navy clearance beforehand or adjust the route as needed. All planned mooring and staging points for barges will be vetted for adequate depth and distance from sensitive military zones, to ensure no interference with naval activities during the project's logistics operations.
	Department of Fisheries & Aquatic Resources	Fisheries Impact Assessment	The EIA should identify any potential impacts on fish stocks or habitats in Dutch Bay due to the project and propose monitoring programs.	The construction plan will include a fisheries impact assessment section prepared in consultation with DFAR, describing how construction (e.g. increased turbidity, noise, vessel traffic) could affect local fish populations or breeding areas. Corresponding aquatic ecology monitoring measures will be outlined, such as periodic water quality and fish catch sampling to detect any project-related changes. This pro-active study and monitoring plan will guide adaptive management to protect the bay's fisheries.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
3.0 Cultural Heritage	Department of Archaeology	Archaeological Survey	Conduct a pre-construction archaeological screening survey of all areas to be disturbed – including turbine sites, access road alignments, cable trench routes, jetty location, and equipment laydown areas. The survey must be led by a qualified archaeologist, and a report with site maps and recommendations submitted to the Department of Archaeology for review before construction begins .	The project will engage a licensed archaeologist to perform a thorough field survey of the entire project footprint prior to any ground breaking. Using techniques like surface inspection and limited test pits, this survey will identify any historical/archaeological artifacts or sites. A report of findings will be provided to the Department well in advance of construction. If the survey uncovers significant artifacts or features, the project will work with the Department to develop site-specific mitigation (avoidance or careful salvage) before proceeding in those areas.
		Chance-Find Procedure	Develop and include a formal chance-find procedure in the project's Environmental Management Plan (EMP) to handle unexpected archaeological finds. The procedure must detail <i>stop-work triggers</i> (when to halt work), immediate notification of the Department, on-site protection measures for the find, and protocols for resuming work after clearance. All workers and contractors are to be briefed on this procedure.	The project will incorporate a Department-approved chance-find protocol into its EMP and train all construction personnel on its implementation. In practice, this means if any crew member encounters something that might be an artifact (e.g. pottery shards, bones, foundations) during excavation, they must stop work in that area and inform site management immediately . The project's environmental manager will then secure the site (marking off the area) and promptly notify the Department of Archaeology. Work will not resume until Department archaeologists have examined the find and given clearance or instructions on how to proceed. This procedure will be covered in worker induction sessions so that everyone on site is aware of the steps to take.
		Protective Buffers	If any archaeological artifacts, features, or sites are identified (either in the pre-construction survey or as a chance-find), appropriate buffer zones must be established around them in consultation with the Department. Construction activities are prohibited from encroaching into these buffer zones.	Should the archaeological survey or any later discovery reveal cultural heritage sites, the project will mark out a no-work buffer around the identified area as directed by archaeologists. The size of the buffer will be determined with Department authorities (ensuring sufficient distance to protect the site from vibration or accidental damage). These exclusion zones will be shown on construction maps and physically fenced off on the ground, and all crews will be made aware that no excavation, material storage, or vehicle movement can occur inside these protected areas.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
		Worker Training & Site Supervision	Project staff should receive basic training on recognizing archaeological remains , and the Department may station officers to monitor excavations in sensitive areas at the proponent’s expense.	Before construction, an archaeologist will conduct toolbox training sessions for site supervisors and machine operators on how to spot potential antiquities or soil changes that could indicate archaeological material. Key personnel will be taught to distinguish common heritage artifacts (pottery, bricks, bones) from ordinary stones or debris. Additionally, the project welcomes Department-appointed archaeological officers to be present during excavation in particularly sensitive or rich areas – and will fund their time as required. This oversight will add an extra layer of protection, as any emerging artifacts can be immediately identified and saved.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
4.0 Infrastructure and Land Use	Urban Development Authority	Zoning and Plan Compliance	The wind farm must conform to the Kalpitiya Local Development Plan and all gazetted zoning and land-use regulations. The UDA requires final site layout drawings showing property boundaries, interfaces with public areas, and all ancillary infrastructure to verify compliance.	The project's final design will fully comply with UDA planning requirements. Prior to construction, the proponent will prepare detailed site layout plans overlaying the project components on land-use zoning maps to ensure consistency with the Kalpitiya Development Plan. These drawings will highlight how turbines, roads, the substation, and other facilities fit within allowed zones and maintain required distances from public lands. The plans (including land parcel details and any designated public open spaces nearby) will be submitted to UDA for endorsement, and UDA's go-ahead is contingent on this demonstrated compliance. No construction will commence until UDA has reviewed the final plans for alignment with all zoning and development guidelines.
	Urban Development Authority	Setbacks from Roads and Settlements	In addition to coastal buffers, the project must maintain UDA-specified setbacks from public roads, dune belts, and settlements . This ensures turbines or tall structures are not too close to roads or villages for safety and visual reasons.	All turbines and structures will be located at safe distances from existing roads and houses as per UDA setback standards. The site layout accounts for a buffer between turbines and the main public road on Kalpitiya Peninsula, and adequate clearance from the nearest residential clusters, to reduce noise and shadow flicker effects on residents. These setbacks will appear on construction drawings and be staked out on the ground before work starts, guaranteeing that the built facilities do not encroach closer than permitted to community areas. (Coastal and dune setbacks are covered under Marine/Coastal measures.)
		Height and Aviation Clearances	The final turbine and mast heights must have approval from the Civil Aviation Authority (CAA), and copies of the height clearance letters should accompany the final plans. UDA will only give its final plan approval after all other relevant authorities (like CAA) have cleared their aspects.	The project will adhere to the maximum turbine tip heights authorized by the Civil Aviation Authority. During the design finalization, the proponent obtained CAA height clearance for the planned turbine model and hub height. These clearance letters (indicating that the structures do not pose risks to air navigation) will be appended to the project's final layout submission to UDA. The project will not exceed the approved heights. UDA's stamping of building plans will be sought only after providing evidence of all such inter-agency clearances, ensuring a coordinated approval process.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
4.0 Infrastructure and Land Use	Urban Development Authority	Visual Impact Mitigation	The EIA should assess visual impacts on key viewpoints and tourism activities (like the kite surfing areas) and propose mitigation such as color schemes or landscaping to preserve important view corridors. UDA may require design adjustments based on this assessment.	A visual impact assessment has been conducted and will be included in the EMP , evaluating how the wind turbines might affect the skyline from popular public vantage points and tourist sites. In line with UDA's guidance, the project will implement measures to minimize aesthetic intrusion – for example, turbines will be painted in neutral, non-reflective colors that blend with the sky, and facilities will have landscape plantings to screen ground structures. If the UDA review identifies any critical view corridor conflicts, the project is prepared to relocate or adjust certain turbines or alter color/finish to mitigate visual impact on tourism. These commitments ensure that the area's scenic and recreational value is maintained as much as possible.
	UDA & CEA	Access and Drainage Planning	The project must ensure that new infrastructure does not obstruct existing access ways, drainage paths, or traditional beach uses , and that stormwater is properly managed. Any new access roads should meet local authority standards and be handed over for public use after construction. A storm-water management plan with necessary drainage structures must be provided (the EIA should not state that no plan is needed).	Project engineers have designed site infrastructure to preserve community access and natural drainage. All internal roads and cable trenches have been aligned to avoid cutting off existing footpaths or blocking village beach entries. Natural drainage channels across the site have been mapped and the design includes culverts and cross-drains so that rainwater can flow unimpeded under new roads. A comprehensive stormwater management plan will be included in the Final EMP, detailing how runoff will be handled to prevent flooding or erosion (correcting any earlier notion that such a plan was unnecessary). This plan specifies where diversion ditches, perimeter drains, and silt traps will be installed to maintain natural flow patterns. The project commits that any new access roads will be engineered to local government specifications (appropriate width, surfacing, and culverts) and, upon completion, formally handed over to the local authority for long-term maintenance and public use. In essence, the infrastructure is being planned not just for the project's needs but also to integrate with and even improve local public infrastructure.
4.0 Regulatory Compliance and Reporting	CEA –	Environmental Protection Licenses (EPLs)	Identify all ancillary facilities (e.g. concrete batching plant, worker camps, and mechanical yards) that require EPLs and obtain these licenses from CEA before construction commences . Include copies of all obtained EPLs in the Final EMP submission.	The project will secure Environmental Protection Licenses for all relevant construction installations prior to breaking ground. A survey has been done to list such facilities: a concrete batching plant and a mechanical workshop have been identified as requiring EPLs. Applications will be submitted to the CEA well ahead of construction, and approval letters/licenses will be appended to the Final EMP submission. All conditions attached to those EPLs (such as pollution control measures or specific monitoring) will be integrated into the construction EMP to ensure compliance from day one.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
4.0 Regulatory Compliance and Reporting	Palmyrah Development Board –	Tree Felling Permits	No palmyrah trees may be removed until the Palmyrah Board issues felling permits for them. Proof of having obtained these permits must be provided.	Before any site clearing that involves palmyrah, the project will obtain the requisite tree felling permission from the Palmyrah Development Board. The permit application will include the inventory of palmyrah trees and justifications for why removal is unavoidable. The Final EMP will include a copy of any felling permit granted, as noted by the Board. By following the formal permit process, the project complies with Sri Lankan regulations on protected palm species and ensures oversight of tree removal.
	NWPEA & DWC	Integration of Environmental Regulations	The EIA's regulatory chapter must cite and comply with the North Western Provincial Environmental Statute and any local by-laws on noise, dust, and waste, as well as all relevant provisions of the national Fauna and Flora Protection Ordinance. It must confirm the project is outside any gazetted wildlife reserves, or else permits would be required.	The project's Final EIA explicitly document compliances with both provincial and national environmental laws. In section 1.5.2 of the EIA, the proponent references the North Western Provincial Environmental Statute and outline how the project meets its requirements (for example, adhering to provincial noise control times, dust suppression standards, and waste disposal rules). It will also state that the wind farm's footprint lies outside any designated wildlife sanctuary or national park , per DWC confirmation, so no special wildlife permit is needed. (If any activity were to impinge on a protected area, the necessary permit under the Flora and Fauna Ordinance would be obtained separately, but currently none is required.) Ensuring these legal references and assurances are in the EIA demonstrates full regulatory compliance at both provincial and national levels.
	Sri Lanka Sustainable Energy Authority	Policy Alignment and Plans	The project must align with national renewable energy policy targets and commitments, including having a decommissioning plan for end-of-project and a pledge for periodic operational reporting.	The project will comply with SLSEA's policy directives by preparing a detailed Decommissioning Plan (outlining how turbines and the Battery Energy Storage System will be dismantled and the site restored at end of life) to include in the EMP . Moreover, the proponent formally commits to submit periodic reports during operation on energy generation and storage performance to SLSEA to help track national renewable energy output. These commitments ensure the project contributes transparently to Sri Lanka's energy goals and will leave no long-term legacy issues at the site.
	Sri Lanka Sustainable Energy Authority	Consolidation of Clearance Conditions:	SLSEA requires that all line-agency clearance conditions be integrated into the final project design and the Final EIA , and that the EIA be finalized (with these conditions) before the proponent applies for the Energy Permit.	The project has compiled every condition from each agency clearance and woven them into the project's design, construction plan, and EIA documentation. The Environmental Management Plan and mitigation tables in the EIA will explicitly reference these clearance requirements (as detailed in this chapter), demonstrating that the project's commitments cover each point raised by the agencies. Only once the EIA has been updated to reflect all such measures will the proponent proceed to seek the Energy Permit from SLSEA, ensuring no permit is sought until regulatory expectations are fully addressed.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Pre - Construction Phase
4.0 Regulatory Compliance and Reporting	North Western PEA	Additional Conditions Adoption:	Any further conditions arising from public consultations or the Final EIA review at provincial level must be added to the project's commitments.	The project will remain flexible and incorporate any new mitigation requirements identified during the final approval process. If, for example, during the NWPEA's review of the EIA or community hearings a need for an additional noise barrier or a specific CSR program is raised, the proponent will formally amend the Environmental Management Plan to include those measures. This adaptive approach is noted in the EIA, affirming that the proponent will comply with any supplemental provincial conditions by integrating them into implementation plans.
	NWPEA	Formal EIA Approval Application	After receiving the central PAC's consent (general approval) for the project, the proponent must apply to the North Western Provincial Environmental Authority for EIA approval , submitting required documents and the prescribed fee.	The proponent will follow through with the provincial EIA clearance process by filing a formal application to NWPEA once the Coast Conservation Department (the project approving agency) issues its decision. All necessary paperwork (Final EIA report, agency clearance letters, environmental management plan, etc.) and the applicable fees will be submitted to NWPEA to obtain the provincial environmental consent letter. This step ensures the project has both national and provincial environmental approvals in place as required by law before construction.
	SLSEA	Energy Permit and SPPA	The in-principle clearance from SLSEA is not a substitute for the Energy Permit or Power Purchase Agreement (SPPA) ; those must be obtained through the usual process after EIA completion.	The proponent acknowledges that an Energy Permit from SLSEA will be obtained only after the EIA is approved and all other pre-requisites (like grid approval and land acquisition) are met. Similarly, the project will enter into a Standardized Power Purchase Agreement with the CEB after tariff finalization, as per Ministry of Energy guidelines. These are administrative follow-up steps ensuring the project's legal and commercial framework is properly secured post-EIA; the commitments in this chapter help pave the way for those final approvals by fulfilling environmental conditions upfront.
	Multi-Agency	Appointment of Focal Points	Each agency letter designates a focal officer for ongoing liaison and expects the project to share contact details of its counterpart officers.	The project has designated specific officers to liaise with each regulatory agency. For example, an environmental manager will be the point of contact with CEA and NWPEA, a biodiversity specialist with DWC, a marine coordinator with the Navy and DFAR, and a social officer for community matters. Their names and contact information will be officially communicated to the respective agency focal points as requested. Regular communication channels (email groups, WhatsApp, meetings) will be established so that each agency's focal officer can easily monitor compliance or provide guidance during the project. This one-on-one coordination framework will facilitate timely reporting and quick resolution of any issues that arise in the field.



Table 5.2: Summary of proposed mitigation commitments for line agency clearances – Construction Phase

Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Construction Phase
1.0 Biodiversity and Ecosystems	Palmyrah Development Board	On-Site Vegetation Protection	During construction, all palmyrah trees identified for preservation will be physically protected and avoided .	Work crews will honor the demarcated buffer zones around palmyrah stands and dune flora. No heavy machinery will be allowed to intrude into these flagged buffer areas, and protective fencing will remain in place to prevent accidental damage. Construction managers will adjust work methods as needed (e.g. hand-clearing near trees) to ensure existing palmyrah and associated dune vegetation are unharmed.
		Tree Replanting Program	For any palmyrah palms removed, the project will carry out the planned replanting program at the 1:10 ratio in coordination with the Palmyrah Board.	<i>After any necessary felling</i> , saplings will be planted at approved sites in the Kalpitiya area to enhance local vegetation cover. The project's environmental team will maintain the new palmyrah plantings (watering, guarding, etc.) as per the maintenance schedule outlined in the replanting plan, ensuring a high survival rate of the young palms.
	Department of Wildlife Conservation	Bird Mortality Curtailment & Monitoring	Once turbines become operational (post-construction), the project will implement active measures to mitigate bird collisions.	Turbine curtailment protocols will be in effect, meaning turbines may be temporarily shut down during peak bird migration periods or adverse weather events known to drive birds to risky altitudes. In addition, a post-construction avian monitoring program will be executed: trained observers will survey bird mortality around turbines and monitor bird activity for at least five years after commissioning , reporting annually to the DWC on bird strike incidents and the effectiveness of mitigation measures. These operational-phase commitments, initiated in the EIA, ensure long-term biodiversity stewardship by the project.
		Habitat Restoration Works	If construction activities result in any damage to natural habitats (e.g. clearing of vegetation outside cultivated lands), the project will promptly implement the DWC-approved habitat restoration plan .	This may include replanting native trees or mangroves, rehabilitating any disturbed wetland areas, or improving alternate habitats to compensate for the loss. Restoration works will commence as soon as practical (even during construction in parallel) to allow new vegetation to establish, and DWC will be updated on progress.
2.0 Marine/Coastal and Fisheries Impacts	DFAR/NAQDA	Regulated Barge Movements	The project will enforce the approved barge and marine logistics plan throughout construction.	All barge trips for equipment delivery will be scheduled according to the agreed timing windows to avoid peak fishing and aquaculture periods. The site marine coordinator (see Navy liaison below) will closely communicate with local fisheries societies and NAQDA officers before each major barge operation. If DFAR or NAQDA request last-minute adjustments (e.g. delaying a barge by a day to avoid a seasonal fishing event), the project will comply to minimize livelihood disruption. All barges will stick to the designated routes and mooring areas defined in the plan, with required navigational lights and markers in place during nighttime or low visibility operations for safety.
	NAQDA	Water Quality Monitoring		A water quality monitoring program will be carried out during construction in Dutch Bay , as committed to NAQDA. Baseline water quality data (turbidity, dissolved oxygen, salinity, etc.) will have been collected prior to starting marine works. During construction, monthly water sampling upstream and downstream of active work sites will be conducted. This will allow early detection of any sediment plumes or contaminant release affecting aquaculture. The



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Construction Phase
2.0 Marine/Coastal and Fisheries Impacts				monitoring results will be compiled and shared with NAQDA and the Coast Conservation Department on a monthly basis to demonstrate compliance with water quality standards.
	NAQDA	Erosion and Pollution Controls	The project will implement strict measures to prevent marine pollution and sedimentation impacts on fisheries and aquaculture.	All vessels and barges will adhere to a zero-discharge policy – no dumping of garbage, no releasing of ballast or bilge water, and no oil/fuel leakage into the bay. Construction teams will install erosion and sediment control structures (such as silt screens or sediment traps) around in-water work areas (e.g. near the jetty construction) to contain any turbidity and protect the clear waters needed for aquaculture. Equipment fueling and maintenance will be conducted on land with secondary containment to avoid spills. The project will also minimize noise and artificial lighting from marine works at night; for example, vessel engines will have mufflers and unnecessary deck lights will be turned off to reduce disturbance to marine mammals and local fishers operating at night. These measures fulfill the Marine Pollution Prevention Act requirements and safeguard the bay’s environmental quality.
	Department of Fisheries & Aquatic Resources	Nearshore Works Mitigation		To reduce interference with nearshore fishing during construction, special procedures will be followed: heavy lifting operations for offshore or intertidal construction (such as lifting turbine components from barges) will be scheduled in the offseason or low-fishing periods whenever possible. The project will deploy escort boats and safety personnel to manage marine traffic when barges are operating, and clearly demarcate temporary exclusion zones in work areas with buoys or flags so that fishing boats can steer clear of construction zones safely. Regular coordination meetings with local fisheries officers will take place (e.g. weekly or biweekly) to discuss upcoming activities and resolve any conflicts or requests from the fishing community.
	Sri Lanka Navy	Advance Notifications:		The project will give the Navy at least 48 hours advance written notice before any major marine operation or heavy load movement by sea. Each notice will include all required details (vessel name, type of cargo, expected time of departure/arrival, route) and contact information for the vessel captain. Simultaneously, the project will coordinate with the Harbour Master to issue formal “Notices to Mariners” if the activity could affect other marine traffic. This early warning system allows the Navy to prepare or request adjustments, and alerts fisherman and boat operators in the area to the forthcoming barge movements.
	Sri Lanka Navy	Liaison and Security Protocols		A Marine Operations Coordinator will be appointed by the project to liaise directly with Sri Lanka Navy operations and harbor authorities throughout the construction period. This coordinator will attend Navy coordination meetings as needed and be on call to respond to any Navy instructions or emergent issues. All construction vessel crews will be briefed on and adhere to Navy communication protocols while in Sri Lankan territorial waters (maintaining radio contact on assigned channels, responding to Navy hails, etc.) . No project vessels will enter restricted naval zones or training areas; the Navy liaison will ensure real-time route adherence and obtain any special clearances if unforeseen deviation is necessary. By maintaining this close operational link, the project will uphold defense security requirements and marine safety throughout its offshore logistics.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Construction Phase
3.0 Cultural Heritage	Department of Archaeology	Enforcement of Chance-Find Protocol		During construction, if any unexpected archaeological object or structure is encountered, the crew will immediately enact the chance-find procedure. Work at that location will stop at once, the find will be left in place and protected (e.g. shaded or covered, and a security perimeter established), and the Department of Archaeology will be contacted right away to inspect the find. Construction in that zone will remain suspended until the Department's experts have evaluated and documented the discovery, and either approve work to resume or implement a recovery plan. This strict adherence to the protocol ensures that heritage resources are not inadvertently destroyed.
		Adherence to Buffers and Monitoring		All established archaeological buffer zones will be honored throughout construction – no personnel, machinery, or construction activity will intrude into demarcated heritage buffers. Site management will regularly inspect buffer fencing to ensure it remains intact and respected. In areas flagged as archaeologically sensitive, the project will coordinate with Department officers who choose to be on site. The construction team will accommodate any instructions from the Department representatives during supervised excavations. If, for example, artifacts are found in a trench, the project will allow time for proper excavation or recording by archaeologists before proceeding. By working hand-in-hand with the Department monitors, the project will ensure that any cultural heritage discovered during works is properly handled and conserved, while avoiding undue construction delays.
4.0 Infrastructure and Land Use	Urban Development Authority	Road Construction & Handover		All new access roads and tracks constructed for the project will be built to the agreed standards and then handed over to the local authorities. During construction, quality control teams will ensure that road base preparation, grading, and surfacing meet the specifications (e.g. adequate gravel thickness, proper drainage). Once turbine installation is complete, the roads will be offered to the local Provincial Council or Pradeshiya Sabha, along with as-built drawings, effectively becoming community assets for long-term use. This provides permanent benefit by improving local connectivity.
		Maintain Public Access and Beach Use		The construction team will keep existing public paths and beach access routes open or provide alternatives whenever possible. For instance, if a cable trench crosses a footpath, a temporary footbridge or bypass route will be provided for residents. Work areas near the shore will not impede traditional beach activities like fishing or recreation – they will be cordoned off in a minimal footprint, and signage will direct the public safely around them. By the end of construction, all original access ways will be restored to equal or better condition.
	Central Environmental Authority	Storm water and Drainage Works		The project will install the planned stormwater drainage measures early in the construction phase. This includes excavating roadside ditches, setting up culverts at crossings, and leveling sites to designed gradients so that rain runoff is effectively channeled. Continuous inspections will ensure that construction debris does not block natural waterways. In the event of heavy rains, the site team will actively manage water flow (with pumps or sandbag diversion) to prevent any induced flooding in adjacent lands. These actions put into practice the stormwater management plan and protect both the project and surrounding lands from water-logging or erosion.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Construction Phase
	Urban Development Authority	Visual Mitigation Implementation		As turbines and facilities are erected, the project will implement the promised visual mitigation measures . All turbines will be painted in the specified low-contrast matte finish to reduce glare. Any required landscaping (such as planting trees or shrubs to screen the substation or depot) will be done at the appropriate stage of construction. Prior to commissioning, the project will invite UDA officials for a site visit to verify that view corridor preservation measures (like the color schemes and landscaping) have been executed in line with the EIA commitments. This will ensure no surprises at operation – the visual profile of the project will match what was agreed upon to protect the area’s aesthetic and tourism appeal.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Construction Phase
5.0 Regulatory Compliance and Reporting	CEA	Wastewater Management Implementation	During construction, the project will put into action the wastewater and sewage management plan developed earlier.	Construction camps and site offices will be equipped with approved sanitation facilities (such as septic tanks or holding tanks for sewage). Wastewater (including from washing or concrete batching) will be collected and treated; for instance, septic tanks will be periodically emptied by gully bowzers and the sewage disposed of at a municipal treatment site per the agreement with the local authority. No effluent will be released to the environment without proper treatment. Additionally, adequate portable toilets will be provided at work fronts (like the turbine sites) and maintained by a hired service, to prevent any ad-hoc defecation or pollution. Site inspections will ensure that wastewater is not leaking or stagnating, thus protecting soil and groundwater quality as well as worker hygiene.
	CEA	Solid Waste Management Implementation	The project will execute the construction solid waste management plan to handle all waste generated on-site.	Separate bins will be placed to segregate recyclables (plastics, metals, paper) from non-recyclable waste at camps and work sites. Waste will be stored securely (in covered areas to prevent scatter by wind or animals) and regularly transported to licensed disposal facilities in Puttalam or designated sites, using authorized waste collectors. Under no circumstances will construction crews dump debris in open lands or burn waste. Especially, any hazardous wastes like used oil, filters, paint, or solvent containers will be collected separately and handed over to a CEA-licensed hazardous waste handler for safe disposal or recycling. The site environmental officer will keep manifests of all waste transfers to ensure traceability and compliance. These practices will minimize pollution and align with national environmental regulations during the construction phase.
	NWPEA	Noise and Dust Control in Practice	The project will continually enforce measures to meet provincial noise and air quality standards at site boundaries.	Noisy works (pile driving, heavy truck traffic) will be restricted to daytime hours (e.g. 7AM–6PM) in or near residential areas, as recommended by NWPEA. High-noise equipment will be fitted with mufflers or temporary noise barriers if needed to stay within decibel limits. Active dust suppression will be used on unpaved roads and construction zones – water bowzers will sprinkle access roads regularly and stockpiles will be covered – especially during dry windy conditions. On-site environmental staff will monitor dust and noise levels with portable devices. If readings approach the threshold set by provincial bylaws, immediate corrective actions (like increasing water spraying, slowing down work, or providing additional enclosures around machinery) will be taken. The project’s adaptive management on noise/dust will thus safeguard community health and comfort.



Sector / Thematic Area	Responsible Agency	Imposed Clearance Condition		Proposed Mitigation Measures
		In Brief	In Detail - Condition	Construction Phase
	NWPEA	Grievance Redress and Reporting:	Throughout construction, a local Grievance Officer will be available to the community to register any complaints, and the project will maintain required compliance logs.	Contact information (phone hotline and an email) for the project's community liaison/grievance officer has been disseminated to local residents and posted publicly. When complaints or concerns are raised (e.g. about excessive dust, noise at night, road inconveniences, etc.), the issue will be logged and addressed promptly – the Grievance Officer will coordinate with construction managers to resolve the matter (for instance, by adjusting work practices or providing remedy) and then communicate the resolution to the complainant. Each week, a summary of any such complaints and the environmental monitoring results will be compiled into a compliance log report and submitted to the North Western Provincial Environmental Authority. This report details monitoring data (noise, dust readings, waste disposal records) and how any community issues were handled, keeping regulators informed of the project's on-ground compliance and social performance.
	CEA & CC&CRMD	Quarterly Environmental Monitoring Reports		The project's environmental team will carry out all monitoring as stipulated (air quality tests, noise measurements, water quality sampling, waste audits, etc.) and prepare quarterly environmental compliance reports during construction. These reports will summarize the mitigation measures implemented and present the monitoring results versus environmental standards. Every three months, reports will be submitted to the Central Environmental Authority and the Coast Conservation & Coastal Resource Management Department. The reports will also note any deviations or incidents and corrective measures taken. By reporting quarterly, the project ensures transparency and allows regulators to track whether the project remains within environmental compliance throughout the build period.
	Multi-Agency	Continued Focal Liaison		The project will maintain active communication with each agency's focal point during construction to report progress and seek guidance. For example, monthly update meetings or teleconferences will be held with CEA's focal officer to review environmental performance, and site visits will be facilitated for the DWC focal officer especially during turbine installation (to observe bird protection measures). The Navy focal officer will receive weekly marine schedules, and the Archaeology focal officer will be alerted ahead of any significant earthworks. This ongoing liaison fulfills the commitment made in clearances to keep agencies in the loop and allows for real-time feedback, thereby ensuring all clearance conditions are effectively implemented on the ground.

Note: All the above commitments are drawn from the specific conditions set by the respective agencies in their TEC comments. Final, formal clearances from these agencies (and the Project Approving Committee) will only be granted once the project demonstrates that these measures have been incorporated into the Final EMP and will be executed accordingly. The project proponent's acceptance of these conditions, as documented in this chapter, forms a binding part of the Environmental Management Plan and assures regulators and stakeholders that environmental, social, and regulatory obligations will be met in full.



6 ENVIRONMENTAL MANAGEMENT AND MONITORING PROGRAMME

The Environmental Management Plan (EMP) is used as a tool for the management of the environmental performance of the project and it is developed and implemented as an important component of project activity. The EMP guides the implementation of Mitigatory Measures and Monitoring throughout the implementation of the project and contributes to the overall process of Project Monitoring and Auditing. The EMP, therefore, presents a consolidation of the recommendations given in the EIA Report, including specific recommendations for environmental mitigation, monitoring and management. In particular, it specifies the mechanisms for the implementation of mitigatory measures and monitoring.

The EMP provides a set of guidelines under which the Project Proponent and the appointed Contractor shall carry out their activities, ensuring that proper attention is paid with respect to implementing the planned impact mitigation measures. It will be the responsibility of the Project Proponent to ensure that the environmental control measures presented in this plan are adhered to by the various parties under his control and that work is progressed in compliance with all relevant environmental regulations.

The main objectives of the EMP are to:

- Define the roles and responsibilities for implementation, including environmental liaison contacts for the Promoter and appointed Contractor.
- Summarize the project-specific activities and planned impact mitigation measures, in line with the EIA report and associated EIA Licence.
- Provide an agreed format by which the environmental performance of the project can be monitored.
- Provide regular status reports to the Project Proponent and concerned authorities, regarding the rate of progress on site.
- Provide regular status reports relating to the discharge of EIA Licence conditions.
- Provide a record of environmental complaints/incidents and actions

This EMP is developed as a part of the project preparation activity. However, prior to the commencement of the construction activities, it has to be updated fully in consultation with the Contractor based on the actual construction methodologies. Discussions with the Contractor(s) are critical because the EMP is part of the relevant contract. The Environmental Management Plan



(EMP) should be seen as a dynamic management tool, which should be reviewed, updated and modified as the project progresses, and additional impacts to be identified even during the construction phase of the project.

This EMP should be a part of the Bidding Documents since the Contractor is responsible for fulfilling EMP requirements during the project and thus their careful consideration on pricing the items of work proposed in the EMP is a mandatory requirement during the bidding process.

6.1 Proposed Environmental Monitoring Programme (EMoP)

The EMP not only facilitates environmental audit of the works but is also a useful tool by which the construction works can be readily audited by the project proponent.

Table 6.1 (Impacts-Mitigation Matrix) summarizes the potential environmental and social impacts of the proposed project as identified and discussed in Chapters 4 and 5, and the mitigation proposed to reduce these impacts to acceptable levels. The table also shows how the mitigation should be implemented, who should be responsible for each action, and where and when the mitigation activities should take place.



Table 6.1: Summary of Anticipated Significant Impacts & Proposed Mitigation Measures

Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
1. Coastal Environment	potential erosion / accretion during the construction phase	provision of sediment traps	No special measures are required.
		construction activities to be carried out during the dry season and calm weather	
		original beach profile be restored after construction activities	
	Degradation of Sea Water Quality	Avoiding spilling overboard during loading, transit and unloading construction materials	No special mitigation measures are required
		provision of sediment traps	
		Any accidental spills or environmental releases that occur should be attended immediately	
2. Noise	Excessive noise generation by the movement of vehicles, construction equipment, operation of vessels/boats/ships and operation of standby generators.	Maintain the noise levels at the boundaries of the Project site during the construction phase under the stipulated levels mentioned in Environmental (Noise Control) Regulations.	As the noise sensitive receptors are located considerable distance away from the tower locations, no special mitigation measures are required
		Restrict construction activities to daytime (between 7 am and 5 pm) especially for the noise-generating activities and use of noise-generating machinery	
		Plan for the alternative routes for transport materials to minimize the noise generated by traffic (e.g. diverting traffic away from noise-sensitive areas using existing alternative/bypass roads)	
		provide appropriate personal protective equipment for the workers who are exposed to a high level of noise for a long period	



Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
3. Vibration	Vibration impacts due to the heavy machineries	Adaptation of construction methods that produce minimal vibrations, especially in sites adjacent to sensitive receptors	No special mitigation measures are required
		Pre-crack surveys will be carried out for any identified sensitive receptors in the area before construction work starts and after the work is complete. For any proven damages, repaired at their own cost reimbursements or compensations will be provided.	
4. Ecological Impacts	Habitat loss due to fragmentation, edge effects and development of invasive species	Stagger construction activities to reduce the extent of active disturbance at any one time	Prepare Landscape Conservation Plan is to conserve and restore ecological integrity within and around the project area
		Use native vegetation for restoration of cleared margins to enhance structural continuity and ecological function, promoting the return of native fauna and flora	
		Material sourcing protocols to ensure soil, gravel, or fill is free from invasive propagules or seeds.	
		Replant native vegetation suited to local soil and climatic conditions, with a focus on species that were originally present or that support local wildlife	
	Impacts on vegetation Direct removal of native vegetation Preventing and managing invasive plant species	Apply a phased clearing strategy, only removing vegetation in areas immediately needed for active construction. This reduces unnecessary loss and gives other parts of the site time to recover or be restored earlier.	
		Promote on-site translocation of rare or slow-growing plant species, where feasible, from high-impact zones to safe, managed receptor sites under ecological supervision	



Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
		Establish an invasive species monitoring program, beginning at the start of construction and continuing into the operational phase, to detect and respond to early invasions rapidly	
		Encourage replanting with native species in disturbed areas as a natural defense against invasive colonization, while simultaneously restoring habitats for dependent fauna	
	Impacts on Fauna Habitat loss of fauna Disturbances during construction direct wildlife mortality water quality and aquatic habitat integrity	Schedule construction activities outside of critical biological periods, such as breeding or nesting seasons for key faunal groups	
		Designate and enforce “no-go” zones within or adjacent to the project area, based on ecological survey data	
		Use low-noise construction equipment	
		Limit worker access and movement in high-sensitivity ecological areas	
		Conduct mandatory environmental awareness and wildlife sensitivity training for all site workers and contractors	
		Avoid or limit artificial lighting during night-time	
		Inspect all work areas before initiation of clearing or excavation, and safely relocate any animals discovered	
		Monitor water quality regularly in adjacent aquatic habitats for signs of turbidity, chemical and oil contamination	



Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
	reducing bird mortality at wind energy facilities	an avifauna detection and control mechanism has been proposed for implementation. A radar-based automated early warning system will be installed to mitigate bird collisions across the entire wind farm area	
5. Hydrological aspects	Possible disturbances to the natural drainage pattern	Drainage Management Plan will be developed for the construction phase of the access roads and road design processes will ensure proper conveyance of flow paths across the roads with necessary hydraulic structures	No special mitigation measures are required
6. Water Quality	Reduction in the water quality of the nearshore water due to high concentration of suspended solids during construction and wash-out of sediments from the material stockpile	The stretch of the sea/lagoon within the construction area at the site may be fenced with a silt trap to limit the spread of sediments away from the working area	No special mitigation measures are required
		Cover stockpiles of soil and sand to minimize exposure to wind and water	
		Creating temporary turbidity barriers along the shoreline to prevent erosion during onshore construction to prevent or lessen the spread of fine material	
	Reduction in the water quality due to fuel or oil spillages	Maintenance of the vehicle should not be carried out on-site but in a proper workshop located far away from the lagoon water body. Refuelling of the equipment should be carefully planned such that no fuel is spilt on the ground, into the lagoon	
		Prepare an emergency preparedness plan for approval by the Engineer in case of a spill of fuel or other fluids from equipment/machinery used or accidents that may lead to fuel spillage of fluids from equipment/machinery on land or in the lagoon	Emergency preparedness plan for oil spill could be applied if any oil spill happened



Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
	Reduction in the water quality due to mixing of wastewater generated during the construction, solid waste generation	Refer to the mitigation measures provided under Solid water and wastewater	
7. Social impacts	Impact on fishing, fish farms, fishing communities, and ongoing aquaculture activities	provide fair and timely compensation program to the affected parties in cases where impacts to infrastructure, private property, or agricultural crops are unavoidable	No special mitigation measures are required
		minimal disruption to the fisheries and beach seine operators will be ensured by meaningful consultations while conducting the loading and unloading operations	No special mitigation measures are required
	Impact on fishing, fish farms, fishing communities, and ongoing aquaculture activities	Material transportation schedules will be planned to avoid beach seine operation times as much as possible	
		Barges transporting materials will maintain a safe and adequate distance from active beach seine operations to prevent interference	
		cost-effective and accessible GRM will be established to address any concerns or complaints from affected communities, helping to mitigate unanticipated impacts	
	Impacts on recreational activities in the area, including kite surfing	Wind monitoring program will be initiated prior to the installation of these turbines to assess any changes in wind patterns and their potential effects on kite surfing conditions. Final turbine siting decisions will be made in consultation with kite surfing operators to ensure that any adverse impacts are avoided or minimized	



Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
	Concerns about the possibility of a gusty effect due to the tower locations T14 and T15		
8. Traffic implications	Impacts due to transportation of material and equipment, and an increase in traffic volumes during the construction period	Issuing advance public notices with transportation schedules	
		Deploying pilot vehicles to alert the public to incoming convoys	
		Establishing temporary bypass routes at key locations to facilitate uninterrupted traffic flow.	
		Transportation activities will be scheduled to avoid peak times, including the start and end times of school sessions, as well as during religious services at the church and mosque.	
		One-way vehicular movement along narrow or constrained sections of the access routes will be carefully managed by deploying trained flagmen equipped with "Stop" and "Go" signboards. These personnel will coordinate traffic flow to ensure the safe and orderly passage of vehicles, minimize delays, and prevent congestion or accidents during material transportation.	
	Impacts on roads, buildings, and other properties during the transportation of material, equipment during the construction period	Contractor will bear full responsibility for rehabilitating and restoring any damaged private properties or public infrastructure to their original or improved condition. To ensure timely restoration, the project proponent will oversee the implementation of these measures and will secure the necessary financial commitments in advance to guarantee prompt action.	



Sector	Impact	Proposed Mitigation Measures	
		Construction Phase	Operation Phase
9. Solid waste and Hazardous waste	Improper discharge of solid waste Washing out waste to sea, blockage of natural drainage paths and spreading of dust could happen	Collect the solid wastes generated in labour camps separately using correctly marked containers (eg. Organic waste, Plastic, Paper, Glass etc.) and disposed of off acceptably to designated places for such disposal.	no waste generation sources during the operational phase of the project
		Avoid open burning of solid waste within the site as this could cause a nuisance to the people in the area	
10. Wastewater	Improper discharge of wastewater/sewage	Provide an adequate number of temporary toilets at worksites;	no waste generation sources during the operational phase of the project
		Provide septic tanks/soakage pits of adequate capacity for all toilets so that it can function properly for the entire duration of the construction phase.	
		Train the construction employees on Project sanitation practices.	
11. Health and safety	Accidents due to site works	Take all precautions as per the Occupational Health and Safety Act.	Take all precautions as per the Occupational Health and Safety Act
		Develop appropriate traffic management schemes during the construction phase to ensure the safety of road users, including pedestrians who transit at the site.	
		Provide appropriate personal protective equipment to workers on site.	



For proper environmental management and effective implementation of the mitigation measures during construction and operation phase of the project, it is essential that an Environmental Monitoring Plan (EMoP) be formulated. Environment monitoring will be focused on the most significant impacts identified in the EIA. The objectives of environment monitoring plan (EMP) is;

- to monitor the performance of a project and the effectiveness of mitigation measures.
- to verify the impact of the project on the environment predicted during Environment Impact Assessment (EIA) studies.

To ensure the effective implementation of the EMP and weigh the efficiency of the mitigation measures, it is proposed to undertake environmental monitoring both during construction and operation period of the proposed coastal protection project.

The proposed environmental monitoring plan is illustrated in **Table 6-2**

Table 6.2: Environmental Management Plan

KEY: Phase: PrC=Pre-Construction; C=Construction; PoC=Post Construction; Responsibility: DEV: Project Developer; CC = Construction Contractor; CCCRM=Department of Coast Conservation & Coastal Recourses Management; MF=Ministry of Fisheries; DFAR= Department of Fisheries and Aquaculture Resources								
Sector	Potential Environmental Impact	Mitigation Actions	Monitoring Parameters / Activities	Monitoring Locations	Phase	Monitoring Frequency	Responsibility	
							Monitor	Supervise
Coastal Environment	potential erosion / accretion during the construction phase	Refer Sector 1 of Table 6.1	Beach profile	Jetty location	C, PoC	One at beginning and one at end of the project	DEV / CC	CC&CRMD
	Degradation of Sea Water Quality		Water quality parameters	Selected five sampling locations within site where required	C	Monthly	DEV / CC	CC&CRMD
Noise and vibration	Excessive noise & vibration generation during the construction	Refer Sector 2 and 3 of Table 6.1	Noise level	Selected five sampling locations within site where required	C	Monthly	DEV / CC	CC&CRMD
	Excessive noise & vibration generation during the construction		Noise level and/or vibration Crack survey	Based on the requirement	C	On a complaint-to-complaint basis	DEV / CC	CC&CRMD
Ecological impacts	Habitat loss due to fragmentation, edge effects and development of invasive species	Refer Sector 4 of Table 6.1	Visual observation of the cleared area and the surrounding	Tower locations Along the roads	C	Monthly	DEV / CC	CC&CRMD
	<u>Impacts on vegetation</u> Direct removal of native vegetation, invasive plant species		Vegetation cover, survival rate of planted species	Tower locations Along the roads	C	Quarterly	DEV / CC	CC&CRMD
	<u>Impacts on Fauna</u>		Wildlife sightings, disturbance events	Tower locations Along the roads	C	Weekly	DEV / CC	CC&CRMD

KEY: Phase: PrC=Pre-Construction; C=Construction; PoC=Post Construction; Responsibility: DEV: Project Developer; CC = Construction Contractor; CCCRM=Department of Coast Conservation & Coastal Recourses Management; MF=Ministry of Fisheries; DFAR= Department of Fisheries and Aquaculture Resources								
Sector	Potential Environmental Impact	Mitigation Actions	Monitoring Parameters / Activities	Monitoring Locations	Phase	Monitoring Frequency	Responsibility	
							Monitor	Supervise
	Habitat loss of fauna Disturbances during construction direct wildlife mortality water quality and aquatic habitat integrity							
	potential bird mortality at wind energy facilities		Mortality rate	Tower locations Along the transmission line	PoC	Daily	DEV	CC&CRMD
Water quality	Reduction in the water quality of the nearshore water	Refer Sector 6 of Table 6.1	Physical and chemical parameters; Turbidity, TSS, DO, NO ₃ ⁻ , PO ₄ ⁻ Bacteriological Parameters BOD, Total Coliform & Faecal Coliforms	Selected five sampling locations within site where required	C	Monthly	DEV / CC	CC&CRMD
Social impacts	Impact on fishing, fish farms, fishing communities, and ongoing aquaculture activities	Refer Sector 7 of Table 6.1	Grievance handling	On requirement	PrC C PoC	On a complaint-to-complaint basis	DEV / CC	DFAR, MF
	Impacts on recreational activities in the area, including kite surfing		Wind speed and direction	Close to T14 and T15	Pre	Daily	DEV	CC&CRMD



6.2 Implementation of Mitigation Measures and Monitoring Program

Concerning Mitigatory Measures (as described in Chapter 5), the EMP sets out the mechanisms for the implementation of such measures for which a prior agreement has to be reached between the Contractor and the Project Proponent (PP). The agreement is required because the EMP is part of the contract. Therefore, the EMP will be used as a means by which the Contractor (and any Sub Contractors) will implement the recommended mitigation measures and achieve the environmental performance standards defined and recommended in environmental legislation, the EIA and the Contract. The primary reason for adopting the EMP approach is to make all parties including the Contractor aware of environmental responsibilities and to be proactive in their commitment to achieving the standards specified.

For fisheries related issues, joint marine resource monitoring committees will be formed including DFAR officers, local '*padu samithiya*' leaders, and project developers. These groups will help document any fish stock changes and ensure participatory mitigation planning.

6.2.1 Implementation of Monitoring Procedures

With respect to Monitoring Procedures, the EMP has set out the relevant mechanisms and institutional arrangements to achieve the objectives of Environmental Monitoring through the Environmental Monitoring Program. As the EMP will form part of the contract there will be provisions to ensure that the Contractor fulfils his obligations regarding the implementation of mitigation measures.

It is recognized that monitoring is required to ensure that the project includes the satisfactory implementation of the EIA recommendations and to confirm that no potential adverse impacts have been excluded from the assessment process. The EIA team believes that the wider participation of all stakeholders is important to achieve this objective. For this purpose, it is important to develop and implement a formal mechanism for such participation and dissemination of information to the general public. If necessary, technical assistance should be provided for an adequate understanding of project interactions with the environmental components and mitigatory actions.

Arising from such activity, the Environmental Management Committee (EMC) in consultation with the Contractor(s) should develop a mechanism to manage, investigate, respond and act upon, any issues raised by the public during construction.



6.2.2 Institutional Arrangement and Implementation Mechanism

6.2.2.1 Responsible Agencies

The EMP sets out the mechanisms for the implementation of actions with respect to the mitigatory measures as described in Chapters 4 and 5 of this EIA. The EMP will be used as means by which the PP and contractors will implement the recommended mitigation measures and achieve the environmental performance standards defined and recommended in Sri Lankan environmental legislation and in the EIA. The primary reason for adopting the EMP approach is to make all parties aware of environmental responsibilities and to be proactive in their commitment to achieve the standards specified.

Effective implementation and supervision of the environmental mitigation measures and monitoring activities identified in this document can only be achieved through a suitable institutional mechanism involving stakeholders of the project. A broad institutional mechanism for environment safeguards associated with the project, roles and responsibilities of various agencies and parties for implementing environment safeguards are provided below.

Further, the need for an institutional mechanism to address possible grievances related to the proposed package during the project implementation period was identified by the EIA team. The detailed description of the Grievance Redress Mechanism is provided in the Chapter 8.



Table 6.3: Organizations Involved in EMP Implementation and their Responsibilities

Agency	Project related responsibilities
Executing Agencies:	
Project Developer	Project Implementing Agency or the Project Proponent (PP). The PP/Developer has the overall responsibility of implementing the recommendations and mitigation measures indicated in this report throughout the project. The PP/Developer will lead in environmental monitoring, implementation of mitigation measures and obtaining environmental clearances. Accordingly, a qualified environment manager and a site environment coordinator will be assigned to ensure effective implementation of the proposed EMP and relevant mitigation measures are in action. Resolving issues that impede the progress of project implementation, through inter-agency coordination.
	Overall project management and coordination: project administration; technical oversight (engineering and environmental); review, supervise and approval of the work of the contractor.
	Responsible for ensuring the project adheres to national law and the policy of the lender, including social and environmental safeguards.
	Ensure inclusion of environmental mitigation measures in design, tender documents and budgets as specified in project EMP. Environmental supervision to ensure that all mitigation is provided as specified in the project EMP (by design consultants, contractor, and sub-contractors etc.). Supervision of the work of the contractor to ensure that the project is constructed as designed.
Main contractor	Responsible of undertake the dredging operations as mentioned;
	Inclusion of environmental mitigation measures in construction activities and budgets as specified in project EMP. Responsible for ensuring the project adheres to national law and the policy of the lender, including social and environmental safeguards. Day-to-day project management and coordination: project administration; technical oversight (engineering and environmental); review, supervise and approval of the work of the sub-contractors. Environmental supervision to ensure that all mitigation is provided as specified in the project EMP by their own staff, sub-contractors, and other service providers. Monitor the environmental compliance of the project according to the EMP.



Agency	Project related responsibilities
Sub-contractors	Construction of specific elements of the project according to the designs and tender documents; Implementation of mitigation and monitoring impacts of construction activities as specified in the EMP, all the conditions of approvals and working instruction of the PP.
Regulatory Agencies:	
CC&CRMD	Functions as the Project Approving Agency (PAA). Overview the entire monitoring plan. Delegating appropriate powers to any other monitoring agencies if required; Responsible for administering the EIA process; including scoping, preparing ToR, public disclosure, review of EIA report, decision-making and compliance monitoring which would involve compliance with conditions and the effectiveness of the mitigatory measures.
Central Environmental Authority (CEA)	Assist the EIA implementation by the PAA, Monitor the quality of the environmental to ascertain the long term effects (if any).

With the approval of the EIA, the PP shall develop a comprehensive Project Environment Management Plan based on the EMP presented here with incorporating the EIA approval conditions and conditions of other clearances etc.

The guidelines for Project EMP are given below as well as the environmental compliance mechanism in place to ensure that the EMP is implemented properly.

1. Project organizational structure showing the implementation, supervision and reporting and responsibilities of key personnel
2. The project program and work activities
3. Specific Management Plans as follows.



Table 6.4: Summary of Management Strategy

Specific Management Plan	Description
Traffic Management Plan	Traffic management during construction
Oil Spill Contingency Plan	Management strategies of accidental spillage of oil during barge operation, ferry operation and construction
Waste Management Plan	Solid waste & other hazardous waste management during construction
Socio-economic Management plan	Socio-economic management plan will address the impacts on fisheries activities during the construction. Proper management scheme will be worked out in detail by the relevant Ministries and organizations /departments involved, with the inputs of community-based organizations (CBOs) such as the fishermen's societies, churches, etc.
Health & Safety Management Plan	It is the Contractors responsibility to implement an Occupational Health and Safety Management System during the construction period to prevent injury to persons or damage to property. The Contractor shall take precautions for protection against risks and shall inspect Health Safety and Environment conditions where the Works are being executed.
Safety and Disaster Management Plan	It is the responsibility of the Project Developer to implement a Safety and Disaster Management System to cover the potential hazards during the operational period to prevent injury to persons or damage to property - Due to the possible blade accidents, fall hazards, fire hazards, etc. - Unexpected situations such as storm surges, cyclones, erosion etc

In addition to the details provided in Table 6.3 and Table 6.4, following Table 6.5 elaborates the Agency Commitments Matrix for the proposed EMP in detail.



Table 6-5: Agency Commitments for Proposed EMP

Category	Agency	Phase	Measure / Condition	KPI	Evidence	Due Date / Frequency	Responsible
Biodiversity	DWC	Pre-construction	Integrate 12-month avifaunal survey and submit a standalone report (methods, seasons, species, and mitigation) to DWC (copy CC&CRMD).	Report submitted and acknowledged; sensitivity maps ready for micro-siting.	Final EIA + Avifauna Report; DWC receipt.	Before Final EIA submission.	EIA Lead; Biodiversity Specialist
	DWC	Pre-construction	Micro-site turbines/roads to avoid mapped migration corridors and roost/nest buffers.	Zero turbines/infrastructure within defined no-go buffers.	IFC layout + GIS overlays; change log.	Before IFC drawings are issued.	Design Engineer; Biodiversity Specialist
	DWC	Pre-construction	Specify bird-friendly obstacle lighting (shielded / activation-on-demand).	Lighting spec meets DWC guidance; sign-off obtained.	Specs; datasheets; DWC concurrence note.	Before turbine procurement.	Electrical Lead; HSE Manager
	DWC	Construction	Implement curtailment protocol during peak migration/adverse weather; set up 5-year post-construction monitoring.	Curtailment logs maintained; annual monitoring reports.	Operational protocol; SCADA logs; annual reports.	Protocol finalized pre-commissioning; reports annually.	Operations Lead; Biodiversity Specialist
	DWC	Construction	Initiate habitat restoration/offset works as approved by DWC.	Milestones met; ≥80% survival at 24 months.	Restoration plan; planting logs; audits.	Start during construction; quarterly tracking.	Biodiversity Specialist; Contractor
Biodiversity/Forestry	Forest Dept.	Pre-construction	Submit GIS/KML for external TL and internal cables; avoid forests/mangroves; justify any constrained segments.	FD concurrence issued; 0 new RoW in gazetted forests.	Cover letter; GIS/KML; justification memo; minutes.	Before TL detailed design / wayleaves.	Transmission Engineer; Permits Coordinator
	Forest Dept.	Pre-construction	Maintain ≥30 m buffer to mangrove stands; include fencing/signage in drawings.	Buffers mapped/staked; compliance verified.	IFC drawings; staking photos; inspection sheets.	Stake prior to site clearance.	Civil Lead; Site Surveyor
	Forest Dept.	Pre-construction	Compensatory planting ≥3:1 (natives) with maintenance & monitoring.	Ratio ≥3:1; ≥80% survival at 24 months.	Approved plan; receipts; survival surveys.	Plan before clearing; implement during construction.	HSE Manager; Landscape Contractor
	Forest Dept.	Pre-construction	Provide certified private land titles/surveys for wind-farm parcels.	100% parcels verified by FD.	Title copies; survey plans; FD email.	Before FD concurrence.	Land/Legal Lead
	Forest Dept.	Construction	Enforce no-go buffers; no machinery within mangrove buffers.	Zero buffer breaches; weekly checks logged.	HSE checklists; geo-tagged photos.	Weekly during construction.	HSE Manager; Site Supervisors
Tourism/Visual	SLTDA	Pre-construction	Protect kite-surfing & water-sports corridors; obtain operator agreements.	MoUs/letters signed; corridors mapped in IFC drawings.	Agreements; corridor maps; MoMs.	Before marine works & IFC issue.	Community Liaison; Planning Lead
	SLTDA	Pre-construction	Complete Visual Impact Assessment and adopt mitigations (neutral finishes, uniform towers, landscaping).	VIA approved; finishes and landscaping in procurement/IFC.	VIA report; paint specs; landscape plan.	Before procurement of turbines/substation finishes.	Design Lead; Architect/Landscape
Access/Coastal	SLTDA	Pre-construction	Protect dunes & public beach access; set dune buffers; designate no-go areas.	Dune buffers shown; access routes preserved.	IFC drawings; access plan; routing notes.	Before mobilization.	Civil Lead; Construction Manager
Tourism/Access	SLTDA	Construction	Keep public access open or provide safe alternatives; minimise footprints and sign work zones.	Zero unmitigated closures; signage in place.	Traffic/pedestrian plan; photo records.	Continuous.	Construction Manager; Community Liaison



Category	Agency	Phase	Measure / Condition	KPI	Evidence	Due Date / Frequency	Responsible
Tourism/Seasonality	SLTDA	Construction	Schedule high-impact works outside peak tourism seasons (Dec–Apr, Jul–Aug) or pre-notify with added mitigations.	No peak-season conflicts; notices ≥7 days in advance.	Work schedules; notices; acknowledgement emails.	Per activity/seasonal.	Construction Manager; HSE
Regulatory/Permitting	SLSEA	Pre-construction	Integrate all line-agency conditions into Final EIA/EMP before Energy Permit application.	Cross-reference matrix complete; SLSEA readiness confirmed.	Updated EIA/EMP; matrix; SLSEA note.	Before Energy Permit application.	EIA Lead; Permits Coordinator
Regulatory/Planning	SLSEA	Pre-construction	Provide decommissioning plan; grid interconnection compliance; local content/employment; periodic ops reporting.	Documents appended to Final EIA; acceptance recorded.	Decom plan; grid compliance; HR/ops templates.	With Final EIA package.	EIA Lead; Electrical/Grid Lead; HR/Community
Coastal Setback	SLSEA	Pre-construction	Confirm ≥60 m coastal setback (consistent with CC&CRMD) in final design and on ground.	All works outside reservation; pegs set and checked.	IFC drawings; pegging records; site checks.	Before mobilization.	Design Lead; Site Surveyor
PAC Coordination	SLSEA	Pre-construction	Coordinate PAC matters; align with CEB/CC&CRMD; consolidate agency conditions for disclosure.	PAC pack issued; consolidated conditions list.	PAC brief; consolidated list; minutes.	Before PAC meeting.	Project Director; Permits Coordinator
Compliance	SLSEA	Construction	Proceed only under approvals granted post-EIA; maintain reporting to SLSEA and PAC as required.	Timely reports; 0 work outside approvals.	Approval register; monthly/quarterly reports.	Monthly/quarterly.	Construction Manager; HSE
Pollution Control	CEA	Pre-construction	Obtain EPLs for batching plant, workshops, camps; embed conditions into EMP.	All EPLs on file; no works without EPL.	EPL copies; EMP annex.	Before facility commissioning.	Permits Coordinator; HSE Manager
Wastewater	CEA	Construction	Provide approved sanitation; collect/treat sewage & greywater; service portable toilets.	Zero uncontrolled discharges; logs up to date.	Service contracts; desludging receipts.	Weekly checks.	HSE Manager; Camp Boss
Solid Waste	CEA	Construction	Segregate; secure storage; licensed disposal; manage hazardous waste via licensed handlers.	100% waste transfers documented; Zero illegal dumping.	Manifests; licenses; photo records.	Weekly; monthly summary.	HSE Manager; Stores Supervisor
Storm-water	CEA	Construction	Install ditches/culverts; keep waterways clear; manage runoff during heavy rains.	Zero off-site flooding; drains functional.	Inspections; rain event logs; photos.	Continuous; intensified pre-monsoon.	Civil Lead; HSE
Compliance/Community	NWPEA	Construction	Meet provincial noise/dust standards; operate grievance mechanism; submit weekly compliance logs.	Noise within limits; grievances closed ≤10 working days.	Monitoring sheets; grievance register; weekly logs.	Weekly during construction.	HSE Manager; Community Liaison
Provincial Approval	NWPEA	Pre-construction	Apply for provincial EIA approval after PAC decision with Final EIA/EMP & fees.	NWPEA approval issued.	Application pack; fee receipt; approval letter.	Immediately after PAC decision.	Permits Coordinator
Marine Security	MOD/Navy	Pre-construction	Submit barge routes & moorings; avoid restricted zones; secure permissions.	Navy concurrence; Zero encroachment.	Charts; Navy letters; permits.	Before first marine move.	Marine Ops Coordinator
Marine Notifications	MOD/Navy	Construction	Issue ≥48-hour notices for major marine ops; coordinate Notices to Mariners.	100% notices on time; NtMs published.	Notice emails; NtMs.	Per operation.	Marine Ops Coordinator
Fisheries Interface	DFAR	Pre-construction	Prepare jetty/barge logistics plan aligned to fishing	Plan approved; Zero vessel discharges.	Plan; skipper briefings; inspection logs.	Before marine mobilization.	Marine Ops Coordinator; HSE



Category	Agency	Phase	Measure / Condition	KPI	Evidence	Due Date / Frequency	Responsible
			seasons/monsoons; enforce no-discharge policy.				
Fisheries Interface	DFAR	Construction	Escort boats; temporary exclusion zones; schedule heavy lifts in low-fishing periods; coordinate with fisheries societies.	Zero conflicts/accidents; minutes filed.	Marine traffic plans; buoy layouts; MoMs.	Per operation; monthly coordination.	Marine Ops Coordinator
Aquaculture Access	NAQDA	Pre-construction	Navigational Management Plan to keep farm waterways open; integrate feeding/harvest schedules.	Zero blocked farm channels; schedule agreed.	Plan; timetables; operator sign-offs.	Before marine mobilization.	Marine Ops Coordinator; Community Liaison
Water Quality	NAQDA	Construction	Baseline then monthly water-quality monitoring (turbidity, DO, salinity) up/downstream of works.	≥3 stations/month; thresholds not exceeded.	Lab results; maps; monthly reports.	Monthly during construction.	Env. Monitoring Team
Cultural Heritage	Dept. of Archaeology	Pre-construction	Full archaeological survey of turbines, access, trenches, jetty & laydown; submit report.	Survey complete; Dept. acknowledgement/clearance.	Survey report; cover letter; maps.	Before ground disturbance.	Archaeologist; EIA Lead
	Dept. of Archaeology	Pre-construction	Include chance-find SOP in EMP; brief all contractors & crews.	100% crews trained; attendance recorded.	EMP SOP; training logs; induction slides.	Before site induction; refresh quarterly.	HSE Trainer; Archaeologist
	Dept. of Archaeology	Construction	Enforce chance-find SOP; protect finds; resume work only after clearance.	Zero SOP breaches; all incidents closed with Dept. sign-off.	Incident forms; Dept. correspondence.	As needed during excavation.	Construction Manager; HSE
Local Services	Pradeshiya Sabha	Construction	Service agreement for construction solid-waste collection/transport/disposal.	Scheduled collection; Zero open dumping/burning.	Service contract; disposal receipts; inspection logs.	Weekly during construction.	Contractor; LA Liaison
	Pradeshiya Sabha	Construction	Road-opening permits; reinstatement deposits; utilities coordination for local roads.	100% permits obtained; reinstatement closed out.	Permits; receipts; reinstatement certificates.	Per work front; close-out end of works.	Site Engineer; Utilities Coordinator
Public Health	Pradeshiya Sabha	Construction	Sanitation facilities plan and service contracts for camps/work fronts.	Adequate facilities per headcount; current service log.	Sanitation plan; service records.	Before mobilization; weekly service.	Camp Boss; HSE
Land/Admin	Divisional Secretariat	Pre-construction	Land schedule with deeds/surveys/consents; GN certifications; wayleave/easement agreements.	100% parcels documented; wayleaves executed.	Land book; GN letters; wayleave deeds.	Before works in each parcel.	Land/Legal Lead
Community	Divisional Secretariat	Pre-construction	Community awareness in affected GN divisions; grievance mechanism & monthly summaries to DS.	≥2 sessions/GN; grievances closed ≤10 working days.	Attendance sheets; hotline records; monthly summary.	Prior to mobilization; monthly during construction.	Community Liaison; HSE

Field	Meaning
Category	Theme (Biodiversity, Transport, etc.).
Agency	Line agency issuing the condition.
Phase	Pre-construction or Construction.
Measure / Condition	Exact requirement / mitigation commitment.
KPI	Measurable indicator of completion/compliance.
Evidence	Documents/records used to prove compliance.
Due Date / Frequency	Deadline or recurrence for the action.
Responsible	Project role(s) accountable for delivery.



7 EXTENDED COST-BENEFIT ANALYSIS

7.1 Introduction

Project Economic Analysis: Project economic analysis is a vital tool used to determine whether an investment is a sound use of a country's scarce resources. It ensures that these resources are allocated efficiently and that the project contributes positively to national development by improving the welfare of the population. All inputs used in a project have an opportunity cost, as they could have been employed in alternative ways to generate value elsewhere in the economy.

An economically viable project must meet three key conditions:

- It must represent the least-cost or most efficient option for achieving the intended development outcomes;
- It must generate an economic surplus over and above the opportunity cost of the resources used; and
- It must have sufficient funding and a robust institutional framework to ensure successful operation and long-term sustainability.

Economic analysis is carried out from the perspective of the entire economy, rather than from the viewpoint of individual stakeholders. It focuses on assessing the project's overall impact on national welfare. Unlike financial analysis, which uses market prices, economic analysis employs economic prices—also known as shadow prices—to capture the true opportunity cost of resources. This approach allows for a more accurate assessment of the real economic value of project costs and benefits, providing a clearer picture of the project's contribution to the country's development objectives¹⁰.

The Project: The Dutch Bay Onshore Wind Farm Project is subject to an Extended Cost Benefit Analysis (ECBA) to assess its overall contribution to national welfare. This approach goes beyond traditional economic analysis by incorporating environmental and social impacts into the economic appraisal. ECBA extends the standard Cost-Benefit Analysis (CBA) by assigning monetary values to these externalities—treating them as either costs or benefits—and analyzing

¹⁰ Asian Development Bank (2017) *Guidelines for the Economic Analysis of Projects*. Mandaluyong City, Philippines: Asian Development Bank.



their effect on the project's viability. The analysis is conducted using a discounted cash flow framework, which allows for a comprehensive evaluation of the project's long-term net benefits.

Key economic indicators used in this process include the Net Present Value (NPV), Economic Internal Rate of Return (EIRR) and Benefit-Cost Ratio (BCR). This rigid economic analysis ensures that the Dutch Bay Onshore Wind Farm Project is not only technically sound and environmentally sustainable, but also economically justifiable for Sri Lanka's development priorities. The Dutch Bay Onshore Wind Farm Project also gives rise to environmental and social impacts that affect its overall contribution to societal well-being. These impacts, whether beneficial or adverse, are systematically identified and evaluated through the Environmental Impact Assessment (EIA) process.

The **Dutch Bay Onshore Wind Farm Project**, proposed by **Envision Energy Co., Ltd, Sri Lanka**, is a large-scale renewable energy initiative located in **Kalpitiya, Puttlam**. The project spans a site area of approximately **0.81 sq. km** with a linear extent of **5.1 km** and is designed to install a total wind power capacity of **150 MW** using **15 wind turbine generator (WTG) units**¹¹, each with a capacity of **10 MW**. The facility is expected to operate for **8,760 hours per year**, delivering a **P50 Net Annual Energy Production (AEP)** of approximately **464,510 MWh/year**¹², which corresponds to **3,098 equivalent full load hours** and a **capacity factor of 35.4%**. A **Battery Energy Storage System (BESS)** with a capacity of **23.8 MWh** will support the grid integration and energy reliability of the wind farm. The construction phase is estimated to last **24 months**, with a planned operational lifespan of **20 years**.

The wind farm will be supported by a 132 kV booster station to facilitate the efficient transmission of electricity to the national grid. The proposed 150 MW wind power generation farm will be connected to the national grid from the wind power collector grid substation at Kalpitiya to the CEB Wariyapola switching station through a 220 kV double circuit overhead transmission line, with an approximate route length of 104.77 km.

The project area enjoys strong wind resources, with an average wind speed of 8.4 m/s, which is favourable for consistent power generation. A wake loss of 11.53% is accounted for in estimating net energy output, reflecting the reduction in efficiency caused by turbine placement. This project represents a significant step toward enhancing Sri Lanka's renewable energy capacity by

¹¹ Please refer pp 4,6,9,39 and 41

¹² Ref: Dutch Bay Onshore Wind Farm Project Feasibility study report (p. 215).



reducing dependency on fossil fuels, strengthening national energy security, and supporting long-term environmental sustainability and regional infrastructure development.

7.2 Demand Analysis

Sri Lanka's long-term electricity demand forecast from 2023 to 2047 underscores a sustained and significant growth direction in energy requirements, reinforcing the need for timely capacity additions and diversification through renewable energy. The net electricity demand is projected to increase from 16,741 GWh in 2023 to 54,315 GWh by 2047, marking a total increase of over 224% in 25 years, with an average annual growth rate of 5.0%. Similarly, net generation requirements are expected to rise from 18,186 GWh in 2023 to 58,560 GWh by 2047—an increase of over 222%—to accommodate both demand growth and transmission and distribution losses, which are estimated to be around 7.25% annually.

Peak electricity demand, which stood at 3,021 MW in 2023, is forecast to reach 9,426 MW by 2047, with a 25-year average growth rate of 4.9%, nearly tripling the peak capacity requirement. Notably, the five-year average growth rate for both net demand and generation is as high as 6.0%, highlighting a more aggressive near-term expansion. It is also projected that, from 2026 onwards, the day peak will surpass the night peak, indicating changing consumption patterns and the increasing role of commercial and industrial load profiles.

Electricity consumption by sector reveals that industrial and commercial users collectively accounted for a larger share than domestic users in 2021. This consumption pattern aligns with Sri Lanka's ambition to transition toward an export-oriented, industrialized economy. Access to electricity has expanded nationally, achieving near-universal coverage. Additionally, transmission and distribution losses have significantly decreased—from 21.4% in 2000 to just 9.46% in 2021—demonstrating improved operational efficiency within the power sector.

Going forward, sustained economic growth and improvements in living standards are expected to drive continued growth in electricity demand. This necessitates strategic investments in dispatchable capacity, grid infrastructure, and renewable energy integration to meet peak demand requirements reliably and cost-effectively. The gradual transition towards indigenous energy sources and natural gas will also play a critical role in mitigating external shocks from



global fuel price fluctuations, thereby enhancing energy security and supporting Sri Lanka's broader development agenda. (CEB 2023)¹³

These given statistics demonstrate a compelling need for additional and sustainable generation capacity. The Dutch Bay Onshore Wind Farm Project is strategically positioned to contribute to this growing demand by supplying clean, renewable energy to the national grid, thereby supporting energy security, reducing dependency on fossil fuels, and aligning with Sri Lanka's climate and energy transition goals.

7.3 Key steps in Extended Benefit Cost Analysis

The key steps involved in this EBCA include:

- 1) Selecting suitable combinations of the 150 MW Dutch Bay Onshore Wind Farm Project components that effectively address and reflect the associated environmental and social impacts.
- 2) Collecting the necessary baseline data on project costs and benefits from the relevant authorities.
- 3) **Identifying environmentally and socially significant impacts** that are economically measurable, and determining whether they constitute net negative impacts (costs) or net positive impacts (benefits) to society and the environment.
- 4) **Acquiring the necessary physical data** related to these impacts (both costs and benefits) through field-level assessments.
- 5) **Valuing the environmental and social impacts** by applying appropriate economic valuation techniques, as carried out by the Economic Expert of the project.
- 6) **Conducting the ECBA** by incorporating an extended scope of costs and benefits, including the computation of key project performance indicators such as ENPV, EIRR and BCR.
- 7) **Incorporating all identified impacts and mitigation measures** outlined in Table 6.4 into the EBCA model.
- 8) **Interpreting the ECBA results** in comparison with the conventional Benefit-Cost Analysis (BCA) conducted during the project feasibility study, in order to evaluate the comprehensive

¹³ Ceylon Electricity Board, 2023. *Long Term Generation Expansion Plan 2023–2042*. Colombo: Ceylon Electricity Board. ISSN 2989-0012.



impact of the 150 MW Dutch Bay Onshore Wind Farm Project, taking into account its environmental and social dimensions.

7.4 Tools Used and Assumptions in ECBA

The identified impacts were quantified using established economic valuation methodologies. All underlying assumptions and reference standards employed in the analysis are detailed in **Table 7-1**.



Table 7-1: Assumptions and standards used in the extended cost-benefit analysis

Parameter	Standard / Assumption	Remarks
Discount Rate	14	Cabinet Memorandum No. 36/2025/P, dated March 18, 2025 ¹⁴ , acknowledges a WACC of 12.10% to account for comprehensive financial risks. However, to appropriately reflect social and environmental risks—which may not be fully captured by the financial WACC—it is reasonable to employ a higher discount rate of 14%.
Evaluation Period	2 years for construction and 20 years for operation.	Cost estimates for the construction were provided for a specified number of years and are consistent with the standard practices applied in similar power supply projects in Sri Lanka and comparable countries. Additionally, the Feasibility Study (FS) report specifies the minimum operational period.
Value	US\$	This is an established standard and is commonly used in similar project economic analysis.
Treatment of Inflation	Constant prices excluding Inflation were used.	Standard practices adopted in the economic analysis.

7.5 Project Costs

7.5.1 Construction cost

The cost items have been categorized under key components, including Engineering, Procurement, and Construction Cost (EPCC), project development and administration fees, and physical contingencies. All transfer payments have been excluded from the calculation of the final economic cost. The total economic cost of the project is estimated at US\$ 200.89 million, expressed in 2026 constant economic prices, as shown in **Table 7-2**. A detailed summary of cost breakdown is provided in **Annexure III M**, Table 1 and Table 2.

¹⁴ The economic analysis for the 150 MW Dutch Bay Onshore Wind Farm Project has been updated to align with key findings from Cabinet Memorandum No. 36/2025/P, dated March 18, 2025. This memorandum, which addresses an average plant capacity of 10 MW, has informed adjustments to the economic assessment, ensuring it accurately reflects the scale and specifics of the Dutch Bay project. The Dutch Bay Onshore Wind Farm will feature 15 Wind Turbine Generators, with a total projected output of 150 MW. Its compatibility with the 10 MW average plant capacity outlined in the Cabinet Memorandum has allowed for a more precise and informed economic evaluation.



Table 7-2: Distribution of Capital Expenditure

Year	Year 1	Year 2	Total
US\$	80,358,651	120,537,977	200,896,628
%	40%	60%	100%

Refer to pages 218 to 242 of the Dutch Bay Onshore Wind Farm Project Feasibility Study Report for a detailed breakdown of the financial cost estimate.

7.5.2 O&M Costs

The **2025 Renewable Energy Tariff Revision Committee** has calculated **Plant Operation and Maintenance (O&M) costs** as a percentage of the previously determined capital costs. Based on these calculations, the committee has set specific O&M costs for various renewable technologies. For **wind technology**¹⁵, the estimated O&M cost is **3% of the capital cost for years 1 to 10**, and **4% of the capital cost for years 11 to 20**. Therefore, the Operation and Maintenance (O&M) cost for the reference project is estimated in accordance with the calculations provided in Table 5 of Cabinet Memorandum No. 36/2025/P, dated March 18, 2025, while reflecting sound project economic analysis principles.

To ensure a robust and consistent economic analysis, the Operation and Maintenance (O&M) cost is estimated at 3.5% of capital cost for the entire operational life cycle. This figure incorporates a 0.25% annual real resource requirement.

The Operation and Maintenance (O&M) cost estimation for the Dutch Bay Onshore Wind Farm Project is based on an annual power generation¹⁶ of 464.51 million kWh, over a 20-year operational lifespan, resulting in a total lifetime energy output of approximately 9.29 billion kWh.

¹⁵ The economic analysis for the 150 MW Dutch Bay Onshore Wind Farm Project has been updated to align with key findings from Cabinet Memorandum No. 36/2025/P, dated March 18, 2025. This memorandum, which addresses an average plant capacity of 10 MW, has informed adjustments to the economic assessment, ensuring it accurately reflects the scale and specifics of the Dutch Bay project. The Dutch Bay Onshore Wind Farm will feature 15 Wind Turbine Generators, with a total projected output of 150 MW. Its compatibility with the 10 MW average plant capacity outlined in the Cabinet Memorandum has allowed for a more precise and informed economic evaluation. (Ref: Cabinet Memorandum No. 36/2025/P, dated March 18, 2025, Table 7)

¹⁶ Ref: Dutch Bay Onshore Wind Farm Project Feasibility Study Report. Table 14.7.21, Page 215.



The total static investment is estimated at US\$ 200.90 million on an economic basis¹⁷. Based on the P50 Net AEP (MWh/year) Most Likely Case, the O&M cost is calculated at 1.47 US cents/kWh (economic) for the year 2025 and adjusted¹⁸ to 1.55 US cents/kWh in 2026 constant economic prices¹⁹. These estimates reflect efficient long-term operation and ensure the project's economic viability over its life cycle. For the project's lifespan, this base year O&M cost was adjusted with annual real resources requirement.

The real resource requirement²⁰ represents the actual expected growth in the quantity or cost of real inputs over time—such as increased maintenance needs due to aging equipment, higher consumption of spare parts, more frequent servicing, and additional labor requirements. Accordingly, a 0.025% annual increase in operational costs has been applied to reflect the projected real growth in resource requirements throughout the project's operational period.

The Feasibility Report for the Dutch Bay Onshore Wind Farm Project examined three scenarios²¹: P50 Net AEP (MWh/year) as the Most Likely case, P75 Net AEP (MWh/year) as the Conservative case, and P90 Net AEP (MWh/year) as the Worst-Case scenario. This analysis utilized the P50 Net AEP (Most Likely) scenario for calculating O&M costs. Each of these three project scenarios corresponds to a different Plant Capacity Factors (PCF). Furthermore, all three results were incorporated into the sensitivity analysis conducted for the ECBA. Please refer to Table 4 of **Annexure III M** for detailed information on the annual O&M costs.

¹⁷ Standard Conversion Factor (SCF) will be used to convert all financial values to economic values.

$CF = BF / DF$, Where; BF – Boarder Price Factor, DF – Domestic Price Factor. $CF = (M + X) / [(M + T_m) + (X - T_x)]$ Where; M – Imports, X – Exports, T_m –Import Tax and T_x – Export Tax. (Please refer to Annex 1, Table 3 for the calculation of the SCF).

¹⁸ **Note:** In the context of Cost-Benefit Analysis (CBA) for infrastructure or energy projects, Operation and Maintenance (O&M) costs are typically treated as separate, recurring annual expenses and are calculated independently of capital cost recovery. This approach is standard practice in economic analysis. Capital cost is considered an upfront investment incurred at the beginning of the project, while O&M costs represent ongoing expenditures necessary to keep the project operational throughout its life cycle. Importantly, capital recovery is a concept relevant to financial accounting or financial analysis and is not included in the economic definition of O&M. Incorporating capital recovery within O&M costs in economic analysis could result in double-counting of capital expenditures, which should be avoided to maintain analytical accuracy.

¹⁹ Ref: Land-Based Wind Market Report: 2022 Edition. August 2022, US Department of Energy, Office of Scientific and Technical Information.

²⁰ Asian Development Bank (2017) *Guidelines for the Economic Analysis of Projects*. . Mandaluyong City, Philippines: Asian Development Bank.

²¹ Please refer to pages 9 and 215 of the Dutch Bay Onshore Wind Farm Project Feasibility Study Report for related information



7.5.3 Risk and environmental mitigation cost

As a mitigation measure against potential turbine failures, the cost-benefit analysis includes annual insurance premiums as part of the project's operating costs. This analysis considers comprehensive coverage through an Operation All-Risk policy²², which includes protection against equipment failure, storms, lightning strikes, and fire after the commissioning stage. Additionally, it accounts for Business Interruption insurance to cover revenue losses resulting from turbine downtime, as well as Third-Party Liability and Environmental Liability coverage to address potential damages to external parties and environmental impacts.

Given the importance of environmental liability coverage—which safeguards against potential claims related to pollution or environmental damage—the annual Risk and Environmental Mitigation Cost is presented as a separate line item, distinct from the general project operating costs. This approach promotes transparency and underscores the project's proactive risk management strategy.

For wind turbine projects, the annual risk premium²³ is typically calculated as a percentage of the total capital investment (CAPEX). Based on industry benchmarks and published literature, this premium generally falls within the range of 0.3% to 0.7% of the total investment cost. In this analysis, a conservative estimate of 0.3% of the total investment cost has been applied.

It is recommended to implement a radar-based bird detection system to help prevent bird collisions with wind turbines. Based on available project information, the estimated cost of such

²² Asian Development Bank (2017) *Guidelines for the Economic Analysis of Projects*. . Mandaluyong City, Philippines: Asian Development Bank.

²³ IEA Wind TCP Task 26 (2020) – “Wind Energy in the United States”

IRENA (2015), Renewable Energy Project Finance

Lazard's Levelized Cost of Energy (LCOE) Analysis (Edition 14, 2020)



a system is approximately²⁴ US\$ 250,000. This cost is treated as a direct project cost and has been incorporated into the ECBA model under the risk and environmental mitigation cost²⁵.

Note: *This is a tentative cost estimate and may be subject to change based on input from Sector Experts. However, as a conservative approach, the above cost (US\$ 250,000) has been used in the ECBA model to ensure more accurate and reliable results.*

7.5.4 Worker's safety and social cost

As part of the socio-economic risk and safety assessment for the Dutch Bay Onshore Wind Farm Project, an insurance premium analysis was conducted to quantify the potential social cost associated with occupational risks during both the construction and operational phases of the project. These costs have been incorporated into the ECBA model under the social cost category.

During the construction phase, the workforce will include approximately 220 (Assume 200 construction workers and 20 technical staff), engaged over a two-year period. The operational phase is expected to span 20 years, employing 10 staff members for long-term operational and maintenance support. The average age of workers is assumed to be 32 years, with a retirement age of 55 years. Key assumptions used in the estimation include a discount rate of 12%, an annual salary increase rate of 5%, and a survival probability of 99% for the relevant age groups.

The assessment distinguishes between two employment categories—construction workers and technical staff—due to differences in wage structures and associated risk profiles. The Value of Statistical Life (VSL) has to be computed based on local benchmarks relevant to the project context. An insurance premium equivalent to 10% of the VSL per annum has been applied separately for both the construction and operational stages, reflecting a precautionary approach to cover potential fatality and injury-related risks.

²⁴ MERLIN™ Bird Control Radar System -<https://detect-inc.com/bird-control-radar-systems/>

Estimated Cost: Around US\$ 250,000 – 300,000

Mount Storm Wind Farm – West Virginia, USA - US\$ 200,000 – 400,000 range.

Thanet Offshore Wind Farm – United Kingdom - US\$ 300,000+

Alta Wind Energy Center – California, USA. US\$ 300,000 – 500,000.

²⁵ The radar-based bird detection system is included under risk and environmental mitigation costs, to be implemented at the commencement of project operations.



These premium values represent the cost of insuring the workforce against occupational hazards and have been accounted for in the ECBA as part of the broader social and safety-related cost considerations of the project. This approach ensures that the project proactively manages and internalizes potential human risk factors, thereby promoting safer working environments and responsible infrastructure development.

7.5.5 Opportunity cost of affected lands uses

Each Wind Turbine Generator, covering the 50-meter diameter circular area. Therefore, the 15 Wind Turbine Generators have a calculated direct impact area of 13.02 hectares. This land parcel includes various land use categories, such as barren land, coconut plantations, unclassified forest areas, unclassified water bodies, lagoons, marshlands, other cultivated lands, sand areas, scrub lands and water holes. (Refer Annexure III M, Table 5 and 6 for details).

These natural and cultivated resources are either fully or partially affected due to the construction and operation of the WTGs. The total economic cost associated with the impacted area has been assessed and incorporated into the ECBA model using a valuation approach based on cost per hectare (US\$/ha). Based on the aforementioned estimation, the total direct impact area spans 13.02 hectares. To quantify the overall economic impact, different values were applied to each land use category. The following table details the direct impact area for each land use category, its estimated value per hectare, and the methodology employed for this estimation.

Table 7-3: Land Use Pattern, Value and Valuation Method

No.	Land use	Economic Value (ha. /US\$/yr)	Valuation Method	Remarks
1	Barren land	0	Opportunity Cost	Already accounted for ECBA
2	Coconut	191	Opportunity Cost	Used for the ECBA
3	Forest - Unclassified	261.53	Env. Damage Cost	Used for the ECBA
4	All Water areas - Unclassified	19.66	Benefit Transfer	Used for the ECBA
5	Lagoon	0	Ecosystem Service	Not taken for the ECBA
6	Marsh lands	350	Benefit Transfer	Used for the ECBA
7	Other cultivation	2938	Opportunity Cost	Used for the ECBA
8	Sand areas	4750	Opportunity Cost	Used for the ECBA
9	Scrub land	950	Opportunity Cost	Used for the ECBA
10	Water holes	19.66	Benefit Transfer	Used for the ECBA



7.5.5.1 Opportunity cost of land - barren land

The 45.74 hectares of barren land could be used for urban or infrastructure projects, which usually bring benefits when the land is in a good location. However, this land is not in a strategic location, so it's not suitable for such developments. Still, it could be used for renewable energy projects like solar or wind farms. If the land is not used for this purpose, the opportunity cost is the missed chance to produce clean energy, which could have brought both economic and environmental benefits.

However, this section does not address environmental benefits, as the Avoided Environmental Damage Cost resulting from reduced fossil fuel use has already been accounted for in the project's benefit stream. To avoid double counting of benefits, this specific advantage is not included in the opportunity cost assessment. The Asian Development Bank's Guidelines for the Economic Analysis of Projects (2017)²⁶ clearly emphasize the importance of avoiding the double counting of benefits. Specifically, the guidelines state: "When project economic benefits are estimated based on measurable impacts, it is important to avoid double counting." This ensures the accuracy and integrity of the economic analysis by preventing the overstatement of a project's benefits.

7.5.5.2 Opportunity cost of land - coconut land

Based on official figures from Sri Lanka's Department of Census & Statistics and the Coconut Research Institute (CRI), a coconut plantation's average yield is 6,601 nuts per hectare. Under full production capacity, this translates to a net revenue of Rs. 190,800 (US\$ 636) per hectare every two months, resulting in an estimated annual revenue of US\$ 3,816 per hectare.

However, the existing coconut lands within the project area are currently unproductive. Due to factors such as fertilizer availability, climatic conditions, and overall land productivity, it's assumed that only 5% of this potential revenue is realized with considering land parcel size. Therefore, the estimated economic price for these lands is US\$ 191 per year.

7.5.5.3 Cost of environmental damage - forest lands

Based on the land use pattern analysis within the 13.02 ha., a small tropical dry forest patch (below 0.5 ha.) was identified. From an economic perspective, all land use patterns are regarded as natural assets that generate a continuous flow of goods and services for society. The vegetation

²⁶ Asian Development Bank (2017) Guidelines for the Economic Analysis of Projects. Manila, Philippines, (p. 16).



within the proposed land parcel contributes to various environmental services and plays a role in sequestering atmospheric carbon. However, placing an economic value on these ecosystem services is a complex task.

According to the Millennium Ecosystem Assessment (MEA), ecosystem services are classified into four main categories: Provisioning, Regulating, Cultural, and Supporting Services. The reference values for these services were obtained from equivalent biomass listed in the TEEB (The Economics of Ecosystems and Biodiversity) Valuation Database for Sri Lanka.

A key source used in this valuation is a 2020 research study (Project Reports. 69) conducted by Portland State University, titled *“Economic Valuation of Ecosystem Services Provided by Forest Ecosystems in Sri Lanka: A Study Based on the 2010 Forest Cover Classification and the TEEB Database.”* The objective of this study was to estimate the Total Economic Value (TEV) of all ecosystem services provided by forest ecosystems in Sri Lanka, using 2019 US\$ values.

According to the study’s findings, the estimated annual unit value of ecosystem services in the Puttlam District for the year 2026 is US\$ 26,153 per hectare per year. Given the modest size of this forest patch, encompassing **0.39 hectares**, a project-specific value of **US\$ 261.53** has been included in the Economic Cost-Benefit Analysis (ECBA) calculations. This figure represents a 1% allocation based on the identified land extent, ensuring the economic assessment accurately reflects the unique scale of this particular area.

7.5.5.4 Environmental value of all water areas - unclassified

To estimate the environmental value of the unclassified water areas within the direct impact zone of the Dutch Bay Onshore Wind Farm Project, the Consultant first assessed the ecological significance of the project site. To quantify the value of the affected unclassified water, the Consultant employed an environmental valuation approach. This approach primarily utilized the **benefit transfer method** as the basis for the valuation.

The **benefit transfer method** is an economic valuation technique used to estimate the monetary value of ecosystem services or environmental benefits by applying results from existing studies conducted in similar contexts. Instead of conducting time-consuming and costly primary research, this method transfers data from one site (the "study site") to another with comparable characteristics (the "policy site"). It is commonly used in environmental impact assessments when primary data is unavailable, allowing decision-makers to approximate the value of non-market environmental goods such as biodiversity, carbon sequestration, or coastal protection.



Förster (2010)²⁷ quantified the direct use value of ecosystem services from irrigation tanks in the Kala Oya region of Sri Lanka at US\$ 2,972 per hectare per year of 2005 price. Based on this, the present study applied the benefit transfer method, adopting 0.25% of the original value to reflect site-specific differences. This results in an adjusted value of US\$ 19.66 per hectare per year, expressed in constant 2026 economic prices. (Some areas may have seasonal or intermittent water sources). This adjusted figure was then incorporated into the ECBA to represent the annual loss of ecosystem service value at the policy site due to project activities.

7.5.5.5 Environmental value of lagoon

The Puttalam Lagoon's mangrove ecosystem is vital to Sri Lanka's coastal zone, offering critical ecological and economic benefits. These mangroves provide essential services, particularly serving as crucial breeding and nursery grounds for various fish species, thereby supporting local fisheries.

The project's total construction period is two years, during which some disturbance to the mangrove ecosystem may occur. However, the total direct impact area is limited to 13.02 hectares, spread across ten different land use patterns. Consequently, the directly affected lagoon area is not significant, and any ecosystem damage will be temporary and limited in extent to the two-year construction period. Therefore, the associated damage cost is considered temporary and insignificant. For these reasons, the environmental damage value related to the mangrove ecosystem has not been incorporated into the ECBA model.

7.5.5.6 Affected flood control area of marsh lands

Marshlands are vital ecosystems that provide flood control, water purification, and habitat for diverse wildlife. They act as natural buffers, absorbing excess rainwater and reducing the impact of floods. Protecting marshlands is essential for maintaining ecological balance and supporting local communities. The economic value assessment of marshland at the Dutch Bay Onshore Wind Farm Project site was conducted using the Millennium Ecosystem Assessment²⁸ (2005) as the reference study. The original study, which focused on the Muthurajawela wetland, estimated the annual local flood control benefit at US\$ 1,750 per hectare per year. Applying a Benefit Transfer

²⁷ Förster, J., 2010. *Valuation of management options for irrigation tanks in Kala Oya, Sri Lanka*. Based mainly on Vindanage et al. (2005). TEEB case: Water tank rehabilitation benefits rural development, Sri Lanka. Version 1.2, December 2010

²⁸ Millennium Ecosystem Assessment, 2005. *Ecosystems and human well-being: synthesis*. Washington, DC: Island Press for the World Resources Institute. (p.56)



method, the economic value was scaled to reflect the policy site conditions at Dutch Bay. The annual environmental value for the Dutch Bay marshland area, which falls within the 13.02-ha. direct impact extent, was determined to be US\$ 350 per ha.

7.5.5.7 Affected other cultivation

The Dutch Bay Onshore Wind Farm Project, located in Kalpitiya, is situated in an area well known for its diverse vegetable cultivation, with farmers growing crops such as chilies, potatoes, and pumpkins. Using official data from the Department of Agriculture, the assessment focused on leading crops—particularly chilies—to estimate cultivation areas. Although land use is not formally classified, the published cost of cultivation data was used to estimate the opportunity cost of land. It is assumed that the lands operate at 75% of the potential achievable under state-of-the-art technology. According to the Department of Agriculture²⁹, the net income from chili cultivation is Rs. 927,697 per season per ha. Based on the 50% productivity assumption, the annual net income—and therefore, the opportunity cost—was estimated at Rs. 881,312 (US\$ 2938) per year, which was incorporated into the ECBA. The total affected cultivation area is assumed to be less than 1 ha. though this figure has been rounded to 1 ha.

7.5.5.8 Affected sand area- opportunity cost of eco tourism - high potential

Kalpitiya is renowned for its high tourism potential, attributed to its scenic beaches, sand dunes, and rich marine biodiversity, which support activities such as dolphin watching, kite surfing, and eco-tourism. According to the Kalpitiya Development Plan³⁰ (2021–2030), Zone 9 is designated as the Eco-Tourism Development Zone, covering an area of 48.18 square kilometers (4,818 hectares). The primary objective of this zone is to protect environmentally sensitive areas within the Kalpitiya Pradeshiya Sabha jurisdiction, thereby ensuring the long-term preservation of the local environment.

The Zone 9 encompasses a variety of land uses and activities, including open spaces, landscaped areas, shrimp and crab farming zones, wind power generation sites, fishing areas, natural parks, and eco-friendly campsites. As part of this zone, it is estimated that approximately 49.89 ha. of sand area may also be included.

²⁹ Department of Agriculture, 2024. *Cost of cultivation of agricultural crops: 2023/24 Maha season*. Peradeniya: Socio Economics & Planning Centre, Department of Agriculture. (p.34).

³⁰ Urban Development Authority – Sri Lanka, 2019. *Kalpitiya Development Plan 2021–2030*. Colombo: Urban Development Authority.(p. 249)



Therefore, the opportunity cost of sand area can be estimated based on the foregone tourism revenue from uses such as camping sites, bird watching, eco-lodges, and other recreational activities. This analysis assumes that waterfront sand area can generate an annual eco-tourism income of US\$ 8,000 per ha., while sand dune areas can yield around US\$ 5,000 per ha. per year. Based on these estimates, lower eco-tourism economic value is calculated at US\$ 4750 per hectare per year, which is used in the ECBA. The total affected sand area is assumed to be less than 1 ha. though this figure has been rounded to 1 ha.

7.5.5.9 Affected scrub land - opportunity cost of eco tourism - low-performing

The majority of the scrubland within the 13.02-hectare direct impact area of the Dutch Bay Onshore Wind Farm Project holds significant eco-tourism potential. This specific area is located within Zone 9 of the Eco-Tourism Development Zone, as delineated in the Kalpitiya Development Plan (2021–2030). Zone 9 itself encompasses a diverse blend of environmentally sensitive and economically valuable land uses, including open spaces, landscaped areas, shrimp and crab farming zones, wind power generation sites, fishing areas, natural parks, and eco-friendly campsites.

The scrub lands, characterized by their natural vegetation and scenic landscapes, offer an ideal setting for low-impact tourism activities such as bird watching, nature trails, Eco-camping, and guided wildlife experiences. Recognizing the ecological and recreational value of these areas, the consultant has adopted a conservative estimate of eco-tourism value, ranging from US\$ 1,000 to US\$ 2,000 per hectare per year, based on constant economic prices and considered lower economic value. This valuation (US\$ 950) has been incorporated into the ECBA as an opportunity cost of scrub land.

7.5.5.10 Affected water holes

In the Kalpitiya Wind Farm Project area, water holes—often seasonal—are vital sources of water for local communities, wildlife, and agriculture, especially where permanent water bodies are limited. In Sri Lanka's dry zones, they play a key role in maintaining biodiversity and supporting livelihoods. Therefore, the Dutch Bay Onshore Wind Farm Project must consider the ecological importance of these water sources during planning.

To estimate the environmental value of the water holes located within the 13.02-hectare direct impact area of the Dutch Bay Onshore Wind Farm Project, the Consultant applied the economic value previously assigned to unclassified water areas. This valuation was based on their ecological significance.



For these water holes, the Consultant adopted the same valuation approach, utilizing a benchmark value of US\$ 19.66 per hectare per year. This method ensures a consistent basis for quantifying the environmental importance of these seasonal water bodies within the project's impact zone.

7.5.6 Unidentified Project Cost Components

The other project costs include several key components: Vibration impact assessments and crack surveys (US\$ 1,000); compensation for private property damages due to vibration impacts (US\$ 43,000); compensation for damages to physical and service infrastructure³¹ (US\$ 21,600); a lump sum allocation for the on-site storm water management plan (US\$ 25,000); a lump sum amount for environmental monitoring across all project stages (US\$ 32,000); and a provision for unidentified other project costs including loss of fisheries income (US\$ 200,000) are top loaded. Accordingly, the total estimated cost under this category amounts to US\$ 322,600 financial cost. It is important to note that this estimate does not include the costs related to human health and safety (workforce safety), which are accounted for separately under the Social Cost category. Please refer to Table 7 of **Annexure III M** for detailed information and consultant's estimate. The following paragraph presents the estimated fisheries income loss at full capacity level.

The 150 MW Dutch Bay Onshore Wind Farm Project is situated in Kalpitiya in Puttalam district, spanning 0.81 square kilometers. According to the Kalpitiya Development Plan 2021-2030, tourism and fisheries activities occur along a 55 km. wide sea belt. The Kalpitiya DS Division comprises 31 GN Divisions, with the project located in Dutch Bay GN Division. The project is expected to directly impact two villages of Dutch Bay GN Division: Mohonthuwaramaa and Kirimundal. Compared to the 55 km. sea belt, the project area is minimal. Of the 31 GN Divisions, only two villages in one GN division (Dutch Bay) are partially affected and estimated fisheries income loss is nearly US\$ 200,000.

7.5.7 Other Key Impacts Identified

A comprehensive assessment of the potential social and environmental impacts associated with both the construction and operational phases of the Dutch Bay Onshore Wind Farm Project has been undertaken. Key areas of impact include traffic, noise, vibration, air quality, water and soil

³¹ Based on the three turbine location maps, a total of 15 Wind Turbine Generator (WTG) units have been identified, and a 0 to 500-meter buffer polygon has been developed accordingly.



quality, solid waste management, landscape, flora and fauna, and overall health and safety of workers and nearby communities. During the construction phase, impacts such as traffic congestion, noise and vibration disturbances, dust emissions, improper sewage discharge, and risk of solid waste accumulation have been identified. These impacts, including potential disaster-related risks and effects on human health, will be mitigated through measures incorporated into the contractor's cost as part of their Environmental Management Plan (EMP).

In the operational phase, potential impacts include noise pollution, vibration effects, waste management at stations and depots, and health and safety risks. These will be addressed through the project developer's long-term operational cost planning. Furthermore, aesthetic impacts on sensitive landscapes and removal of trees will be considered during the detailed design stage and incorporated into the project cost. The cost of all mitigation measures has been allocated accordingly—either under the contractor's responsibilities during construction or the developer's responsibilities during operation—ensuring that environmental and social safeguards are fully integrated into the project implementation framework.

The detailed project cost stream is presented in Table 8 of **Annexure III M**.

7.6 Project Benefits

The primary benefit of the proposed 150 MW Dutch Bay Onshore Wind Farm Project is the incremental electricity supply to the national grid, contributing approximately 464,510 MWh of additional power annually through clean, renewable wind energy.

The project's benefits can be broadly categorized into three key areas:

- 1) Increment supply
- 2) Environmental benefits
- 3) Social Benefits

7.6.1 Incremental Supply

The project will significantly enhance the country's energy security by increasing the availability of affordable and sustainable electricity, reducing dependence on fossil fuels and imported energy sources.



The P50 Annual Energy Production (AEP) represents the most probable or expected estimate of the annual energy output (464,510 MWh/year) from Dutch Bay Onshore Wind Farm Project³². In probabilistic terms, it indicates that there is a 50% chance that the actual energy production in a given year will be equal to or greater than this value. This makes it the most likely estimate used for financial and performance modeling of wind energy projects.

The P50 AEP calculation takes into account several critical factors that influence actual energy output, including: Expected operational uptime of the wind turbines throughout the year, Wake losses, Losses occurring in cables, transformers, and other electrical components within the system and site-specific wind conditions etc. By integrating these variables, the P50 AEP offers a balanced and realistic prediction of wind energy output, supporting investment decisions and economic analysis for Dutch Bay Onshore Wind Farm Project.

To supply the generated wind energy to the national grid, the project developer negotiated a rate of Rs. 21 per kWh with the Ceylon Electricity Board. This financial price was then converted to a shadow price of Rs. 19.95 per kWh.

This shadow price has been applied to estimate the economic value of the incremental annual electricity supply generated by the project. The total incremental economic benefit to the country was calculated over a 20-year project economic life, consistent with standard economic appraisal methodologies. It is noted that the technical life of the wind farm infrastructure is longer than the assumed economic life³³, offering potential for extended benefits beyond the appraisal period.

Based on the P50 AEP scenario—representing the most likely case—the estimated net value of annual energy generation is US\$ 30,889,915. This figure reflects the projected incremental benefits from electricity generated by the Dutch Bay Onshore Wind Farm Project under typical wind conditions and has been used as a key input in the ECBA model to evaluate the project's overall economic viability.

7.6.2 Environmental Benefits

By replacing fossil fuel-based power generation, the Dutch Bay Onshore Wind Farm Project will contribute to substantial reductions in greenhouse gas (GHG) emissions and other air pollutants,

³² Ref: Feasibility Study Report (p.215). This estimate is used in the ECBA model to assess the value of the incremental power supply to the economy.

³³ Asian Development Bank (2017) *Guidelines for the Economic Analysis of Projects*. . Mandaluyong City, Philippines: Asian Development Bank. (p.14)



supporting the country's climate change mitigation goals and commitments under international environmental agreements.

In assessing the environmental implications of energy generation, it is essential to consider projected greenhouse gas (GHG) emissions from different fuel sources. According to the Ceylon Electricity Board's *"Long Term Generation Expansion Plan 2023-2042"* (CEB, 2023³⁴), the projected CO₂ emissions from fuel oil-fired power plants in 2026 are estimated at approximately 0.35 kg. per kWh. This emission factor highlights the relatively high carbon intensity of thermal power generation using fuel oil, underscoring the need to transition towards cleaner, low-emission energy sources as part of Sri Lanka's sustainable development goals and climate commitments.

Based on the annual energy output of the Dutch Bay Onshore Wind Farm Project, estimated at 464,510 MWh per year, a CO₂ emission factor of 0.35 kg per kWh—as provided in the Ceylon Electricity Board's Long Term Generation Expansion Plan 2023-2042—was applied to quantify the avoided greenhouse gas emissions. Accordingly, the project is expected to result in an annual reduction of approximately 162,579 metric tons of CO₂, contributing significantly to national efforts to reduce emissions from fossil fuel-based power generation.

The CO₂ emission cost associated with fuel-burning power generation is considered high due to the substantial volume of GHGs released per unit of electricity produced, particularly from oil- and coal-based thermal plants. According to the World Bank's 2024 publication³⁵ *"The Shadow Price of Carbon in Economic Analysis: 2024 Guidance Note"*, the estimated shadow price of CO₂ emission is US\$ 91 per mt. for the year 2026, increasing to US\$ 146 per mt. by 2047.

Given that the Dutch Bay Onshore Wind Farm Project is expected to operate from 2026³⁶ through 2047, the 2026 base year shadow price of US\$ 91 per mt. was assumed to estimate the economic value of avoided CO₂ emissions. Based on the project's projected annual CO₂ emission reduction of approximately 162,579 metric tons, the corresponding avoided environmental damage cost is estimated at US\$ 14,794,644 per year. This value has been incorporated into the project's benefit

³⁴ Ceylon Electricity Board, 2023. *Long Term Generation Expansion Plan 2023-2042*. [pdf] Colombo: Ceylon Electricity Board, pp.11-18. ISSN 2989-0012.

³⁵ World Bank (2024) 'The Shadow Price of Carbon in Economic Analysis: 2024 Guidance Note', Washington, DC: World Bank Group. (p.7)

³⁶ 2026 -2027 - Project construction phase and 2028 -2047 is project operational phase.This study used 2026 constant economic prices for the analysis.



stream, recognizing the environmental and economic advantages of displacing fossil fuel-based power generation.

7.6.3 Direct Social Benefits:

The project is expected to generate employment opportunities during both the construction and operational phases, thereby contributing to local economic development and supporting improvements in rural infrastructure. Moreover, enhanced access to a reliable electricity supply will help improve the quality of life for surrounding communities.

During the construction phase, approximately 220 direct employment opportunities will be created, while the 20-year operational phase is expected to generate 10 long-term employment opportunities. Based on an estimated annual income of US\$ 6,181.82 per employee, the total annual income benefit during the construction phase is valued at approximately US\$ 1,292,000, and during the operation phase, it is approximately US\$ 58,727 annually at 2026 constant economic price.

These employment benefits have been incorporated into the project's ECBA using shadow pricing. An annual income growth rate of 5% has been applied to reflect expected long-term real wage³⁷ increases, consistent with economic development trends and labor market projections.

7.6.4 Indirect Social Benefits

The implementation of the Dutch Bay Onshore Wind Farm Project is expected to generate substantial annual social benefits for the surrounding communities. During the construction phase, the project will significantly improve local infrastructure, including the reconstruction of 16 km of internal roads, the construction of 3.4 km of new roads, and the development of 268 meters of 5.5-meter-wide concrete pavements, thereby enhancing local connectivity and accessibility. These improvements not only facilitate the transportation of wind turbine components, but also benefit local residents by improving mobility and road safety.

The rehabilitation, reconstruction, and construction of new road sections associated with the Dutch Bay Onshore Wind Farm Project are expected to generate significant annual indirect benefits. These include reductions in vehicle operating costs (VOC), savings in local travel time, and a decrease in emissions-related costs. While these benefits are largely intangible and not

³⁷ Please note that, this is the real wage rate



easily quantified, based on the professional experience of the consultants and comparable projects, it is estimated that the project will yield a minimum of US\$ 2,500,000 in indirect annual benefits. Accordingly, this economic value has been incorporated into the ECBA model to reflect the broader socio-economic advantages brought about by the project.

Additionally, the project is ready to stimulate the local economy by creating indirect employment opportunities, encouraging local business and tourism development, and supporting community-based initiatives and research. As one of the most significant capital works projects in the area, it will contribute to the diversification of the local industrial structure and support long-term economic growth. Moreover, by contributing to the country's renewable energy capacity, the wind farm will strengthen national energy security, further underpinning its broad social and economic value. Please refer detailed project benefit stream is presented in Table 9 of **Annexure III M**.

7.7 Project Cost Benefit Analysis

This section presents the detailed economic cost-benefit analysis of the proposed project. The evaluation is based on three widely accepted decision-making criteria: Net Present Value (NPV), Internal Rate of Return (IRR), and Benefit-Cost Ratio (BCR). These indicators provide a comprehensive assessment of the project's economic viability.

For the 150 MW Dutch Bay Onshore Wind Farm Project, the analysis has been conducted using a discount rate of 14%, which is consistent with national and international funded project economic analysis guidelines and best practices. The base case economic analysis yields the following results:

- Economic Internal Rate of Return = 18.07%
- Net Present Value @ 14% discount rate = US\$ 48.95 million
- Benefit Cost Ratio = 1.21

These results indicate that the project is economically viable under the assumed discount rate. The expected economic benefits are significantly exceeding the associated costs, thereby supporting the case for the project's implementation from an economic viewpoint. The project cash flow stream is presented in Table 10 of **Annexure III M**.

7.8 Sensitivity Analysis

To further validate the robustness of the economic analysis, sensitivity tests were carried out. These tests assess the impact of variations in key parameters on the project's economic viability,



thereby reinforcing the reliability of the analysis and supporting informed decision-making. The results confirm that the 150 MW Dutch Bay Onshore Wind Farm Project remains economically viable across all tested cases (Table 7-4).

Table 7-4: Sensitivity Analysis Results

Sc.No.	Scenario	EIRR %	NPV @ 14% (US\$ mn.)	BCR
	Base Case - P50 AEP (MWh/year) - Most Likely	18.07	48.96	1.21
1	Cap Ex Increased by 10%	16.33	30.35	1.12
2	Benefit Declined by 10%	15.74	20.46	1.10
3	Cap Ex Increased +10% and Benefits Declined -10%	14.15	1.85	1.01
4	P75 AEP (MWh /year) - Conservative	16.35	27.81	1.12
5	P90 Net AEP (MWh/year) - Worst Case	14.34	3.98	1.02

When the **capital expenditure (CapEx)** was increased by 10%, the **Economic Internal Rate of Return (EIRR)** slightly decreased to 16.33%, while the **Net Present Value (NPV)** declined to **US\$ 30.35 million** with a **Benefit-Cost Ratio (BCR)** of **1.12**. In the scenario where **benefits were reduced by 10%**, the EIRR fell to **15.74%**, with an NPV of **US\$ 20.46 million** and a BCR of **1.10**. The combined case of a **10% increase in CapEx and 10% decrease in benefits** still showed economic feasibility, with an EIRR of **14.15%**, NPV of **US\$ 1.85 million**, and BCR of **1.01**.

Further, the analysis incorporated conservative energy yield scenarios. Under the **P75 net Annual Energy Production (AEP)** scenario, representing a conservative output estimate, the EIRR stood at **16.35%**, NPV at **US\$ 27.81 million**, and BCR at **1.12**. Even in the **P90 net AEP scenario**, representing the worst-case energy output assumption, the project yielded an EIRR of **14.34%**, NPV of **US\$ 3.98 million**, and BCR of **1.02**.

These results underscore the project's resilience and economic soundness, even under adverse conditions, further reinforcing its justification for implementation.

7.9 Conclusion

The economic analysis of the **150 MW Dutch Bay Onshore Wind Farm Project** demonstrates strong viability under the base case scenario, with favorable outcomes across all key economic indicators—**Net Present Value (NPV)**, **Internal Rate of Return (IRR)**, and **Benefit-Cost Ratio (BCR)**—when evaluated using a **14% discount rate**.



The project's robustness was further confirmed through a comprehensive sensitivity analysis, which tested the impacts of increased capital costs, reduced benefits, and varying levels of energy production. In all tested scenarios, the project remained economically feasible, with EIRRs well above the discount rate and positive ENPVs and BCRs above 1.0.

These findings clearly indicate that the Dutch Bay Onshore Wind Farm Project is a sound investment from an economic standpoint, capable of delivering substantial benefits even under conservative or adverse conditions. Therefore, the project is recommended for implementation, as it promises to contribute significantly to the country's renewable energy goals while remaining economically sustainable.



8 GRIEVANCE REDRESS MECHANISM

This section describes the grievance redress framework (informal and formal channels) setting out the time frame and mechanism to resolve complaints about environmental performance.

Establishing and effectively communicating an adequate, project-specific Grievance Redress Mechanism (GRM) is a key requirement of the national Environmental Impact Assessment (EIA) process. Accordingly, the project proponent will implement a time-bound and transparent GRM to enable Project Affected Parties (PAPs) to voice their concerns. The mechanism will be designed to receive, facilitate, assess, and resolve grievances related to the environmental and social safeguards of the project during both the implementation and operational phases.

8.1 Objectives of the Grievance Redress Mechanism (GRM):

The primary objective of establishing a project-specific Grievance Redress Mechanism (GRM) is to create a transparent, accessible, and responsive platform through which project-affected persons (PAPs) and other stakeholders can raise concerns or grievances related to the implementation of project activities. These activities may include land clearing, land filling, transportation of materials (by sea and land), installation of wind towers, and the associated influx of labor to the project site. The specific objectives of the GRM are as follows:

- To provide a formal avenue for affected parties and stakeholders to express their comments, concerns, or grievances related to project activities in a timely and constructive manner.
- To ensure that affected persons have access to a simple, efficient, and cost-effective mechanism for lodging grievances, regardless of their socioeconomic status or literacy level.
- To promote fair, transparent, and impartial resolution of grievances in line with national regulatory requirements and relevant international standards or project reference frameworks.
- To enhance stakeholder engagement and build trust by demonstrating accountability and responsiveness to community concerns throughout the project lifecycle.
- To ensure that all stakeholders have access to accurate and timely information about the GRM process, their rights, and the steps involved in the resolution of grievances.



- To reduce the risk of conflicts and delays by addressing grievances proactively, thereby supporting smooth project implementation and fostering positive community relations

Likewise, the project aims to establish an effective and responsive approach to addressing the concerns of affected individuals and communities. In doing so, it seeks to foster trust and build confidence among community members regarding the project's intentions and operations. This proactive engagement not only enhances transparency and accountability but also contributes to the development of constructive and collaborative relationships between the project team and all relevant stakeholders. Ultimately, these efforts help create a more inclusive, participatory, and socially sustainable project environment.

The interventions associated with the project, such as land clearance, grading, filling, transportation of materials (by both sea and land), installation of wind towers, and the influx of labor, have the potential to give rise to a range of social, economic, environmental, and cultural challenges as well as user conflicts. These impacts can be complex and may significantly alter the livelihoods, living conditions, and cultural practices of people directly or indirectly affected by the project.

Recognizing the likelihood of such issues emerging, particularly across different phases of the project planning, construction, and operation, a formally structured and time-bound Grievance Redress Mechanism (GRM) will be established. This mechanism will serve as a critical tool for identifying, receiving, addressing, and resolving grievances in a timely and equitable manner. Special emphasis will be placed on the concerns and complaints of project-affected persons, ensuring that their voices are heard and their issues are addressed with fairness and transparency.

The proposed GRM will be applicable throughout the entire lifecycle of the project, serving as a consistent platform for stakeholder engagement and conflict resolution and contributing to the project's social acceptability and long-term sustainability.

8.2 Concerns Raised During Primary Stakeholder Consultation

The following concerns emerged during the primary stakeholder consultation conducted by the consultants through Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), and a Social Survey:

- Possibility of blade accidents, fall hazards, and property damage.



- Impacts on existing fishing activities, especially for beach seine operation due to material transportation by sea.
- Negative impacts on kite surfing activities due to the possibility of emerging gusty effect (which can cause instability, sudden power surges, or loss of control while kite surfing).

In addition to the broader community concerns, the following prevailing issues were identified through the social survey and Focus Group Discussions (FGDs) conducted in the coastal villages of Mohottuwarama and Kirimundal located within the Dutch Bay Grama Niladhari Division (GND), the Area of Influence (AOI) of the proposed project. These issues highlight the need to mitigate adverse impacts on social receptors while enhancing community benefits:

- **Inadequate Transport Infrastructure:** Poorly maintained access roads and an unreliable transport system, particularly the dependence on boats traveling from Thoradiya to Mohututuwarem pose significant challenges to both the fishing communities and students. These constraints limit access to education, markets, and essential services, thereby impeding social and economic development.
- **Severe Coastal Erosion:** The ongoing and intensifying coastal erosion along the shorelines of Mohottuwarama, Kirimundal, and Uchchamunai villages has severely impacted fishing operations and related livelihoods. The loss of land and damage to marine infrastructure have heightened the vulnerability of these communities.
- **Limited Access to Safe Drinking Water:** The lack of access to safe and affordable drinking water remains a pressing concern. Many households are compelled to purchase drinking water from external sources, increasing household expenditures and financial strain.
- **High Unemployment Rate:** Unemployment remains a critical socio-economic issue in the project area. As of 2021, approximately 33.2% of the labor force within the Dutch Bay GND is unemployed and actively seeking suitable employment opportunities.

8.3 Key Grievances to be considered for Redressal from the Inception Stages of the Project

The Community Consultations, Focus Group Discussions (FGDs), Key Informant Interviews (KIIs), and the sample social survey conducted in the villages of Mohottuwarama and Kirimundal revealed that slightly over 82% of the households within the project Area of Influence (AOI)



perceived the proposed project as generally positive. However, the consultations also brought to light several concerns and grievances that require special attention. These include, but are not limited to, the following areas:

- **Potential Impacts on Beach Seine Operations:** The transportation of construction materials via sea routes may pose potential impacts on six registered and three unregistered beach seine operations located in the nearshore areas of Mohottuwarama and Kirimundal, within the Dutch Bay Fisheries Inspector Division (FID).
- **Impacts on Kite Surfing Activities During Wind Turbine Operation:** The operation of wind turbines may affect kite surfing activities in areas commonly used for water sports.
- **Siting and Operation of Temporary Labour Camps During Construction:** The establishment and operation of temporary labour camps during the construction period could result in environmental and social impacts.
- **Safety and Security of School Children and Devotees Using Interior Roads in Dutch Bay GND:** The interior roads within the Dutch Bay Grama Niladhari Division (GND) are frequently used by school children and religious devotees of the following social receptors;
 - Avey Maria Catholic church- Kirimundal
 - Catholic church Uchchamunai
 - Mari Amman Hindu Kovil – Mohottuwarama
 - Masjid Rahuman Mosque – Mohottuwarama
 - Mohottuwarama Muslim Vidyalaya
 - Penapidugama Sinhala Vidyalaya
 - Uchchamunai Tamil/Sinhala Vidyalaya

8.4 Grievance Redress Process

In alignment with the regional and local administrative structure, as well as the socio-economic context of the project site and its surrounding areas, the Grievance Redress Mechanism (GRM) is structured to operate effectively across three tiers. Each tier is designed to address grievances based on their nature, complexity, and significance, ensuring that affected parties have access to timely and fair resolution channels.



Tier 1: Site-Level Redress

At the first level, grievances are handled directly at the project site. A Grievance Redress Committee (GRC) composed of relevant members — including local project representatives, community leaders, and other stakeholders — is established to receive and resolve complaints. This tier is intended to address routine or minor issues that can be resolved quickly through local engagement and direct communication with the complainant.

Tier 2: Regional-Level Redress

Grievances that are more complex or beyond the resolution capacity of the site-level committee are escalated to the regional level. At this tier, a regional GRC comprising senior project personnel and regional administrative representatives assesses and addresses complaints that require broader coordination or policy-level interpretation but are still manageable without national involvement.

Tier 3: National-Level Redress

Highly significant, complex, or unresolved grievances are referred to the national-level GRC, involving senior officials such as the Secretary or Additional Secretary of the project-executing agency. This tier is designed to handle cases requiring institutional-level decisions, legal interpretations, or substantial policy interventions to ensure equitable outcomes for all parties involved.

To ensure efficiency and consistency across all levels, Grievance Redress Committees (GRCs) will be formally established at each tier. Each committee will be responsible for reviewing, investigating, and resolving complaints in accordance with clearly defined procedures, timelines, and escalation protocols. The goal is to promote transparency, accountability, and trust throughout the grievance redress process.

8.5 Composition of the Grievance Redress Committee

8.5.1 Tier One – Site level

- Grama Niladhari (GN) of the Dutch Bay GND (Chairman)
- Representative of Envision Pvt. Ltd.
- Representative of the Supervision consultant
- Representative of the contractor



- Representative of Ekamuthu fisheries society Mohottuwarama Village
- Representative of the Fisheries society – Kirimundal
- Representative from existing beach seine operators
- Women representatives from the local community

8.5.2 Tier Two – Regional Level

- Divisional Secretary, Kalpitiya DSD (Chairman)
- Representative of Supervision consultant
- Representative of contractor
- Representative of Envision Pvt. Ltd
- Representative of local CBO/Religious Leader
- Women representatives from the local community

8.5.3 Tier Three - National Level

- Secretary, Ministry of Power and energy (Chairman)
- Director – Envision Pvt. Ltd.
- Representative of Project Implementing Consultant
- Representative of a Community-Based Organization
- Women representatives from the local community

The Grievances/complaints submitted related to the project interventions shall be treated equally irrespective of gender or social status of the person. To ensure gender responsive GRM process, the GRC established at all three levels will be represented by a female member to signify the local community.

In considering the multi-ethnic social profile of the project influence area, all communication matters pertaining to grievances will be conducted in all three languages to ensure ethnic and cultural harmony.



8.6 Complaint Reception Channels

Affected individuals may submit their complaints or grievances in writing to the Project Management Unit (PMU), which acts as the primary point of contact for receiving, acknowledging, and processing such submissions. Complaints will be referred to the Project-Level Grievance Redress Committee (PGRC) and escalated to higher levels if necessary.

Complaints can be submitted through any of the following channels:

1. **By post** (ordinary, registered, or courier) addressed to the Project Grievance Redress Manager.
2. **Online** via the project grievance portal [insert URL once created].
3. **Email** to the Project Grievance Redress Manager at [insert email ID once created].
4. **Hand delivery** to the PMU office.
5. **Complaint boxes** are located at the Kalpitiya Divisional Secretariat Division (DSD) and the PMU site.

The PMU will acknowledge receipt of complaints within **one week** of submission. Acknowledgement will be sent via registered post or another appropriate recorded delivery method, along with a **Registration ID** that the complainant may use in all further communications with the PMU. The GRM procedure and timeline is illustrated in the following table. Where possible, acknowledgement will also be provided via SMS and/or email if such contact information is available.

If the complainant wishes to remain anonymous, they may submit under the "Tagged-Anonymous" category. In such cases, all personal information—including name and contact details—will be kept strictly confidential. Further investigation into the complaint will proceed on a special-case basis without requiring the complainant's physical presence



8.7 Grievance Redress Procedure and Time frame

The time frame to be adopted on grievance redress procedures is depicted in **Table 8-1**.

Table 8-1: The Time Frame to be adopted on Grievance Redress Procedure

Process	Description	Time frame
	Submission of grievance either orally or one to one meeting, letter, e-mail, SMS, recorded at stakeholder meetings and other interaction	1 day
	Level of significance assessed and recorded	4-7 days
	Acknowledgement of grievance through appropriate means	7-14 days
	Grievance assigned to appropriate/responsible party to resolve	4-7 days
	Response formulated with input from management and other stakeholders	7-14 days
	Redress action approved at appropriate level	4-7 days
	Redress action implemented and update of progress on resolution and communicate to complainants	10-14 days
	Redress action recorded in grievance logbook confirming that grievance can be closed or what follow-up action is required	4-7 days
	Record final signed off on grievance. If grievance cannot be resolved direct to appropriate parties.	4-7 days

8.8 GRM Database

GRM database will be developed to collect, manage and monitor data and information grievances as depicted in **Table 8-2**. Updating the database will be responsibility of Solar Forge Pvt. Ltd stakeholder engagement team. The team responsible for;

- Documenting and maintain records of all the grievances received from stakeholders through any medium;
- Responding to all the grievances received in a timely and transparent manner; and,
- Ensuring the dissemination of “grievance mechanism point of contacts” appropriately, among affected parties and stakeholders.



Table 8-2: Grievances Database Template

Nature of the grievance	Complainant ID	Date grievance received	Date grievance resolved	Final outcome



9 CONCLUSION AND RECOMMENDATIONS

This Final EIA report is prepared covering core environmental, ecological and socio – economic aspects strictly adhering to the provided detailed final ToR by CC&CRMD while fulfilling the regulatory compliances for proposed Envision Dutch Bay 150 MW + 23.8 Bess Renewable Energy Wind Power Project in Kalpitiya.

The report elaborates the details on:

- The compliance of EIA study with the given ToR by CC&CRMD
- Information on detailed methodologies followed to complete the comprehensive study including the results of the conducted field investigations, field measurements etc.
- Detailed Descriptions on the proposed project
- Potential environmental impacts associated with the project development during construction and operation of the project, required mitigation measures to mitigate the significant impacts, monitoring, and management of the environmental impacts to ensure compliance with environmental legislation and standards.
- Grievance Redress Mechanism
- Extended Cost Benefit analysis

The proposed 150 MW wind power generation farm will be connected to the national grid from the wind power collector grid substation at Kalpitiya to the CEB Wariyapola switching station through a 220 kV double circuit overhead transmission line, with an approximate route length of 105 km. It should be noted that the EIA component related to this external transmission line will be carried out separately and hence, not included in this report.

The impact identification and assessment exercise were carried out for the project during both construction and operational phases. The impact significance was assessed based on the intensity of the impacts and the likelihood of occurrence for ecology including avifaunal movements, biodiversity, socio-economic impacts, noise, water quality, air quality, and health and safety, wastewater and solid waste management, mitigation measures have been formulated to reduce or limit the impacts identified.

The proposed construction activities of wind farm will not cause significant environment impacts and most of the potential environment impacts are temporary in nature mainly restricted to the construction period, which can be mitigated through application of best engineering and



environmental practices and working in accordance to the proposed Environmental Management Plan (EMP) etc.

During operation, the project impacts are mostly positive as it provides renewable energy and does not generate any substantial issues. However, the potential impact on the migratory birds is a prime concern of the project, and therefore an avifauna detection and control mechanism has been proposed for implementation. Further, a radar-based automated early warning system will be installed to mitigate bird collisions across the entire wind farm area.

Further, EIA team recommended to remove or relocate the Turbine No 1, as the tower location is identified as a significant location for bird migration. For relocation, an alternative location closer to the area between Turbines 13 and 15 (T13–T15) is recommended, where bird activity appears to be less critical.

All impact mitigation and control measures proposed herein are reliable and effective and should be taken during the construction and operation period thereby the proposed project will just have an insignificant negative impact on the surrounding environment. The project promoter is committed for developing the project while considering the EIA study. However, the proponent will require this EIA approval to go ahead with the development facility taking a significant step towards a sustainable energy future for Sri Lanka, aiming to harness the abundant local wind energy resources to generate 150 MW of clean electricity daily for the national grid. It is therefore recommended that the proposed project is approved for an EIA licence under the EPA 2002.



10 ANNEXURES

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